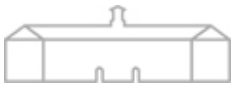
Fraunhofer-Institut für Optronik, Systemtechnik und Bildauswertung IOSB



TRM4.v3

Parameter Description and
Model Documentation

Fraunhofer IOSB Technical Report 2021/04

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1 Introduction

1.1 The TRM4 modelling approach

TRM4 is an analytical model for the assessment of the device and range performance of electro-optical systems operating in wavelength bands from the visible to the thermal IR. The model is based on the perception of a standard 4-bar test pattern, and uses modified Johnson criteria to calculate ranges.

TRM4 uses the AMOP figure of merit (Average Modulation at Optimum Phase) for assessing the spatial transfer characteristics of the imager [1]. AMOP replaces the system MTF (Modulation Transfer Function) used in previous performance models. It was originally developed and validated [2] to characterize undersampled (thermal) imagers for which no system MTF is defined. The AMOP concept can be applied to all kinds of imagers, whether undersampled or not.

The overall figure of merit for characterizing imagers that operate in the thermal IR is the MTDP (Minimum Temperature Difference Perceived). Similarly, for other imager types the MDSP figure of merit (Minimum Difference Signal Perceived) is introduced, where the difference signal is expressed in number of signal carriers.

TRM4 is an extension of the previous TRM3 model [3]. TRM3 was developed for thermal imagers only, whereas TRM4 allows to model various kinds of electro-optical systems operating in the visible and IR spectrum. In TRM4, imagers are classified in three different categories:

- *Thermal* (blackbody emissivity only),
- *Visible/NIR/SWIR* (reflectivity only), and
- *General* (reflectivity and emissivity).

The modelling of the imager category *Thermal* is fully compatible with TRM3 and adds features like

- Optional 3D modelling of targets,
- Horizontal and slant path target observation,
- Specification of spectral MTF data for optics and detectors,
- Assessment of uncooled detectors.

The modelling of the imager categories *Visible* and *General* provides for the specific features of these systems. This includes:

- For imagers of category *Visible* radiation reflected by target and background is considered, rather than emitted radiation as for thermal imagers.
- For imagers of category *General* both emitted and reflected radiation are considered to contribute to the system performance.
- Spectral input to characterize target and background reflectivities / emissivities.
- Two different irradiation sources (illuminators) can be specified either as blackbodies or via their spectral irradiance. One of them may be specified to be on the same platform as the imager.
- The influence of atmospheric turbulence and path radiance can be modelled as well.
- Spectral atmospheric transmission data may be specified.

1.2 The TRM4 software

The TRM4 software is intended for the calculation of imager performance. Generally, there are two types of performance data:

- data that can be measured in the laboratory or in the field by using specific test patterns, and
- range data.

To characterize imager performance, TRM4 calculates and provides a large number of performance parameters such as:

- General system performance parameters,
- MTFs/AMOP and related parameters,
- Overall performance measures like MTDP for thermal imagers, MCP (Minimum Contrast Perceived) for imagers of category *Visible/NIR/SWIR* and MDSP,
- Signal, noise and signal-to-noise ratio (SNR) data as function of target-imager distance,
- Task related performance measures like MNTD/ETD (Minimum Necessary Temperature Difference/Effective Temperature Difference) for thermal imagers and MNDS/EDS (Minimum Necessary Difference Signal/Effective Difference Signal) for other imager types,
- Target detection, recognition, and identification ranges.

Moreover, the TRM4 software provides means to assess imager performance in aerial imaging applications by calculating NIIRS values (National Imagery Interpretability Rating Scale).

The software provides tools to specify imager and task where the task consists of perceiving a target (e.g. a vehicle) in a given scenario (target aspect, line-of-sight angles, altitude etc.) and environment (e.g. atmosphere, irradiation). The parameters required to specify the system are described in Chapter 2. The modelling assumptions and the formulas used to calculate imager performance are described in Chapter 3.

The TRM4 software package runs on PC platforms under all current Microsoft Windows operating systems. The package contains templates for different imager types and fixed built-in data sets for characterizing radiation sources, target and background features, etc. This data can be used for reference purposes. The handling of the software package using its graphical user interface is described in [4].

1.3 Validity – TRM and *nominal* range performance data

TRM4 is supplied with several “standard” sets of data specifying targets, backgrounds, tasks, atmospheres, and scenarios for the *Thermal* category. Using these standard sets nominal range performance data is calculated. This issue is addressed in NATO STANAG 4347 [5]:

> “.. Validity
>
> There are many variables which affect the performance of thermal imaging systems. In order that different systems may easily be compared, a standard set of conditions need to be defined as basis on which performance calculations can be made. “Nominal range performance” is the term applied to the range calculated under these standard conditions. The standard conditions have been chosen to be reasonably representative of conditions experienced in the field.
>
> The task considered is that of static performance, no search …”

For better assessing the range performance against actual, real targets in a given scenario the standard assumptions (target dimensions, temperature difference target/background, atmosphere, etc.) need to be replaced by actual data.

2 System Characterization

2.1 Introduction

This chapter describes the input parameters that are required to specify imager and performance task[1].

As already mentioned, TRM4 distinguishes between three different imager categories:

- *Thermal* imagers (blackbody emissivity only),
- Imagers operating in the *Visible/NIR/SWIR* (reflectivity only), and
- *General* imagers (reflectivity and emissivity).

Each imager, irrespective of its specific type, can be assigned to one of these three categories. Imager types differ either with respect to the image formation technique used (staring/micro-scanned, scanning), with respect to the sensing technology applied (Thermal category detector types: cooled, uncooled, integrating, non-integrating, sampled) or otherwise.

The *Modify Individual Parameters* dialog of the software (see Section 2.4 and the *TRM4 User Manual* [4]) gives access to various dialogs that allow specification of components or features of the imager and of the task. The parameters available in those dialogs are described in this chapter.

Dialogs for the specification of the imager are:

- Basic Imager Data,
- Optics,
- Image Intensifier (II + Coupling)
- Detector (Geometry + Sensitivity),
- Analog Signal Processing,
- Digital Signal Processing,
- AD Converter MTF,
- Video Signal Processing,
- Display and Observation,
- Stabilization,
- Additional Imager MTFs,
- Prefilter MTF (or MTDP/MCP).

Dialogs for the specification of the task and environment are:

- Target, Scenario and Task,
- Irradiance,
- Atmosphere,
- Turbulence.

[1] The units of the parameters and conversion factors between them are summarized in Appendix A.

It should be noted that the lay-out and the parameters in some dialogs depend on imager category and type. Also, not all dialogs are accessible for all imager types and settings.

Specified input data must be consistent. The TRM4 software has a specific *Diagnostics* tool that analyzes the consistency of data and calculates basic imager features. If data is dubious or inconsistent, warnings and/or fatal messages are displayed. This feature is described in detail in [4].

2.2 Imager category

The three imager categories in TRM4 are distinguished by the type of radiation that is used to sense the scene. Imagers of category *Thermal* sense the radiation emitted by the target, the background, and the air path between target and imager due to their absolute temperatures; target and background are assumed to be blackbodies and the air temperature is assumed to be equal to the background temperature.

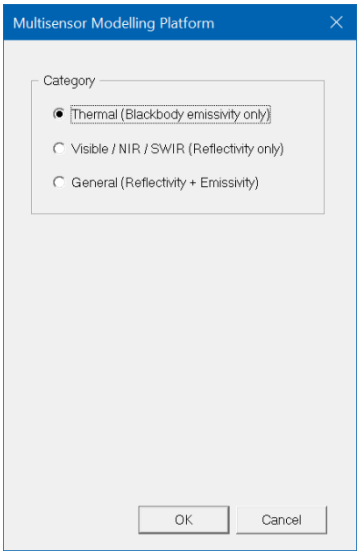


Figure 2.1: Imager category selection dialog.

Imagers of category *Visible/NIR/SWIR* sense only radiation that is reflected by the target, the background and the air path between target and background. Scene irradiation originates from either natural or artificial radiation sources. Path radiance and air turbulence may have a significant impact on range performance. TRM4 provides a number of preset data sets for the characterization of irradiation sources and target/background spectral reflectivity properties (see also Appendix C).

Imagers of category *General* sense both reflected radiation and radiation emitted by the target, the background, and the air path. Targets and background need not be blackbodies and the air temperature needs not be equal to the background temperature. Imager performance data are described in difference signal units (number of carriers) by default.

2.3 Imager type

The major components of an imager are shown in Figure 2.2. While the stabilization platform and irradiation source are not part of the imager itself, they are considered to be part of the overall imaging system and are therefore modelled by TRM4.

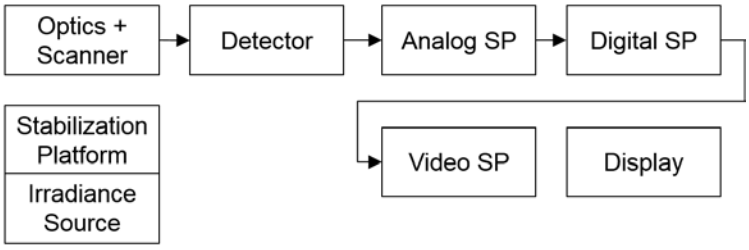


Figure 2.2: Major components of an electro-optical system.

2.3.1 Imager category *Thermal*

In the category *Thermal* five different imager types can be specified as shown in Figure 2.3.

Thermal imagers with continuous analog (non-integrating) detector signals are called *1st generation imagers*. Thermal imagers with a focal plane integrating detector array (FPA) are called *2nd generation imagers*. In these, detector signals are integrated, sampled and multiplexed on the chip.

A scanning imager is an imager with opto-mechanical scan. Scanning is always assumed to be in the horizontal direction. In the following paragraphs the characteristic features of these five basic imager types are summarized.

1st generation (scanning imager) without horizontal sampling

This is a classical thermal imager with analog detector signal (usually these detectors are of the photoconductive type) and purely analog signal processing. The imager may have either serial or parallel scan. A typical representative is the US Common Module. No signal storage or reformatting is needed in order to display the image. Signal processing, no matter how complex it may be in reality, can simply be modelled by an analog signal processing component. It is assumed that the analog signal processing acts on each detector signal individually. No electronic sampling is applied and the continuous analog signal is fed directly to the display.

1st generation scanning imager with horizontal sampling

1st generation imagers with signal reformatting component(s) belong to this imager type. Signal reformatting can be done either in a digital or analog manner. Typical representatives are devices which have serial/parallel scan or which contain a digital scan converter.

It is assumed that the analog signal processing acts on each detector signal individually whereas video processing acts on the serialized signal. Sampling occurs in both directions. The term “digital” includes cases with sampled analog signals.

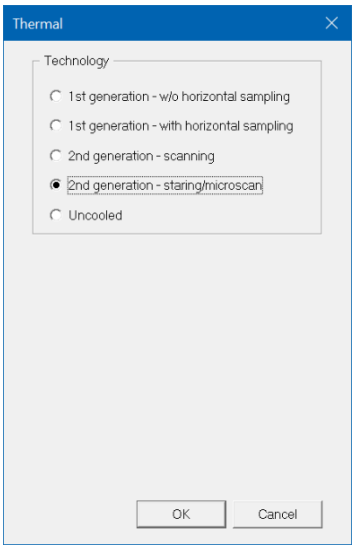


Figure 2.3: Technology selection dialog for thermal imagers.

2nd generation scanning imager

2nd generation scanning imagers are those with a linear focal plane array (FPA) (usually of the photovoltaic type). In most cases the signals of several adjacent horizontal detectors are integrated (TDI) in order to increase the thermal sensitivity. Detector signals are sampled in both directions and multiplexed on the chip. No analog signal processing is applied before the signals are reformatted. Digital signal processing is used to process the signals and to reformat them in order to obtain an analog video signal. “Digital” here includes cases with sampled analog signals.

2nd generation staring or micro-scanned imager

This type of imager contains a two-dimensional focal plane array which permits operation in staring and/or micro-scanned mode. Micro-scan may be used to increase the sampling frequency of the imager and thus remove undersampling artifacts.

Uncooled imagers

The *Uncooled* imager type corresponds to thermal detectors that can be operated at (or near) ambient temperature. The most important representatives are microbolometers, but pyroelectric detectors belong to this type as well.

These detectors are assumed to be two-dimensional focal plane arrays making them a sub group of 2nd generation staring/micro-scanned imagers.

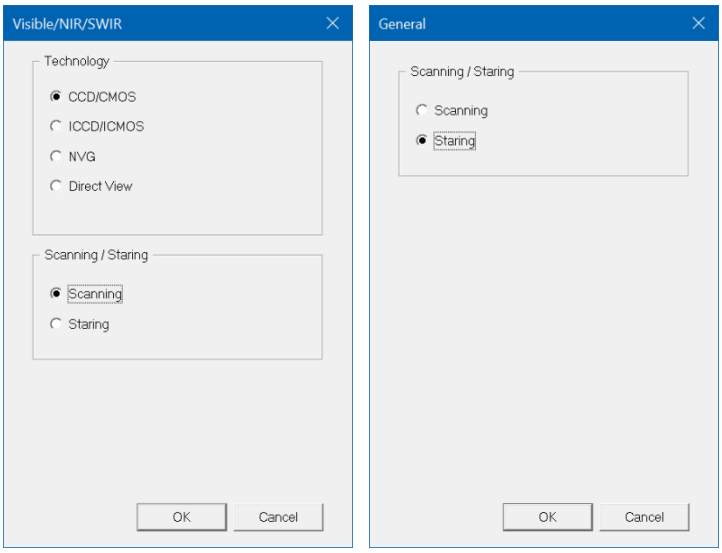


Figure 2.4: Technology selection for imagers of categories *Visible* (left) and *General* (right).

2.3.2 Imager category *Visible/NIR/SWIR*

CCD/CMOS

This imager type refers to scanning and starring systems with a CCD or CMOS image sensor. Scanning includes serial, serial-parallel and parallel scanning with or without mechanical interlace. Staring includes micro-scanned (dithered) modes.

ICCD/ICMOS

An ICCD/ICMOS consists of an image intensifier coupled by fiber optics or relay optics to a CCD/CMOS sensor. Only starring operation without micro-scan is consider for this type of imager.

NVG

The NVG type refers to standard night vision goggles using an image intensifier tube to magnify the incoming radiation. Range performance calculation for the NVG type is based on laboratory measured MCP (Minimum Contrast Perceived) or MRC (Minimum Resolvable Contrast) data.

Direct View

Direct View refers to purely optical systems without any conversion of incoming photons into electrons (e.g. telescopic sights). Range performance calculation for this type is based on laboratory measured MCP (Minimum Contrast Perceived) or MRC (Minimum Resolvable Contrast) data.

2.3.3 Imager category *General*

For the *General* category, TRM4 distinguishes between scanning and staring systems only. Scanning includes serial, serial-parallel and parallel scanning with or without mechanical interlace. Staring includes micro-scanned (dithered) modes.

2.4 Modify individual parameters

This dialog (Figure 2.5) displays basic information about the project in the *Info* box and provides access to all parameter group dialogs used for specifying the imager, the observation task, and environmental conditions. In addition, the options *TRM/NIIRS*, *Field/Lab*, and *Video/Still* can be specified here.

Modelling

If *TRM* is selected, the range performance as well as many other device performance parameters are calculated.

If *NIIRS* is selected, NIIRS (National Imagery Interpretability Rating Scale) values are calculated for the specified aerial imaging scenario. Only those parameter dialogs and options in the *Target, Scenario and Task* dialog are active that are required for NIIRS calculation according to the General Imager Quality Equation version 5, see Section 3.13.

Field/Lab

Some performance parameters like MDSP depend on environmental conditions such as irradiance sources and atmospheric path. They may, therefore, be different whether measured in the lab or in the field. If measured in the field, they may also depend on the distance between target and imager, due to varying path radiance, turbulence, atmospheric transmission, etc. TRM4 takes this into account by distinguishing between lab and field conditions.

If *Laboratory performance* is selected, TRM4 does not calculate range data but figures of merit (FOMs) that can be measured in the laboratory. Examples are MTFs, MTDP (category *Thermal*), MCP (category *Visible/NIR/SWIR*), and MDSP (category *General*). These calculations are done for zero *Reference range* and by assuming radiation/irradiation sources as specified, but no atmosphere.

If *Field performance* is chosen, TRM4 calculates all ranges, FOMs, and data for the task defined in the *Target, Scenario and Task* dialog. FOMs such as the MTDP/MDSP, MTFs, and system performance data are calculated and displayed for the specified *Reference range*. In case no *Reference range* is provided by the user a zero reference range is used to evaluate FOMs. Notice that the specified reference range has no effect on the actual range and NIIRS calculation.

> **Note:** In the *Thermal* category, MTDP and MTFs are independent of range. Hence, the *Reference range* check box is disabled.

Video/Still

TRM4 assesses the imager performance for either an image sequence (option *Video*) or a single frame (option *Still*). In the latter case the calculation does not take into account the eye integration time.

NIIRS values are evaluated for single images. Thus, the option *Still* is automatically used in case *NIIRS* is specified.

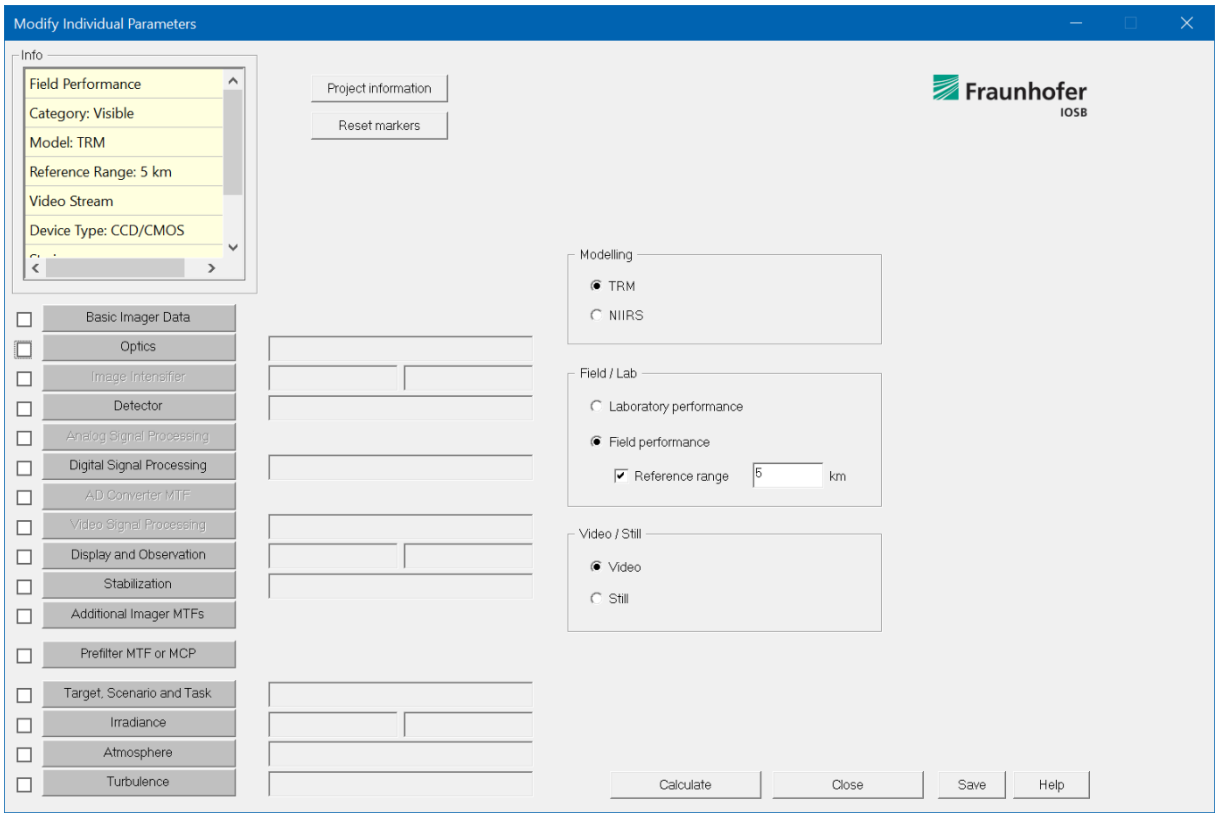


Figure 2.5: Modify Individual Parameter dialog.

2.5 Basic imager data

There are two slightly different *Basic Imager Data* dialogs in TRM4, one for scanning and one for staring imagers. They are depicted in Figure 2.6.

2.5.1 Field of view (FOV)

Calculate FOV

With the exception of serial scanned imagers, for which the field of view (FOV) needs to be specified, TRM4 can calculate the FOV if desired. For scanning imagers, the FOV aspect ratio needs to be specified, for staring/micro-scanned devices the FOV is calculated by using the specified optics and detector data.

Measured FOV

Here you can specify the FOV of the imager. It should be noted that even if the *Measured FOV* is provided by the user, TRM4 calculates the FOV whenever possible, by using other specified imager data[2]. The equations to calculate the FOV are given in Section 3.2.3.

[2] FOV specifications are often not exact. If, for instance, the exact FOV is 5.021° the manufacturer specifies a value of 5°. TRM4 accepts deviations between the calculated and specified (measured) data as long as the deviation is less than 10 % (for deviations greater than 2 % a warning is issued). Performance calculations, however, are made with the exact (calculated) FOV.

FOV Units

The FOV can be specified in either [mrad] or [°]. If you change the unit, all displayed FOV values are re-calculated automatically.

FOV aspect ratio

The FOV aspect ratio is defined as:

$$\alpha_{\mathrm{FOV}} = \frac{\mathrm{FOV_h}}{\mathrm{FOV_v}}. \tag{2.1}$$

2.5.2 Frame rate

The frame rate is the rate at which one full frame of the image is captured in the focal plane. This (imager) frame rate may be different from the (video) frame rate at which the image is displayed. TRM4 checks whether the two frame rates are compatible with each other, see Section 3.2.13.

2.5.3 Sampling

Number of samples per line/ Number of horizontal samples

The number of samples per line (scanning imagers) or the number of horizontal samples (staring imagers) needs to be specified for imagers that are sampled in horizontal direction.

It is the number of independent horizontal data samples taken by the imager over the specified horizontal FOV in order to obtain a horizontal line of the image. For staring/micro-scanned imagers it depends on the number of horizontal detectors used to form the image, and on the specific micro-scan factor.

The number of samples per line/number of horizontal samples should not be confused with the number of horizontal pixels displayed in the output image, which may be higher, e.g. due to electronic zoom, see Section 3.2.9.

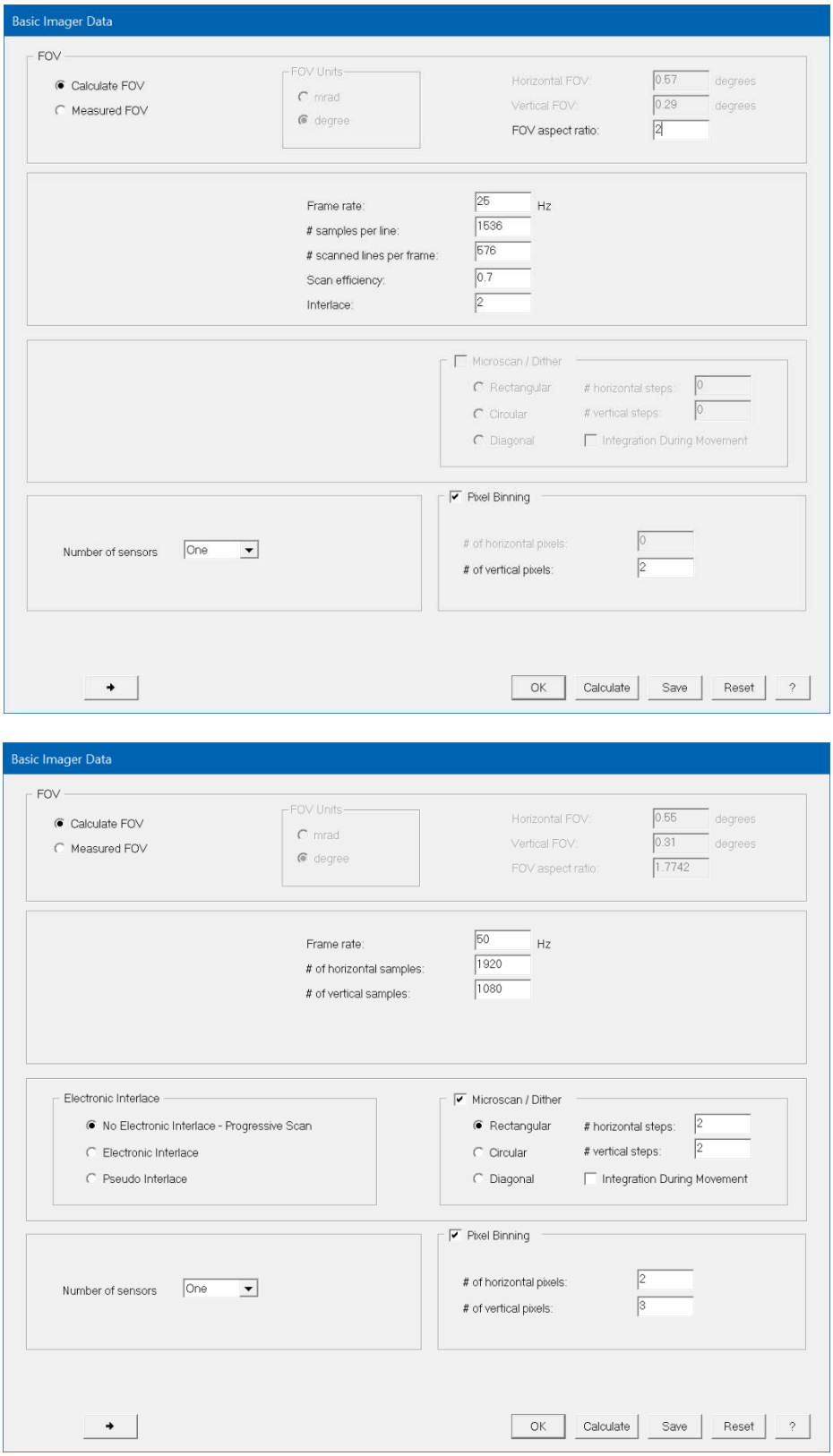


Figure 2.6: Basic imager properties definition for scanning (top) and staring (bottom) systems.

Number of scanned lines per frame / Number of vertical samples

This is the number of independent vertical samples (a scanned line in scanning imagers is equivalent to a vertical sample) taken by the imager over the specified vertical FOV in order to obtain one frame of the image.

It should not be confused with the number of lines displayed in the output image, which may be different, e.g. due to electronic zoom.

2.5.4 Scanning

Scan efficiency

This parameter needs to be specified for scanning imagers only. It is the time per frame during which the detector is “looking” into the scene, multiplied by the imager frame rate.

Interlace

This parameter describes the opto-mechanical interlace applied in order to increase the number of scanned lines. The parameter has to be specified for serial/parallel or parallel scanned imagers only, and is used for the calculation of the FOV.

2.5.5 Electronic interlace (staring imagers only)

TRM4 provides three different schemes of how to generate one frame of the image by manipulating the existing detectors signals: *Progressive Scan*, *Electronic Interlace* or *Pseudo Interlace.* Electronic interlace is usually only applied in the visible.

No Electronic Interlace - Progressive Scan

In this mode no interlace is used. No interlace means that all detector signals are used and displayed in every field, i.e. each field is a frame.

Electronic Interlace

Electronic interlace indicates that standard (TV) interlace is used, with two fields per frame. Alternating rows of detectors are displayed for each field (horizontal rows 1, 3, 5 … for the first field and rows 2, 4, 6 … for the second field).

Standard interlace reduces the sensitivity of the imager, because light is only collected and displayed by half of the detectors at any given time.

Pseudo Interlace

This technique uses two fields to form a frame. The first line in field one consists of the added signals from detector rows 1 + 2, the second line in field one consists of the added signals from rows 3 + 4, etc. The first line in field two is formed from the added detector rows 2 + 3, the second line in field two consists of the added signals from rows 4 + 5, etc.

Pseudo interlace reduces the spatial resolution in vertical direction by approximately a factor of two because the signals of two detectors are added thus increasing the spatial dimension over which light is collected. Since, however, the vertical sampling interval remains unchanged, the undersampling in the vertical direction is reduced.

Pseudo interlace has the advantage of not reducing the sensitivity of the imager.

2.5.6 Micro-scan/Dither

Micro-scan or dither is in many ways analogous to the interlace mechanism in scanned imagers. Three types of micro-scan patterns can be modelled in TRM4:

- rectangular micro-scan,
- circular (elliptical) micro-scan, and
- diagonal micro-scan.

Number of micro-scan/dither steps

The number of steps needs to be specified for rectangular micro-scan only (see Figure 2.7). In case of a circular micro-scan (depicted in Figure 2.8), TRM4 assumes that two steps are made in both directions, which gives a total number of four steps per frame. For devices with diagonal micro-scan, the total number of steps is assumed to be two per frame (shown in Figure 2.7).

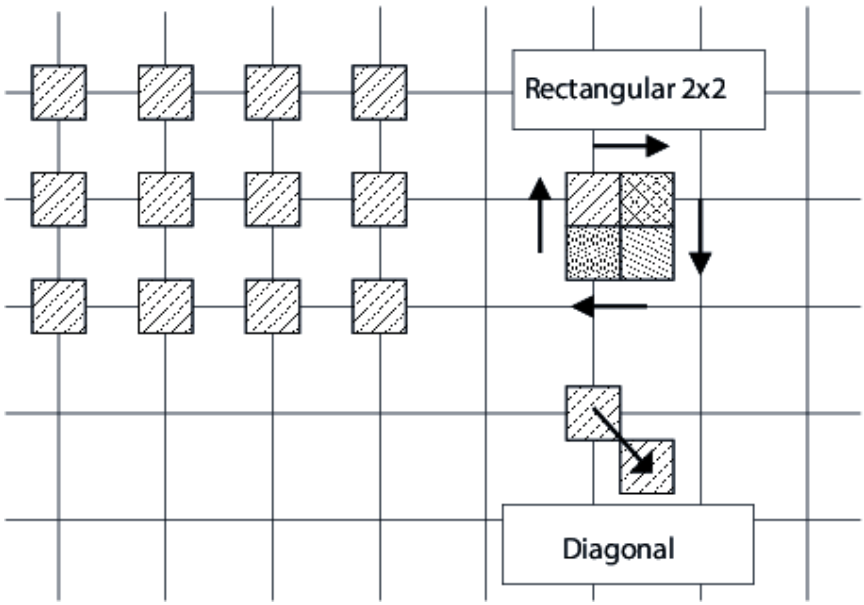


Figure 2.7: 2 × 2 rectangular and diagonal micro-scan pattern.

Integration during movement

This option is active only for staring imagers with micro-scan/dither. Integration during movement of the micro-scan leads to an MTF degradation.

Possible reasons to use this approach may be the need to use the maximum integration time available, and/or simplification of the micro-scan mechanism.

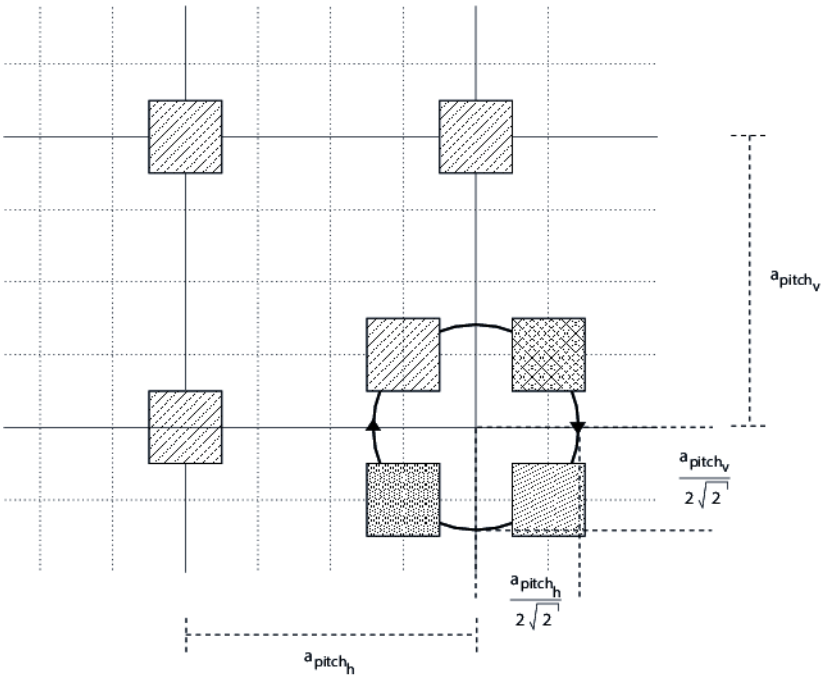


Figure 2.8: Circular micro-scan pattern.

2.5.7 Pixel binning

Pixel binning is a means to increase the sensitivity of the imager at the expense of its spatial resolution. This is obtained by summing the signals of several neighboring detector elements to form one single output.

The effective sensitive area of a detector element is increased by a factor of $n_{\text{bin}_h} \cdot n_{\text{bin}_v}$. The effective detector pitch is assumed to be

$$a_{\text{pitch_binned}} = n_{\text{bin}} \cdot a_{\text{pitch}} \,, \tag{2.2}$$

where n_{bin} is the number of binned pixels in that direction.

> **Note:** The effective number of detector elements in each direction is reduced by the number of binned detector elements in the specific direction. This affects the specification of the number of samples, the number of displayed pixels, etc.

Number of binned pixels

This is the number of detector elements that are binned in horizontal and vertical directions for staring or micro-scanned imagers. In case of a scanning system, only vertical binning is considered.

2.5.8 Number of sensors

The number of sensors can be chosen as one or two. Normally only one sensor is used to observe the scene with one or two eyes. If two sensors are specified the sensors are assumed to be identical. Two sensors increase sensitivity (an example for use of this setup is the ANVIS aviator goggles).

2.6 Optics

2.6.1 Type of aperture

Specific aperture

Different types of apertures can be specified:

- Circular apertures,
- Circular aperture with (circular) central obscuration (cco),
- Circular aperture, truncated (symmetrical or asymmetrical),
- Circular aperture, truncated with cco.

A schematic and the data of the currently specified aperture appear on the upper right side of the *Optics* dialog as shown in Figure 2.9. The proper specification is made in a separate sub-dialog (*Aperture Geometry*), which opens after a click on the *Change specification* button.

General (aperture)

If the optics MTF is specified via external data and the design parameters of the optics are unknown or of no interest, the *General* type of aperture should be selected. In this case only the focal length, the effective F-number, and the transmission of the optics need to be specified.

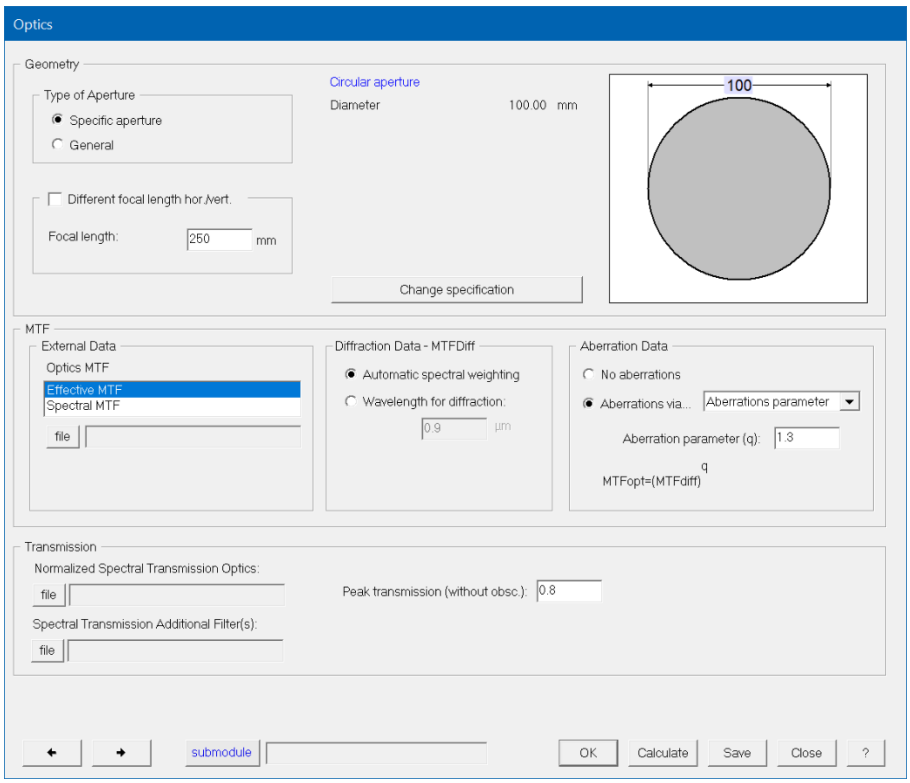


Figure 2.9: System optics definition dialog.

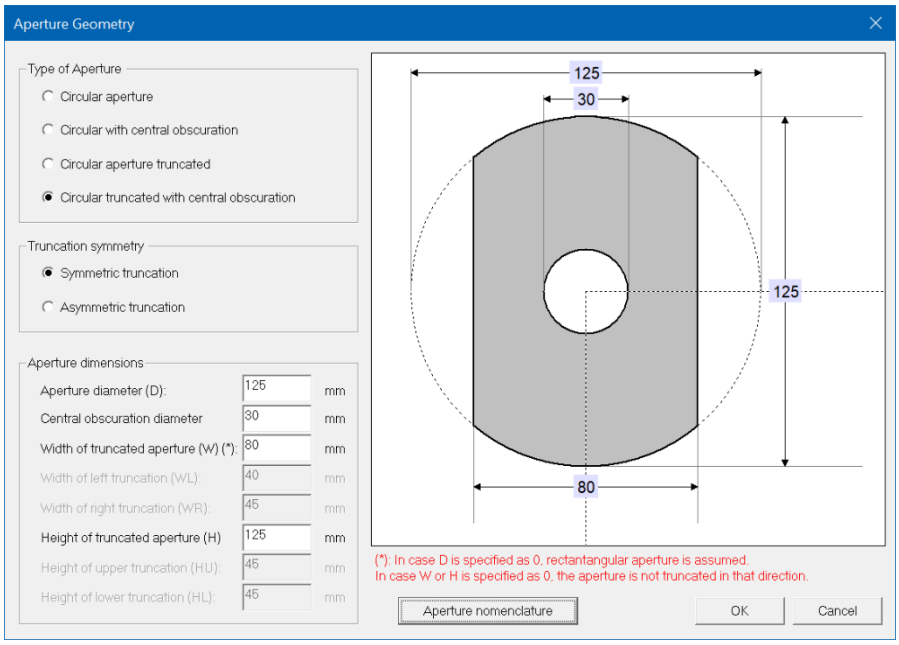


Figure 2.10: Aperture shape definition dialog.

2.6.2 Aperture dimensions

To specify or change type of aperture and/or aperture dimensions, use the *Aperture Geometry* subdialog. To access this subdialog, click on the *Change Specification* button in the *Optics dialog/Geometry group*.

On the right-hand side of this dialog a drawing of the specified aperture is shown. The drawing is automatically updated when aperture values or settings are changed.

In case of invalid or inconsistent input data, the aperture drawing is replaced by an error message. If you need help in specifying values, click on the *Aperture nomenclature* button. You will get a schematic drawing which explains the use of the parameter variables.

Any new aperture specification becomes effective only after you click on the OK buttons in both the *Aperture Geometry* and the *Optics* dialogs. Otherwise, the previous specification will stay in effect.

A simplified procedure to specify an aperture truncated in only one direction is the following one:

- For symmetric truncations: Specify the width or height of the truncation, as appropriate. Specify 0 (zero) for the non-truncated direction. TRM4 automatically substitutes the zero value with the specified aperture diameter.
- For asymmetric truncations: Specify the truncation dimensions. Specify zero for the direction where the pupil is not truncated (both left and right (width) or upper and lower (height) should be set to zero). TRM4 will automatically substitute the zero values with the specified aperture radius.

Rectangular apertures can be considered as symmetrically truncated circular apertures. To specify a rectangular aperture, select *Symmetric truncation* and set the aperture diameter to zero. Width and height of the rectangular aperture are then specified in the fields *Width of truncated aperture* and *Height of truncated aperture*. Figure 2.11 shows an example.

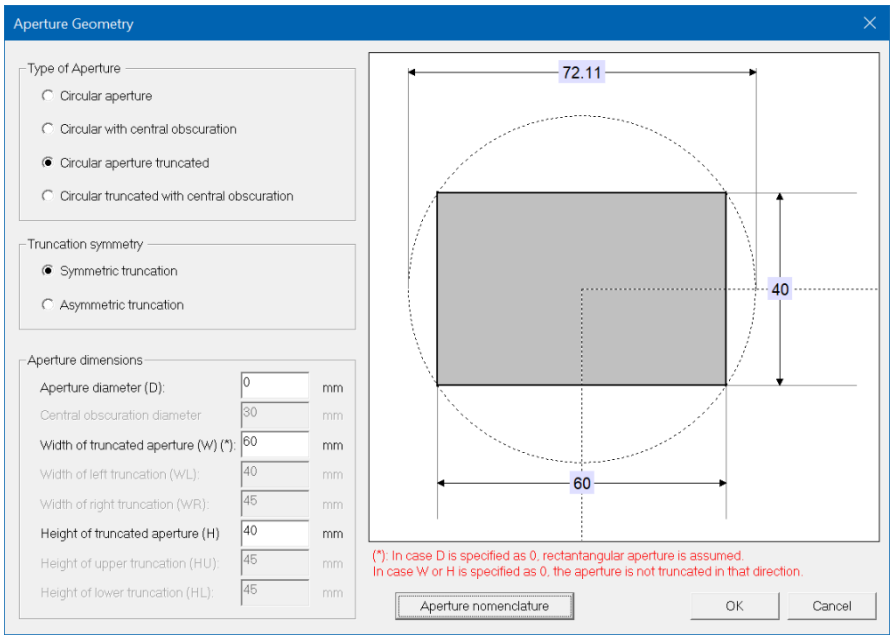


Figure 2.11: Example for the definition of a rectangular aperture.

2.6.3 Different focal length horizontal/vertical direction

Some imagers have different focal lengths in the horizontal and vertical directions. If this case is specified, the resulting *effective focal length* is then given by:

$$f_{\mathrm{opt}} = \sqrt{f_{\mathrm{opt_h}} \cdot f_{\mathrm{opt_v}}}\,, \tag{2.3}$$

where $f_{\mathrm{opt_h}}$ and $f_{\mathrm{opt_v}}$ are the horizontal and vertical focal lengths, respectively.

2.6.4 Effective F-number

The effective F-number of the front optics needs to be specified only if the *General* type of aperture is selected. For all other apertures TRM4 calculates the effective F-number from the specified input data. The result is displayed in the *Diagnostics* message box.

The effective F-number is given by

$$F_{\mathrm{eff_opt}} = \frac{f_{\mathrm{opt}}}{D_{\mathrm{eff}}} \;\text{ with }\; D_{\mathrm{eff}} = \sqrt{\frac{4A_{\mathrm{pup}}}{\pi}}\,, \tag{2.4}$$

where A_{pup} is the clear pupil area of the front optics.

2.6.5 Diffraction MTF

The optics MTF is calculated unless measured data is specified via the external data group. Two options are available for the MTF calculation.

Automatic spectral weighting

If this option is selected, the diffraction MTF is calculated for each wavelength and then spectrally weighted to form the effective optics MTF. The weighting takes into account the spectral scene irradiation/radiation, spectral characteristics of the optics, warm filter, detector response, and spectral atmospheric transmission, see Section 4.1.2.

Wavelength for diffraction MTF

If this option is selected, the diffraction MTF is calculated for a specified single wavelength only. After closing the parameter editing session, TRM4 checks whether the specified value is within the operating spectral band of the imager.

> **Note:** In some cases, the MTF of aberrated optics can approximately be modelled by specifying a diffraction limited MTF with a *Wavelength for diffraction* longer than the operating spectral band of the imager. This is allowed in TRM4. In those cases, a Warning message is issued in the *Diagnostics* message box, but can be safely ignored.

2.6.6 Aberration

No aberration

If this option is selected, TRM assumes that the MTF of the front optics is diffraction limited.

Aberration via …

The influence of aberrations on the optics MTF can be modelled either via an *Aberration parameter* or via *Gaussian blur*.

The *aberration parameter* affects the optics MTF as follows:

$$\mathrm{MTF}_{\mathrm{opt}}(r) = [\mathrm{MTF}_{\mathrm{diff}}(r)]^{q} \ \text{ with } \ q > 1, \tag{2.5}$$

where $\mathrm{MTF}_{\mathrm{diff}}$ denotes the (spectrally weighted) diffraction-limited MTF and q the *aberration parameter*.

A *Gaussian blur* affects the optics MTF as follows:

$$\mathrm{MTF}_{\mathrm{opt}}(r) = \mathrm{MTF}_{\mathrm{diff}}(r) \cdot \exp\left[-2\left(\pi \cdot r \cdot \frac{\sigma_{\mathrm{blur}}}{f_{\mathrm{opt}}}\right)^{2}\right]. \tag{2.6}$$

2.6.7 External Optics MTF data – Effective MTF

The direct specification of (modelled by other means or measured) MTF data is optional. It is mandatory in case *General* type of aperture is specified. The external specification of the optics MTF then replaces the internal MTF calculation.

The effective MTF may be specified for both directions or for a single direction. If specified for a single direction, TRM calculates the MTF for the other direction by using other specified optics data. If the *General* type of aperture is specified, the MTF must be specified for both vertical and horizontal directions.

The external MTF data may refer to either object (target) space or focal plane. In case of object space, the spatial frequency unit should be [$mrad^{-1}$], otherwise the unit should be [mm^{-1}]. The appropriate unit needs to be specified together with the MTF data in the *Optics MTF* data dialog, see also [4].

2.6.8 External Optics MTF data – Spectral MTF

The same rules hold for spectral MTFs as for effective MTFs. In addition, TRM verifies the compatibility of the specified spectral band to other spectral bands specified for the imager and scene.

2.6.9 Transmission

Optics transmission is affected by the transmission of the front optics, window materials, reflectivities of the mirrors, anti-reflection coatings, and additional filters. Normally, only the effective transmission of the optics is supplied and thus can be specified. If available, spectral transmission data can also be specified in TRM4 via external data.

External data: Normalized spectral transmission optics

The spectral transmission of the front optics, $\tau_{\mathrm{opt_norm}}(\lambda)$, can be specified here. Transmission data must be normalized (TRM4 will automatically normalize the data if you ask it to do so). If no spectral transmission is specified, TRM4 assumes that the transmission has a flat spectral response.

Peak transmission (without obscuration)

This is the peak transmission, $\tau_{\mathrm{opt_max}}$, through the clear pupil area. The actual spectral transmission of the optics is hence given by

$$\tau_{\mathrm{opt}}(\lambda) = \tau_{\mathrm{opt_max}} \cdot \tau_{\mathrm{opt_norm}}(\lambda). \tag{2.7}$$

If no spectral transmission of the front optics is specified the *average (effective) transmission* through the clear pupil area of the front optics needs to be specified here.

> **Note:** The effect of central obscuration (and truncations) on the overall transmission of the optics, is considered in TRM by calculating the effective optic area corresponding to the specified aperture geometry and shouldn't be taken into account here. In case of *General* aperture specified, the overall effective transmission should be specified here.

External data: Spectral transmission of additional filter(s):

Some devices allow the insertion of filters into the optical path. If a filter is used, its spectral transmission should be specified here.

Note:	Contrary to the spectral transmission of the optics, the input of spectral transmission of additional filter(s), $\tau_{\mathrm{filter}}(\lambda)$, is not normalized.

Note:	In case of thermal imagers, the additional filters are regarded as “warm” filters. Warm filters transmit, absorb, reflect, and **emit** radiation and must be distinguished from “cold” filters which only transmit, reflect and absorb radiation. Cold filters are considered part of the detector assembly and should be specified there.

2.7 Image Intensifier / II

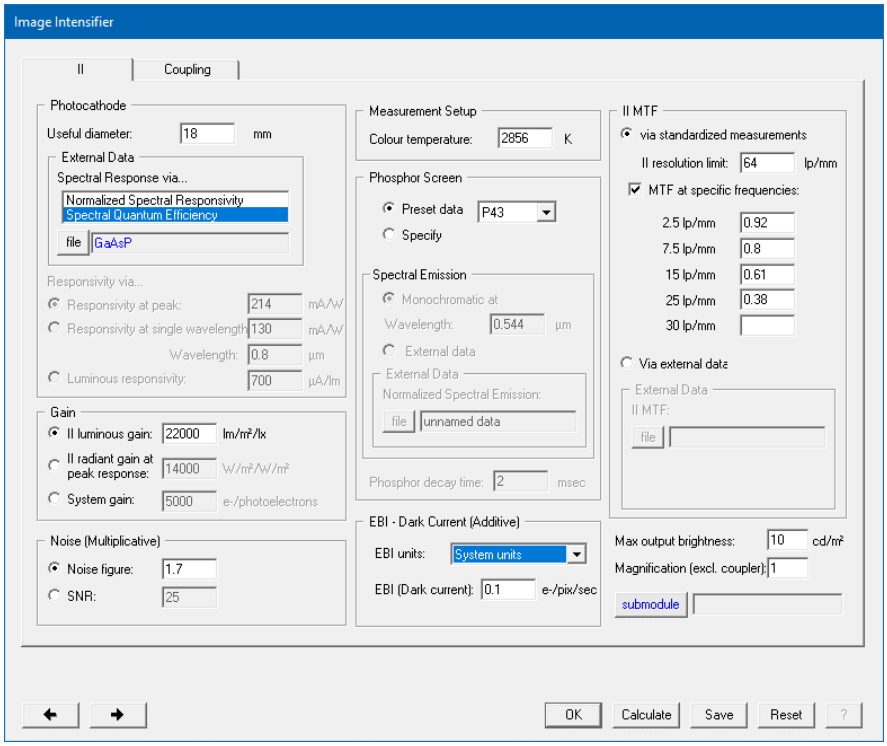


Figure 2.12: Image intensifier definition dialog.

2.7.1 Photocathode

Useful diameter

This is the input diameter of the photocathode. It is used to determine the shape of the FOV for ICCD/ICMOS cameras.

External Data: Spectral response via…

The specification of either the *Normalized Spectral Responsivity* of the photocathode, $R_{\mathrm{PC_norm}}(\lambda)$, or the *Spectral Quantum Efficiency* of the photocathode, $\eta_{PC}(\lambda)$, is mandatory.

Responsivity via…

If you specify the *Normalized Spectral Responsivity* of the photocathode, you have to specify the responsivity of the photocathode in one of three options.

Responsivity at peak

This is the responsivity at the wavelength of maximum photocathode responsivity.

Responsivity at single wavelength

This is the responsivity at a specific wavelength, specified in *Wavelength*.

Luminous Responsivity

The luminous responsivity is defined as the responsivity in the photopic spectral band. The spectral responsivity of the photocathode is derived by assuming the luminous responsivity is measured using a blackbody source with the *Colour temperature* defined in the *Measurement Setup* box.

Note: *Normalized Spectral Responsivity* and *Responsivity* of the photocathode must be consistent. If the specified values result in quantum efficiencies larger than 100 % within the specified spectral band, the specification is rejected and an error message is displayed.

2.7.2 Gain

II luminous gain

The image intensifier luminous gain is defined as the ratio between phosphor screen luminous emittance and the illuminance incident to the photocathode.

II radiant gain at peak response

The image intensifier radiant gain at peak response is defined as the ratio between phosphor screen emittance and the irradiance incident to the photocathode at the wavelength of maximum photocathode responsivity.

System gain

The system gain is defined as the number of electrons collected at the detector per photoelectron emitted from the photocathode. It includes effects of the image intensifier, the coupling element and the detector responsivity. If *System gain* is specified, the *Collection Efficiency* inputs in the *Coupling* dialog are disabled as well as the detector responsivity inputs in the *Detector Sensitivity* dialog.

Note: If system gain is specified, the spectral radiance from the phosphor screen and the corresponding output brightness cannot be calculated. Thus, the phosphor screen *Spectral Emission* and the *Max. output brightness* inputs are disabled. If phosphor screen is specified via *Preset data*, only the corresponding phosphor decay time will be relevant.

2.7.3 Noise (Multiplicative)

Noise figure and SNR

The additional multiplicative noise introduced by the image intensifier is defined by its noise figure. Typical noise figures are 1.6 for Gen II and filmless Gen III image intensifier tubes and 2 for filmed Gen III image intensifier tubes [6]. If *SNR* is specified, it is assumed that the SNR measurement is done in accordance with the military standard [7], but with a colour temperature specified in the *Measurement Setup* box. The image intensifier noise figure is calculated from the SNR as described in Section 3.5.8.

2.7.4 Measurement Setup

Colour temperature

Colour temperature of the illumination source of the measurement setup is needed in case one or more inputs are specified in luminous units or if *SNR* is specified. Note that the colour temperature is assumed to be the same for all measurements of image intensifier quantities.

2.7.5 Phosphor Screen

Preset data

Preset data is available for four commonly used phosphor screen types. In this case, TRM4 uses normalized spectral emission and temporal phosphor screen MTF from preset data.

Specify

If this option is selected, the phosphor screen *Spectral Emission* and *Phosphor decay time* has to be specified by the user.

Spectral Emission: Monochromatic and External Data

The spectral emission of the phosphor screen can be specified as monochromatic at a specific wavelength or as broadband. In case of broadband spectral emission, external data is used to specify normalized spectral emission of the phosphor screen.

Phosphor decay time

The phosphor decay time is defined as the time at which the signal is 10% of the initial value. This input is needed for the calculation of the noise bandwidth, the temporal noise and the stabilization MTF due to linear motion.

2.7.6 EBI – Dark Current (Additive)

The EBI (Equivalent Background Illumination) defines the dark emission of the photocathode and is needed to calculate the dark current due to the photocathode. It can be specified in one of three options:

- in *Luminous units*,
- as EBI at wavelength of photocathode peak responsivity in radiometric units,
- in *System units*. In this case, the number of electrons per detector pixel and second generated by the photocathode dark emission is specified.

2.7.7 II MTF

The MTF of the image intensifier can be either specified by external MTF data or by using measured data

- of the resolution limit of the image intensifier. TRM4 assumes that the MTF value at the resolution limit is 4.5%;
- of the resolution limit of the image intensifier plus a standard set of MTF values at spatial frequencies of 2.5, 7.5, 15, 25 and 30 lp/mm. MTF values have to be specified for at least the first three spatial frequencies.

All spatial frequencies in the *II MTF box* refer to the photocathode plane.

Note that even in case a device prefilter MTF is specified in the *Prefilter MTF or MCP* dialog, II MTF has to be specified in order to calculate the spatial filtering of the temporal noise.

Max output brightness

The maximum output brightness of the phosphor screen has to be specified. It is assumed that an automatic brightness control circuit limits the average output brightness of the phosphor screen to be not higher than the specified maximum output brightness.

Magnification (excl. coupler)

The image intensifier magnification is the magnification between the photocathode plane (focal plane of the optics) and the phosphor screen output plane.

2.8 Image Intensifier / Coupling

This dialog (Figure 2.13) serves to specify the coupling between the image intensifier and the solid-state detector array. The coupling element can be either lenses or fiber optic taper.

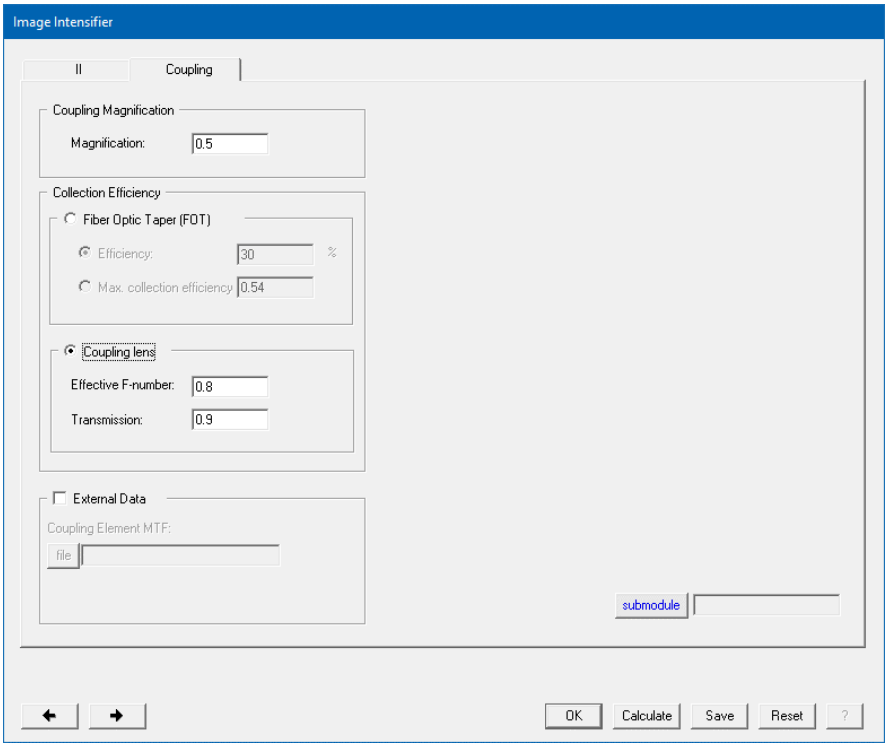


Figure 2.13: Coupling element definition dialog.

2.8.1 Coupling Magnification

The coupling magnification is the magnification of the coupling element between the phosphor screen output plane and the detector plane.

2.8.2 Collection Efficiency

Fiber Optic Taper (FOT)

If FOT is selected as coupling element, the collection efficiency can be specified by one of two options:

1. Direct specification of the *Efficiency*.

2. Specification of the *Maximum collection efficiency*, which is the product of the overall FOT transmission (for collimated light) and the square of the maximum numerical aperture. For demagnifying FOTs, the collection efficiency is given by (maximum collection efficiency)*(coupling magnification)^2, otherwise it equals the maximum collection efficiency.

Coupling lens

If coupling lens is selected as coupling element, the effective F-number and the transmission of the coupling lens have to be specified.

2.8.3 External Data: Coupling Element MTF

The specification of coupling element MTF is optional. Note that even in case a prefilter MTF is specified in the “Prefilter MTF or MCP” dialog, the coupling MTF can be specified in order to calculate the spatial filtering of the temporal noise.

2.9 Detector/Geometry

Two different versions exist for the *Detector Geometry* dialog. One for the imager categories *Visible* and *General* and another for the imager category *Thermal*. They are shown in Figure 2.14.

2.9.1 Detector shape

Four different detector element shapes can be specified for all imager types (Figure 2.15). A special detector type is the SPRITE (Signal PRocessing In The Element) [8], which is used in 1st generation thermal imagers only. Its geometry is depicted in Figure 2.16.

Horizontal detector dimension (active)

This is the dimension of the active area of a rectangular or rhombic detector element in the horizontal direction. The data is used to calculate the detector’s sensitive area and its horizontal MTF.

Vertical detector dimension (active)

This is the dimension of the active area of a rectangular, rhombic or SPRITE detector element in the vertical direction. The data is used to calculate the detector’s sensitive area and its vertical MTF.

Detector diameter (active)

This is the active diameter of a single circular detector element. The values are used in order to calculate the detector’s sensitive area and its MTF.

Diameter of hole

This parameter needs to be specified for circular loophole detector elements only.

Read-out zone length, diffusion length

These two parameters need to be specified for SPRITE detectors only. They are required for the calculation of detector area and detector MTF. In this type of detector, the read-out zone length, together with the vertical detector dimension, is used for the calculation of the detector geometrical MTF. The diffusion length degrades the horizontal MTF.

2.9.2 Focal plane geometry

Number of horizontal/serial (TDI) detectors

For scanning imagers, this parameter serves to specify the number of detectors per row used for time delayed integration (TDI). For staring or micro-scanned devices, this parameter specifies the number of detectors per row.

Number of vertical detectors

For scanning imagers this is the number of detectors in the vertical direction, irrespective of staggering depicted in Figure 2.17. For staring or micro-scanned imagers, the number of detectors per column is specified here.

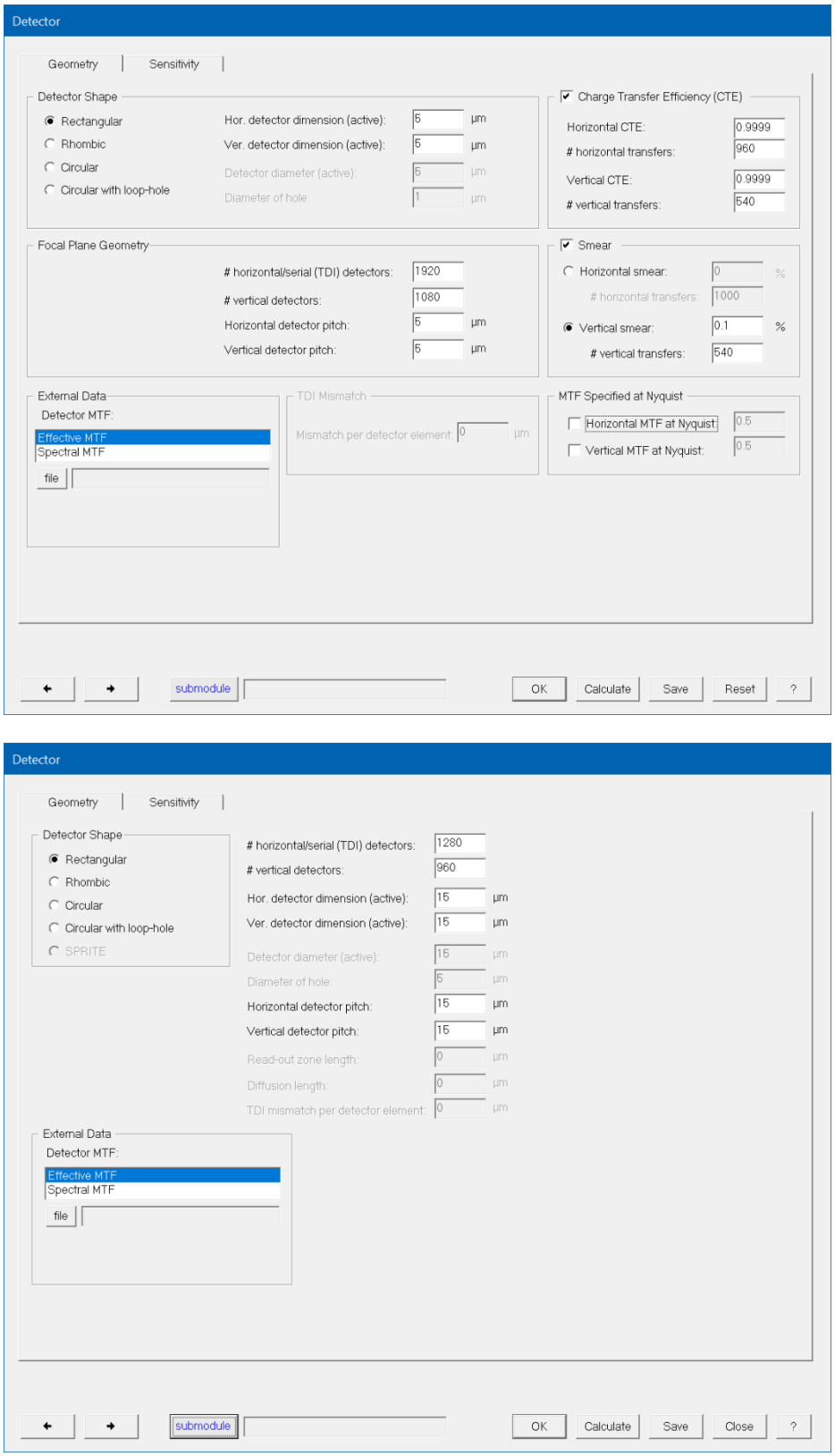


Figure 2.14: Detector geometry definition dialogs for general/visible systems (top) and thermal systems (bottom).

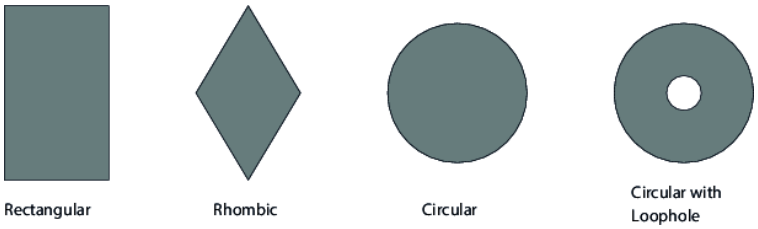


Figure 2.15: Detector shapes available in TRM4. The grey area corresponds to the sensitive area of a single detector element.

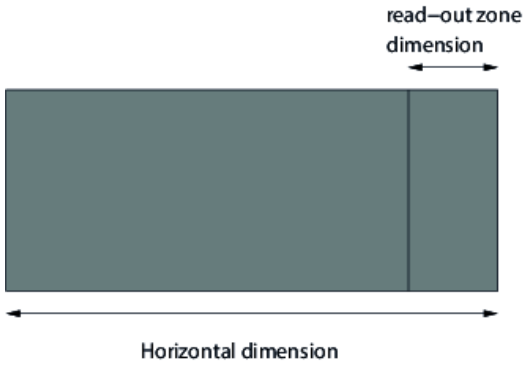


Figure 2.16: Geometry of a SPRITE detector.

Detector pitch

The detector pitch needs to be specified only if the FOV is to be calculated by TRM4. Otherwise, the specification is optional. A pitch value of zero is valid and means “not specified”. If specified, TRM4 uses the data for a FOV consistency test.

> **Note:** The consistency test is always performed once pitch values larger than zero are specified. To avoid conflicts, either zero values or correct (exact) pitch data should be specified.

For a scanning imager, only the vertical detector pitch needs to be specified. For staggered detector arrays, the vertical pitch is defined as shown in Figure 2.17.

2.9.3 TDI mismatch per detector element

This parameter needs to be specified for scanning imagers where the number of serial detectors (TDI) is larger than one.

2.9.4 Charge transfer efficiency (CTE)

Charge loss in the process of signal read-out causes spatial signal degradation. The degradation can be described by an MTF. This MTF depends on the number of charge transfers and hence on the location in the FOV[3]. For example, for a 1000×1000 array 500 horizontal and 500 vertical transfers may be needed to read out the central element or the center of the FOV.

[3] This means that, strictly speaking, an MTF cannot be defined for the CTE. Care must thus be taken in applying this option. Using the number of the transfers for the center of the FOV, however, should be a sensible compromise for the assessment of the performance impact of CTE.

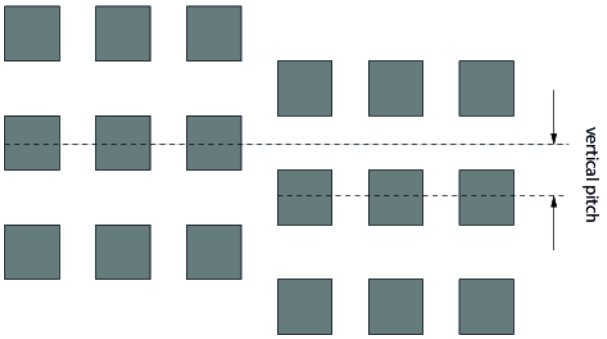


Figure 2.17: Pitch definition in a staggered array.

Note: In TRM4 Charge Transfer Efficiency is specified as the efficiency to transfer the signal from one detector element to its neighbor. If there are phases involved and only the phase transfer efficiency is known, the user should calculate the transfer efficiency between two elements (involves usually 1 to 4 phases) and specify this value as the CTE.

Horizontal CTE

This is the horizontal charge transfer efficiency between two detector elements. Typical values are 0.9999 and higher.

Number of horizontal transfers

The horizontal MTF associated with the CTE depends on the number of horizontal transfers. This number cannot be higher than the number of horizontal detector elements.

Vertical CTE

This is the vertical charge transfer efficiency between two detector elements. Typical values are 0.9999 and higher.

Number of vertical transfers

The vertical MTF associated with the CTE depends on the number of vertical transfers. This number cannot be higher than the number of vertical detector elements.

2.9.5 Smear

In frame transfer CCDs, the sensitive detector elements themselves are used to transfer the entire frame to the output. The transfer is done with the sensitive detector elements still picking up signal from the scene, causing image smear. Although the transfer is fast, image smear is caused.

Smear leads to an increased background signal and may affect the noise. The effect of any smear on the displayed image depends on the scene content and on the sensor read-out technology (interline-transfer, frame transfer). Smear has no effect on the MTF.

With respect to smear, TRM4 assumes a homogeneous (uniform) scene where the signal is given by the specified background.

Smear may be specified in the horizontal or vertical directions but not in both directions simultaneously.

Horizontal smear

This is the smear per transfer in the horizontal direction for maximum integration time (duty cycle of 1). TRM assumes that at each horizontal transfer the background signal is increased according to the specified value.

Number of horizontal transfers

This number corresponds to the number of light-sensitive elements that are passed in horizontal direction during signal read-out. For full frame transfer arrays, the number of transfers should be chosen to be half the number of pixels in the horizontal direction.

Vertical smear

This is the smear per transfer in the vertical direction for maximum integration time (duty cycle of 1). TRM assumes that at each vertical transfer the background signal is increased according to the specified value.

Number of vertical transfers

This number corresponds to the number of light-sensitive elements that are passed in vertical direction during signal read-out. For full frame transfer arrays, the number of transfers should be chosen to be half the number of pixels in the vertical direction

2.9.6 MTF specified at Nyquist

TRM4 uses this parameter together with the ideal detector area MTF (calculated internally by TRM4) to assess the MTF degradation factor caused by crosstalk at frequencies other than Nyquist. The calculation of the corresponding crosstalk MTF and diffusion constant are given in Equations (4.29) and (4.30). Note that the specified value should be the static detector MTF without binning, CTE, and pseudo interlace.

Horizontal MTF at Nyquist

This parameter is used to calculate horizontal detector crosstalk.

Vertical MTF at Nyquist

This parameter is used to calculate vertical detector crosstalk.

2.9.7 External data: Detector MTF

The direct specification of (e.g. measured) detector MTF data is optional. The external specification replaces the internal MTF calculation. It can be specified for both directions, or for a single direction. If specified for a single direction, TRM calculates the MTF for the other direction by using other specified detector data.

External detector MTF data must refer to the detector plane. Correspondingly, the spatial frequency unit is [mm^{-1}].

Effective MTF

The effective MTF is a broad-band MTF, independent of wavelength.

Spectral MTF

If spectral MTFs are specified, TRM checks the compatibility of the specified spectral band to other spectral bands specified for the imager and scene.

2.10 Detector/Sensitivity – 1st and 2nd generation imagers of category *Thermal*

The *Detector Sensitivity* dialog serves to specify data required to assess the thermal sensitivity of the imager. Most parameters of this dialog are disabled if measured temporal noise (Noise Equivalent Temperature Difference, see Section 3.9) is specified.

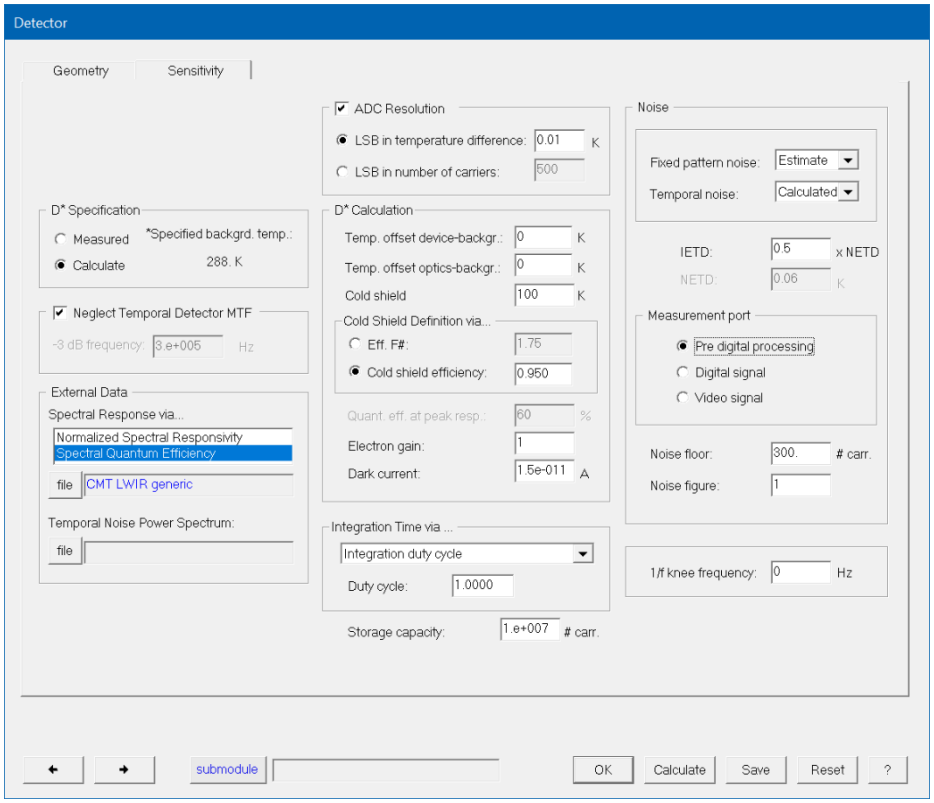


Figure 2.18: Detector sensitivity dialog of 1st and 2nd generation thermal imagers.

2.10.1 D^* specification

In TRM4, D^* designates the peak detectivity, D^*_{peak}, of a single detector element. It can either be directly specified or calculated.

Measured

If this option is selected, a measured D^* value needs to be specified. Unless the detector is *detector noise limited*, the detectivity depends on imager design features. Care should be taken that the specified measured D^* value is valid for the actual device configuration and the specified background temperature. As a reminder to the user, the background temperature as specified in the *Target, Scenario and Task* dialog (Section 2.24) is displayed in the D^* *Specification* group.

Calculate

If this option is selected TRM4 calculates D^* for the specified background temperature and with the parameters specified in the D^* *Calculation* group of the dialog.

2.10.2 D^* calculation

Temperature offset device-background

If the cold shield efficiency (see Equation (2.10)) is smaller than one, energy emitted by the inner housing of the imaging device contributes to the background irradiance in the focal plane (see Section 3.7).

TRM4 assumes that the inner housing is a blackbody at a uniform temperature of T_{device}. Given the scene background temperature T_{scene} (specified in the *Target, Scenario and Task* dialog, see Section 2.24), the *Temperature offset device-background* is defined as

$$T_{\mathrm{off_dev}} = T_{\mathrm{device}} - T_{\mathrm{scene}}. \tag{2.8}$$

Temperature offset optics-background

If the transmission of optics and “warm” filter (specified in the *Optics* dialog) are not equal to one, the clear pupil and/or the filter contribute to the irradiance in the focal plane (Section 3.7).

TRM4 assumes that the lenses/mirrors that constitute the clear pupil and the “warm” filter are greybodies at a uniform temperature of T_{opt}. Given the scene background temperature T_{scene} (specified in the *Target, Scenario and Task* dialog, see Section 2.24), the *Temperature offset optics-background* is defined as

$$T_{\mathrm{off_opt}} = T_{\mathrm{opt}} - T_{\mathrm{scene}}. \tag{2.9}$$

Cold shield temperature

The cold shield also emits thermal radiation if its temperature is higher than 0 K. This radiation contributes to the background signal (and thus to detector noise, see Section 3.7).

Cold shield definition via ...

Either the *effective F-number* of the cold shield or the *cold shield efficiency* should be specified. This input is required to calculate the amount of radiation emitted by the inner housing of the imager and sensed by the detector. Once one of the two parameters is specified, TRM4 calculates the other parameter and displays the result in the *Diagnostics* message box and in the *System Performance Data* message box.

The effective F-number of the cold shield and the cold shield efficiency, η_{cs}, are related by:

$$\eta_{\mathrm{cs}} = \frac{4F_{\mathrm{eff_cs}}^2 + 1}{4F_{\mathrm{eff_opt}}^2 + 1}. \tag{2.10}$$

If the cold shield efficiency is equal to one, the cold shield is perfectly matched to the optics aperture. If the cold shield efficiency is larger than one, part of the scene radiation is obscured by the cold shield. If the cold shield efficiency is smaller than one, energy emitted by the inner housing of the imaging device reaches the focal plane.

Note: TRM4 does not permit cold shield efficiencies >1. In such cases, a fatal error message is issued. The inconsistency must be corrected before resuming editing.

Note: For truncated optics or optics with a central obscuration, a cold shield efficiency larger than

$$\eta_{\mathrm{cs}} = \frac{4F_{\mathrm{opt}}^2 + 1}{4F_{\mathrm{eff_opt}}^2 + 1} \tag{2.11}$$

(where $F_{\mathrm{opt}} = f_{\mathrm{opt}}/D_{\mathrm{opt}}$) can be obtained only if a special radiation shield with a shape similar to the front optics (scaled) is used.

Quantum efficiency at peak responsivity

The *Quantum efficiency at peak responsivity* needs to be specified in combination with the *Normalized spectral responsivity*. It is not required if the spectral quantum efficiency is directly specified via external data. Quantum efficiency is specified in [%].

Note: This parameter describes the intrinsic quantum efficiency of the active detector area and not an “effective” quantum efficiency taking into account the detector pitch and the fill factor.

Electron gain

Electron Gain may be used in order to enhance the signal from the scene and thus diminish the effect of read-out noise on the signal-to-noise ratio. Electron gain mechanisms may be associated with additional multiplicative and/or additive noise, which can be specified via the parameters *Noise floor* and *Noise figure*, see Section 2.10.8.

> **Note:** In practice, electron gain amplifies both the detector signal and the detector dark current. As a result, electron gain does not reduce the effect that the dark current has on the overall noise and the SNR.

Dark current

This parameter refers to a single detector element. The dark current has to be specified in [A]. Dark current refers to the input of the detector (this is important to keep in mind in case of electron gain mechanisms).

2.10.3 Integration time via ...

The integration time is used for the calculation of signal and noise. It is also required for the calculation of the MTF, which might include image smear[4].

Three options exist for specifying the detector integration time in a 2nd generation imager:

- *Direct specification*,
- via *Integration duty cycle*,
- via *Well-fill*.

The integration duty cycle, η_{cyc}, is the actual integration time, t_{int}, divided by the maximum theoretically available integration time (Section 3.5.7).

The *Well-fill* option (= % of the storage capacity utilized at the specified background temperature) can be selected only if D^* is calculated by TRM4 and a *Storage capacity* >0 is specified (see below).

> **Note:** If the *Well-fill* option is specified, TRM4 calculates the integration time that is necessary to fill the storage gate (well) up to the specified value. If this integration time exceeds the maximum possible integration time (corresponding to a duty cycle of unity), TRM4 displays a fatal error message after closing the specification session.

> **Note:** TRM4 calculates the integration time if the integration duty cycle is specified or the duty cycle if the integration time is specified. Calculated data is listed in the *Diagnostics* and in the *System Performance Data* message boxes.

[4] Image smear occurs if the image is moving on the detector focal plane during signal integration. This is the case of a 2nd generation scanning or micro-scanned imager for which *Integration during movement* is specified (in the *Basic Imager Data* dialog).

2.10.4 Temporal detector MTF

The *temporal detector MTF* is assumed to act like a first order Butterworth low-pass filter characterized by its *-3dB cut-off frequency*. This MTF contribution is relevant for scanning imagers only, particularly for 1st generation thermal imagers.

2.10.5 Storage capacity

Storage capacity is relevant for 2nd generation imagers only. The storage capacity is defined as the number of carriers that can be stored in one detector element. A storage capacity value of zero means “not specified”.

In case of calculated overflow in the detector well, a runtime warning message is issued but calculation is not aborted.

Note:	In TRM4 the storage capacity refers to a single detector element. For 2nd generation linear FPAs with TDI, the manufacturer usually specifies the storage capacity with respect to the integrated detector signal.

2.10.6 ADC resolution

The quantization noise, N_{quant}, associated with the analog-to-digital conversion process may be specified in this group. There are two options to specify the quantization noise via the *Least Significant Bit* (LSB):

- The actual *Number of carriers* or, for 1st generation imagers, the current in [A] that is associated with the LSB. In the latter case, the quantization noise current i_{quant} is used in the noise calculations.
- The *temperature difference* that is equivalent to the LSB.

The root mean square (RMS) quantization noise then follows as

$$N_{\mathrm{quant}} = \frac{\mathrm{LSB}}{\sqrt{12}}\ , \tag{2.12}$$

where LSB in the equation stands for the actual number of carriers, the current, or the temperature difference associated with the LSB. It is important to emphasize again that, in case of scanning devices with TDI, the LSB should be specified for the individual detector element and not for the signal of the integrated output after the TDI process (this is irrelevant, however, for the specification via temperature differences).

2.10.7 External data

Spectral response via ...

The specification of the *Normalized Spectral Responsivity*, $R_{\mathrm{DA_norm}}(\lambda)$, or of the *Spectral Quantum Efficiency*, $\eta(\lambda)$, is mandatory unless a measured NETD is specified (and no other spectral data is specified anywhere else).

Note: As mentioned earlier, the data should include the spectral response of a cold filter if part of the detector/dewar assembly.

Note: Normalized spectral response and peak quantum efficiency must be consistent. If both are specified, TRM4 calculates the spectral quantum efficiency of the detector assembly. If the specification leads to quantum efficiencies greater than 100 % anywhere along the specified spectral band, the specification is rejected and an error message is displayed.

Note: *Spectral quantum efficiency* describes the intrinsic quantum efficiency of the active detector area and not an "effective" quantum efficiency taking into account the detector pitch and fill factor.

Temporal noise power spectrum

The specification of the temporal noise power spectrum is optional. This is a normalized power spectrum. It should have the value of 1.0 at the measurement frequency of D^*, or at mid-range frequency in case of calculated D^*. If no data is specified, white noise plus the specified $1/f$ noise contribution (Section 2.10.9) will be assumed.

2.10.8 Noise group

Fixed pattern noise

Fixed pattern noise in scanning, staring or micro-scanned imagers is characterized by the IETD (Inhomogeneity Equivalent Temperature Difference, Section 3.10). It affects the MTDP and hence the range performance of the imager.

IETD describes the contribution of randomly distributed fixed pattern noise to the overall system noise. It is defined as the temperature difference which gives a signal equal to the RMS value of the fixed pattern noise. Non-randomly distributed fixed pattern noise is not assessed.

Three options are available to characterize fixed pattern noise:

- *No noise*,
- *Measured* (to be specified in temperature units),
- *Estimated* (to be specified as fraction of NETD, see below).

Temporal noise

Two options are available in assessing the temporal noise of the imager:

- *Measured* and
- *Calculated*.

If the *Measured* option is selected, most parameter inputs in the *Detector/Sensitivity* dialog are disabled.

Measurement port

Depending on the imager type, NETD and IETD can be measured at several different locations in the system. If measured IETD and/or NETD data is specified it must be indicated where the measurement was made. This can be at the

- output of the analog signal processing / Pre ADC / Pre digital processing,
- output of the digital signal processing, or at the
- output of the video signal processing.

Noise figure and noise floor

Noise sources other than the detector itself, like read-out electronics, preamplifiers, etc., may contribute to the overall noise.

If the additional noise is additive in nature, it is defined as *Noise floor*, N_{floor}. The noise floor is assumed to be white and is characterized by the RMS number of electrons which constitute this noise (for 1st generation imagers it is specified in [A], and the noise floor current, i_{floor}, is used for noise calculations). Multiplicative noise is specified by the *Noise figure* NF. The contribution of these parameters to the overall temporal noise is described in Section 3.8.1.

> **Note:** The *Noise floor* refers to a single detector element. The overall noise floor of n detector elements in series is assumed to be $\sqrt{n}$ times higher.

2.10.9 $1/f$ knee frequency

This parameter is relevant for scanning imagers only. It is used to specify any low frequency $1/f$ noise contribution to the white noise power spectrum (see Section 3.9). If no $1/f$ noise is present, the parameter value must be set to zero.

2.11 Detector/Sensitivity – Uncooled detector of category *Thermal*

TRM4 assumes for the assessment of the sensitivity of uncooled detectors that the detector (or device) NETD is measured and the data is available to the user. If a device measured NETD is specified most parameters in the dialog are disabled.

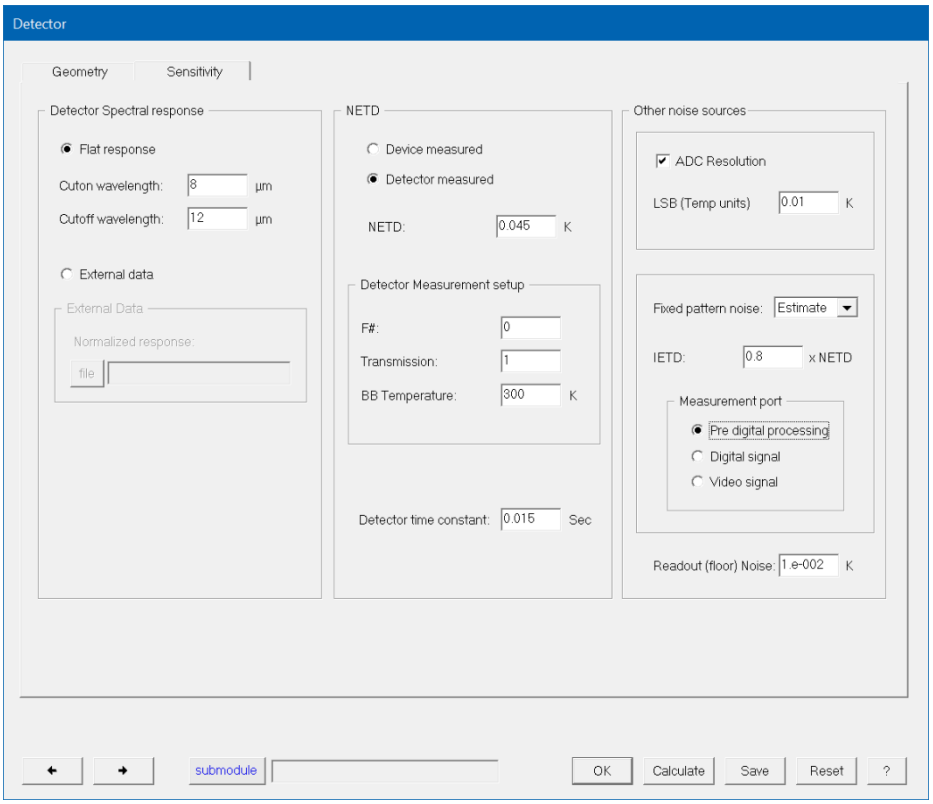


Figure 2.19: Detector sensitivity definition dialog for uncooled detectors.

2.11.1 Detector spectral response

The *Detector Spectral response* has to be provided in three cases:

- *Detector measured* NETD is specified,
- Spectral detector and/or optics MTF data is specified (and no external prefilter is specified), and
- Automatic spectral weighting of the optics MTF is selected (and no external prefilter is specified).

There are two ways of specifying the spectral response of the detector: Either by choosing a *Flat response* or by providing the *Normalized response* as *External data*.

Flat response

For this option, the detector response is assumed to be nonzero and constant only inside the spectral range defined by the *Cuton wavelength* and *Cutoff wavelength*.

External data

User-defined or measured detector response may be provided as external data.

2.11.2 NETD

It has to be specified whether the provided value of the NETD refers

- to the whole device (*Device measured*) or
- to the detector alone (*Detector measured*).

In case of a *Detector measured* NETD, TRM4 needs additional information about the *Detector Measurement setup*, the *Detector Spectral response* and several other noise sources in order to properly calculate the NETD of the device for the conditions specified in the project.

For a *Device measured* NETD, all options and parameters except the *detector spectral response and Fixed pattern noise* become inactive.

Detector Measurement setup

The following parameters, which describe the configuration with which the NETD was measured, have to be specified:

- *F-number* used in the measurement setup,
- *Transmission* of the optics of the measurement setup, and
- *Background blackbody temperature* of the target used in the measurement setup.

TRM4 uses these parameter values to scale the specified *Detector measured* NETD to the corresponding parameter values defined in the specified project.

> **Note:** If a parameter of the *Detector Measurement setup* is set equal to zero it is assumed that the value of this parameter is identical to the value specified in the TRM4 project under consideration. In case of the F-number, the effective F-number as defined by Equation (2.4) is used.

Detector time constant

The thermal time constant of the detector, t_{th}, describes the temporal response of an uncooled detector to a change in the input radiation. For a bolometer, it is the temperature change of the absorptive element. Considering an input signal step at time $t = 0$, the change in the signal at the detector output, $S(t)$, is assumed to have the following temporal behavior

$$S(t) \propto 1 - \exp\left(-\frac{t}{t_{\mathrm{th}}}\right) \quad \text{for} \quad t > 0. \tag{2.13}$$

The thermal time constant is required to calculate the noise equivalent bandwidth and (linear motion) stabilization error MTF of uncooled detectors. Not considered are theoretically possible effects due to temporal noise correlations between frames in case of frame integration and spatial noise correlations in case of micro-scan, which might occur for a thermal time constant of the order or larger than the inverse frame rate.

2.11.3 Other noise sources

ADC Resolution

The quantization noise, N_{quant}, associated with the analog-to-digital conversion process is specified by the temperature difference that is equivalent to the *Least Significant Bit* (LSB). The RMS quantization noise is given by Equation (2.12).

Fixed pattern noise

The fixed pattern noise is characterized by the IETD (Inhomogeneity Equivalent Temperature Difference, see Section 3.10). The IETD describes the contribution of randomly distributed fixed pattern noise to the overall system noise. It is defined as the temperature difference that gives a signal equal to the RMS value of the fixed pattern noise. Non-randomly distributed fixed pattern noise is not assessed in TRM4.

Three options are available to specify fixed pattern noise:

- *No noise*,
- *Measured* (to be specified in temperature units), and
- *Estimated* (to be specified as fraction of the NETD).

Measurement port

The NETD and IETD can be measured at different ports of the system. You must specify whether the measurement or estimation was made at the

- output of the detector,
- output of the digital signal processing, or at the
- output of the video signal processing.

For the option *Device measured NETD* it is assumed that the NETD and the IETD are measured at the same port. For the case of a *Detector measured NETD,* TRM4 assumes that the NETD is measured at the detector output, while the IETD may be specified at any of the three ports.

Read-out noise

The contribution of the read-out electronics to the overall temporal noise can be specified here. This noise is additive in nature and assumed to be white. It is characterized by an equivalent temperature difference that results in a signal equal to the RMS value of the read-out noise. This option is only available for *Detector measured* NETD.

2.12 Detector/Sensitivity – *Visible* category

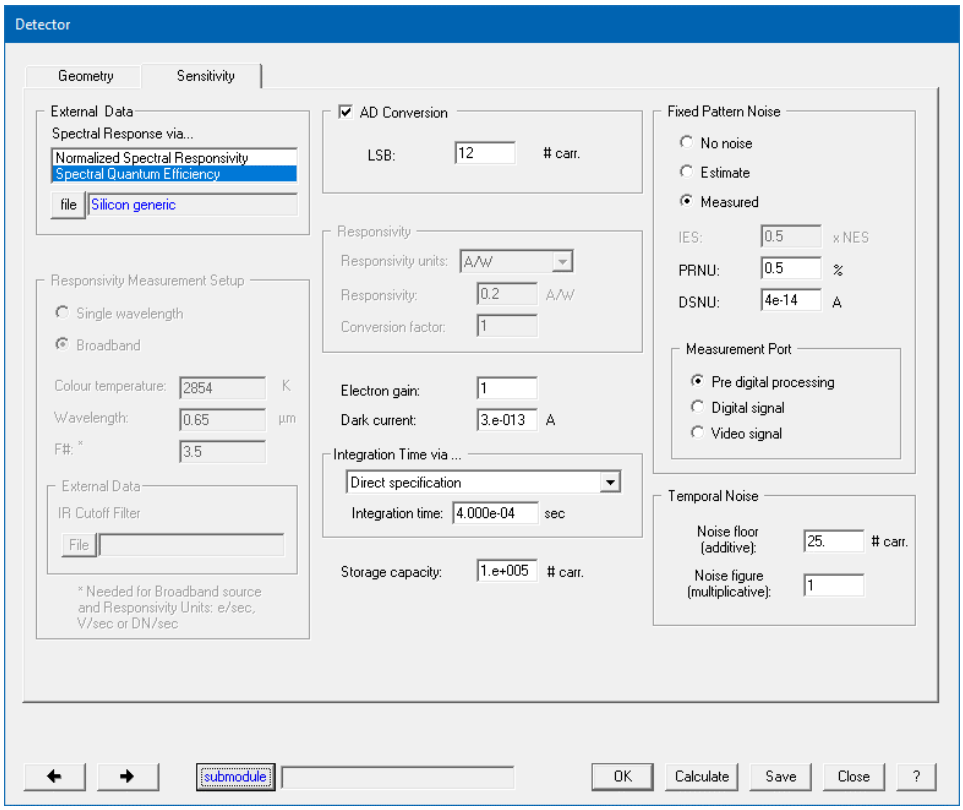


Figure 2.20: Detector sensitivity definition dialog for visible systems.

2.12.1 External Data: Spectral response via ...

The specification of either the *Normalized Spectral Responsivity*, $R_{\mathrm{DA_norm}}(\lambda)$, or of the *Spectral Quantum Efficiency*, $\eta(\lambda)$, is mandatory.

Many parameters of this dialog are disabled if the *Spectral Quantum Efficiency* is specified. If possible, it is recommended to specify spectral quantum efficiency data rather than normalized spectral response data.

> **Note:** Spectral quantum efficiency describes the intrinsic quantum efficiency of the active detector area and not and “effective” quantum efficiency taking into account the detector pitch and fill factor.

2.12.2 Responsivity measurement setup

If you specify the *Normalized Spectral Responsivity*, you also have to specify the peak *Responsivity* and how the responsivity was measured since responsivity data strongly depends on the measurement conditions. Usually, sensor manufacturers specify a setup. TRM4 needs to know these conditions in order to properly calculate the detector responsivity for the conditions specified in the project.

Specify whether the measurement was done with a monochromatic or with a *Broadband* irradiation source.

Single wavelength

In case of a *monochromatic* source, the *Wavelength* needs to be specified. In addition, the *Irradiance* has to be specified when the *Responsivity* is given in [Unit/sec], where [Unit] stands for [V], [e], or [DN].

Broadband

For a broadband illumination source, the *Colour temperature* hat to be specified. If *Responsivity* is given in [Unit/sec], where [Unit] stands for [V], [e], or [DN], the F-number F# of the measurement set-up needs to be specified, too.

External Data: IR Cut-off Filter

In many cases, an IR cut-off filter is used in sensor responsivity measurements. The spectral transmission of this filter needs to be specified.

2.12.3 Responsivity

The entire group is only active if the *Normalized Spectral Responsivity* is specified.

Responsivity Units

There are 10 optional units by which you may specify the *Responsivity*:

- A/W
- V/(µJ/cm^2)
- V/(Lux sec)
- V/sec
- e/(µJ/cm^2)
- e/(Lux sec)
- e/sec
- DN/(µJ/cm^2)
- DN/(Lux sec)
- DN/sec

Here, DN stands for Digital Numbers.

Conversion factor

A responsivity specification in voltage or DN needs to be complemented by specifying the conversion factor [µV/e] or the # of electrons per digital number [e/DN], as appropriate.

2.12.4 ADC resolution

The quantization noise, N_{quant}, associated with the analog-to-digital conversion process is calculated by specifying the number of carriers per *Least Significant Bit* (LSB). The quantization noise is then given by

$$N_{\mathrm{quant}} = \frac{\mathrm{LSB}}{\sqrt{12}}\ , \tag{2.14}$$

It is important to remember that in case of scanning devices with TDI, the LSB should be specified for the individual detector element and not for the signal of the integrated output after the TDI process.

2.12.5 Electron gain

Electron Gain is used in certain CCDs (EMCCD) in order to enhance the signal from the scene and thus reduce the effect of read-out noise on the SNR. Electron gain mechanisms may be associated with additional multiplicative and/or additive noise (specified as *Noise floor* and *Noise figure*, see Section 2.12.10).

Note: In practice, electronic gain mechanisms amplify the detector signal as well as the detector dark current. As a result, electron gain does not reduce the effect the dark current has on the overall noise.

2.12.6 Dark current

This parameter refers to a single detector element. The dark current has to be specified in [A]. If electron gain is used, the specified dark current should be referred to the input of the electron gain mechanism (the detector).

2.12.7 Integration time via ...

The integration time is needed for sensitivity calculations. It is also required for the calculation of the MTF possibly including image smear[5].

Three options exist for specifying the detector integration time:

- *Direct specification*,
- via *Integration duty cycle*,
- via *Well-fill*.

The *Integration duty cycle*, η_{cyc}, is the actual integration time, t_{int}, divided by the maximum theoretically available integration time (see Section 3.5.7).

The *Well-fill* option (= % of the storage capacity utilized at the specified scenario) can be selected only if the *Storage capacity* is specified as >0.

[5] Image smear occurs if the image is moving on the detector focal plane during signal integration. This is the case with scanning imagers and with micro-scanned imagers with the *Integration during movement* specified (in the *Basic Imager Data* dialog).

Note: If the *Well-fill* option is specified, TRM4 calculates the integration time that is necessary for the background signal to fill the storage gate (well) up to the specified value. The calculation is done for each considered target distance and for all specified atmospheres. If the calculated integration times exceed the maximum possible integration time (corresponding to a duty cycle of unity), TRM4 uses the maximum possible integration time and displays a warning message to inform the user.

Note: TRM4 calculates the integration time if the integration duty cycle is specified or the duty cycle if the integration time is specified. Calculated data is listed in the *Diagnostics* and *System Performance Data* message boxes.

2.12.8 Storage capacity

The *Storage capacity* is the number of carriers, which can be stored on the focal plane per detector element before being saturated. A storage capacity value of zero means “not specified”.

In case of calculated overflow in the detector well, a runtime warning message is issued but calculation is not aborted.

Note: The storage capacity specified in TRM4 refers to a single detector element. In case of scanning devices with TDI, the manufacturer usually specifies the storage capacity with respect to the integrated detector signal (“channel”).

2.12.9 Fixed pattern noise

Fixed pattern noise is characterized by two parameters:

- PRNU (Photo Response Non-Uniformity) and
- DSNU (Dark Signal Non-Uniformity).

Specified PRNU and DSNU define the contribution of randomly distributed, white fixed pattern noise to the overall system noise [9]. Non-randomly distributed fixed pattern noise is not modelled in TRM4.

TRM4 provides three options to specify fixed pattern noise:

- *No noise*,
- *Measured noise* (expressed via PRNU and DSNU), and
- *Estimated noise* (as fraction of the Noise Equivalent Signal (NES)).

PRNU

PRNU describes the RMS gain variation between individual detectors. Hence, it is of multiplicative nature. In TRM4, PRNU is specified as a percentage of the average signal.

DSNU

DSNU describes the RMS offset variations between individual detectors (dark current). It is specified in [A].

Note: For EMCCD and ICCD/ICMOS, TRM4 assumes that DSNU and PRNU values are measured for the gain settings of the imager as specified in the TRM4 project.

Note: In case of binning, specified DSNU and PRNU refer to binned detector pixels. If only values without binning are available, it is suggested to scale these values using the equations

$$\mathrm{PRNU}_{\mathrm{binned}} = \frac{\mathrm{PRNU}_{\mathrm{unbinned}}}{\sqrt{n_{\mathrm{bin_h}} \cdot n_{\mathrm{bin_v}}}}, \tag{2.15}$$

$$\mathrm{DSNU}_{\mathrm{binned}} = \mathrm{DSNU}_{\mathrm{unbinned}} \cdot \sqrt{n_{\mathrm{bin_h}} \cdot n_{\mathrm{bin_v}}}\,. \tag{2.16}$$

Measurement port

Depending on the type of imager, fixed pattern noise can be measured at several points. You must specify whether the measurement or estimation was made at the

- pre digital processing,
- output of the digital signal processing, or at the
- output of the video signal processing.

2.12.10 Temporal noise

Noise sources other than the detector itself, like read-out electronics, preamplifiers, etc., may contribute to the overall temporal noise. The two parameters *Noise floor* and *Noise figure* describe these additional noise contributions.

Noise floor and noise figure

If the additional noise is additive in nature, it is defined as *Noise floor*. The noise floor, N_{floor}, is assumed to be white and is characterized by the RMS number of electrons which constitute this noise. Multiplicative noise is specified by the *Noise figure* NF. The contributions of these parameters to the temporal noise are described in Section 3.8.1.

Note: The *Noise floor* refers to a single detector element. The overall noise floor of n detector elments in series (TDI) is assumed to be $\sqrt{n}$ times higher.

2.13 Detector/Sensitivity – *General* category

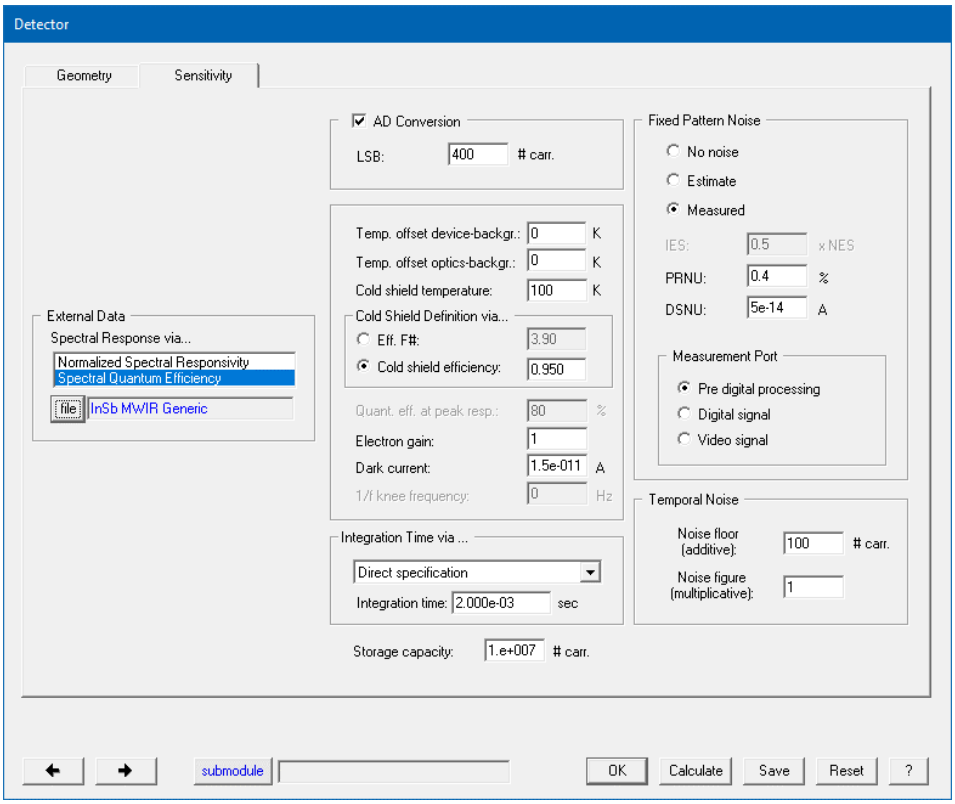


Figure 2.21: Detector sensitivity definition dialog for general systems.

2.13.1 External data: Spectral response via ...

The specification of either the *Normalized Spectral Responsivity*, $R_{\mathrm{DA_norm}}(\lambda)$, or of the *Spectral Quantum Efficiency*, $\eta(\lambda)$, is mandatory.

Note: As mentioned earlier, the data should include the spectral response of a cold filter if it is part of the device.

Note: Normalized spectral response and peak quantum efficiency must be consistent. If both are specified, TRM4 calculates the spectral quantum efficiency of the detector assembly. If the specification leads to quantum efficiencies greater than 100 % anywhere along the specified spectral band, the specification is rejected and an error message is displayed.

Note: *Spectral quantum efficiency* describes the intrinsic quantum efficiency of the active detector area and not and “effective” quantum efficiency taking into account the detector pitch and fill factor.

2.13.2 ADC resolution

The quantization noise, N_{quant}, associated with the analog-to-digital conversion process is calculated by specifying the number of carriers per *Least Significant Bit* (LSB). The quantization noise is then given by

$$N_{\mathrm{quant}} = \frac{\mathrm{LSB}}{\sqrt{12}}. \tag{2.17}$$

It is important to remember that in case of scanning devices with TDI, the LSB should be specified for the individual detector element and not for the signal of the integrated output after the TDI process.

2.13.3 Detector temporal noise calculation

Temperature offset device-background

If the cold shield efficiency (see Equation (2.20)) is smaller than one, energy emitted by the inner housing of the imaging device contributes to the background irradiance in the focal plane (see Section 3.7).

TRM4 assumes that the inner housing is a blackbody at a uniform temperature of T_{device}. Given the scene background temperature T_{scene} (specified in the *Target, Scenario and Task* dialog, see Section 2.24), the *Temperature offset device-background* is defined as

$$T_{\mathrm{off_dev}} = T_{\mathrm{device}} - T_{\mathrm{scene}}. \tag{2.18}$$

Temperature offset optics-background

If the transmission of optics and “warm” filter (specified in the *Optics* dialog) are not equal to one, the clear pupil and/or the filter contribute to the irradiance in the focal plane (Section 3.7).

TRM4 assumes that the lenses/mirrors that constitute the clear pupil and the “warm” filter are greybodies at a uniform temperature of T_{opt}. Given the scene background temperature T_{scene} (specified in the *Target, Scenario and Task* dialog, see Section 2.24), the *Temperature offset optics-background* is defined as

$$T_{\mathrm{off_opt}} = T_{\mathrm{opt}} - T_{\mathrm{scene}}. \tag{2.19}$$

Cold shield temperature

The cold shield also emits thermal radiation if its temperature is higher than 0 K. This radiation contributes to the background signal (and thus to detector noise, see Section 3.7).

Cold shield definition via ...

Either the *effective F-number* of the cold shield or the *cold shield efficiency* should be specified. This input is required to calculate the amount of radiation emitted by the inner housing of the

imager and sensed by the detector. Once one of the two parameters is specified, TRM4 calculates the other parameter and displays the result in the *Diagnostics* message box and in the *System Performance Data* message box.

The effective F-number of the cold shield and the cold shield efficiency, η_{cs}, are related by:

$$\eta_{\mathrm{cs}} = \frac{4F_{\mathrm{eff_cs}}^2 + 1}{4F_{\mathrm{eff_opt}}^2 + 1}. \tag{2.20}$$

If the cold shield efficiency is equal to one, the cold shield is perfectly matched to the optics aperture. If the cold shield efficiency is larger than one, the cold shield obscures part of the scene radiation. If the cold shield efficiency is smaller than one, energy emitted by the inner housing of the imaging device reaches the focal plane.

Note: TRM4 does not permit cold shield efficiencies >1. In such cases, a fatal error message is issued. The inconsistency must be corrected before resuming editing.

Note: For truncated optics or optics with a central obscuration, a cold shield efficiency larger than

$$\eta_{\mathrm{cs}} = \frac{4F_{\mathrm{opt}}^2 + 1}{4F_{\mathrm{eff_opt}}^2 + 1} \tag{2.21}$$

(where $F_{\mathrm{opt}} = f_{\mathrm{opt}}/D_{\mathrm{opt}}$) can be obtained only if a special radiation shield with a shape similar to the front optics (scaled) is used.

Quantum efficiency at peak responsivity

The *Quantum efficiency at peak responsivity* needs to be specified in combination with the *Normalized spectral responsivity*. It is not required if the spectral quantum efficiency is directly specified via external data. Quantum efficiency is specified in [%].

Note: This parameter describes the intrinsic quantum efficiency of the active detector area and not an “effective” quantum efficiency taking into account the detector pitch and the fill factor.

Electron Gain

Electron Gain may be used in order to enhance the signal from the scene and thus diminish the effect of read-out noise on the signal-to-noise ratio. Electron gain mechanisms may be associated with additional multiplicative and/or additive noise, which can be specified via the parameters *Noise floor* and *Noise figure*, see Section 2.13.7.

Note: In practice, electron gain amplifies both the detector signal and the detector dark current. As a result, electron gain does not reduce the effect that the dark current has on the overall noise and the SNR.

Dark current

This parameter refers to a single detector element. The dark current has to be specified in [A]. Dark current refers to the input of the detector (this is important to keep in mind in case of electron gain mechanisms).

2.13.4 Integration time via ...

The integration time is used for the calculation of signal and noise. It is also required for the calculation of the MTF, which might include image smear[6].

Three options exist for specifying the detector integration time in a 2nd generation imager:

- *Direct specification*,
- via *Integration duty cycle*,
- via *Well-fill*.

The *Integration duty cycle*, η_{cyc}, is the actual integration time, t_{int}, divided by the maximum theoretically available integration time (see Section 3.5.7).

The *Well-fill* option (= % of the storage capacity utilized at the specified scenario) can be selected only if the Storage capacity is specified.

Note: If the *Well-fill* option is specified, TRM4 calculates the integration time that is necessary for the background signal to fill the storage gate (well) up to the specified value. The calculation is done for each considered target distance and for all specified atmospheres. If the calculated integration times exceed the maximum possible integration time (corresponding to a duty cycle of unity), TRM4 uses the maximum possible integration time and displays a warning message to inform the user.

Note: TRM4 calculates the integration time if the integration duty cycle is specified or the duty cycle if the integration time is specified. Calculated data is listed in the *Diagnostics* and *System Performance Data* message boxes.

2.13.5 Storage capacity

The *Storage capacity* is the number of carriers, which can be stored on the focal plane per detector element. A storage capacity value of zero means “not specified”.

Note: The storage capacity to be specified in TRM4 refers to a single detector element. In case of scanning devices with TDI, the manufacturer usually specifies the storage capacity with respect to the integrated detector signal (“channel”).

[6] Image smear occurs if the image is moving on the detector focal plane during signal integration. This is the case of a scanning or micro-scanned imager for which *Integration during movement* is specified (in the *Basic Imager Data* dialog).

2.13.6 Fixed pattern noise

Fixed pattern noise is characterized by two parameters:

- PRNU (Photo Response Non-Uniformity) and
- DSNU (Dark Signal Non-Uniformity).

PRNU and DSNU specify the contribution of randomly distributed, white fixed pattern noise to the overall system noise. Non-randomly distributed fixed pattern noise is not modelled.

Three options are available to characterize fixed pattern noise:

- *No noise*,
- *Measured noise* (expressed via PRNU and DSNU), and
- *Estimated noise* (as fraction of the Noise Equivalent Signal (NES)).

PRNU

PRNU describes the RMS gain variation between individual detectors. Hence, it is of multiplicative nature. In TRM4 PRNU is specified as a percentage of the average signal.

DSNU

DSNU describes the RMS offset variations between individual detectors (dark current). It is specified in [A].

Measurement port

Depending on the type of imager, fixed pattern noise can be measured at several points. You must specify whether the measurement or estimation was made at the

- pre digital processing,
- output of the digital signal processing, or at the
- output of the video signal processing.

2.13.7 Temporal noise

Noise sources other than the detector itself, like read-out electronics, preamplifiers, etc., may contribute to the overall temporal noise. The two parameters *Noise floor* and *Noise figure* describe these additional noise contributions.

Noise floor and noise figure

If the additional noise is additive in nature, it is defined as *Noise floor*. The *Noise floor*, N_{floor}, is assumed to be white and is characterized by the RMS number of electrons which constitute this noise. Multiplicative noise is specified by the *Noise figure* NF. The contributions of these parameters to the temporal noise are described in Section 3.8.1.

Note: The *Noise floor* refers to a single detector element. The overall noise floor of n detector elements in series (TDI) is assumed to be $\sqrt{n}$ times higher.

2.13.8 $1/f$ knee frequency

This parameter is relevant for scanning imagers only. It is used to specify any low frequency $1/f$ noise contribution to the white noise power spectrum (see Section 3.9). If no $1/f$ noise is present, the parameter value must be set to zero.

2.14 Analog signal processing

In TRM4, the term “analog signal processing” refers to signal processing applied to individual detector signals (or channels after the TDI process), before multiplexing or sampling. This type of signal processing is performed only in 1st generation imagers where the analog detector signals are read out directly and each detector/channel signal is amplified/processed individually in parallel. One or more of the filters described below may be specified.

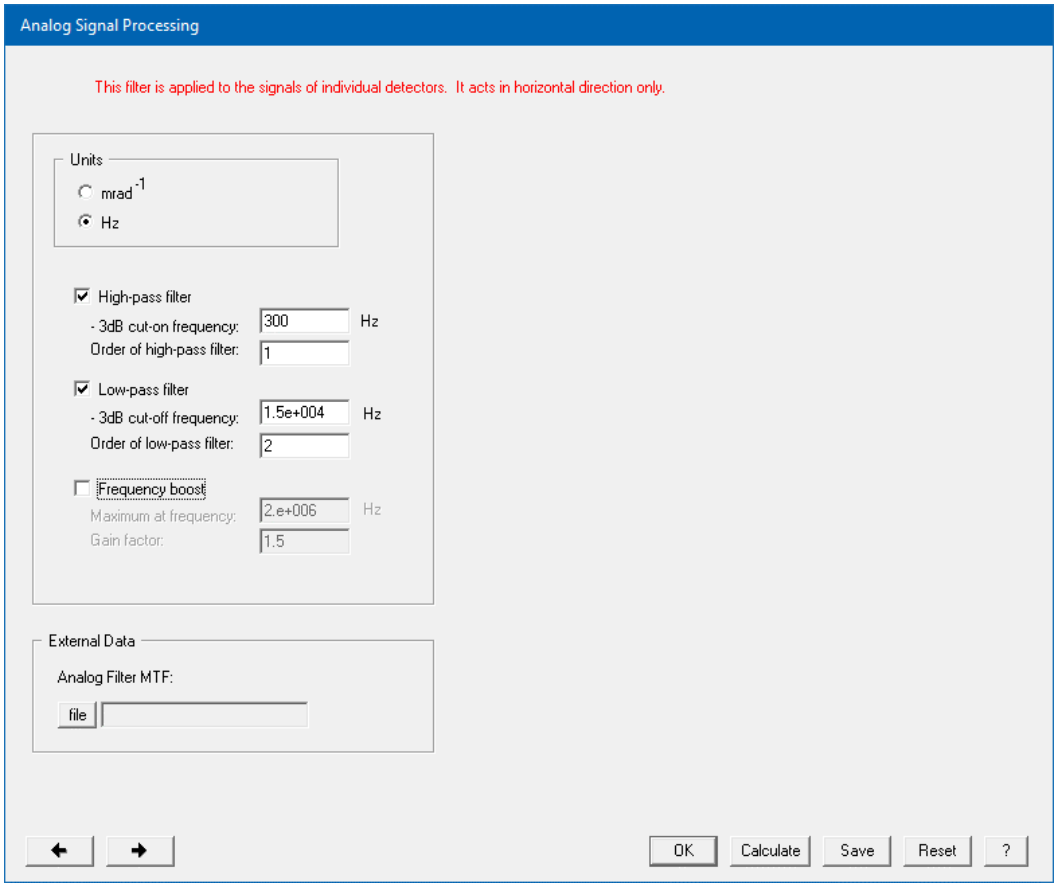


Figure 2.22: Analog signal processing dialog (only available for 1st generation thermal systems).

2.14.1 High-pass filter

This is a traditional AC coupling filter, which is only scarcely used in modern imagers. It affects very low frequencies only. TRM4 assumes that the high-pass filter is a Butterworth filter. It is characterized by its *-3dB cut-on frequency* and its *Order*. It is used in noise calculations only (and not in signal MTF calculations).

2.14.2 Low-pass filter

A low-pass filter is used for eliminating high frequency noise and minimizing aliasing effects in sampled imagers. TRM4 assumes that the low-pass filter is a Butterworth filter. It is characterized by its *-3dB cut-off frequency* and its *Order*.

2.14.3 High frequency boost

High frequency boost is used in order to enhance the high frequency information, attenuated by the optics and detector MTFs. TRM4 assumes that this filter has an MTF value of unity at zero spatial frequency. It is characterized by the frequency at which it reaches its maximum response *(Maximum at frequency)* and by the *Gain factor* at this frequency.

> **Note:** It must be emphasized that this is a pre-sampling filter that is only scarcely used in modern imagers. For the same reason, it should not be confused with post-sampling boost (usually digital), which is sometimes used in modern imagers.

2.14.4 External data: Analog filter MTF

The external analog filter MTF is an optional input. If specified, the other input parameters in this dialog become inactive.

2.15 Digital signal processing

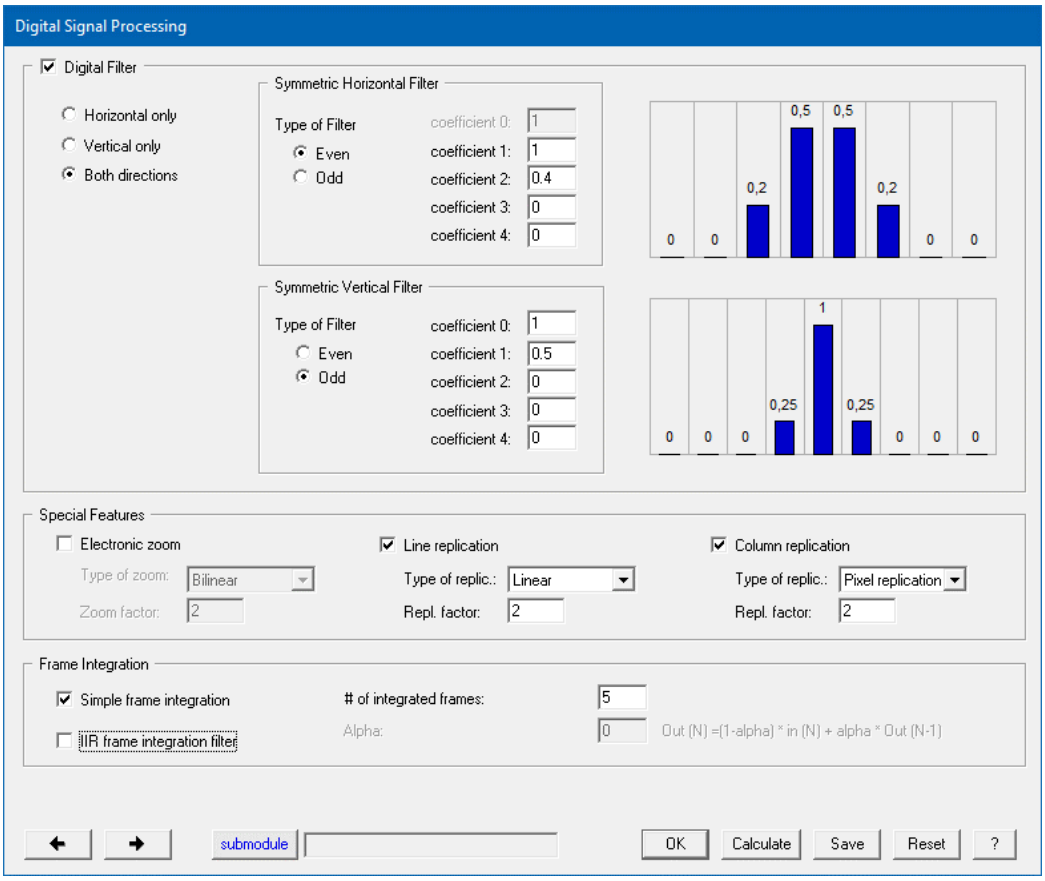


Figure 2.23: Digital signal processing dialog.

The *Digital Signal Processing* dialog (Figure 2.23) is the same for all imager categories and is active for all imager types, except 1st generation thermal imagers without sampling.

The dialog may also be used to specify imagers with sampled analog signal processing. In TRM4, analog sampled imagers with analog memories and analog finite impulse response (FIR) filters are treated as imagers with digital signal processing.

Pixel distances are assumed to be equal to the sampling distances.

> **Note:** All digital filters in TRM4 are one-dimensional filters with one-dimensional kernels. Filter operations are done separately for each direction.

2.15.1 Digital filters

TRM4 can model linear phase, symmetrical odd or even FIR filters in one or both directions.

Filter coefficients

A maximum number of five filter coefficients can be specified. The relation between filter coefficients $c_{\mathrm{dig},n}$ (input parameters) and impulse response coefficients, h_n, is given by

$$h_0 = c_{\mathrm{dig},0} \text{ and } h_{\pm n} = \frac{1}{2} c_{\mathrm{dig},n} \text{ for } n = 1, \dots, 4. \tag{2.22}$$

TRM4 graphically displays the corresponding impulse response, i.e. the filter kernel, during the coefficient specification process. The sum over the specified filter coefficients $c_{\mathrm{dig},n}$ has to be positive in order for the corresponding MTFs to be well-defined.

2.15.2 Special features

Electronic zoom

Electronic zoom magnifies the displayed image. Zoom acts in both horizontal and vertical directions simultaneously and is characterized by two parameters:

- *Type of zoom* and
- *Zoom factor.*

Two different types of zoom can be specified:

- *Bilinear interpolation* (*bilinear* here simply indicates that the process is applied both for the horizontal and vertical directions), or
- *Pixel replication*.

For *Bilinear interpolation*, the signals of additional pixels in vertical and horizontal directions are calculated by linearly interpolating between the neighboring vertical and horizontal pixels.

> **Note:** TRM4 requires that all specified imager parameters are compatible with the selected zoom, line replication, and column replication factors. This should be kept in mind when changing these factors.

Line replication

Line replication acts only in the vertical direction. It is sometimes used to avoid geometric distortions in the displayed image. Line replication may become necessary if the number of samples in the vertical direction is not compatible with the video standard or the display.

Line replication is characterized by two parameters:

- *Type of replication* and
- *Replication factor.*

Two different types of line replication can be specified: Either *linear interpolation* or *pixel replication*. In case of linear interpolation, the signals of additional vertical pixels are calculated by linearly interpolating between the neighboring vertical pixels.

Column replication

Column replication acts only in the horizontal direction. It is sometimes used to avoid geometric distortions in the displayed image. Column replication may become necessary if the number of samples in the horizontal direction relative to the number of samples in the vertical direction is not compatible with the video standard or the display.

Column replication is characterized by two parameters:

- *Type of replication* and
- *Replication factor.*

Two different types of column replication can be specified: Either *linear interpolation* or *pixel replication*. In case of linear interpolation, the signals of additional horizontal pixels are calculated by linearly interpolating between the neighboring horizontal pixels.

2.15.3 Frame integration

In many devices, frame integration is used to enhance the signal-to-noise ratio in the displayed image. Frame integration may also be used to convert imager frame rate to the (standard) video frame rate. In TRM4, two frame integration algorithms can be specified and may be applied simultaneously:

- *Simple frame integration* and
- *IIR (infinite impulse response) frame integration filter.*

Number of integrated frames

This parameter needs to be specified if the *Simple frame integration* option is selected.

Alpha

The alpha value must be specified if the *IIR Frame integration filter* option is selected. With this filter the output signal is given by

$$S_{\text{out}}(N) = (1-\alpha)\cdot S_{\text{in}}(N) + \alpha \cdot S_{\text{out}}(N-1), \tag{2.23}$$

where N is the current, and $N-1$ the previous frame.

2.16 Analog-Digital converter

This dialog is enabled only if a 1st generation thermal imager with horizontal sampling is specified.

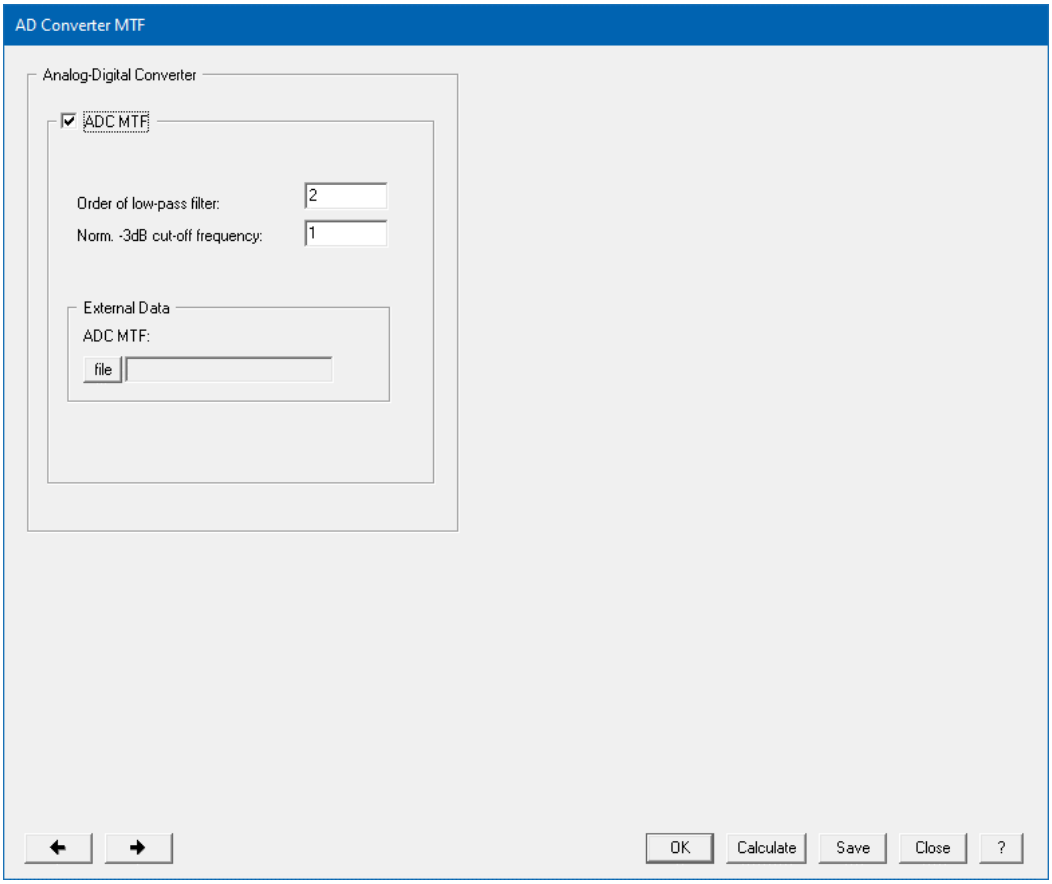


Figure 2.24: Analog-Digital converter dialog (only available for 1st generation thermal systems with horizontal sampling).

2.16.1 Analog-Digital converter

Analog-Digital converters (ADCs) contain a low-pass filter, which acts on the signal before sampling. The filter is used to minimize unwanted aliasing effects in the sampled signal.

TRM4 assumes the detector signals to be sampled in horizontal direction. Consequently, the ADC low-pass filter contributes to the horizontal prefilter MTF. The filter is assumed to be a low-pass Butterworth filter of nth order.

The filter is characterized by two parameters:

- *Order of the filter* and
- *Normalized cut-off frequency.*

The cut-off frequency should be specified in units normalized with respect to the sampling frequency.

2.16.2 External data: ADC MTF

The externally specified low-pass filter ADC MTF is an optional input. The external specification replaces the internal MTF calculation. If specified, other input parameters in this dialog become inactive.

2.17 Video signal processing

This dialog is only relevant and enabled if a cathode ray tube (CRT) display is specified. CRT displays require an analog video signal which frequently complies with a commercial standard format.

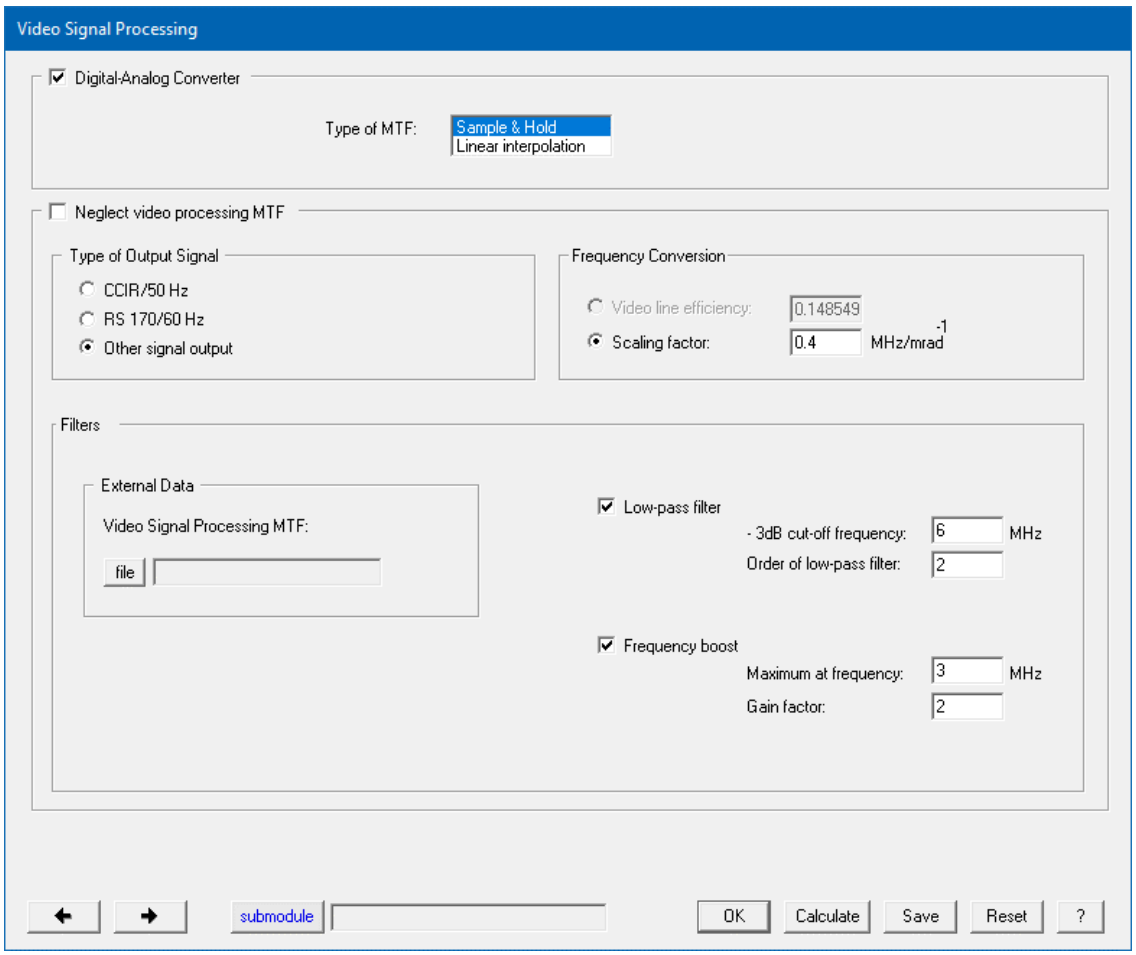


Figure 2.25: Video signal processing dialog (only available for CRT monitors).

2.17.1 Digital-Analog converter

The Digital-Analog (DA) converter converts the digital data into a serial analog signal. TRM4 permits the specification of two types of DA converters with different signal reconstruction functions:

- rectangular reconstruction (*Sample & Hold* technique) and
- triangular reconstruction (*Linear interpolation* between subsequent digital data).

The signal reconstruction corresponds to MTFs of type:

- sinc (for rectangular reconstruction) and
- sinc^2 (for triangular reconstruction).

The pixel distances are given by the sampling distances, divided by the zoom and/or line/column replication factors.

2.17.2 Type of output signal

TRM4 can model both standard video formats and user-defined signal formats:

- European TV standard *CCIR/50 Hz,*
- American TV standard *RS-170/60 Hz,* and
- *Other signal output* (user-defined).

If the CCIR or RS-170 options are selected, TRM4 assumes the image to be displayed on a standard monitor with a 4:3 aspect ratio.

2.17.3 Frequency conversion

The video frequency data in this dialog has to be specified in temporal frequency units ([MHz]), except for the video MTF defined via *External Data*.

TRM4 offers two alternatives for the conversion of temporal frequency values into spatial frequencies:

- *Video line efficiency* and
- *Scaling factor.*

Video line efficiency

Many imagers use part of the video signal for the presentation of auxiliary information. The *video line efficiency* is the time required to display one line of the horizontal FOV (pure image data), $t_{\mathrm{line_FOV}}$, divided by the active line time, $t_{\mathrm{line_active}}$:

$$\eta_{\mathrm{video}} = \frac{t_{\mathrm{line_FOV}}}{t_{\mathrm{line_active}}}. \tag{2.24}$$

Video line efficiency can be specified only if CCIR or RS-170 standard video is selected. The active line time for CCIR signals is 51.7 μs and for RS-170 signals 52.09 μs.

Scaling factor

The scaling factor is given by

$$\mathrm{scf} = \frac{\nu_{\mathrm{video}}}{r}, \tag{2.25}$$

where ν_{video} is the temporal video frequency in [MHz] and r is the corresponding spatial frequency in $[\mathrm{mrad}^{-1}]$, referring to object space. This is a conversion factor from the units used in the dialog [MHz] to the units in object space used by TRM $[\mathrm{mrad}^{-1}]$.

2.17.4 Filters

Two types of filters can be specified:

- *Low-pass* and
- *Frequency boost.*

Low-pass filter

The *low-pass filter* is used to eliminate high frequency noise. The filter is assumed to be a Butterworth filter, characterized by its *-3dB cut-off frequency* and by its *Order*.

High frequency boost

A high frequency boost is used to enhance the medium and high frequency content in the signal. TRM4 assumes that this filter has an MTF value of unity at zero spatial frequency. The boost is characterized by the frequency at which it reaches its maximum response *(Maximum at frequency)* and by the *Gain factor* at this frequency.

2.17.5 External data: Video signal processing MTF

This is an optional input to specify the MTF of filters used in the video signal processing. The external specification replaces the internal MTF calculation. Hence, if specified, the corresponding filter input parameters in this dialog become inactive.

Notice: In case the units used to specify the Video MTF frequencies are temporal (Hz, kHz, MHz), the *Type of Output Signal* group is active. If spatial units are selected $[\mathrm{mrad}^{-1}]$ it is deactivated.

2.18 Display and observation

This dialog is almost the same for all imager categories.

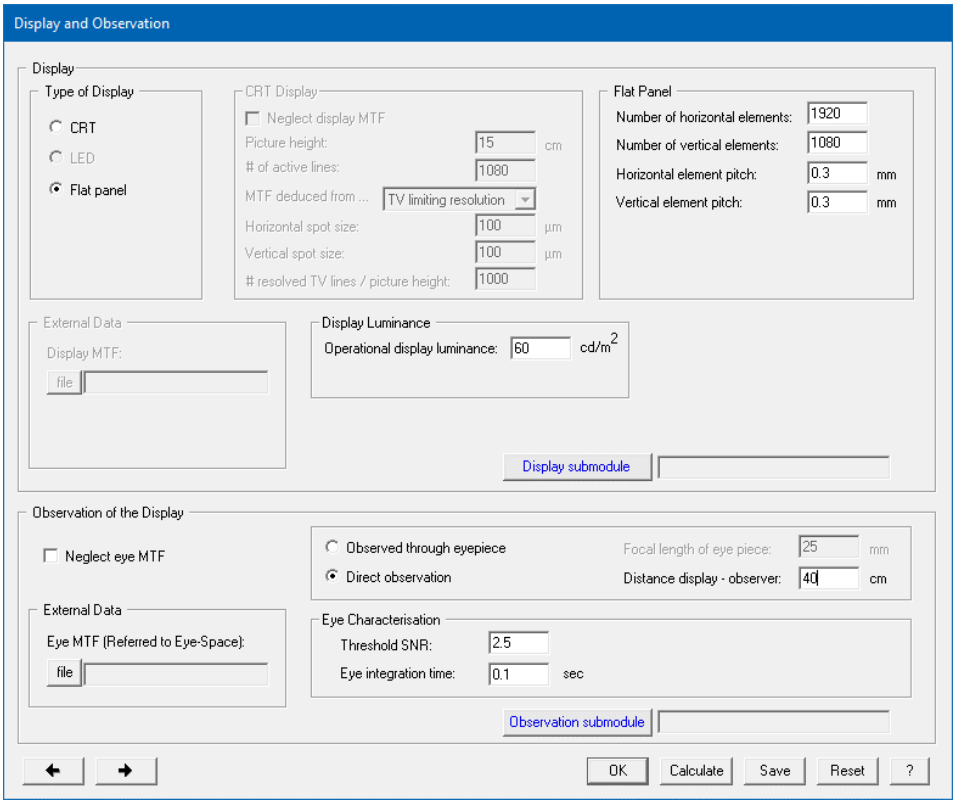


Figure 2.26: Definition of display and properties of (spatial and temporal) eye integration.

2.18.1 Type of display

Three different options are available:

- *CRT* display,
- *LED* display, and
- *Flat panel* display.

CRTs can be specified for all types of imagers. The *LED* display option is only available for 1st generation thermal imagers without horizontal sampling and the *Flat panel* display is available for all except 1st generation thermal imagers without horizontal sampling

TRM4 models the flat panel displays to be directly linked to the digital output of the imager (no video processing). It is further assumed that its display elements have a fill factor of 100 %.

Note: Do not mistake the LED here for a modern Flat panel LED display. The LED display specified herein is a simple device with few LEDs corresponding to the detector channels where the signals from the detectors were fed directly to the LEDs and a scanning mechanism was used to reproduce the image and display it to the observer. Modern Flat panel LEDs are represented by the *Flat panel* option.

2.18.2 CRT display

Picture height

The *Picture height* is the height of the image when all active lines of the CRT display are used to display the scene (in other words, this is the maximal height that may be used to display an image).

> **Note:** If CCIR or RS-170 video standards are specified TRM4 assumes the display to have a 4:3 aspect ratio. Sometimes, however, a commercial TV monitor is used where the active lines are squeezed in vertical direction to obtain an aspect ratio different from 4:3.

> **Note:** Some imagers use only the central part of the display to present the image and the upper and/or lower parts of the display are used to present auxiliary data like, e.g., information about the FOV, zoom factor, etc.

Number of active lines

The *number of active lines* is the total number of lines in the display. It should not be confused with the number of lines required to display the image. Typically, the number of active lines is 576 in CCIR compatible and 485 in RS-170 compatible displays.

> **Note:** TRM4 calculates the number of video lines required to display the image and its actual height on the monitor. The actual height is used to calculate CRT and observer MTFs.

> **Note:** TRM4 uses the *Picture height* and the *number of active lines* to calculate the displayed vertical image height and the displacement between subsequent image lines.

MTF deduced from ...

There are two options for specifying the MTF of a CRT display:

- *Spot size* and
- *TV limiting resolution.*

Spot size

The *Spot size* is defined as half the distance between the points at which the point spread function (PSF) reaches $\exp(-1/2)$ of its maximum value. Since the PSF is assumed to have a Gaussian beam profile

$$\mathrm{PSF}_{\mathrm{CRT}}(x) \propto \exp\left(-\frac{x^2}{2{\sigma_{\mathrm{CRT}}}^2}\right) \tag{2.26}$$

the half width corresponds to the standard deviation σ_{CRT}. Different spot sizes may be specified for the vertical and horizontal directions.

Number of resolved TV lines

The *Number of resolved TV lines per picture height* is a wide-spread method for the specification of CRTs display resolution. It is defined as the maximal number of lines that can be resolved in the vertical direction.

At the resolution limit, the modulation between the lines corresponds to an MTF of approximately 2.8 %. If $N_{\mathrm{disp_res}}$ is the number of resolved TV lines per picture height, H_{picture}, the spot size σ_{CRT} is then given by

$$\sigma_{\mathrm{CRT}} = \frac{H_{\mathrm{picture}}}{1.18 \cdot N_{\mathrm{disp_res}}}. \tag{2.27}$$

2.18.3 LED display

The *LED* display option is available for 1st generation thermal imagers without sampling only.

Horizontal/vertical element dimensions

Display elements are assumed to have a rectangular shape. The horizontal and vertical (active) dimensions of the elements must be specified.

Vertical element pitch

The vertical distance between two displayed lines is to be specified here.

2.18.4 Flat panel display

Number of horizontal/vertical elements

The flat panel should have sufficient elements to display all image pixels. Otherwise, TRM4 issues a fatal error message after closing the project specification session.

Horizontal/vertical element pitch

This is the distance between the centers of adjacent display elements. Since TRM4 assumes a fill factor of 100 %, element pitches and element dimensions are identical.

2.18.5 Display luminance

This is the luminance of the display at background signal. The *Operational display luminance* must be specified for eye MTF calculations (see Section 4.2.12). The parameter is disabled if the eye MTF is specified by external data, or if the eye MTF is neglected.

2.18.6 External data: Display MTF

This is an optional input for LED and CRT displays. The display MTF may be specified for one or both directions. The external specification replaces the internal MTF calculation. If an external display MTF is specified, the corresponding input parameters in the dialog become inactive.

2.18.7 Observation of the display

TRM offers two alternatives to observe the display:

- *Direct observation* and
- *Observation through eyepiece.*

Focal length of the eyepiece

This parameter needs to be specified if the display is observed through an eyepiece. From it, the size of the image as perceived through the eyepiece is derived. This parameter is disabled if the eye MTF is neglected.

Distance display-observer

If the display is watched directly, the distance between observer and display needs to be specified in order to scale the eye MTF. This parameter is disabled if the *Neglect eye MTF* option is specified.

2.18.8 Eye characterization

Threshold SNR

The threshold signal-to-noise ratio must be specified (see also Section 3.11.2). A value of 2.5 is recommended.

Eye integration time

The model assumes that the eye integrates over subsequent frames. An eye integration time of 0.1 sec is recommended.

2.18.9 External data: Eye MTF

This is an optional input. The external specification replaces the internal eye MTF calculation. The data is assumed to be valid for both the horizontal and the vertical direction.

> **Note:** The spatial frequency scale of external eye MTF data should refer to eye space and not to object space.

2.19 Stabilization

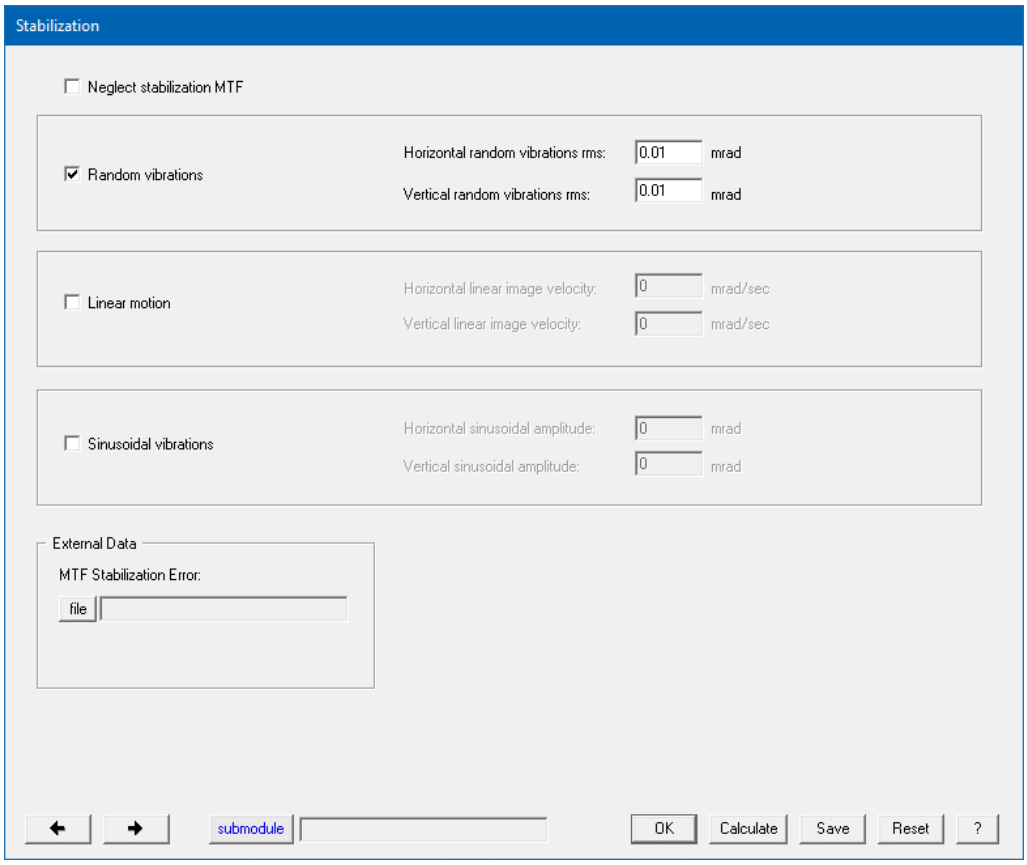


Figure 2.27: Definition of residual imager motion.

2.19.1 General comments

Stabilization errors are usually caused by mechanical vibrations and/or drift of a platform. The imager itself may also have inherent stabilization errors (e.g., vibration of the detector array caused by the cooler, etc.).

Stabilization errors lead to MTF degradation and/or to distorted images. Only the first effect, MTF degradations, is modelled in TRM4. The modelling is basic and special care must thus be taken in specifying stabilization errors.

When specifying stabilization errors, their temporal frequency spectrum has to be considered. Low frequency stabilization errors, e.g., usually result in a drift which often has no degrading effect on image perception. In some cases, it may be helpful to split vibrations/movements into their low- and high-frequency components and to model each component independently.

Different types of stabilization errors can be modelled:

- *Random vibrations*,
- *Sinusoidal vibrations*, and
- *Linear motion*.

TRM4 allows for all types of errors to be present simultaneously.

2.19.2 Types of stabilization errors

Random vibrations

High frequency random platform vibrations are assumed to have a Gaussian amplitude distribution with zero mean. The *random vibrations rms value* is defined as the value of the amplitude where the probability density function drops to exp(-1/2) of its maximum value.

Linear motion

The *image velocity* describes the linear movement of the imager platform. This type of error can only be specified accurately for staring imagers and for single frames. In scanning imagers, the MTF degradation due to a linear motion during a single scan is usually negligible and is not modelled in TRM4.

The MTF degradation due to a linear motion might be significant, however, if frame integration algorithms and/or interlace/micro-scan mechanisms are used in the imager. This case is treated by TRM4.

It is further assumed that the eye does not contribute to the smearing of targets due to the eye integration time.

Sinusoidal vibrations

The *sinusoidal amplitude* describes the amplitude of a sinusoidal vibration of the platform.

2.19.3 External data: MTF stabilization error

This is an optional input to specify the stabilization error MTF. The external specification replaces the internal MTF calculation. It may be specified for one or both directions. If specified, the corresponding input parameters in this dialog become inactive.

2.20 Additional imager MTFs

This dialog is the same for all imager categories. It permits the specification of additional MTFs that cannot be attributed to one of the components already described in the previous sections. Additional vertical and horizontal MTFs can be specified for both the prefilter and for the postfilter.

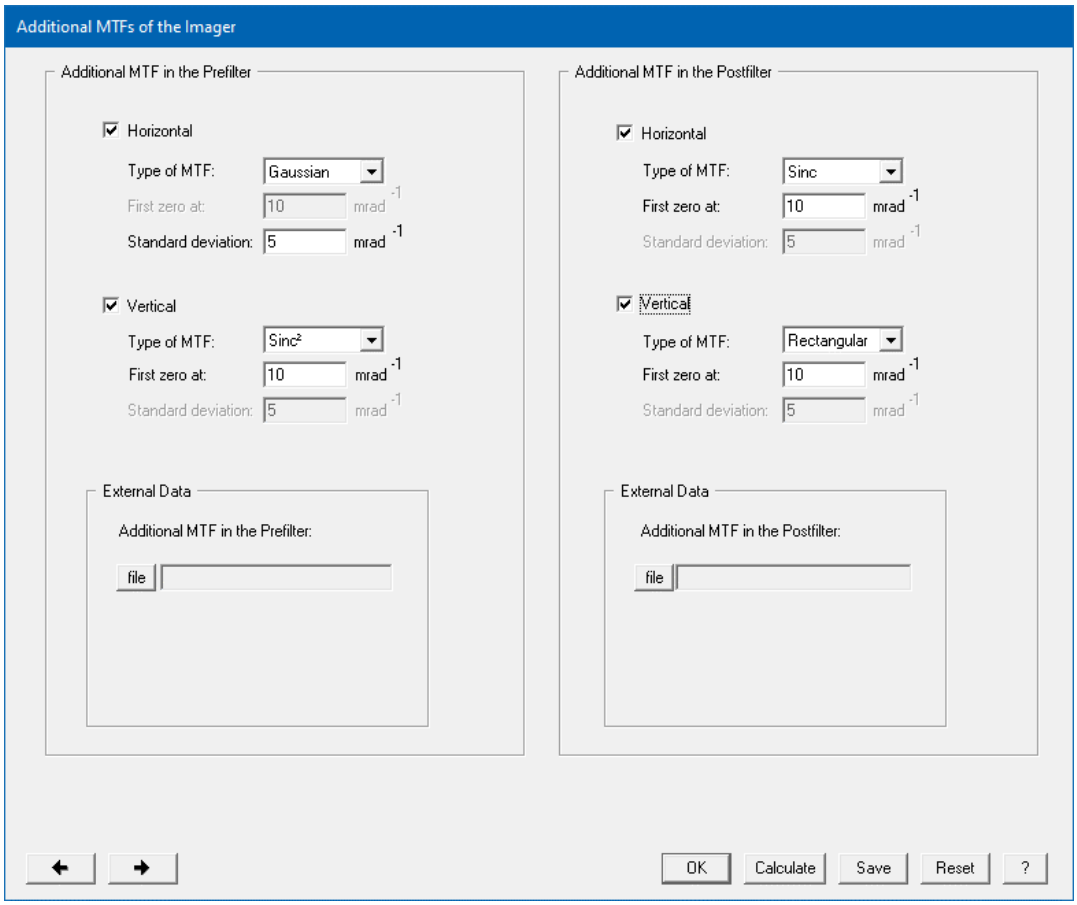


Figure 2.28: Definition of additional MTFs for prefilter and postfilter calculations.

2.20.1 Type of MTFs

The MTF can be calculated internally if it belongs to one of the four following types:

- sinc,
- sinc^2,
- *Gaussian*, and
- *Rectangular.*

2.20.2 External data: Additional MTFs

External MTF data may also be added to both the pre- and postfilter MTFs.

Note: Externally specified MTFs do not replace the internal MTFs. They may be used simultaneously with internally calculated additional filters.

2.21 Prefilter MTF or MTDP – *Thermal* category

This dialog is available for imagers of category *Thermal* only.

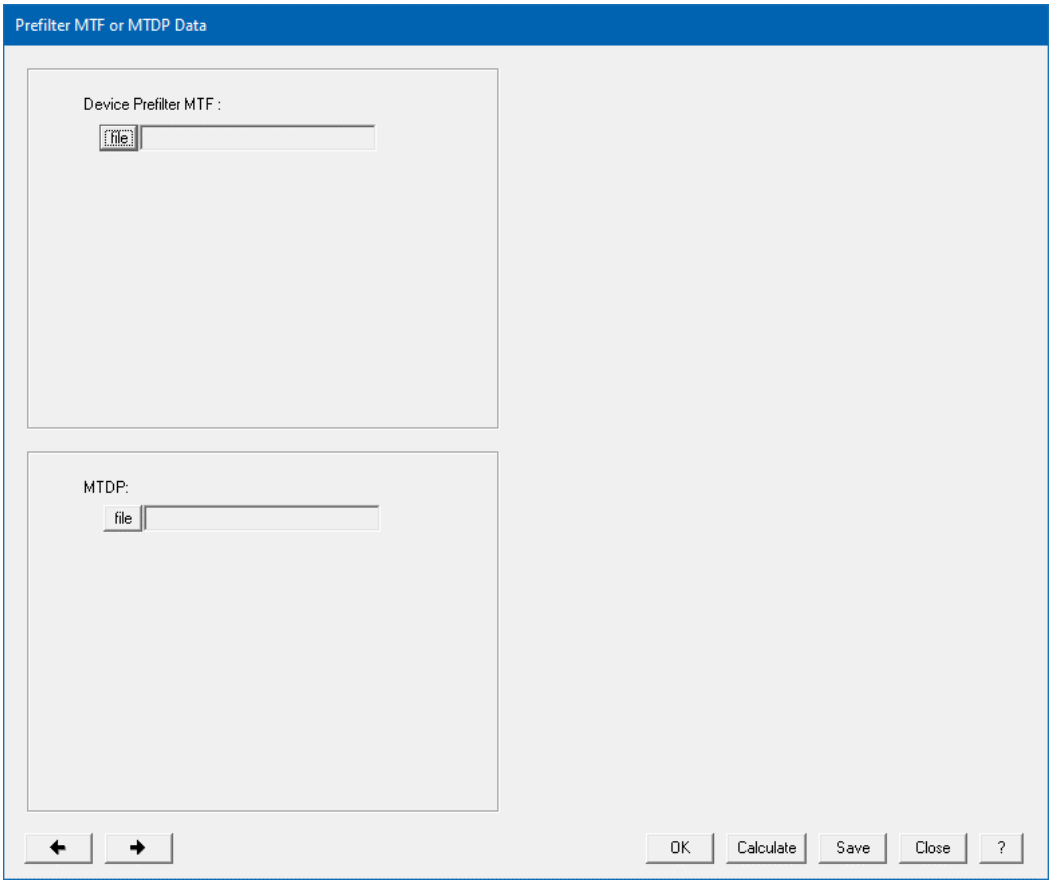


Figure 2.29: For thermal imagers, the MTDP can be defined explicitly.

2.21.1 Device prefilter MTF

The optional external device prefilter MTF must be specified in both directions. If specified, the external prefilter MTF replaces all MTFs that contribute to the device prefilter (optics, detector, etc.) specified elsewhere.

Note: The stabilization MTF is not considered to be part of the device prefilter MTF and can still be specified independently.

2.21.2 MTDP

If MTDP (Minimum Temperature Difference Perceived) data is specified in this dialog, all imager dialogs are disabled. The MTDP needs to be specified for both directions.

2.22 Prefilter MTF or MCP – *Visible* category

This dialog is available for imagers of category *Visible* only.

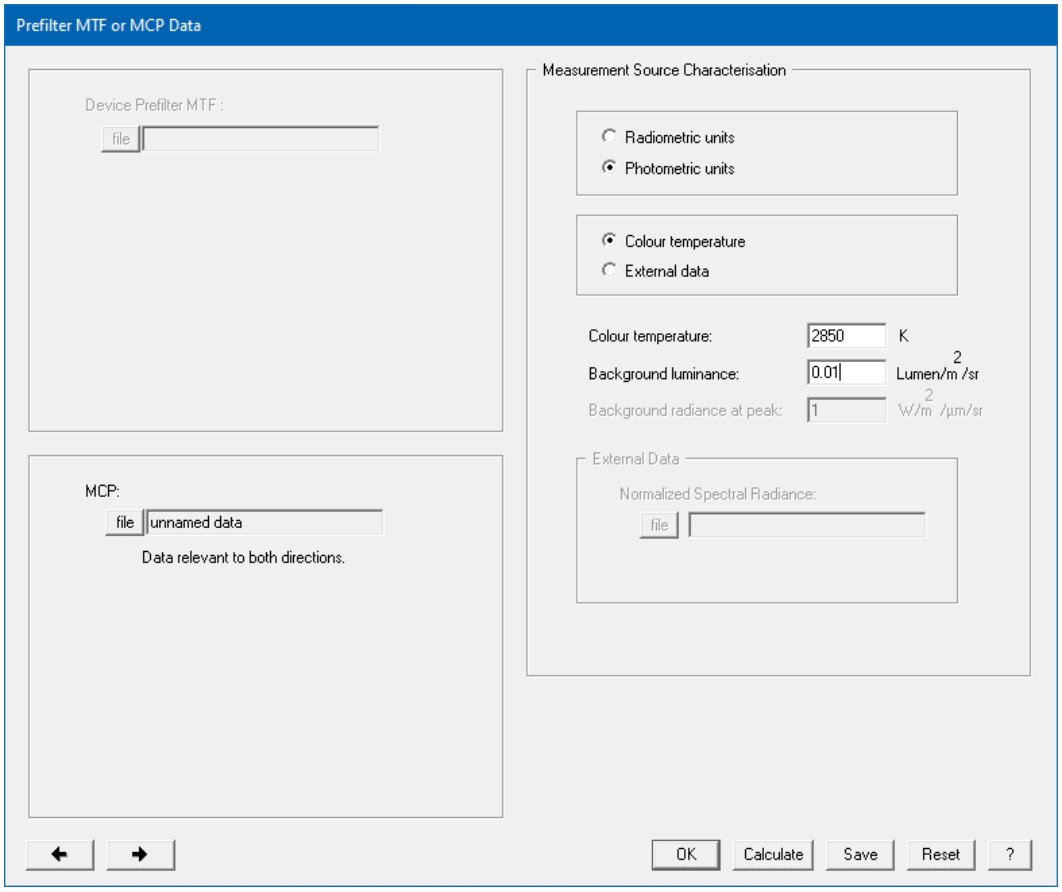


Figure 2.30: For visible imagers, measured device prefilter MTF or MCP (Minimum Contrast Perceived) data can be defined explicitly.

2.22.1 Device prefilter MTF

The optional external device prefilter MTF must be specified in both directions. If specified, the external prefilter MTF replaces all MTFs that contribute to the device prefilter (optics, detector, etc.) specified elsewhere.

> **Note:** The stabilization MTF and air turbulence MTF are not considered to be part of the device prefilter MTF and can still be specified independently.

2.22.2 MCP

MCP (Minimum Contrast Perceived) needs to be specified for both directions. If MCP data is specified in this dialog, all imager dialogs are disabled except for *Optics*, *Detector/Sensitivity* or *Image Intensifier*.

> **Note:** While TRM4 uses MCP as device figure of merit for *Visible/NIR/SWIR* imagers, the user might insert measured MRC values as well if range performance calculation should be based on common MRC measurements [10].

2.22.3 Measurement source characterization

The illumination source used for the MCP measurement can be specified here. The data is used to compare the illumination of the MCP measurement with the illumination level of the corresponding scenario. The result is given in the *Effective Laboratory Background Radiance* output window. With the help of this output window the user can work out whether the illumination level in the scenario for the calculated range agrees with the light level in the measurement. Note that MCP may heavily depend on the background illumination level.

Preferably, radiometric units should be used to characterize illumination sources. For sensors operating in the visible spectral band, however, photometric units are more commonly available. TRM4 offers the following alternatives for measurement source characterization:

- *Radiometric units,* or
- *Photometric units*.

With respect to the specification of the source spectrum, TRM4 offers the following options:

- *Colour temperature*, and
- *External data.*

Colour temperature and *External data* options are available for specifying the spectral features of broadband sources.

Colour temperature

This parameter needs to be specified if the *Blackbody* or the *Broadband/Colour temperature* options are selected to characterize the spectral behavior of the source. TRM4 uses the *Colour temperature* to derive the source spectrum from Planck's Law.

Background luminance

This parameter needs to be specified if a *Broadband* source is selected, and *Radiometric* units are used to characterize the source. The *Irradiance at peak* refers to the wavelength where the spectral irradiance is maximal.

Background radiance at peak

This parameter describes the intensity per steradian of a *Broadband Point source* illuminator that is *Fixed to the imager* in *Radiometric* units.

External data – Normalized spectral radiance

This option allows for the specification of spectral irradiation sources other than blackbodies

2.23 Prefilter MTF – *General* category

This dialog is available for the imager category *General* only.

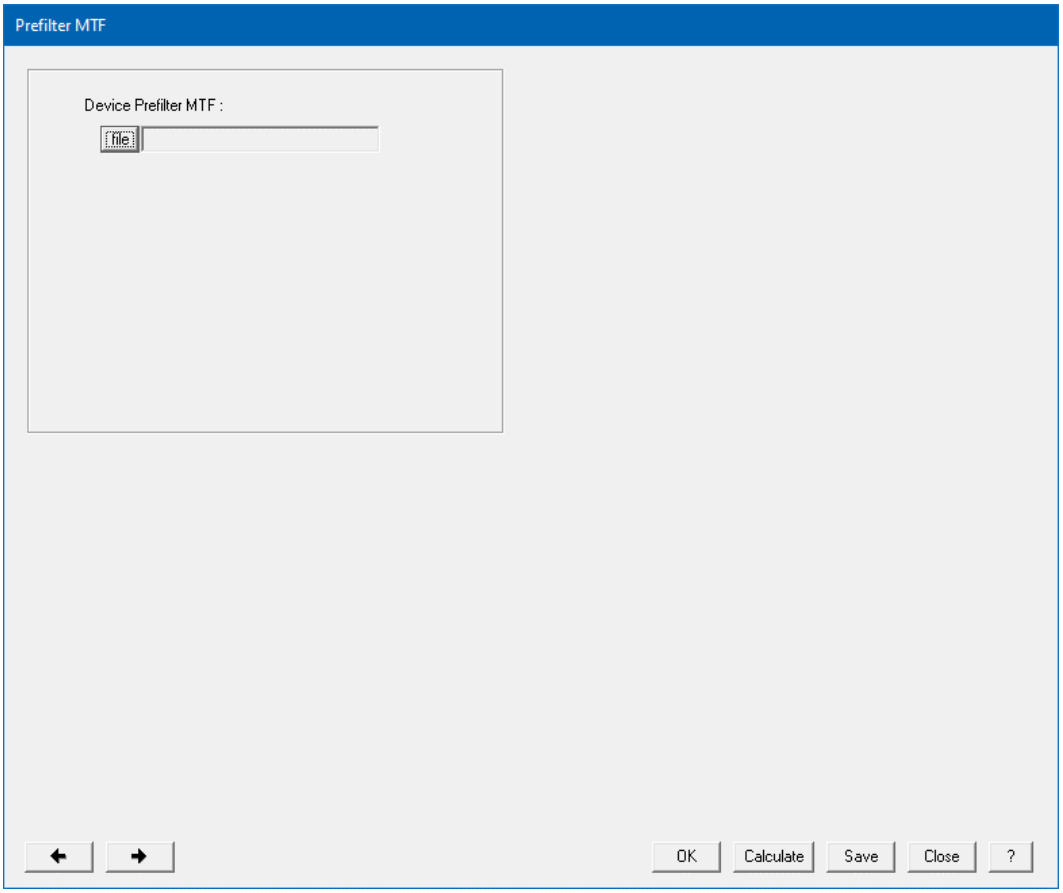


Figure 2.31: Measured device prefilter MTFs can be included in this dialog.

The optional external device prefilter MTF must be specified in both directions. If supplied, the external device prefilter MTF replaces all MTFs that contribute to the device prefilter (optics, detector, etc.) specified elsewhere.

Note: The stabilization MTF and the air turbulence MTF are not considered to be part of the device prefilter MTF and can still be specified independently.

2.24 Target, scenario and task – *General* and *Thermal* categories

There are some minor differences in the dialogs between the two imager categories *Thermal* and *General*. They will be addressed in the text whenever necessary.

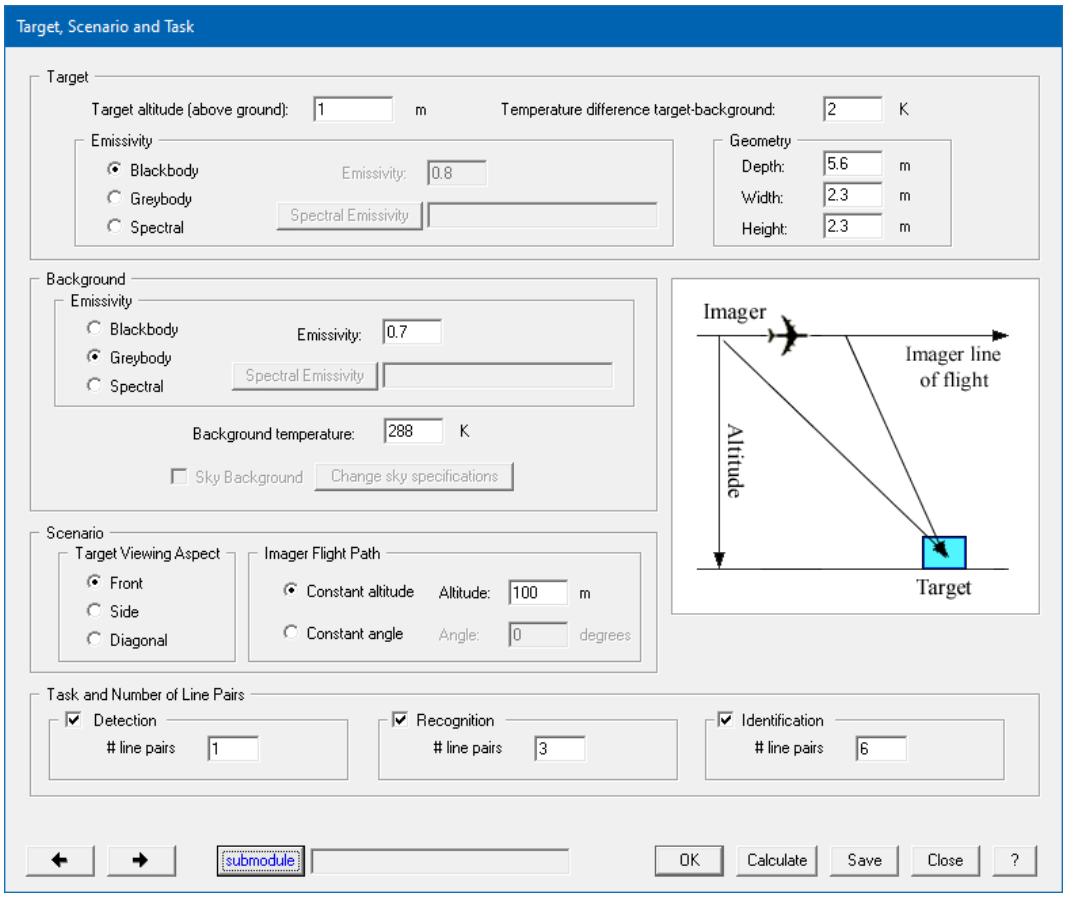


Figure 2.32: *Target, Scenario, and Task* dialog for category *General.*

2.24.1 Target

TRM4 provides several features for user defined targets.

Target altitude (above ground)

This input parameter is used to calculate the turbulence MTF along the line-of-sight in case that C_n^2 is defined by a simplified height dependence (Section 2.28.2). For category *Thermal*, it is used for reference purposes only and does not affect range calculation.

Temperature difference target background

Positive as well as negative values may be defined.

Emissivity

For the imager category *General*, the target emissivity can be specified as:

- *Blackbody,*
- *Greybody,* or
- *Spectral.*

For the imager category *Thermal*, the target is always assumed to be a blackbody, thus the *Greybody* and *Spectral* radio buttons are disabled.

For greybodies, the *Emissivity* of the greybody needs to be specified.

If the option *Spectral* is selected, the *Spectral Emissivity* of the target must be supplied via an external data dialog.

> **Note:** TRM4 provides a set of spectral emissivity data for different target types in its internal data library (see Appendix C.5).

Geometry

In TRM4 the target can be modelled in three dimensions as a rectangular cuboid with specified *Height, Width,* and *Depth*. This permits performance calculations for aspects and/or illuminations from directions other than the front of the target (including views from above and below).

2.24.2 Background

With respect to the background emissivity, TRM4 offers the same options as for the target emissivity. In the *General* category, the *Sky Background* option is available. A detailed description of the sky background specifications is given in Section 2.25.1.

> **Note:** TRM4 provides a set of spectral emissivity data for different background types in its internal data library (see Appendix C.5).

> **Note:** For imager flight altitude below the target or negative observation angle and one or two illumination sources specified, the background emissivity inputs in *General* category are disabled. In this case, *Sky Background* has to be specified.

Background temperature

The *Background temperature* is required for signal, noise, spectral weighting, and well-fill calculations

> **Note:** In the *General* category, it is assumed that target and background reflectivities are related to the corresponding emissivities by emissivity + reflectivity = 1.

> **Note:** The sensitivity and MTDP/MDSP of the imager may strongly depend on the background temperature, especially for imagers operating in the 3-5 µm wavelength range. Thus, care must be taken that specified and measured D^* or NETD values of thermal imagers are valid for the specified background temperature.

2.24.3 Scenario

Target viewing aspect

The target viewing aspect determines which lateral surfaces of the target (rectangular cuboid) are seen by the imager. The user can choose between three different target aspect angles

- *Front* (the front surface specified by *Width* and *Height* is seen by the imager),
- *Side* (the side surface specified by *Depth* and *Height* is seen by the imager), or
- *Diagonal* (the front as well as the side surface are seen by the imager and the line of sight is perpendicular to the bottom face diagonal of the target).

This choice affects the actual target area presented to the imager.

Imager flight path

Two options are available:

- *Constant altitude* and
- *Constant angle.*

If the *Constant altitude* option is selected, TRM4 needs to be supplied with the *Altitude* of the imager. This is the altitude relative to the target. *Altitude* may also take negative values which means that the imager is at a lower altitude than the target. An altitude of zero meters corresponds to a horizontal line of sight.

If the *Constant angle* option is selected, TRM4 needs to be supplied with the observation angle. For an upward-looking imager, the observation angle is negative. The angle is measured from the horizontal plane, meaning that an angle of 0° corresponds to a horizontal line of sight.

Note: For *Constant altitude*, the observation angle changes with target distance unless the *Altitude* is specified as zero meters. Therefore, the target area presented to the imager may increase or decrease, depending on altitude, target geometry, and target distance.

Note: *Altitude* is a relative altitude (as depicted in the figure in the dialog). The absolute altitude is only of interest when modelling the atmosphere, more specifically the turbulent atmosphere. If required, the absolute altitude can be specified in the *Turbulence* dialog (Section 2.28).

Note: For horizontal observations, the background is assumed to be on a plane perpendicular to the line of sight of the imager. For cases where the line of sight is not horizontal, the background is assumed to be on the horizontal plane which includes the target.

Note: For the calculation of ranges, only the movement of the imager (and not the one of the target) can be specified. This has to be remembered when specific tactical situations are to be modelled.

2.24.4 Task and number of line pairs

Three different perception tasks can be specified simultaneously: one for *Detection, Recognition,* and *Identification*. Only the *number of line pairs* is relevant for TRM4 calculations. The titles (*Detection, Recognition*, and *Identification*) are auxiliary information for the user.

2.24.5 Standard target and perception models (*Thermal* category only)

For the assessment of thermal imagers NATO standards are available to characterize target and background features as well as atmospheric conditions. Such standards do not exist for other types of imagers.

The STANAG 4347 [5] is the standard for defining the range performance for thermal imagers. In addition to STANAG 4347, TRM4 provides two new target models: One model for the perception of handheld objects and another for persons. The features and standard values of these models are summarized in Table 2.1.

All three models are part of the TRM4 internal public library and can be imported via the *Submodule* button, see Figure 2.32. Any change in a feature of the model or in the *Scenario* group will automatically remove the submodule name from the submodule edit field.

Feature	**STANAG 4347**	**Handheld objects**[7]	**Persons**
Width	2.3 m	0.08 m	0.6 m
Height	2.3 m	0.70 m	1.7 m
Depth	---	0.08 m	0.4 m
Background temperature	288 K	288 K	288 K
Temperature difference target-background	2 K	1 K	2 K
Detection	1 lp	2.5 lp	1 lp
Recognition	3 lp	3.5 lp	2.5 lp
Identification	6 lp	5.4 lp	---
Line of sight	horizontal	horizontal	horizontal
Atmosphere 1[8]	$\sigma = 0.2 \,/\, \mathrm{km}$	---	---
Atmosphere 2[8]	$\sigma = 1.0 \,/\, \mathrm{km}$	---	---

Table 2.1: TRM4 target models for imagers of the category *Thermal.*

[7] Data extracted from Uwe Adomeit, 'Infrared Detection, Recognition and Identification of Handheld Objects', Proc. SPIE 8541 (2012)

[8] STANAG 4347 defines also atmospheric conditions. This needs to be specified in the *Atmosphere* dialog/submodule.

2.25 Target, scenario and task – *Visible* category

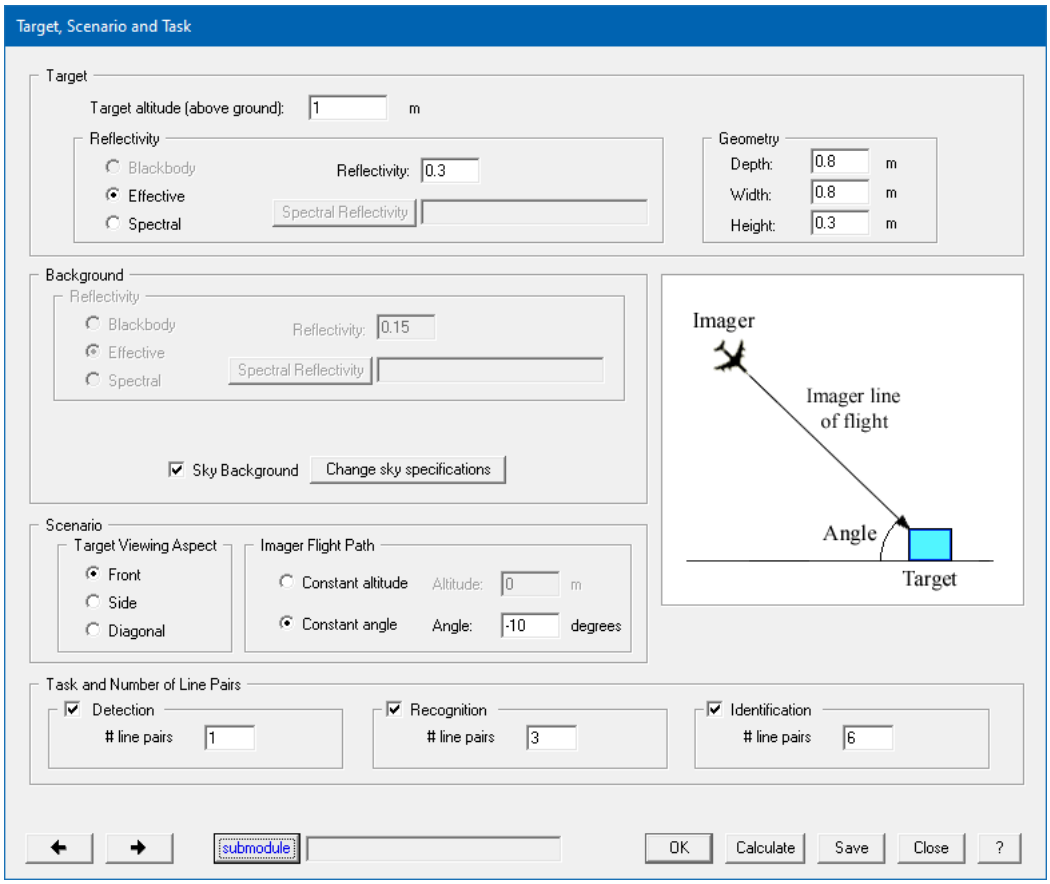


Figure 2.33: *Target, Scenario, and Task* dialog for the *Visible/SWIR/NIR* imager category.

2.25.1 Target and background

At present, there are no generally accepted standard targets for assessing imagers in the *Visible/NIR/SWIR*.

Reflectivity

TRM4 offers two options for both target and background reflectivity specification:

- *Effective,* or
- *Spectral.*

If the option *Effective* is selected, TRM4 asks for the effective *Reflectivity* value (i.e. the reflectivity of the object as a fraction of the perfect reflectivity). If the option *Spectral* is selected, the *Spectral Reflectivity* of the target needs to be specified via an external data dialog.

Note: It is assumed that the target surface is ideal diffusely reflecting and can be modeled as a Lambertian surface.

Note: TRM4 provides a set of spectral reflectivity data for different target and background types in its internal data library (see Appendix C.5).

Note:	For imager flight altitude below the target or negative observation angle, the background reflectivity inputs are disabled. In this case, Sky Background has to be specified.

Geometry

The target is modelled as a three-dimensional rectangular cuboid, specified by its *Height, Width,* and *Depth*.

Sky Background (available also in *General* category)

In case the imager is located at a smaller altitude than the target (i.e. a negative *Constant altitude* or negative observation angle is specified in the *Imager Flight Path* box), TRM4 assumes that the target is seen against the sky and the *Sky Background* option has to be selected. For horizontal observation (i.e. zero *Constant altitude* or zero observation angle) the specification of *Sky Background* is optional and for other Imager Flight Path specifications it is disabled.

The specification of the sky background is done in the *Sky Background Radiance* subdialog shown in Figure 2.34. To access this subdialog, click on the *Change sky specifications* button.

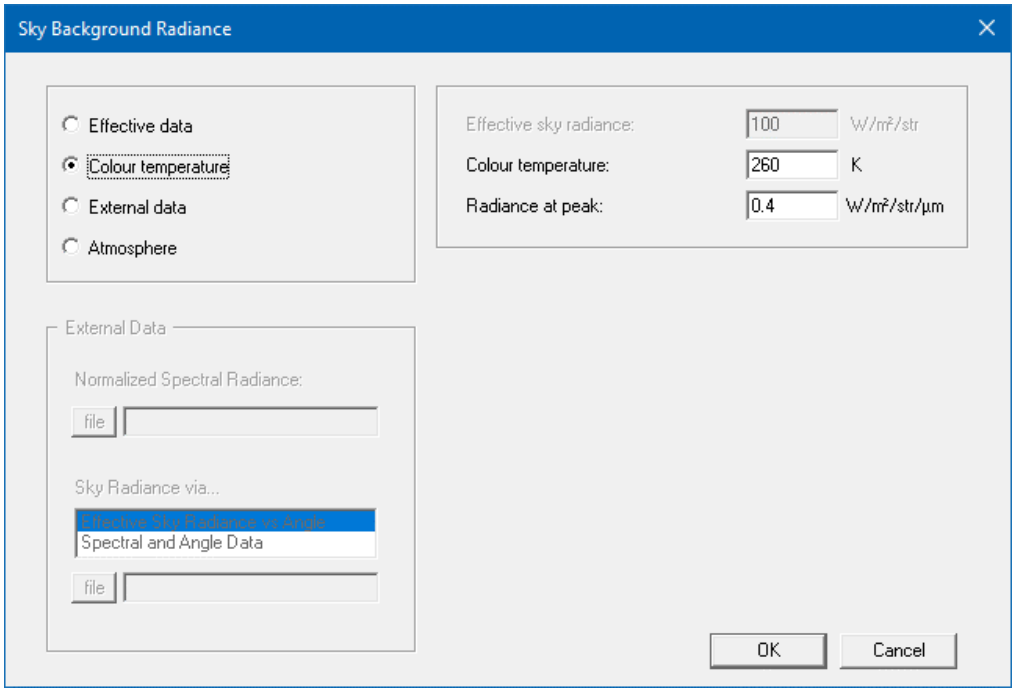


Figure 2.34: *Sky Background Radiance* dialog.

TRM4 offers the following options for sky background specification:

- *Effective data,*
- *Colour temperature,*
- *External data,* and
- *Atmosphere.*

The *Effective data* option assumes that the sky radiance spectrum is flat in the operational spectral band of the imager. Sky background with wavelength-dependent radiance spectrum can be defined by using the *Colour temperature* option or by specifying *Normalized Spectral Radiance* external data.

Specification of sky radiance via *Effective Sky Radiance vs. Angle* or *Spectral and Angle Data* is available when the imager flight path is specified at *Constant altitude* below zero (relative to the target). Such data may be obtained from MODTRAN or similar software packages.

When the *Atmosphere* option is selected, sky background radiance is derived from specified atmospheric data. For each atmosphere, the sky radiance is set to the path radiance at infinite distance using the path radiance factor specified in the *Atmosphere* dialog. If external path radiance data is specified, the sky radiance will be taken as the path radiance at maximum specified range.

Note: If two atmospheres are specified in the *Atmosphere* dialog, the only way to specify different sky backgrounds for the different atmospheres is by using the *Atmosphere* option. In all other cases the sky background signal will be the same for the two atmospheres.

Note: If path radiance at maximum range is greater than the specified sky radiance, a runtime warning message will be pop up during calculations. This scenario is not possible if *Atmosphere* option is used for specification of sky background.

Note: If *Atmosphere* option is used, the atmospheric absorption and path radiance have to be larger than zero for all atmospheres specified in the *Atmosphere* dialog. If zero atmospheric absorption or zero path radiance is found in one or more atmospheres, an error message will be shown.

Note: If *Constant altitude* below zero (relative to the target) is specified, it is recommended to use external data to specify angle dependent sky radiance in order to take into account different sky radiance values for different line of sights.

2.25.2 Scenario and task

The definitions of the scenario and the task are done in the same manner as for the categories *General* and *Thermal* (see Section 2.24).

2.26 Irradiance

The *Irradiance* dialog is available for the *General* and *Visible* imager categories only. For imagers of the *Thermal* category, TRM4 always assumes targets and backgrounds to be (purely emissive) blackbodies. Thus, for that category, irradiation sources are irrelevant. If targets and/or backgrounds observed by a thermal system are not blackbodies the imager category *General* should be selected for specifying this system.

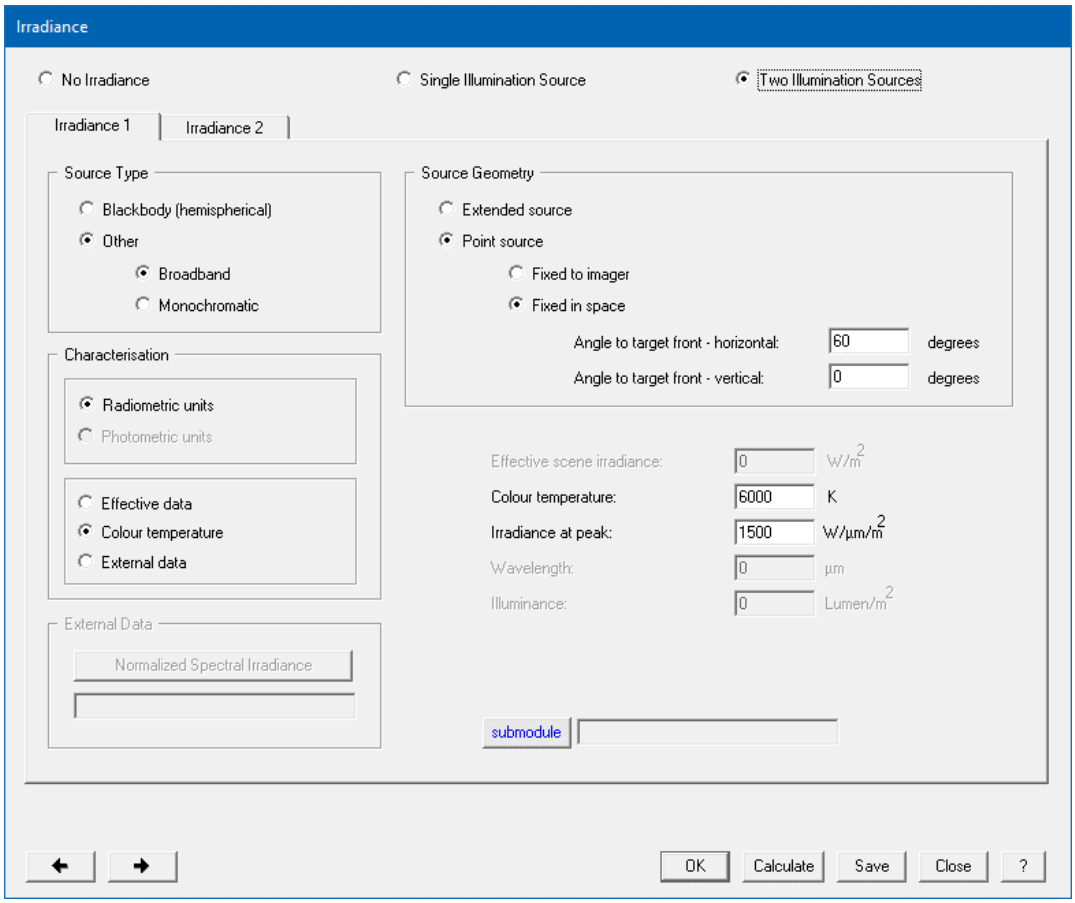


Figure 2.35: Irradiation source definition dialog.

Note: Many parameters and parameter names in this dialog change according to the selections made with respect to the selected data characterization and the source geometry. The general description in this section uses the following settings: *Source fixed in space* and specified in *Radiometric Units*.

Note: For all sources except those fixed to the imager, irradiation should be specified at the target/background plane (i.e., after transmission through the atmosphere). This is especially important to remember when specifying hemispherical blackbody sources. In this case, a pure blackbody with a constant emissivity of one will be considered to illuminate the target/background plane without any attenuation due to atmospheric absorption.

2.26.1 Illumination sources

There are three options for irradiance:

- *No Irradiance* (only for category *General*),
- *Single Illumination Source,* and
- *Two Illumination Sources.*

No irradiance

This option is only available for the *General* imager category. If selected, TRM4 assumes that only thermally emitted radiation contributes to the signal.

Single illumination source

This setting can be used as a standard when the sun, moon, etc. is the only illumination source or if a single artificial illuminator is used.

If the *Single Illumination Source* option is chosen, only the *Irradiance 1* tab is active.

Two illumination sources

Two illumination sources can be used for situations where, an additional spot light is used to illuminate the target along with a natural illumination source. The *Irradiance 2* dialog is nearly identical to the *Irradiance 1* dialog with the following exceptions: illuminators *fixed to imager and/or monochromatic* sources (such as lasers) cannot be specified in the *Irradiance 2* dialog.

2.26.2 Source Type

Blackbody (hemispherical)

For this type of source, illumination is considered homogeneous from all directions of the hemisphere. A typical representative of this type of source is an overcast sky. The blackbody is characterized by its temperature.

Other – Broadband

Several options are available to specify a Broadband illumination source:

- *Effective data,*
- *Colour temperature,* and
- *External data.*

The simplest way to characterize a broadband source is via *Effective data*. In this case it is sufficient to specify the *Effective scene irradiance* (Section 2.26.6). In that case, TRM4 assumes that the radiation spectrum is flat over the specified spectral band of the imager.

Spectral features of a broadband source may also be specified by its *Colour temperature* or through *External data* (normalized). In both cases, the *Target irradiance at peak* wavelength needs to be specified as well.

Other – Monochromatic

Monochromatic or narrow-band illumination sources are assumed to be point sources. A monochromatic source is specified by its *Wavelength* and the *Scene irradiance*.

2.26.3 Source geometry

The illumination sources may be either:

- *(Spatially) Extended sources,* or
- *Point sources.*

Extended Sources

The irradiance at the target/ background plane has to be specified. A typical representative of this type of source is an overcast sky.

Point sources

TRM4 permits the modelling of two types of point sources. Sources that are

- *Fixed to the imager,* and those that are
- *Fixed in space.*

Point source – Fixed to imager

The radiant intensity of *Fixed to imager* source should be specified. In this case, the irradiance on the target is a function of range and the atmospheric attenuation of the light beam on its way to the target is taken into account. It is assumed that the target and its background are homogeneously irradiated.

Point source – Fixed in space

The angle of the illuminating beam relative to the front surface normal of the target needs to be specified. This situation is depicted in Figure 2.36. Additionally, the irradiance on the target plane needs to be specified (more specifically, the irradiance perpendicular to the irradiating beam and not the actual irradiance on the target surface).

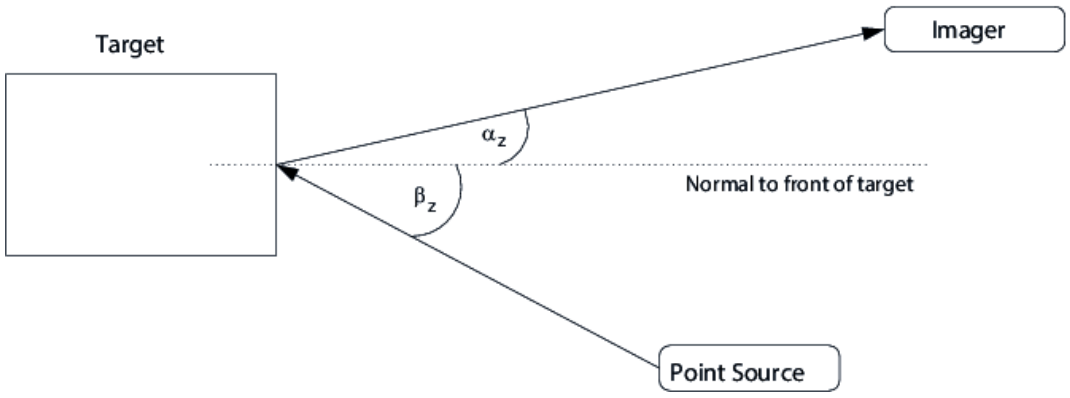


Figure 2.36: Point source (fixed in space) irradiance geometry. The angle between the front surface normal of the target and the source, β_z, and the angle between the line of sight and the surface normal, α_z, are shown.

Note: For illumination sources fixed in space, irradiance should be specified at the target/background plane, this means that the user has to take into account for the atmospheric attenuation between the source and the target/background when specifying the irradiance.

Note: Sources fixed to the imager are specified by their radiant intensity [W/sr], which is an inherent property of the source. In these cases, the attenuation of the irradiance between the source and the target is calculated by TRM4.

Note: The source geometry is inherently three-dimensional. Both the vertical angle β_{v} and the horizontal angle β_{h} with respect to the front surface normal of the target need to be given.

2.26.4 Characterization

Preferably, radiometric units should be used to characterize illumination sources. For sensors operating in the visible spectral band, however, photometric units are more commonly available. TRM4 offers the following alternatives for source characterization:

- *Radiometric units,* or
- *Photometric units* (for imager category *Visible*).

With respect to the specification of the source spectrum, TRM4 offers the following options:

- *Effective data,*
- *Colour temperature*, and
- *External data.*

The *Effective data* option assumes that the source spectrum is flat in the operational spectral band of the imager. The option is only available in combination with radiometric units. For photometric units *illuminance* replaces *Effective data*.

Colour temperature and *External data* options are available for specifying the spectral features of broadband sources.

Note: It is preferable to specify the spectrum of the illumination source(s) in order to better assess the spectral characteristics of energy from the target, background, etc. and to perform appropriate spectral weighting. The use of effective radiance data should be considered a compromise.

2.26.5 External data – Normalized spectral irradiance

This option allows for the specification of spectral illumination sources other than blackbodies.

Note: It is important to remember that all data entered via the external data option is normalized. Thus, the *irradiance at peak* (or illuminance) has to be specified as well. This is also true for the preset data supplied with TRM4 (see Section 0).

2.26.6 Other parameters

Effective scene irradiance

This parameter needs to be specified for a broadband source if the *Effective data* option is selected and the irradiance is expressed in *Radiometric units*.

Effective radiant intensity

This parameter describes the effective intensity per steradian of a *Broadband* point source illuminator *Fixed to the imager* in *Radiometric* units.

Colour temperature

This parameter needs to be specified if the *Blackbody* or the *Broadband/Colour temperature* options are selected to characterize the spectral behavior of the source. TRM4 uses the *Colour temperature* to derive the source spectrum from Planck's Law.

Irradiance at peak

This parameter needs to be specified if a *Broadband* source is selected, and *Radiometric* units are used to characterize the source. The *Irradiance at peak* refers to the wavelength where the spectral irradiance is maximal.

Radiant intensity at peak

This parameter describes the intensity per steradian of a *Broadband Point source* illuminator that is *Fixed to the imager* in *Radiometric* units.

Illuminance

This parameter describes the irradiance of an *Extended source* or of a *Point source/Fixed in space* at the target plane in *photometric units*.

Note:	In case of a *broadband* point source, *Illuminance* is defined as the illuminance perpendicular to the irradiating beam and not the actual illuminance on the target surface.

Scene irradiance

This parameter describes the irradiance at the target plane originating from a monochromatic point source *Fixed in space*. More specifically, it is the irradiance perpendicular to the irradiating beam and not the actual irradiance on the target surface.

Wavelength

The wavelength of a monochromatic illuminator/spot light source is specified here.

Radiant intensity/Luminous intensity

This parameter describes the intensity per steradian of a *monochromatic* illuminator that is *Fixed to the imager* in *Radiometric* units and of a *broadband* illuminator that is *Fixed to the imager* in *Photometric* units, respectively.

2.27 Atmosphere

The layout of the *Atmosphere* dialog is very similar for all three imager categories. Depending on the actual category, however, various parameters/options are either disabled or not shown. Figure 2.37 depicts the dialog for imager category *General*.

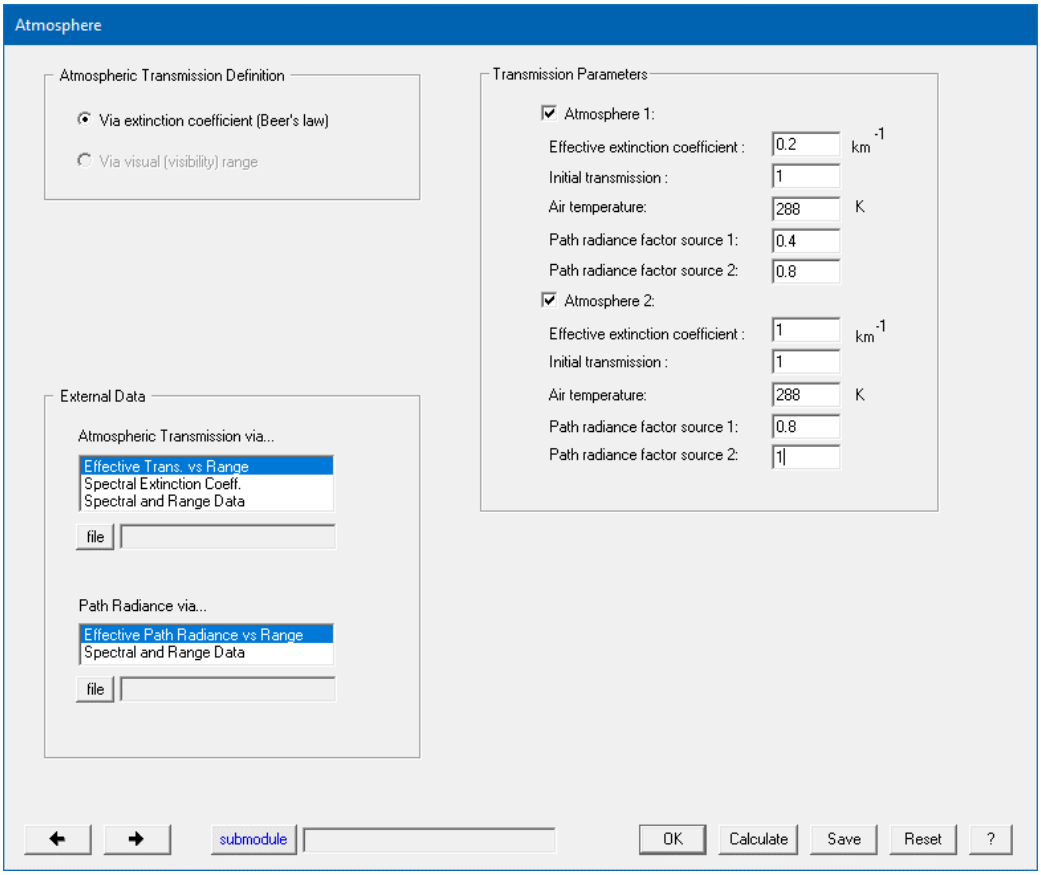


Figure 2.37: Atmosphere dialog for the imager category *General*.

2.27.1 Atmospheric transmission definition

Two options are available for imager category *Visible* to calculate the atmospheric transmission:

- *Via extinction coefficient (Beer's Law)* and
- *Via Visual range (Visibility).*

Transmission calculations via the (atmospheric) extinction coefficient (Beer's Law) can be specified for all imager categories. The *Visibility* option is restricted to imagers operating in the Visible spectrum.

Extinction coefficient σ_{atm} and visibility V are related by the Koschmieder equation:

$$\sigma_{\mathrm{atm}} = \frac{3.912}{V}. \tag{2.28}$$

2.27.2 Transmission parameters

Up to three different atmospheric conditions *(Atmospheres)* can be specified simultaneously. Each condition is characterized by one or several of the following parameters.

Effective extinction coefficient

Assuming an exponential transmission as a function of range (Beer-Lambert law), the extinction coefficient (σ_{atm}) is defined in Equation (2.29).

Note: For thermal imagers, STANAG 4347 provides two standard atmospheres with different extinction coefficients:

- average weather condition: $\sigma_{\mathrm{atm}} = 0.2\ \mathrm{km}^{-1}$, and
- bad weather condition: $\sigma_{\mathrm{atm}} = 1\ \mathrm{km}^{-1}$.

These standardized atmospheres might be selected by choosing the preset STANAG 4347 in the submodule. For the imager categories *General* and *Visible*, standard values are not available.

Initial transmission

If $\tau_{\mathrm{atm_nul}}$ designates the *Initial transmission*, the overall transmission at range R is given by:

$$\tau_{\mathrm{atm_eff}}(R) = \tau_{\mathrm{atm_nul}} \cdot e^{-\sigma_{\mathrm{atm}} R}. \tag{2.29}$$

Note: In the thermal spectral band, especially in the $3-5\ \mu\mathrm{m}$ band, a rapid drop in transmission is typical at very short ranges due to strong CO_2 absorption. *Initial transmission* takes care of this short distance drop in the signal strength. An initial transmission value of 0.8 is recommended for $3-5\ \mu\mathrm{m}$ and a value of 1.0 for $8-12\ \mu\mathrm{m}$ thermal imagers.

Visual range (*Visible* category only)

Visibility is defined as the longest range at which a black target can be perceived against a bright white background.

Air temperature

This parameter is available for imagers of category General only. For imagers of category Thermal, the air temperature is assumed to be equal to the background temperature. For imagers of category Visible, the air temperature is not relevant.

TRM4 uses the specified air temperature to calculate radiation emitted by the air. This radiation contributes to the path radiance (Section 3.7.3). If the air temperature is equal to the background temperature, the atmosphere emits as much radiation as it absorbs. In this case the measured background radiation is independent of range (this is true for purely emissive scenarios only).

Path radiance factor

Path radiance due to reflected radiation can be specified for the imager categories *General* and *Visible* only.

The *Path radiance factor* F_{PR} is used to calculate the path radiance L_{PR} as function of the range R:

$$L_{\mathrm{PR}}(R) = F_{\mathrm{PR}} \cdot [1 - \tau_{\mathrm{atm}}(R)] \cdot \frac{I_{\mathrm{source}}}{\pi}, \tag{2.30}$$

where I_{source} is the irradiance perpendicular to the target surface and $\tau_{\mathrm{atm}}(R)$ is the atmospheric transmission.

For each of the two optional irradiation sources, a different path radiance factor can be specified.

Note: Equation (2.30) describes the path radiance due to radiation reflected by the air particles.

Note: The relation between the sky-to-ground ratio, SGR, and the path radiance factor is

$$\mathrm{SGR} = \frac{F_{\mathrm{PR}}}{\rho_{\mathrm{B}}}, \tag{2.31}$$

where ρ_{B} is the reflectivity of the background. If the path radiance factor is set equal to the background reflectivity, the energy that reaches the detector from the background is independent of range (in cases where path radiance consists of reflected energy only).

2.27.3 External data – Atmospheric transmission via

The external specification replaces the internal calculation of the atmospheric transmission.

Note: External transmission data applies to all specified atmospheres.

Note that at least one atmosphere needs to be activated in the *Transmission Parameters* group of the dialog for the specified external data to be used in calculations. There are two exceptions:

- *Visible and General* categories: when both atmospheric transmission and path radiance are specified via external data, all check boxes are disabled, and
- *Thermal category*: all checkboxes are disabled by default.

Effective transmission vs. range

The *Effective transmission* data for the specified spectral response of the imager has to be supplied by the user.

> **Note:** This option is the only method to specify external atmospheric transmission for the *Thermal* category.

Spectral extinction coefficient

If you specify *Spectral extinction coefficients*, TRM4 assumes that Beer's Law applies to each wavelength separately. TRM4 calculates the effective extinction coefficient by automatically weighting the signal over the specified spectrum.

Spectral and range data

This is the most detailed way to specify the atmospheric transmission. Data can be obtained from MODTRAN or similar software packages.

2.27.4 External data – Path radiance via

Path radiance can be also specified via external data. The external specification replaces the internal calculation of the path radiance.

Path radiance data can be specified for imager categories *General* and *Visible* only. If you want to specify the path radiance for a thermal imager, you have to select the imager category *General*.

Effective path radiance vs. range

The effective path radiance should take into account the spectral response of all relevant components.

Spectral and range data

Detailed spectral path radiance data as function of range can be obtained from MODTRAN and similar software packages.

2.28 Turbulence

The Turbulence dialog is active for imager categories *Visible* and *General* only. It serves to specify the data required to calculate the turbulence MTF. This MTF is calculated for the path specified in the *Target, Scenario and Task* dialog.

> **Note:** The MTF used to describe turbulence effects depends on the distance between imager and target. It equals 1 for zero range. (In case the reference range is set to 0 (*Modify Individual Parameters* dialog) no turbulence MTF will be displayed in the MTF I or MTF II output window.)

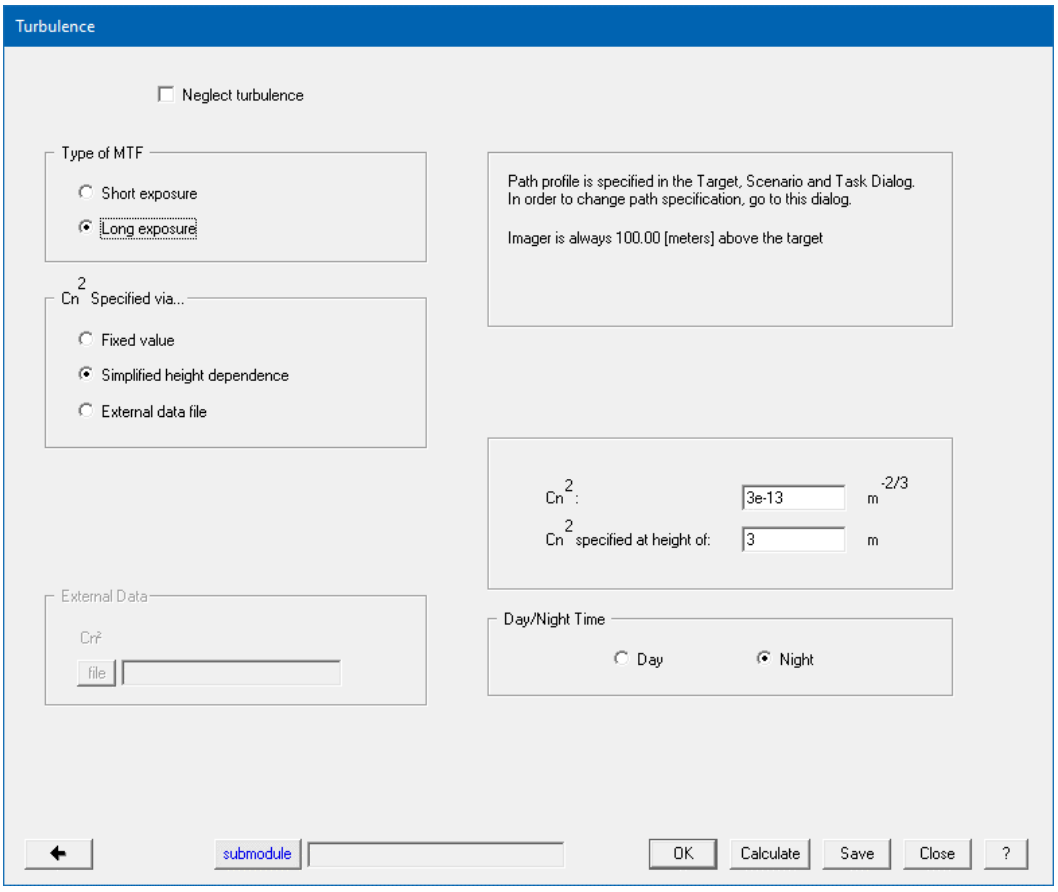


Figure 2.38: The turbulence definition dialog.

2.28.1 Type of MTF

The effects of turbulence on imaging systems are highly anisoplanatic, i.e., they depend on the actual position of an object within the field of view. The overall degrading influence, however, can be modelled by using a time-averaged MTF.

Long exposure

The *long exposure* MTF takes into account all effects of image degradation by turbulence, and leads (depending on factors like turbulence strength, IFOV, etc.) to a severe blurring of the image.

Short exposure

The *short exposure* MTF is the tip-tilt reduced variant of the long exposure MTF. Tip-tilt refers to full image motion in the horizontal and vertical direction.

Note: Since a human observer is capable of disregarding such tip-tilt movement in most cases and imagers are usually recording at a sufficiently high frame rate, thus avoiding co-integration of shifted frames, it is recommended to use the *short exposure* MTF as a default.

2.28.2 C_n^2 specified via ...

Fixed value

If selected, the *External Data*, C_n^2 *specified at height of*, and *Day/Night Time* groups of the dialog will be deactivated and a fixed C_n^2 value will be used for all calculations. This setting is recommended for horizontal lines of sight. Typical ground level values of C_n^2 are listed in Table C.2 of Appendix C.

Simplified height dependence

If selected, a C_n^2 value and a reference height need to be given. C_n^2 values along the momentary line of sight are calculated using the height dependence

$$C_n^2(h) = C_{n,\mathrm{ref}}^2 \cdot \left(\frac{h}{h_{\mathrm{ref}}}\right)^{-4/3} \tag{2.32}$$

for the setting *Day* and

$$C_n^2(h) = C_{n,\mathrm{ref}}^2 \cdot \left(\frac{h}{h_{\mathrm{ref}}}\right)^{-2/3} \tag{2.33}$$

for the setting *Night*, where $C_{n,\mathrm{ref}}^2$ is the reference turbulence strength, and h_{ref} the reference measurement altitude.

These profiles are a simplification and valid for the planetary boundary layer only. It is thus recommended to use this option up to maximum target altitudes of 100 m or less.

External data file

If this option is selected, the user can explicitly specify the altitude dependence of turbulence. Some standard profiles are part of the TRM4 distribution (see Appendix C).

3 Performance Assessment

3.1 Introduction

This chapter presents the concept and the equations that TRM4 uses to assess imager performance. At the heart of the range performance calculation is the Minimum Difference Signal Perceived (MDSP) figure of merit, which assesses the perception of a 4-bar target by an observer. Using modified Johnson criteria, the Minimum Necessary Difference Signal (MNDS) to solve the specified observation task can be deduced from the MDSP. Finally, the maximal distance at which this observation task can be successfully performed is found by comparing the Effective Difference Signal (EDS) between target and background with the MNDS.

Sections 3.2 to 3.5 give the equations that are used to derive fundamental properties of the imaging system, including scanning properties, display effects, sampling, focal length, etc. Of special importance is the calculation of the (spectral) sensitivity of various detector types treated in Section 3.5.

Irradiation sources illuminating the target and background are treated in Section 3.6. Depending on which imager category was chosen, target and background are either purely emissive or reflect light from illumination sources.

The signal sensed by the detector is only partly due to the photons emanating from the target and background. Additional scene contributions include the signal modification by the atmosphere and the radiance of the air path. In addition to the carrier flux induced by sources external to the imager, there are contributions of internal sources (e.g. warm optics, device walls, etc.) and (noise) contributions of the imager and detector itself. The necessary calculations and equations are listed in Section 3.7.

Section 3.8 describes the difference signal and the noise at the detector assembly output.

Temporal and spatial noise contributions are closely examined in Sections 3.9 and 3.10, respectively. These contributions are calculated for various stages within the system, namely the pre-digital, digital, and video signal.

In Section 3.11, signal, noise, and other system properties (e.g. MTFs, see Chapter 4) are brought together to form the equation for the overall figure of merit used in TRM4: the MDSP. This equation also treats the spatial and temporal noise averaging properties of the human eye.

With the MDSP defined, Section 3.12 explains how TRM4 arrives at a range estimate for observer tasks by taking into account the MDSP, the observer task, and the effective signal reaching the imager.

Section 3.13 is devoted to the calculation of NIIRS (National Imagery Interpretability Rating Scale) values for aerial imaging scenarios.

Symbols and nomenclature used in this chapter are listed in Appendix D. Appendix A summarizes the units of system component parameters and corresponding conversion factors. MTF equations are given in Chapter 4 and the derivations of the optics diffraction limited MTFs are presented in Appendix B.

3.2 Basic imager features

3.2.1 Pixel binning

> **Note:** In the following, the subscript z stands for either the horizontal or the vertical direction.

Pixel binning reduces the effective number of detector elements available to

$$n_z = \mathrm{floor}\left(\frac{n_{\mathrm{det}_z}}{n_{\mathrm{bin}_z}}\right), \tag{3.1}$$

where n_{dez_z} = number of actual detector elements in z-direction,
n_{bin_z} = number of binned detector elements in z-direction, and
$\mathrm{floor}(x)$ = mathematical function returning the largest integer smaller or equal x.

Pixel binning also changes the effective sensitive detector area

$$A_{\mathrm{det_binned}} = A_{\mathrm{det}} \cdot n_{\mathrm{bin_h}} \cdot n_{\mathrm{bin_v}}\,, \tag{3.2}$$

where A_{det} is the sensitive area of a single (unbinned) detector element, the effective storage capacity

$$N_{\mathrm{stor_binned}} = N_{\mathrm{stor}} \cdot n_{\mathrm{bin_h}} \cdot n_{\mathrm{bin_v}}\,, \tag{3.3}$$

where N_{stor} is the storage capacity of a single (unbinned) detector element, and the effective detector pitch

$$a_{\mathrm{pitch_binned}_z} = a_{\mathrm{pitch}_z} \cdot n_{\mathrm{bin}_z}\,, \tag{3.4}$$

where a_{pitch_z} is (unbinned) detector pitch dimension in z direction.

The effective detector dimension is given by

$$a_{\mathrm{det_binned}_z} = a_{\mathrm{det}_z} + \left(n_{\mathrm{bin}_z} - 1\right) \cdot a_{\mathrm{pitch}_z}, \tag{3.5}$$

where a_{det_z} denotes the (unbinned) active detector dimension in z-direction.

> **Note:** If pixel binning is specified, A_{det}, N_{stor}, a_{pitch_z} and a_{det_z} have to be replaced by the corresponding binned versions in all equations of Chapter 3 below.

3.2.2 Electronic pseudo interlace

Electronic pseudo interlace is in many ways similar to vertical binning of two detector elements. However, the detector pitch is unchanged in case of electronic pseudo interlace.

The effective sensitive detector area becomes

$$A_{\mathrm{det_epi}} = 2 \cdot A_{\mathrm{det}} \,, \tag{3.6}$$

while the effective detector vertical dimension is given by

$$a_{\mathrm{det_epi_v}} = a_{\mathrm{pitch_v}} + a_{\mathrm{det_v}} \,. \tag{3.7}$$

The effective storage capacity is in this case

$$N_{\mathrm{stor_epi}} = 2 \cdot N_{\mathrm{stor}} \,. \tag{3.8}$$

Note: If electronic pseudo interlace is specified, A_{det}, N_{stor}, $a_{\mathrm{pitch_z}}$ and $a_{\mathrm{det_z}}$ have to be replaced by the corresponding electronic pseudo interlace versions in all equations of Chapter 3 below.

3.2.3 Field of view (FOV)

Imager with opto-mechanical scan

For a parallel or serial/parallel scanned imager, the vertical FOV is given by

$$\mathrm{FOV_v} = \frac{\left(\frac{N_{\mathrm{scl}} - 1}{n_{\mathrm{inter}}}\right) \cdot a_{\mathrm{pitch_v}} + a_{\mathrm{det_v}}}{f_{\mathrm{opt_v}}} \,, \tag{3.9}$$

where N_{scl} = number of scanned lines per frame,
n_{inter} = opto-mechanical scan interlace factor, and
$f_{\mathrm{opt_v}}$ = vertical optics focal length.

The horizontal FOV for scanning imagers is derived from the vertical FOV and the FOV aspect ratio α_{FOV}. It is given by

$$\mathrm{FOV_h} = \alpha_{\mathrm{FOV}} \cdot \mathrm{FOV_v} \,. \tag{3.10}$$

Staring or micro-scanned imagers

For a staring or micro-scanned imager, the FOV in z-direction is given by

$$\mathrm{FOV}_z = \frac{\left(\frac{N_{\mathrm{sam}_z} - 1}{N_{\mu\mathrm{step}_z}}\right) \cdot a_{\mathrm{pitch}_z} + a_{\mathrm{det}_z}}{f_{\mathrm{opt}_z}} \,, \tag{3.11}$$

where N_{sam_z} denotes the number of samples in z-direction and $N_{\mu\mathrm{step}_z}$ is the number of micro-scan steps in z-direction (equal to one for pure staring system).

ICCD/ICMOS

The FOV of ICCD/ICMOS imagers depends on the optics focal length, the input diameter of the photocathode as well as the size of the part of the CCD/CMOS sensor, which is used for displaying the image on the screen. The shape of the FOV can be

- rectangular,
- circular or
- circular truncated.

The ratio between the size of the part of the CCD/CMOS sensor used for displaying the image on the screen and the dimension of image intensifier image in the sensor plane is

$$\mathrm{F}_{\mathrm{ccd}_z} = \frac{\left(N_{\mathrm{sam}_z} - 1\right) \cdot a_{\mathrm{pitch}_z} + a_{\mathrm{det}_z}}{D_{\mathrm{pc}} \cdot M_{\mathrm{ICCD}}} \,, \tag{3.12}$$

where D_{pc} is the input diameter of the photocathode and M_{ICCD} is the magnification of the ICCD/ICMOS.

If $\mathrm{F}_{\mathrm{ccd}_z} < 1$ for both directions, the image intensifier image in the CCD/CMOS sensor plane is larger than the used CCD/CMOS sensor dimensions in both directions. In this case, the FOV is determined by the CCD/CMOS dimensions and is given by:

$$\mathrm{FOV}_z = \mathrm{F}_{\mathrm{ccd}_z} \cdot \frac{D_{\mathrm{pc}}}{f_{\mathrm{opt}_z}} . \tag{3.13}$$

The FOV has a rectangular shape if the diagonal of the sensor array used for displaying the image is smaller than the diameter of the image stemming from the image intensifier. Otherwise the FOV shape is circular with truncations in one or two dimensions.

If $\mathrm{F}_{\mathrm{ccd}_z} \geq 1$ for one or both directions, the image intensifier image in the CCD/CMOS sensor plane is smaller or equal than the used CCD/CMOS sensor dimensions. In this case, the FOV shape is circular. This scenario is not allowed in TRM and an error message is displayed.

3.2.4 Imager magnification

The angular magnification of the imager, M_{sys}, is defined as the angular subtense of the image as seen by the observer, divided by the field of view of the imager.

For CRT displays it is given by

$$M_{\mathrm{sys}} \approx \frac{N_{\mathrm{FOV_v}} \cdot H_{\mathrm{picture}}}{\mathrm{FOV_v} \cdot D_{\mathrm{obs}} \cdot n_{\mathrm{active}}} , \tag{3.14}$$

where
$N_{\mathrm{FOV_v}}$ = number of video lines covering the vertical FOV of the imager,
H_{picture} = displayed image height,
D_{obs} = distance observer display, and
n_{active} = number of active lines of the display.

For LED displays, it is given by

$$M_{\mathrm{sys}} \approx \frac{N_{\mathrm{scl}} \cdot \mathrm{dis}_{\mathrm{pitch_v}}}{\mathrm{FOV_v} \cdot n_{\mathrm{inter}} \cdot D_{\mathrm{obs}}} \approx \frac{\mathrm{dis}_{\mathrm{ptich_v}}}{a_{\mathrm{s_v}} \cdot n_{\mathrm{inter}} \cdot D_{\mathrm{obs}}}, \tag{3.15}$$

where $\mathrm{dis}_{\mathrm{pitch_v}}$ is the pitch between display elements (LEDs) and $a_{\mathrm{s_v}}$ is the vertical angular sampling interval (Section 3.3.1).

For Flat panel displays, it is given by

$$M_{\mathrm{sys}} \approx \frac{N_{\mathrm{FOV_v}} \cdot \mathrm{dis}_{\mathrm{pitch_v}}}{\mathrm{FOV_v} \cdot D_{\mathrm{obs}}}, \tag{3.16}$$

where $N_{\mathrm{FOV_v}}$ is the number of display pixels covering the vertical FOV of the imager.

Note: The equations above are valid for direct observation of the display. If an eye-piece is used, D_{obs} is replaced by the focal length of the eye piece, f_{eyepiece}.

For ICCD/ICMOS, the magnification between the optics and the detector elements is calculated from the image intensifier tube magnification, M_{II}, and the magnification of the coupling element, M_{coup}:

$$M_{\mathrm{ICCD}} = M_{\mathrm{II}} \cdot M_{\mathrm{coup}}\,. \tag{3.17}$$

3.2.5 Detector angular subtense

The detector angular subtense in z-direction, DAS_z, is given by

$$\mathrm{DAS}_z = \frac{a_{\mathrm{det}_z}}{f_{\mathrm{opt}_z}}, \tag{3.18}$$

with the exception of ICCD/ICMOS where it reads

$$\mathrm{DAS}_z = \frac{a_{\mathrm{det}_z}}{M_{\mathrm{ICCD}} \cdot f_{\mathrm{opt}_z}}\,. \tag{3.19}$$

3.2.6 Detector pitch dimensions

Imagers with opto-mechanical scan

For imagers with opto-mechanical scan the vertical detector pitch, $a_{\mathrm{pitch_v}}$, can be expressed as

$$a_{\mathrm{pitch_v}} = \frac{(f_{\mathrm{opt_v}} \cdot \mathrm{FOV_v} - a_{\mathrm{det_v}}) \cdot n_{\mathrm{inter}}}{N_{\mathrm{scl}} - 1}\,. \tag{3.20}$$

Note: For scanning imagers with a staggered detector array, the vertical pitch, $a_{\mathrm{pitch_v}}$, is defined as the smallest vertical distance between two detectors (Figure 2.17).

Staring or micro-scanned imagers

For staring or micro-scanned imagers, both horizontal and vertical detector pitches can be derived from other imager parameters. The pitch in z-direction, a_{pitch_z}, is given by

$$a_{\mathrm{pitch}_z} = \frac{(f_{\mathrm{opt}_z} \cdot \mathrm{FOV}_z - a_{\mathrm{det}_z}) \cdot N_{\mu\mathrm{step}_z}}{N_{\mathrm{sam}_z} - 1}, \tag{3.21}$$

with the exception of ICCD/ICMOS where it is

$$a_{\mathrm{pitch}_z} = \frac{(f_{\mathrm{opt}_z} \cdot M_{\mathrm{ICCD}} \cdot \mathrm{FOV}_z - a_{\mathrm{det}_z}) \cdot N_{\mu\mathrm{step}_z}}{N_{\mathrm{sam}_z} - 1}. \tag{3.22}$$

3.2.7 Type of opto-mechanical scan

Serial Scan

When the number of detector elements in vertical direction is one, the imager is said to have a serial scan.

Serial/parallel scan

If the number of detector elements in the vertical direction $n_{\mathrm{det_v}}$ is greater than one and $x > 1$, with

$$x = \frac{N_{\mathrm{scl}}}{n_{\mathrm{det_v}} \cdot n_{\mathrm{inter}}}, \tag{3.23}$$

then the imager is said to have serial/parallel scan.

The number of blocks, m_{blocks}, required to form one frame of the image is given by

$m_{\mathrm{blocks}} = x$ if x is an integer and
$m_{\mathrm{blocks}} = \mathrm{floor}(x) + 1$ if x is not an integer.

If Equation (3.23) leads to an x-value which is not an integer then not all vertical detector elements are used to from the image (at least not in all the blocks).

Parallel scan

If Equation (3.23) results in an x-value $x \leq 1$, the imager is said to have a parallel scan. If $x < 1$, not all vertical detector elements are used to form the image.

3.2.8 Scan velocity

The (angular) scan velocity of scanning imagers with parallel or serial/parallel scan is given by

$$v_{\mathrm{scan}} = \frac{n_{\mathrm{inter}} \cdot F_{\mathrm{r,imager}} \cdot m_{\mathrm{blocks}} \cdot \mathrm{FOV_h}}{\eta_{\mathrm{scan}}}, \tag{3.24}$$

where $F_{\mathrm{r,imager}}$ is the imager frame rate and η_{scan} denotes the scan efficiency.

If the scanning imager has serial scan, the (angular) scan velocity is determined by

$$v_{\text{scan}} = \frac{N_{\text{scl}} \cdot F_{\text{r,imager}} \cdot \text{FOV}_{\text{h}}}{\eta_{\text{scan}}}. \tag{3.25}$$

3.2.9 Number of displayed pixels

Scanning imagers

The number of display pixels required to cover the FOV is given in the horizontal direction (Flat panel display only) by

$$N_{\text{FOV}_{\text{h}}} = N_{\text{sam}_{\text{h}}} \cdot M_{\text{zoom}} \cdot M_{\text{crep}} \tag{3.26}$$

and in the vertical direction by

$$N_{\text{FOV}_{\text{v}}} = N_{\text{scl}} \cdot M_{\text{zoom}} \cdot M_{\text{lrep}}\,, \tag{3.27}$$

where M_{zoom} = zoom factor,
M_{lrep} = line replication factor, and
M_{crep} = column replication factor.

Staring imagers

The number of display pixels required to cover the FOV in the horizontal direction, $N_{\text{FOV}_{\text{h}}}$, is given by Equation (3.26), for the vertical direction, however, it is

$$N_{\text{FOV}_{\text{v}}} = N_{\text{sam}_{\text{v}}} \cdot M_{\text{zoom}} \cdot M_{\text{lrep}}\,. \tag{3.28}$$

Number of lines required to display the image

The number of lines required to display the image on a CRT is determined by Equation (3.27).

Number of required display elements

The number of display elements required to display the image on a flat panel display is given for scanning images by Equations (3.26) and (3.27) for staring/micro-scanning imagers by Equations (3.26) and (3.28).

3.2.10 Video line efficiency and scaling factor

Video line efficiency, η_{video}, and scaling factor, scf, are related to each other by

$$\eta_{\text{video}} = \frac{t_{\text{line_FOV}}}{t_{\text{line_active}}} = \frac{\text{FOV}_{\text{h}}}{\text{scf} \cdot t_{\text{line_active}}}, \tag{3.29}$$

where $t_{\text{line_active}}$ = 51.7 μsec for CCIR video standard,
52.09 μsec for RS170 video standard,
scf = scaling factor (video signal), and
$t_{\text{line_FOV}}$ = time to display the horizontal FOV in one video line.

Note: The units to be used are [μsec] for all times t, [mrad] for $\mathrm{FOV_h}$, and [MHz*mrad] for scf.

3.2.11 Raster modulation

Displayed CRT/LED raster modulation

The line raster in CRT/LED displays is described by the raster modulation, which is given by

$$M_{\mathrm{raster,display}} = \frac{S_{\max} - S_{\min}}{S_{\max} + S_{\min}}, \tag{3.30}$$

where $S_{\max}$ = maximum signal (brightness) in a displayed line, and
$S_{\min}$ = minimum signal (brightness) between two displayed lines.

The signal is given by

$$S(y) = \left(\sum_{n=-\infty}^{\infty} \delta\,(y - n\Delta) \right) \otimes \mathrm{LSF_{display}}(y) \tag{3.31}$$

with $\Delta = \alpha_{s_\mathrm{v}}/(M_{\mathrm{zoom}} \cdot M_{\mathrm{lrep}})$ for CRT and $\Delta = \alpha_{s_\mathrm{v}}$ for LED displays. $\mathrm{LSF_{display}}(y)$ denotes the vertical line spread function of the display beam or LED element and α_{s_v} is the vertical angular sample distance, see Section 3.3.1.

The horizontal CRT/LED displayed raster modulation is always equal to zero.

Flat panel display raster modulation

In TRM4, the flat panel display modulation in both directions is always zero as rectangular display elements with 100% fill factor are assumed.

Perceived raster modulation

The raster modulation perceived by the eye is given by

$$M_{\mathrm{raster,eye}} = \frac{S_{\max} - S_{\min}}{S_{\max} + S_{\min}}, \tag{3.32}$$

where $S_{\max}$ = maximum *perceived* signal in a displayed line, and
$S_{\min}$ = minimum *perceived* signal between two displayed lines.

The perceived signal is given by

$$S(y) = \left(\sum_{n=-\infty}^{\infty} \delta\,(y - n\Delta) \right) \otimes \mathrm{LSF_{display}}(y) \otimes \mathrm{LSF_{eye}}(y)\,, \tag{3.33}$$

where $\mathrm{LSF_{eye}}(y)$ is the line spread function of the eye in vertical direction. The perceived modulation in horizontal direction is always zero.

3.2.12 Geometrical distortion

CRT display

In case standard video is specified (CCIR or RS-170) TRM checks the deviation of the displayed image from the standard 4:3 aspect ratio.

Unless special measures are taken (e.g. anamorphic eyepiece), the aspect ratio of the image, α_{image}, displayed on a TV monitor with a 4:3 aspect ratio is given by

$$\alpha_{\mathrm{image}} = \frac{4}{3} \cdot \frac{n_{\mathrm{active}} \cdot \eta_{\mathrm{video}}}{N_{\mathrm{FOV_v}}}, \tag{3.34}$$

where n_{active} is the number of active lines in the monitor.

The displayed image is geometrically distorted if α_{image} differs from the FOV aspect ratio of the imager α_{FOV} (Section 3.2.3). This geometrical distortion is given by

$$D_{\mathrm{dist}} = \left| \frac{\alpha_{\mathrm{image}} - \alpha_{\mathrm{FOV}}}{\alpha_{\mathrm{FOV}}} \right|. \tag{3.35}$$

Flat panel display

For a flat panel display, the distortion can be expressed as

$$D_{\mathrm{dist}} = \left| \frac{\alpha_{\mathrm{s_h}} \cdot \mathrm{dis}_{\mathrm{pitch_v}} \cdot M_{\mathrm{lrep}}}{\alpha_{\mathrm{s_v}} \cdot \mathrm{dis}_{\mathrm{pitch_h}} \cdot M_{\mathrm{crep}}} - 1 \right|, \tag{3.36}$$

where $\alpha_{\mathrm{s_z}}$ denotes the angular sampling interval in z-direction (Section 3.3.1).

3.2.13 Mismatch of frame rates

Imager frame rate and (if specified) video frame rate may differ. Special techniques can be used to convert signals recorded at the imager frame rate into signals at video frame rate. These techniques involve electronic zoom, line and column replication, frame integration, as well as different read/write rates of the sampled data from/to the digital memory. A combination of the mentioned techniques may be used to fix incompatibilities.

The image can be displayed without loss of information if

$$F_{\mathrm{r,video}} \geq \frac{F_{\mathrm{r,imager}}}{M_{\mathrm{zoom}}^2 \cdot M_{\mathrm{lrep}} \cdot M_{\mathrm{crep}} \cdot N_{\mathrm{frames}}}, \tag{3.37}$$

where $F_{\mathrm{r,video}}$ is the display frame rate and N_{frames} is the number of integrated frames.

3.3 Sampling characteristics

3.3.1 Sampling distances

For 1st generation scanning imager without horizontal sampling (*Thermal category*), a sampling distance is defined for the vertical direction only; for all other imager types, sampling distances exist for both horizontal and vertical directions.

Scanning imagers

The vertical sampling distance for an imager with serial scan is given by

$$\alpha_{s_v} = \frac{\mathrm{FOV_v} \cdot f_{opt_v} - a_{det_v}}{f_{opt_v} \cdot (N_{scl} - 1)}. \tag{3.38}$$

For other scanning systems it is

$$\alpha_{s_v} = \frac{a_{pitch_v}}{f_{opt_v} \cdot n_{inter}}. \tag{3.39}$$

The horizontal sampling distance for scanning imagers with horizontal sampling is given by

$$\alpha_{s_h} = \frac{\mathrm{FOV_h} \cdot f_{opt_h} - a_{det_h}}{f_{opt_h} \cdot (N_{sam_h} - 1)}. \tag{3.40}$$

Staring or micro-scanned imagers

For staring and micro-scanned imagers, the sampling distance in z-direction is defined as

$$\alpha_{s_z} = \frac{a_{pitch_z}}{f_{opt_z} \cdot N_{\mu step_z}}, \tag{3.41}$$

with the exception of ICCD/ICMOS where it reads

$$\alpha_{s_z} = \frac{a_{pitch_z}}{f_{opt_z} \cdot M_{ICCD} \cdot N_{\mu step_z}}. \tag{3.42}$$

3.3.2 Sampling frequency and Nyquist frequency

Sampling frequencies are the inverse of the angular sampling distance in any given direction:

$$r_{s_z} = \frac{1}{\alpha_{s_z}}. \tag{3.43}$$

The Nyquist frequency is half of the sampling frequency:

$$r_{N_z} = \frac{1}{2 \cdot \alpha_{s_z}}. \tag{3.44}$$

3.4 Focal length and F-number

3.4.1 Effective focal length

If the focal length is different for the horizontal and vertical directions, the *effective focal length* is defined by

$$f_{\mathrm{opt}} = \sqrt{f_{\mathrm{opt_h}} \cdot f_{\mathrm{opt_v}}}\,. \tag{3.45}$$

3.4.2 Effective F-number of the front optics

For frontal optics with a simple circular aperture, the *F-number* is given by

$$F_{\mathrm{opt}} = \frac{f_{\mathrm{opt}}}{D_{\mathrm{opt}}}, \tag{3.46}$$

where D_{opt} is the aperture diameter of the front optics.

For complex optical systems, e.g. for those with truncated pupils, an *effective F-number* is defined by

$$F_{\mathrm{eff_opt}} = \frac{f_{\mathrm{opt}}}{D_{\mathrm{eff}}} \tag{3.47}$$

with

$$D_{\mathrm{eff}} = \sqrt{\frac{4A_{\mathrm{pup}}}{\pi}}, \tag{3.48}$$

where A_{pup} denotes the clear pupil area of the front optics.

For rectangular apertures, the aperture diameter can be calculated as follows

$$D_{\mathrm{opt}} = \sqrt{W_{\mathrm{trunc_h}}^2 + W_{\mathrm{trunc_v}}^2}\,, \tag{3.49}$$

where $W_{\mathrm{trunc_h}}$ = width of the truncated aperture in the horizontal direction, and
$W_{\mathrm{trunc_v}}$ = width of the truncated aperture in the vertical direction.

3.4.3 Effective cold shield F-number and cold shield efficiency (categories *Thermal* and *General*)

The effective F-number of the cold shield, $F_{\mathrm{eff_cs}}$, and the cold shield efficiency, η_{cs}, are related by

$$\eta_{\mathrm{cs}} = \frac{4F_{\mathrm{eff_cs}}^2 + 1}{4F_{\mathrm{eff_opt}}^2 + 1}\,. \tag{3.50}$$

For truncated optical systems, or for those with a central obscuration, a cold shield efficiency greater than a critical cold shield efficiency, $\eta_{\mathrm{cs,critical}}$, given by

$$\eta_{\mathrm{cs,critical}} = \frac{4F_{\mathrm{opt}}^2 + 1}{4F_{\mathrm{eff_opt}}^2 + 1} \tag{3.51}$$

can only be reached if a special radiation shield is used.

The minimum value for the cold shield efficiency, is defined by the hemisphere and given by

$$\eta_{\mathrm{cs,min}} = \frac{1}{4F_{\mathrm{eff_opt}}^2 + 1}. \tag{3.52}$$

3.5 Sensitivity

3.5.1 Spectral response of the imager

The imager spectral response is given by

$$R_{\mathrm{sys}}(\lambda) = \tau_{\mathrm{opt}}(\lambda) \cdot \tau_{\mathrm{filter}}(\lambda) \cdot R_{\mathrm{DA}}(\lambda)\,, \tag{3.53}$$

where $\tau_{\mathrm{opt}}(\lambda) =$ spectral transmission of the front optics,
$\tau_{\mathrm{filter}}(\lambda) =$ spectral transmission of additional filters, and
$R_{\mathrm{DA}}(\lambda) =$ spectral responsivity of the detector assembly.

The normalized spectral response of the imager, $R_{\mathrm{sys_norm}}(\lambda)$, is given by

$$\begin{aligned} R_{\mathrm{sys_norm}}(\lambda) &= \frac{R_{\mathrm{sys}}(\lambda)}{\max_\lambda\,(R_{\mathrm{sys}}(\lambda))} \\ &= \frac{\tau_{\mathrm{opt}}(\lambda) \cdot \tau_{\mathrm{filter}}(\lambda) \cdot R_{\mathrm{DA}}(\lambda)}{\max_\lambda\,(R_{\mathrm{sys}}(\lambda))} \\ &= \frac{\tau_{\mathrm{opt_norm}}(\lambda) \cdot \tau_{\mathrm{filter_norm}}(\lambda) \cdot R_{\mathrm{DA_norm}}(\lambda)}{R_{\max}}, \end{aligned} \tag{3.54}$$

where the subscript *norm* denotes normalized spectral data and

$$R_{\max} = \max_\lambda(\tau_{\mathrm{opt_norm}}(\lambda) \cdot \tau_{\mathrm{filter_norm}}(\lambda) \cdot R_{\mathrm{DA_norm}}(\lambda))\,. \tag{3.55}$$

Thus, $R_{\max}$ is the maximum value of the multiplied normalized spectral data.

3.5.2 Spectral windows

During the actual calculation, three different spectral windows are used by TRM: the detector, the imager (device), and the (overall) system spectral window.

Detector spectral window

The detector spectral window is determined by the detector spectral responsivity. Photons with energies within the detector spectral window may originate from background, target, or path radiance, but are also emitted by the optics, device inner walls, and the cold shield. These photons are detected and converted into carriers and therefore contribute to background and noise.

Imager spectral window

The combination of detector responsivity, spectral transmission of the front optics, and optical filters determine the imager spectral window. It indicates the maximum spectral range for photons coming from outside.

Overall spectral window

The overall spectral window is determined by the combination of imager spectral window, spectral illumination, spectral reflectivity and/or emissivity of target and background, and the atmospheric spectral transmission. The overall spectral window may change with range.

Note:	The spectral windows may consist of two or more sub-windows separated by region(s) with zero responsivity.

Note:	If external spectral MTFs are specified (optics and/or detector), the spectral window within which they are specified takes part in setting the overall spectral window.

Note:	TRM is not, in general, carrying out extrapolations on external data. In case specified external spectral data is limited to a certain spectral window, no data outside of that window will be calculated. An exception: In case spectral reflectivity of the target and/or the background is specified (*Visible* category only), the spectral reflectivity outside the band specified in the external data is assumed to be 0.

3.5.3 Effective scene radiance and differential radiance (*Thermal* category only)

The effective radiance of a blackbody at background temperature T is given by

$$L_{\text{eff}}(T) = \int_0^{\infty} R_{\text{sys_norm}}(\lambda) \cdot L(\lambda, T)\, \mathrm{d}\lambda \,, \tag{3.56}$$

where $L(\lambda, T)$ is the Planck blackbody spectral radiance:

$$L(\lambda, T) = \frac{2hc^2}{\lambda^5} \cdot \frac{1}{\exp\left\{\frac{hc}{\lambda k T}\right\} - 1} \,. \tag{3.57}$$

Small changes in radiance due to small changes in temperature ΔT are calculated by using the first order linear approximation

$$L_{\mathrm{eff}}(T + \Delta T) = L_{\mathrm{eff}}(T) + W_{\mathrm{eff}}(T) \cdot \Delta T, \tag{3.58}$$

where $W_{\mathrm{eff}}(T)$ is the effective differential radiance of a blackbody at temperature T. It is given by

$$W_{\mathrm{eff}}(T) = \left.\frac{\mathrm{d}L_{\mathrm{eff}}(T')}{\mathrm{d}T'}\right|_{T} = \int_0^{\infty} R_{\mathrm{sys_norm}}(\lambda) \left.\frac{\partial L(\lambda, T')}{\partial T'}\right|_{T} \mathrm{d}\lambda\,. \tag{3.59}$$

Note: In several places in TRM, the exact difference $L_{\mathrm{eff}}(T_2) - L_{\mathrm{eff}}(T_1)$ is calculated.

3.5.4 Normalized spectral responsivity from quantum efficiency

If the spectral quantum efficiency is specified, the normalized spectral responsivity will be calculated from the specified data.

As a first step, the spectral responsivity of the detector assembly $R_{\mathrm{DA}}(\lambda)$ is calculated. It is given by

$$R_{\mathrm{DA}}(\lambda) = \frac{e \cdot \lambda}{h \cdot c} \cdot \frac{\eta(\lambda)}{100}, \tag{3.60}$$

where $\eta(\lambda)$ is the spectral quantum efficiency of the detector specified in [%].

From $R_{\mathrm{DA}}(\lambda)$, the wavelength of peak responsivity λ_{p}, and the quantum efficiency at λ_{p}, $\eta_{\lambda_{\mathrm{p}}}$, are derived. The normalized spectral responsivity is then obtained as

$$R_{\mathrm{DA_norm}}(\lambda) = \frac{\eta(\lambda) \cdot \lambda}{\eta_{\lambda_{\mathrm{p}}} \cdot \lambda_{\mathrm{p}}} \tag{3.61}$$

and the peak responsivity $R_{\mathrm{DA_max}}$ of the detector assembly is given by

$$R_{\mathrm{DA_max}} = \frac{e \cdot \lambda_p}{h \cdot c} \cdot \frac{\eta_{\lambda_{\mathrm{p}}}}{100}. \tag{3.62}$$

Note: The calculated values of λ_p and $\eta_{\lambda_{\mathrm{p}}}$ are reported in the *System Performance Data* dialog.

3.5.5 Detectivities (*Thermal* category)

For a long time, the peak detectivity D^*_{peak} was used as a defining property for the performance of thermal imagers. Thus, for backward-compatibility, TRM will calculate the actual peak detectivity of a thermal imager for a given detector-dewar assembly at a given background temperature. Two other variants of the D^*_{peak} value, the optimized peak detectivity, and the BLIP (Background Limited Performance) peak detectivity are also calculated at the specified background temperature. The calculated values are reported in the *System Performance Data* dialog.

Peak detectivity

The peak detectivity for non-integrating detectors is given by

$$D^*_{\mathrm{peak}}(T) = \frac{R_{\mathrm{DA_max}} \cdot (\Delta f \cdot A_{\mathrm{det}})^{1/2}}{\sigma_{\mathrm{det_noi}}}, \tag{3.63}$$

where $R_{\mathrm{DA_max}}$ = peak responsivity of the detector assembly,
Δf = noise bandwidth, and
$\sigma_{\mathrm{det_noi}}$ = (non-integrating) detector noise (Equation (3.137)).

For integrating detectors, it is given by

$$D^*_{\mathrm{peak}}(T) = \frac{R_{\mathrm{DA_max}} \cdot (t_{\mathrm{int}} \cdot A_{\mathrm{det}})^{1/2}}{\sqrt{2} \cdot e \cdot \sigma_{\mathrm{det_int}}}, \tag{3.64}$$

where t_{int} is the integration time and $\sigma_{\mathrm{det_int}}$ is the (integrating) detector noise (Equation (3.130)).

Note: The apparent inconsistency between the units in the two equations above owes to the fact that noise is defined in number of carriers for integrating detectors and in current units for non-integrating detectors.

Optimized D^*_{peak}

The optimized D^*_{peak} is the peak detectivity obtained for an optimized imager. An imager is called “optimized” when the transmission of the optics as well as the cold shield efficiency are equal to one and the cold shield temperature is 0 Kelvin. For an optimized imager, the only radiative contribution to the signal is due to scene radiation.

Background limited performance (BLIP) D^*_{peak}

The BLIP peak detectivity is the peak detectivity obtained for the actual imager with an idealized detector. An idealized detector is a detector without dark current, no additive noise floor as well as a noise figure value of one and no quantization noise.

3.5.6 Responsivity calculations (*Visible* category)

For the Visible category, responsivity can be entered in several ways. The peak responsivity is then calculated from the values given. In the following [Unit] can stand for [Volts], [DN] (Digital Numbers), or [e] (electrons). $R_{\mathrm{DA_max}}$ units are [Unit/J].

Responsivity R_{sp} specified in [Unit/(J/m²)]

In this case, peak responsivity R_{DA_max} is given by

$$R_{DA_max} = R_{sp} \cdot \frac{\int_0^\infty L(\lambda) \cdot \tau_{IR_cut}(\lambda)\, d\lambda}{A_{det} \cdot \int_0^\infty L(\lambda) \cdot \tau_{IR_cut}(\lambda) \cdot R_{DA_norm}(\lambda)\, d\lambda}, \tag{3.65}$$

where $\tau_{IR_cut}(\lambda)$ = transmission of IR cut-off filter used for responsivity measurements and
$L(\lambda)$ = radiance of the broadband responsivity measurement source, defined by its color temperature.

For a measurement setup with a single wavelength λ_0 this reduces to

$$R_{DA_max} = \frac{R_{sp}}{A_{det} \cdot R_{DA_norm}(\lambda_0)}. \tag{3.66}$$

Responsivity R_{sp} specified in [Unit/(lux sec)]

If the responsivity is specified in the data sheet in [Unit/(lux sec)], the peak responsivity R_{DA_max} is given by

$$R_{DA_max} = R_{sp} \cdot \frac{683 \cdot \int_0^\infty L(\lambda) \cdot V(\lambda) \cdot \tau_{IR_cut}(\lambda)\, d\lambda}{A_{det} \cdot \int_0^\infty L(\lambda) \cdot \tau_{IR_cut}(\lambda) \cdot R_{DA_norm}(\lambda)\, d\lambda}, \tag{3.67}$$

where $V(\lambda)$ is the CIE photopic luminous efficiency function.

For measurement setups with a single wavelength λ_0 this reduces to

$$R_{DA_max} = R_{sp} \cdot \frac{683 \cdot V(\lambda_0)}{A_{det} \cdot R_{DA_norm}(\lambda_0)}. \tag{3.68}$$

Responsivity R_{sp} specified in [Unit/sec]

In this case, peak responsivity R_{DA_max} is given by

$$R_{DA_max} = R_{sp} \cdot \frac{(4F_{meas}^2 + 1)}{\pi \cdot A_{det} \cdot \int_0^\infty L(\lambda) \cdot \tau_{IR_cut}(\lambda) \cdot R_{DA_norm}(\lambda)\, d\lambda} \tag{3.69}$$

or, for a single wavelength λ_0 by

$$R_{DA_max} = R_{sp} \cdot \frac{(4F_{meas}^2 + 1)}{\pi \cdot A_{det} \cdot L(\lambda_0) \cdot \tau_{IR_cut}(\lambda_0) \cdot R_{DA_norm}(\lambda_0)}, \tag{3.70}$$

where F_{meas} denotes the F-number of the measurement setup.

Responsivity R_{sp} specified in [A/W]

Here, peak responsivity $R_{\mathrm{DA_max}}$ is given by

$$R_{\mathrm{DA_max}} = R_{\mathrm{sp}} \cdot \frac{\int_0^{\infty} L(\lambda) \cdot \tau_{\mathrm{IR_cut}}(\lambda)\, \mathrm{d}\lambda}{\int_0^{\infty} L(\lambda) \cdot \tau_{\mathrm{IR_cut}}(\lambda) \cdot R_{\mathrm{DA_norm}}(\lambda)\, \mathrm{d}\lambda}, \tag{3.71}$$

or, for a single wavelength λ_0:

$$R_{\mathrm{DA_max}} = \frac{R_{\mathrm{sp}}}{R_{\mathrm{DA_norm}}(\lambda_0)}. \tag{3.72}$$

Conversion factor

For further calculation, TRM4 converts the peak responsivity from their specified units to [A/W] and designates the value by r_{max}. Depending on how the responsivity was specified, a conversion factor C_{f} is needed:

- Unit in electrons or Amperes: $C_{\mathrm{f}} = 1$ and $r_{\mathrm{max}} = R_{\mathrm{DA_max}}$,
- Unit in Volts, C_{f} in [V/e]: $r_{\mathrm{max}} = R_{\mathrm{DA_max}} \cdot \frac{e}{C_{\mathrm{f}}}$,
- Unit in Digital Numbers, C_{f} in [e/DN]: $r_{\mathrm{max}} = R_{\mathrm{DA_max}} \cdot C_{\mathrm{f}} \cdot e$.

3.5.7 Integration time

The integration time is interrelated with the integration duty cycle and the well-fill.

Scanning imagers

The relation between detector integration time and integration duty cycle, η_{cyc}, is given by

$$t_{\mathrm{int}} = \frac{\eta_{\mathrm{scan}} \cdot \eta_{\mathrm{cyc}}}{F_{\mathrm{r,imager}} \cdot N_{\mathrm{sam_h}} \cdot m_{\mathrm{blocks}} \cdot n_{\mathrm{inter}}}. \tag{3.73}$$

Staring or micro-scanned imagers

For staring or micro-scanned imagers (except diagonal scan), the detector integration time is given by

$$t_{\mathrm{int}} = \frac{\eta_{\mathrm{cyc}}}{F_{\mathrm{r,imager}} \cdot N_{\mu\mathrm{step_h}} \cdot N_{\mu\mathrm{step_v}}} \tag{3.74}$$

and for diagonal micro-scan it is

$$t_{\mathrm{int}} = \frac{\eta_{\mathrm{cyc}}}{F_{\mathrm{r,imager}} \cdot 2}. \tag{3.75}$$

Note: If electronic interlace or pseudo-interlace is switched on, the integration time t_{int} must be divided by a factor of two.

Integration time from well-fill

Alternatively, integration time may be specified by the desired well fill p (in percent). In that case, the integration time required to fill the storage gate to the desired percentage p of its capacity is given by (see also Section 3.8.2)

$$t_{\mathrm{int}}(R) = \frac{p \cdot N_{\mathrm{stor}}}{\left(j_{\mathrm{total_B}}(R) + j_{\mathrm{dark}}\right) \cdot A_{\mathrm{det}} \cdot E_{\mathrm{gain}} \cdot 100}, \tag{3.76}$$

where p = desired well fill [in percent],
E_{gain} = electronic gain factor,
$j_{\mathrm{total_B}}(R)$ = carrier flux density due to the background (Section 3.7.1), and
j_{dark} = flux density of dark carriers (Section 3.7.5).

In case of ICCD/ICMOS systems, $j_{\mathrm{ICCD_total_B}}(R)$ and $j_{\mathrm{ICCD_dark}}$ have to be used instead of $j_{\mathrm{total_B}}(R)$ and j_{dark}, respectively.

3.5.8 Image intensifier noise figure from measured SNR

TRM4 assumes that the specified SNR of image intensifiers is measured in accordance with the military specification [7], but possibly with a different color temperature of the measurement source. The image intensifier noise figure is calculated from the specified SNR by

$$\mathrm{NF_{II}} = \frac{\sqrt{\frac{\pi}{2} \cdot 10^{-9}[\mathrm{m^2 s}] \cdot j_{\mathrm{pc}}\left[\frac{1}{\mathrm{m^2 s}}\right]}}{\mathrm{SNR}}, \tag{3.77}$$

with the photocathode carrier flux density

$$j_{\mathrm{pc}} = \frac{1}{e} \cdot \frac{108\,\mu\mathrm{lux}}{K_m} \cdot \int \frac{\lambda}{hc} \cdot \frac{\eta_{\mathrm{pc}}(\lambda)}{100} \cdot \frac{L\left(\lambda, T = T_{\mathrm{spec}}\right)}{\int \mathrm{d}\lambda'\, L\left(\lambda', T = T_{\mathrm{spec}}\right) \cdot V(\lambda')} \mathrm{d}\lambda\,, \tag{3.78}$$

where $K_m = 683\ \mathrm{lm/W}$, the maximum luminous efficacy for photopic vision,
$V(\lambda)$ = the spectral luminous efficiency function for photopic vision [11],
$\eta_{\mathrm{pc}}(\lambda)$ = the spectral quantum efficiency of the photocathode, and
$L(\lambda, T)$ = the blackbody spectral radiance of a source with temperature T.

In case the photocathode is sensitive in the visible spectral band only and the user has specified the luminous responsivity, the following simple formula is used:

$$\mathrm{NF_{II}} = \frac{1.03 \cdot \sqrt{R_l}}{\mathrm{SNR}}\,. \tag{3.79}$$

Here, R_l is the luminous responsivity in $\frac{\mu\mathrm{A}}{\mathrm{lm}}$ units and it is assumed that R_l is measured using a source with the same color temperature as in the SNR measurement.

3.6 Scene illumination and target orientation

3.6.1 Effective target dimensions

The decisive angles for the calculation of effective target areas and target dimensions are the target viewing aspect (front/side/diagonal) and the imager LOS (line-of-sight) to target (constant angle/constant altitude). These angles correspond to α_z in Figure 2.36 with z denoting the horizontal and vertical directions, respectively.

Effective target area

Given a target of width W, height H, and depth D, the effective areas as seen by the imager are

$$\begin{aligned}
\text{Front:} \quad & A_{\text{front}} = W \cdot H \cdot \cos\alpha_{\text{h}} \cdot \cos\alpha_{\text{v}}\,, \\
\text{Side:} \quad & A_{\text{side}} = D \cdot H \cdot \sin\alpha_{\text{h}} \cdot \cos\alpha_{\text{v}}\,, \\
\text{Top:} \quad & A_{\text{top}} = W \cdot D \cdot \sin|\alpha_{\text{v}}| \;\text{ for }\; \alpha_{\text{v}} > 0 \;\text{ else }\; A_{\text{top}} = 0, \\
\text{Bottom:} \quad & A_{\text{bottom}} = W \cdot D \cdot \sin|\alpha_{\text{v}}| \;\text{ for }\; \alpha_{\text{v}} < 0 \;\text{ else }\; A_{\text{bottom}} = 0,
\end{aligned} \tag{3.80}$$

where α_z is measured from the normal to the front of the target. The resulting total (projected) target area is

$$A_{\text{tar}} = (W \cdot \cos\alpha_{\text{h}} + D \cdot \sin\alpha_{\text{h}}) \cdot H \cdot \cos\alpha_{\text{v}} + W \cdot D \cdot \sin|\alpha_{\text{v}}|\,. \tag{3.81}$$

Note: While $\alpha_{\text{h}} = 0°$ for frontal aspect, and $\alpha_{\text{h}} = 90°$ for side aspect, the angle depends on the actual target dimensions for diagonal aspect, and can be found from

$$\alpha_{\text{h}} = \cos^{-1}\left(\frac{W}{\sqrt{W^2 + D^2}}\right). \tag{3.82}$$

Effective target horizontal dimension

The effective horizontal dimension of the target is given by

$$\alpha_{\text{tar}_{\text{h}}} = W \cdot \cos\alpha_{\text{h}} + D \cdot \sin\alpha_{\text{h}}\,. \tag{3.83}$$

Effective target vertical dimension

The effective vertical dimension of the target is given by

$$\alpha_{\text{tar}_{\text{v}}} = H \cdot \cos\alpha_{\text{v}} + \frac{W \cdot D \cdot \sin|\alpha_{\text{v}}|}{W \cdot \cos\alpha_{\text{h}} + D \cdot \sin\alpha_{\text{h}}}\,. \tag{3.84}$$

3.6.2 Emitted and reflected radiance

TRM4 assumes targets to be Lambertian reflectors or emitters. In the *Thermal* category, the background is also assumed to be a Lambertian emitter. In the *Visible/NIR/SWIR* and *General* categories, on the other hand, the background can be either a Lambertian reflector/emitter or a sky background, depending on the specified scenario:

- For *slant observation* from above, the background is modelled as Lambertian reflector/emitter plane below the target;
- For *slant observation* from below, sky background has to be specified;
- For *horizontal observation*, the user can choose whether to specify the background as a Lambertian reflector/emitter or as sky background. In case a Lambertian reflector/emitter is specified, the background is assumed to be a plane behind the target.

For Lambertian reflectors/emitters, the power per unit surface area and unit solid angle emitted by the target and background surface or reflected from it is given by Equations (3.85) and (3.86), respectively. For sky background, the sky radiance is given by Equations (3.87) to (3.91).

Note: For slant path observation from above, only point source fixed in space with a vertical angle $\beta_v \geq 0$ can be specified in TRM4.

Emitted radiance ERI

The emitted spectral radiance $[\mathrm{W}/(\mathrm{m}^2 \cdot \mathrm{sr} \cdot \mu\mathrm{m})]$ of the background/target surfaces is

$$\mathrm{ERI}_{\mathrm{facet}}(\lambda) = \varepsilon(\lambda) \cdot \frac{2c^2 h}{\lambda^5 \left(\mathrm{e}^{\frac{hc}{\lambda k T}} - 1\right)} \cdot \cos\theta\,, \tag{3.85}$$

where $\varepsilon(\lambda)$ = spectral emissivity of the target/background,
T = target/background temperature, and
θ = angle measured from the surface normal.

Reflected radiance RRI

$$\mathrm{RRI}_{\mathrm{facet}}(\lambda) = \frac{(1 - \varepsilon(\lambda)) \cdot J_{\mathrm{facet}}(\lambda) \cdot \cos\theta}{\pi}, \tag{3.86}$$

where $J_{\mathrm{facet}}(\lambda)$ is the spectral irradiance incident on the actual area of a facet of the target.

Sky radiance

If sky background is specified by *Effective sky radiance*, the sky radiance is given by

$$L_{\mathrm{sky}}(\lambda) = \frac{\mathrm{L}_{\mathrm{sky_eff}}}{\Delta\lambda}, \tag{3.87}$$

where $\mathrm{L}_{\mathrm{sky_eff}}$ is the specified effective sky radiance and Δλ is the spectral width of the imager spectral response.

For sky background defined by *Color temperature* and *Radiance at peak*, the sky radiance is given by

$$L_{\mathrm{sky}}(\lambda) = \frac{L\left(T_{\mathrm{sky}}, \lambda\right)}{\max_{\lambda}\left(L\left(T_{\mathrm{sky}}, \lambda\right)\right)} \cdot L_{\mathrm{max}}\,, \tag{3.88}$$

where $\mathrm{L}\left(\mathrm{T}_{\mathrm{sky}}, \lambda\right)$ is the spectral radiance of a blackbody with specified color temperature $\mathrm{T}_{\mathrm{sky}}$ and L_{max} is the specified radiance at peak.

If sky radiance is specified by *Normalized Spectral Radiance,* $\mathrm{L}_{\mathrm{sky_norm}}(\lambda)$, and *Radiance at peak*, L_{max}, then the sky radiance is given by

$$L_{\mathrm{sky}}(\lambda) = \mathrm{L}_{\mathrm{sky_norm}}(\lambda) \cdot L_{\mathrm{max}}\ . \tag{3.89}$$

In case the *Atmosphere* option is used to define sky background and no external path radiance data is specified, then the sky radiance is calculated as the path radiance for an infinite path length. The contribution of air emission is given by

$$L_{\mathrm{sky_E}}(\lambda) = L(T_{\mathrm{air}}, \lambda) \cdot (1 - \tau_{\mathrm{atm}}(\lambda, R \to \infty))\,, \tag{3.90}$$

where T_{air} is the air temperature and $\tau_{\mathrm{atm}}(\lambda, R)$ is the atmospheric transmission at distance *R*. The contribution of light reflected from atmosphere is determined via

$$L_{\mathrm{sky}_R}(\lambda) = I(\lambda) \cdot F_{\mathrm{PR}} \cdot \left(1 - \tau_{\mathrm{atm}}(\lambda, R \to \infty)\right), \tag{3.91}$$

where $I(\lambda)$ is the spectral irradiance of the illumination source and F_{PR} is the path radiance factor defined in Equation (2.31). In case that external path radiance data is specified, the sky radiance is assumed to be the path radiance at maximum specified range.

3.6.3 Correction factor for reflected radiance

For directional radiation, the actual irradiance on the facets of the target and background is determined by the two angles β_{h} and β_{v}. These are the horizontal and vertical angles of the irradiating beam with respect to the normal to the front of the target (see Figure 2.36).

Irradiance on target

The irradiance on the different surfaces of the target box are given by

$$\begin{aligned}
&J_{\mathrm{front}}(\lambda) = I(\lambda) \cdot \cos\beta_h \cdot \cos\beta_v\,,\\
&J_{\mathrm{side}}(\lambda) = I(\lambda) \cdot \sin\beta_{\mathrm{h}} \cdot \cos\beta_{\mathrm{v}}\,,\\
&J_{\mathrm{top}}(\lambda) = I(\lambda) \cdot \sin|\beta_v| \text{ for } \beta_v > 0 \text{ else } J_{\mathrm{top}}(\lambda) = 0,\\
&J_{\mathrm{bottom}}(\lambda) = I(\lambda) \cdot \sin|\beta_v| \text{ for } \beta_v < 0 \text{ else } J_{\mathrm{bottom}}(\lambda) = 0,
\end{aligned} \tag{3.92}$$

where $I(\lambda)$ is the spectral irradiance of the illuminating source at target in a plane perpendicular to the direction of beam propagation. Spectral irradiance is given in $[\mathrm{W}/(\mathrm{m}^2 \cdot \mu m)]$.

Irradiance on background

For *horizontal observation*, the background is assumed to be a flat plane behind the target, which is normal to the imager line of sight to the target. Thus, the irradiance on the background is given by

$$J_{\mathrm{back}}(\lambda) = I(\lambda) \cdot \cos(\beta_{\mathrm{h}} - \alpha_h) \cdot \cos \beta_{\mathrm{v}} \,. \tag{3.93}$$

For *slant observation* from above, the background is assumed to be a flat horizontal plane below the target (flat ground). The irradiance on the background thus depends on the irradiation angle and is given by

$$J_{\mathrm{back}}(\lambda) = I(\lambda) \cdot \sin \beta_v \,. \tag{3.94}$$

For *slant observation* from below, sky background has to be specified.

Effective reflected radiance from target

The actual radiant intensity reflected from the target in the direction of the imager can now be calculated:

$$\begin{aligned} E_{\mathrm{total,r}}(\lambda) &= E_{\mathrm{front}}(\lambda) + E_{\mathrm{side}}(\lambda) + E_{\mathrm{top/bottom}}(\lambda) \\ &= W \cdot H \cdot \mathrm{RRI}_{\mathrm{front}}(\lambda) + D \cdot H \cdot \mathrm{RRI}_{\mathrm{side}}(\lambda) + W \cdot D \cdot \mathrm{RRI}_{\mathrm{top/bottom}}(\lambda) \\ &= \frac{I(\lambda)\left(1 - \varepsilon(\lambda)\right)}{\pi} (A_{\mathrm{front}} \cos \beta_{\mathrm{h}} \cos \beta_{\mathrm{v}} + A_{\mathrm{side}} \sin \beta_{\mathrm{h}} \cos \beta_{\mathrm{v}} + \delta\, A_{\mathrm{top}} \sin|\beta_{\mathrm{v}}|). \end{aligned} \tag{3.95}$$

The factor δ in front of A_{top} is added in order to take into account scenario where reflected light from target top does not reach the imager. $\delta = 1$ in case of slant observation from above and directional irradiation from above or slant observation from below and directional irradiation from below. $\delta = 0$ for slant observation from below and directional irradiation from above.

This radiant intensity is related to an effective (flat) target perpendicular to the line of sight with the same reflectivity as the original target and a projected area given by Equation (3.81). This leads to the total reflected spectral radiance

$$\begin{aligned} \mathrm{RRI}_{\mathrm{tar}}(\lambda) &= \frac{E_{\mathrm{total,r}}\,(\lambda)}{A_{\mathrm{tar}}} \\ &= \frac{A_{\mathrm{front}} \cos \beta_{\mathrm{h}} \cos \beta_{\mathrm{v}} + A_{\mathrm{side}} \sin \beta_{\mathrm{h}} \cos \beta_{\mathrm{v}} + \delta\, A_{\mathrm{top}} \sin|\beta_{\mathrm{v}}|)}{(W \cdot \cos \alpha_{\mathrm{h}} + D \cdot \sin \alpha_{\mathrm{h}}) \cdot H \cdot \cos \alpha_{\mathrm{v}} + W \cdot D \cdot \sin \alpha_{\mathrm{v}}} \cdot \frac{I(\lambda)(1 - \varepsilon(\lambda))}{\pi} \\ &= F_{\mathrm{irr,T}} \cdot \frac{I(\lambda) \cdot \left(1 - \varepsilon(\lambda)\right)}{\pi}, \end{aligned} \tag{3.96}$$

where $F_{\mathrm{irr,T}}$ is the target irradiative correction factor.

Effective reflected radiance from background

Similarly, a correction factor for the background is used with directional sources. For horizontal observation, this factor is given by

$$F_{\mathrm{irr,B}} = \cos(\beta_{\mathrm{h}} - \alpha_{\mathrm{h}}) \cdot \cos\beta_{\mathrm{v}}\,, \tag{3.97}$$

for slant observation paths from above by

$$F_{\mathrm{irr,B}} = \sin\beta_{\mathrm{v}}\,. \tag{3.98}$$

Note: For laboratory measurement calculations, the target and background are both assumed to be perpendicular to the (horizontal) line of sight between imager and target and

$$F_{\mathrm{irr,B}} = F_{\mathrm{irr,T}} = \cos\beta_{\mathrm{h}} \cdot \cos\beta_{\mathrm{v}}\,. \tag{3.99}$$

3.7 Focal plane irradiance and carrier flux density

3.7.1 Effective irradiance in the focal plane

The irradiance in the focal plane is composed of five contributions:

$$J_{\mathrm{fp}}(\lambda, R) = J_{\mathrm{fp_scene}}(\lambda, R) + J_{\mathrm{fp_optics}}(\lambda) + J_{\mathrm{fp_device}}(\lambda) + J_{\mathrm{fp_cs}}(\lambda) + J_{\mathrm{fp_path}}(\lambda, R)\,, \tag{3.100}$$

where
$J_{\mathrm{fp_scene}}(\lambda, R) =$ irradiance due to light emitted and reflected from the target/background,
$J_{\mathrm{fp_optics}}(\lambda) =$ irradiance due to emission of the (warm) optics,
$J_{\mathrm{fp_device}}(\lambda) =$ irradiance due to emission of the device walls,
$J_{\mathrm{fp_cs}}(\lambda, R) =$ irradiance due to emission of the cold shield,
$J_{\mathrm{fp_path}}(\lambda, R) =$ irradiance due to light emitted or reflected from the path between the target and the imager.

It is in general dependent on the range R from the target.

Note: Internal reflected radiation from the optics, device walls, and cold shield walls is not taken into account in TRM4.

It is assumed that for imagers in the *Thermal* category:

$$J_{\mathrm{fp_scene}}(\lambda, R) + J_{\mathrm{fp_path}}(\lambda, R) = J_{\mathrm{fp_scene}}(\lambda, 0) = J_{\mathrm{fp_scene}}(\lambda). \tag{3.101}$$

Hence, the total effective irradiance on the focal plane for the *Thermal* category is independent of range.

Total carrier flux density

The spectral irradiance on the focal plane can be translated into the carrier flux density $j_{\text{total_}i}$ by using the (specified or calculated) quantum efficiency:

$$j_{\text{total_}i}(\lambda, R) = j_{\text{e/r_}i}(\lambda, R) + j_{\text{optics}}(\lambda) + j_{\text{device}}(\lambda) + j_{\text{cs}}(\lambda) + j_{\text{path}}(\lambda, R). \tag{3.102}$$

The final step consists of eliminating the explicit dependence on wavelength by integrating over the relevant spectral window(s):

$$j_{\text{total_}i}(R) = \int_0^\infty j_{\text{total_}i}(\lambda, R)\, \mathrm{d}\lambda\,. \tag{3.103}$$

Note: The index i refers to either the target or the background.

Note: The contributions described in Sections 3.7.2 and 3.7.3 are valid for both the target and the background signal.

3.7.2 Contributions of optics, device, and cold shield (*Thermal* and *General* categories)

In *Thermal* and *General* category imagers, the optics, the device, and the cold shield emit photons in the spectral range of the detector. These photons contribute to the total carrier flux.

The carrier flux density due to the optics and filters is given by

$$j_{\text{optics}}(\lambda) = \frac{\lambda}{hc} \cdot \frac{\pi}{4F_{\text{eff_opt}}^2 + 1} \cdot L(T_{\text{opt}}, \lambda) \cdot \frac{(1 - \tau_{\text{opt}}(\lambda) \cdot \tau_{\text{filter}}(\lambda)) \cdot \eta(\lambda)}{100}, \tag{3.104}$$

due to the device by

$$j_{\text{device}}(\lambda) = \frac{\lambda}{hc} \cdot \frac{\pi}{4 \cdot F_{\text{eff_opt}}^2 + 1} \cdot \frac{(1 - \eta_{\text{cs}})}{\eta_{\text{cs}}} \cdot L(T_{\text{device}}, \lambda) \cdot \frac{\eta(\lambda)}{100}, \tag{3.105}$$

and due to the cold shield by

$$j_{\text{cs}}(\lambda) = \frac{\lambda \cdot \pi}{hc} \cdot \left(1 - \frac{1}{4 \cdot F_{\text{eff_cs}}^2 + 1}\right) \cdot L(T_{\text{cs}}, \lambda) \cdot \frac{\eta(\lambda)}{100}. \tag{3.106}$$

3.7.3 Contribution of path radiance

The path radiance contributes with two types of components to the carrier flux density: one is due to emission by the air itself and the other is due to scattering of light from an illumination source into the line of sight.

In case sky background is specified, no path radiance term is used in the total carrier flux density of the background signal $j_{\text{total_}B}(\lambda, R)$.

Emitted component (*General* and *Thermal* categories)

With the assumption that the air has an emissivity of $(1-\tau_{\mathrm{atm}}(\lambda,R))$, where $\tau_{\mathrm{atm}}(\lambda,R)$ is the wavelength and distance dependent transmission coefficient of the atmosphere, the carrier flux density caused by air emission is given by

$$j_{\mathrm{path_E}}(\lambda,R)=\frac{\lambda}{hc}\cdot\frac{\pi}{4F_{\mathrm{eff_opt}}^2}\cdot L(T_{\mathrm{air}},\lambda)\cdot(1-\tau_{\mathrm{atm}}(\lambda,R))\cdot\frac{\tau_{\mathrm{opt}}(\lambda)\cdot\tau_{\mathrm{filter}}(\lambda)\cdot\eta(\lambda)}{100}. \quad (3.107)$$

Note: In the *Thermal* category, T_{air} is assumed to equal the background temperature.

Reflected component (*General* and *Visible* categories)

The following relation describes the contribution of reflected source light:

$$j_{\mathrm{path_R}}(\lambda,R)=\frac{\lambda}{hc}\cdot\frac{1}{4F_{\mathrm{eff_opt}}^2}I(\lambda)\cdot F_{\mathrm{PR}}\cdot\left(1-\tau_{\mathrm{atm}}(\lambda,R)\right)\frac{\tau_{\mathrm{opt}}(\lambda)\cdot\tau_{\mathrm{filter}}(\lambda)\cdot\eta(\lambda)}{100}, \quad (3.108)$$

where F_{PR} is the path radiance factor defined in Equation (2.31).

Note: F_{PR} is specified in such a way that for cases where $F_{\mathrm{irr,B}}=1$ and F_{PR} set equal to the background reflectivity ρ_{B} the accumulated carrier flux on the focal plane due to irradiation from the background and path radiance is independent of range.

3.7.4 Contribution of target and background

The contribution of target and background consists of two components: carrier flux densities due to emitted and reflected radiation. It reads

$$j_{\mathrm{e/r_}i}(\lambda,R)=j_{\mathrm{e_}i}(\lambda,R)+j_{\mathrm{r_}i}(\lambda,R)\,, \quad (3.109)$$

where $j_{\mathrm{e_}i}$ = carrier flux density caused by emitted radiation and
$j_{\mathrm{r_}i}$ = carrier flux density caused by reflected radiation.

Here, as in the following, the index i stands for either target or background.

Scene emission (*Thermal* and *General* categories)

For both *Thermal* and *General* category imagers, the carrier flux density caused by radiation emitted from the scene is calculated using

$$j_{\mathrm{e_}i}(\lambda,R)=\frac{\lambda\pi}{hc}\cdot\frac{1}{4F_{\mathrm{eff_opt}}^2}L(T_i,\lambda)\cdot\varepsilon_i(\lambda)\cdot\tau_{\mathrm{atm}}(\lambda,R)\frac{\tau_{\mathrm{opt}}(\lambda)\cdot\tau_{\mathrm{filter}}(\lambda)\cdot\eta(\lambda)}{100}, \quad (3.110)$$

where T_i denotes the target/background temperature and $\varepsilon_i(\lambda)$ is the (wavelength-dependent) emissivity of target/background.

Scene reflection (*General* and *Visible/NIR/SWIR* categories)

For imagers of categories *General* and *Visual/NIR/SWIR*, the carrier flux density due to radiation of an external source reflected by the scene is given by

$$j_{\mathrm{r}_i}(\lambda, R) = \frac{\lambda}{hc} \cdot \frac{1}{4F_{\mathrm{eff_opt}}^2} I(\lambda) \cdot F_{\mathrm{irr},i} \cdot \rho_i(\lambda) \cdot \tau_{\mathrm{atm}}(\lambda, R) \frac{\tau_{\mathrm{opt}}(\lambda) \cdot \tau_{\mathrm{filter}}(\lambda) \cdot \eta(\lambda)}{100}, \quad (3.111)$$

where $I(\lambda)$ is the incident spectral irradiance as defined in Section 3.6.3 and $\rho_i(\lambda)$ denotes the reflectivity of the target/background.

Sky background emission (*General* and *Visible/NIR/SWIR* categories)

If sky background is specified for imagers of categories *General* and *Visual/NIR/SWIR*, the carrier flux density caused by radiation emitted from the sky background is given by

$$j_{\mathrm{e}_B}(\lambda, R) = j_{\mathrm{sky}}(\lambda, R) = \frac{\lambda}{hc} \cdot \frac{\pi}{4F_{\mathrm{eff_opt}}^2} \cdot L_{\mathrm{sky}}(\lambda) \cdot \frac{\tau_{\mathrm{opt}}(\lambda) \cdot \tau_{\mathrm{filter}}(\lambda) \cdot \eta(\lambda)}{100}. \quad (3.112)$$

3.7.5 Detector dark current

Detector dark current is specified in TRM4 as current per detector element, i_{dark}, and not as carrier flux density. Thus, the contribution of detector dark current to the carrier flux density can be written as

$$j_{\mathrm{dark}} \cdot A_{\mathrm{det}} = N_{\mathrm{dark}} = \frac{i_{\mathrm{dark}}}{e}. \quad (3.113)$$

3.7.6 Carrier flux density for ICCD/ICMOS and effective gain

Carrier flux density in photocathode plane

The carrier flux density generated at the photocathode is calculated in the same way as described above for imagers of the category *Visible/NIR/SWIR*, using the photocathode quantum efficiency, $\eta_{\mathrm{pc}}(\lambda)$, instead of the detector quantum efficiency:

$$\begin{aligned} j_{\mathrm{pc_total}_i}(\lambda, R) &= j_{\mathrm{pc}_i}(\lambda, R) + j_{\mathrm{pc_path_R}}(\lambda, R) \\ &= \frac{\lambda}{hc} \cdot \frac{1}{4F_{\mathrm{eff_opt}}^2} I(\lambda) \cdot F_{\mathrm{irr},i} \cdot \rho_i(\lambda) \cdot \tau_{\mathrm{atm}}(\lambda, R) \frac{\tau_{\mathrm{opt}}(\lambda) \cdot \tau_{\mathrm{filter}}(\lambda) \cdot \eta_{\mathrm{pc}}(\lambda)}{100} \\ &+ \frac{\lambda}{hc} \cdot \frac{1}{4F_{\mathrm{eff_opt}}^2} I(\lambda) \cdot F_{\mathrm{PR}} \cdot \left(1 - \tau_{\mathrm{atm}}(\lambda, R)\right) \frac{\tau_{\mathrm{opt}}(\lambda) \cdot \tau_{\mathrm{filter}}(\lambda) \cdot \eta_{\mathrm{pc}}(\lambda)}{100}. \end{aligned} \quad (3.114)$$

The contribution of the photocathode dark emission to the dark current is specified in TRM4 by EBI (Equivalent Background Illumination). The photocathode carrier flux density due to dark emission is given by:

$$j_{\mathrm{pc_dark}} = \int_0^\infty \frac{\lambda}{hc} \cdot \frac{\eta_{\mathrm{pc}}(\lambda)}{100} \cdot \mathrm{EBI}(\lambda)\, \mathrm{d}\lambda\,. \quad (3.115)$$

Note: The calculations described in this paragraph assume that EBI given in radiometric units. If EBI is specified in photometric units, it is converted to radiometric units using the colour temperature of the illumination source specified in the *Measurement Setup* box.

Note: If photocathode spectral response is specified via *Normalized Spectral Responsivity* and *Responsivity at peak* or *Responsivity at single wavelength,* the corresponding quantum efficiency is calculated by the relations given in Section 3.5.4. In case luminous responsivity is specified, responsivity is converted to radiometric units using the colour temperature of the illumination source specified in the *Measurement Setup* box.

Spectral radiance from phosphor screen

The spectral radiance from the phosphor screen is calculated using the idea of the monochromatic power gain method [12]. If luminous gain is specified, the spectral radiance from the phosphor screen due to target/background radiation and path radiance is given by:

$$L_{\mathrm{phos_}i}(\lambda,R) = G_l \cdot \frac{\int_0^\infty V(\lambda')\cdot L\left(\lambda',T_{\mathrm{spec}}\right)\mathrm{d}\lambda'}{\int_0^\infty \frac{\lambda'}{hc}\cdot\frac{\eta_{\mathrm{pc}}(\lambda')}{100}\cdot L\left(\lambda',T_{\mathrm{spec}}\right)\mathrm{d}\lambda'}\cdot\frac{\frac{1}{\pi}\cdot I_{\mathrm{phos}}(\lambda)\cdot j_{\mathrm{pc_total_}i}(R)}{\int_0^\infty V(\lambda')\cdot I_{\mathrm{phos}}(\lambda')\,d\lambda'}, \tag{3.116}$$

where G_l = image intensifier luminous gain,
$V(\lambda)$ = spectral luminous efficiency function for photopic vision [11],
$L\left(\lambda,T_{\mathrm{spec}}\right)$ = spectral radiance of a blackbody with temperature T_{spec} (specified color temperature of the illumination source used for measuring the image intensifier parameters), and
$I_{\mathrm{phos}}(\lambda)$ = normalized spectral emission of the phosphor screen.

In the same way, the spectral radiance from the phosphor screen due to photocathode dark emission is given by:

$$L_{\mathrm{phos_dark}}(\lambda) = G_l \cdot \frac{\int_0^\infty V(\lambda')\cdot L\left(\lambda',T_{\mathrm{spec}}\right)\mathrm{d}\lambda'}{\int_0^\infty \frac{\lambda'}{hc}\cdot\frac{\eta_{\mathrm{pc}}(\lambda')}{100}\cdot L\left(\lambda',T_{\mathrm{spec}}\right)\mathrm{d}\lambda'}\cdot\frac{\frac{1}{\pi}\cdot I_{\mathrm{phos}}(\lambda)\cdot j_{\mathrm{pc_dark}}}{\int_0^\infty V(\lambda')\cdot I_{\mathrm{phos}}(\lambda')\,d\lambda'}. \tag{3.117}$$

In case the *II radiant gain at peak response,* G_R, is specified, the corresponding equations for the spectral radiance from the phosphor screen are:

$$L_{\mathrm{phos_}i}(\lambda,R) = \frac{G_R\cdot hc\cdot 100}{\lambda_p\cdot\eta_{\mathrm{pc}}(\lambda_p)}\cdot\frac{\frac{1}{\pi}\cdot I_{\mathrm{phos}}(\lambda)}{\int_0^\infty I_{\mathrm{phos}}(\lambda')\,d\lambda'}\cdot j_{\mathrm{pc_total_}i}(R)\,, \tag{3.118}$$

$$L_{\mathrm{phos_dark}}(\lambda) = \frac{G_R\cdot hc\cdot 100}{\lambda_p\cdot\eta_{\mathrm{pc}}(\lambda_p)}\cdot\frac{\frac{1}{\pi}\cdot I_{\mathrm{phos}}(\lambda)}{\int_0^\infty I_{\mathrm{phos}}(\lambda')\,d\lambda'}\cdot j_{\mathrm{pc_dark}} \tag{3.119}$$

Note: In case *System gain* is specified, the spectral radiance from the phosphor screen is not calculated because the ICCD/ICMOS effective gain is obtained directly from the system gain (see below). The spectral emission of the phosphor screen is not used in calculations and thus inputs related to phosphor screen emission are disabled.

Phosphor screen output brightness and automatic brightness control

The phosphor screen output brightness is defined as the average luminance from the phosphor screen due to background radiation and dark emission. It is obtained by converting the calculated spectral radiance from the phosphor screen to luminance units (cd/m^2) for each distance:

$$L^{v}_{\mathrm{phos}}(R) = K_m \cdot \int_0^{\infty} (L_{\mathrm{phos_B}}(\lambda, R) + L_{\mathrm{phos_dark}}(\lambda)) \cdot V(\lambda)\,\mathrm{d}\lambda\,, \tag{3.120}$$

where $K_m = 683\ \mathrm{lm/W}$ is the maximum luminous efficacy for photopic vision.

If the specified luminous or radiant gain results in a phosphor screen output brightness that is larger than the specified maximum output brightness, then TRM4 assumes that an automatic brightness control (ABC) technique is used to reduce the image intensifier gain. The effective luminous or radiant gain (including the ABC effect) is given by:

$$G_{\mathrm{R}/l}(R) = G^{\mathrm{spec}}_{\mathrm{R}/l} \cdot \frac{L^{v}_{\mathrm{max}}}{L^{v}_{\mathrm{phos}}(R)}, \tag{3.121}$$

where $G^{\mathrm{spec}}_{R/l}$ is the specified luminous or radiant gain and L^{v}_{max} is the specified maximum output brightness.

Note: TRM4 assumes an ideal automatic brightness control, which does not affect the gain as long as the output luminance does not exceed the maximum output brightness. This means that for range dependent background radiance the effective gain may be affected by the ABC below (or above) a certain range. The *II effective gain* (at the reference range), $G_{\mathrm{R}/l}$ / $G^{\mathrm{spec}}_{\mathrm{R}/l}$, is given as an output data in the *System Performance* window.

ICCD/ICMOS effective gain

The effective gain of the ICCD/ICMOS is defined as the number of electrons collected by the detector per photoelectron emitted from the photocathode. If luminous gain is specified, the effective gain is calculated by

$$G_{\mathrm{ICCD}} = G_l\, \gamma_{\mathrm{coup}}\, {M_{\mathrm{II}}}^2 \frac{\int_0^{\infty} V(\lambda) \cdot L\left(\lambda, T_{\mathrm{spec}}\right) \mathrm{d}\lambda}{\int_0^{\infty} \lambda \cdot \eta_{\mathrm{pc}}(\lambda) \cdot L\left(\lambda, T_{\mathrm{spec}}\right) \mathrm{d}\lambda} \; \frac{\int_0^{\infty} I_{\mathrm{phos}}(\lambda) \cdot \lambda \cdot \eta(\lambda)\,\mathrm{d}\lambda}{\int_0^{\infty} V(\lambda) \cdot I_{\mathrm{phos}}(\lambda)\,\mathrm{d}\lambda}, \tag{3.122}$$

where γ_{coup} is the coupling efficiency of the coupling element and M_{II} is the image intensifier magnification.

If *II radiant gain at peak response* is specified, then the ICCD/ICMOS effective gain is given by:

$$G_{\mathrm{ICCD}} = \frac{G_R \cdot \gamma_{\mathrm{coup}} \cdot {M_{\mathrm{II}}}^2}{\lambda_{\mathrm{p}} \cdot \eta_{\mathrm{pc}}(\lambda_{\mathrm{p}})} \cdot \frac{\int_0^\infty I_{\mathrm{phos}}(\lambda) \cdot \lambda \cdot \eta(\lambda)\, \mathrm{d}\lambda}{\int_0^\infty I_{\mathrm{phos}}(\lambda')\, \mathrm{d}\lambda'}, \tag{3.123}$$

where λ_{p} is the wavelength of peak response.

In case *System gain* is specified, the ICCD/ICMOS effective gain equals the specified system gain, G_{sys}:

$$G_{\mathrm{ICCD}} = G_{\mathrm{sys}}\,. \tag{3.124}$$

Note: The effective gain of the ICCD/ICMOS is given as output data in the *System Performance* window.

Carrier flux density at the detector plane

The carrier flux density in the CCD/CMOS sensor plane due to the spectral radiance from the scene and path radiance can be expressed by the photocathode carrier flux density, the effective gain and the magnification of the ICCD/ICMOS:

$$j_{\mathrm{ICCD_total_}i}(\lambda, R) = \frac{j_{\mathrm{pc_total_}i}(\lambda, R)}{{M_{\mathrm{ICCD}}}^2} \cdot G_{\mathrm{ICCD}}\,. \tag{3.125}$$

In the same way, the dark carrier flux density in the CCD/CMOS sensor plane given by:

$$j_{\mathrm{ICCD_dark}} = j_{\mathrm{dark}} + j_{\mathrm{pc_dark}} \cdot G_{\mathrm{ICCD}}\,, \tag{3.126}$$

where the first term is the dark carrier flux density of the CCD/CMOS array and the second term is due to the dark emission of the photocathode.

3.8 Difference signal and noise at the detector output

3.8.1 Detector assembly signal and noise

The signal and noise at the detector assembly output are different between non-integrating and integrating detectors.

Integrating detectors (except ICCD/ICMOS)

The number of carriers in the storage gate of a detector element due to background or target is given by

$$\begin{aligned} N_{\mathrm{carrier_}i}(R) &= N_{\mathrm{total_}i} + N_{\mathrm{dark}} \\ &= t_{\mathrm{int}} \cdot \left(j_{\mathrm{total_}i}(R) + j_{\mathrm{dark}}\right) \cdot A_{\mathrm{det}} \cdot E_{\mathrm{gain}}\,, \end{aligned} \tag{3.127}$$

where t_{int} = integration time,
N_{total_i} = total number of target/background carriers,
j_{total_i} = total carrier flux density (target/background),
N_{dark} = number of detector dark carriers,
j_{dark} = detector dark carrier flux density, and
E_{gain} = electronic gain factor.

At the detector output, the difference signal is given by

$$\mathrm{DS}(\lambda, R) = \left[j_{\mathrm{total_T}}(\lambda, R) - j_{\mathrm{total_B}}(\lambda, R)\right] \cdot A_{\mathrm{det}} \cdot t_{\mathrm{int}} \cdot E_{\mathrm{gain}} \tag{3.128}$$

where $j_{\mathrm{total_T}}(\lambda, R)$ = spectral carrier flux density due to target emission and reflection, and
$j_{\mathrm{total_B}}(\lambda, R)$ = spectral carrier flux density due to background emission and reflection.

Integrating this equation over all wavelengths yields the difference signal at the detector output

$$\mathrm{DS}(R) = \int_0^{\infty} \mathrm{DS}(\lambda, R)\, \mathrm{d}\lambda \tag{3.129}$$

The noise can be calculated as

$$\sigma_{\mathrm{det_int}}(R) = \left[\left(\mathrm{NF}^2 \cdot \frac{j_{\mathrm{total_T}}(R) + j_{\mathrm{total_B}}(R)}{2} + j_{\mathrm{dark}}\right) \cdot A_{\mathrm{det}} \cdot E_{\mathrm{gain}}^2 \cdot t_{\mathrm{int}} + N_{\mathrm{quant}}^2 + N_{\mathrm{floor}}^2\right]^{1/2}, \tag{3.130}$$

where NF = noise figure of the detector,
N_{floor} = noise floor of the detector, and
N_{quant} = the quantization noise.

ICCD/ICMOS

For ICCD, the number of carriers in the storage gate due to background or target radiation is given by

$$N_{\mathrm{carrier}_i}(R) = t_{\mathrm{int}} \cdot \left(j_{\mathrm{ICCD_total}_i}(R) + j_{\mathrm{ICCD_dark}}\right) \cdot A_{\mathrm{det}}, \tag{3.131}$$

where $j_{\mathrm{ICCD_total}_i}$ is the total carrier flux density due to the target/background at the CCD/CMOS sensor plane and $j_{\mathrm{ICCD_dark}}$ is the dark carrier flux density due to photocathode dark emission and CCD/CMOS dark current.

The difference signal is given by

$$\mathrm{DS}(\lambda, R) = \left[j_{\mathrm{ICCD_total_T}}(\lambda, R) - j_{\mathrm{ICCD_total_B}}(\lambda, R)\right] \cdot A_{\mathrm{det}} \cdot t_{\mathrm{int}}\,, \tag{3.132}$$

where $j_{\mathrm{ICCD_total_T}}(\lambda, R)$ = spectral carrier flux density at the detector due to target reflection, and
$j_{\mathrm{ICCD_total_B}}(\lambda, R)$ = spectral carrier flux density at the detector due to the background.

The total temporal noise of ICCD/ICMOS is composed of noise due to the signal amplified by the image intensifier, dark noise due to the photocathode, and inherent noise of the CCD/CMOS sensor:

$$\sigma_{\mathrm{ICCD}}(R) = (\sigma_{\mathrm{II}}(R)^2 + {\sigma_{\mathrm{CCD}}}^2)^{1/2}. \tag{3.133}$$

Here, σ_{II} is the noise from the scene signal and dark emission of the image intensifier:

$$\sigma_{\mathrm{II}}(R) = \mathrm{NF}_{\mathrm{II}} \cdot \left[\left(\frac{j_{\mathrm{pc_total_T}} + j_{\mathrm{pc_total_B}}}{2} + j_{\mathrm{pc_dark}} \right) \cdot 2\Delta f_{\mathrm{DA}} \cdot 2\Delta f_{\mathrm{h}} \cdot 2\Delta f_{\mathrm{v}} \right]^{1/2} \cdot G_{\mathrm{ICCD}} \cdot t_{\mathrm{int}} \cdot \frac{A_{\mathrm{det}}}{M_{\mathrm{ICCD}}^2}, \tag{3.134}$$

where
$\mathrm{NF}_{\mathrm{II}}$ = noise figure of the image intensifier,
G_{ICCD} = effective gain of the ICCD/ICMOS,
$j_{\mathrm{pc_total_T}}$ = total carrier flux density at the photocathode due to radiation reflected at the target and path radiance
$j_{\mathrm{pc_total_B}}$ = total carrier flux density at the photocathode due to background and path radiance
$j_{\mathrm{pc_dark}}$ = total carrier flux density at the photocathode due to dark emission of the photocathode,
Δf_{DA} = temporal noise bandwidth of the ICCD/ICMOS in the pre-digital signal [Equation (3.151)], and
$\Delta f_{\mathrm{h/v}}$ = spatial noise bandwidth of the ICCD/ICMOS in the pre-digital signal [Equation (3.152)].

The inherent noise of the CCD/CMOS is given by:

$$\sigma_{\mathrm{CCD}} = \left[j_{\mathrm{dark}} \cdot t_{\mathrm{int}} \cdot A_{\mathrm{det}} + N_{\mathrm{floor}}^2 + N_{\mathrm{quant}}^2 \right]^{1/2}. \tag{3.135}$$

Non-integrating detectors (1st generation thermal imagers only)

For a non-integrating detector, the signal of a single detector element is given by

$$i_{\mathrm{det}} = A_{\mathrm{det}} \cdot R_{\mathrm{DA_max}} \cdot J_{\mathrm{fp}} + i_{\mathrm{dark}}, \tag{3.136}$$

where
A_{det} = sensitive detector area,
J_{fp} = effective integrated irradiance in the focal plane, and
i_{dark} = dark current (specified in Amperes).

Finally, the temporal noise is calculated using

$$\sigma_{\mathrm{det_noi}} = \left[2 \cdot \left(\mathrm{NF}^2 \cdot k \cdot \frac{i_{\mathrm{det}} - i_{\mathrm{dark}}}{e \cdot \Delta f} + \frac{i_{\mathrm{dark}}}{e \cdot \Delta f} + \left(\frac{i_{\mathrm{floor}}}{e \cdot \Delta f} \right)^2 \right) + \left(\frac{i_{\mathrm{quant}}}{e \cdot \Delta f} \right)^2 \right]^{1/2}, \tag{3.137}$$

where
i_{floor} = current due to noise floor,
i_{quant} = current due to quantization error,
k = constant depending on detector type; TRM always assumes k=1, and
Δf = detector noise bandwidth.

3.8.2 Storage gate fill

Storage gate fill (in percent) is found by using

$$p = 100 \cdot \frac{t_{\mathrm{int}} \cdot \left(j_{\mathrm{total_B}}(R) + j_{\mathrm{dark}}\right) \cdot A_{\mathrm{det}} \cdot E_{\mathrm{gain}}}{N_{\mathrm{stor}}}. \tag{3.138}$$

In case of ICCD/ICMOS systems, $j_{\mathrm{ICCD_total_B}}(R)$ and $j_{\mathrm{ICCD_dark}}$ have to be used instead of $j_{\mathrm{total_B}}(R)$ and j_{dark}, respectively.

If $p > 100\%$, the user will be notified of the storage gate overflow but all calculations will be done.

Integration time required for 50 % and 100 % of maximum storage capacity (*Thermal* category only)

For *Thermal* category imagers, the storage well fill due to background photons is not range dependent. The integration times that result in a storage gate with 50 % and 100 % filling are calculated using Equation (3.76).

The results are reported in the *System Performance Data* message box.

3.9 Noise equivalent signal and temporal noise

Noise Equivalent Temperature Difference (NETD) gives a rough indication of the imager sensitivity. NETD is exclusively used in the *Thermal* category. There is no equivalent to the NETD in the *Visible* and *General* categories. In these categories, the noise power spectrum and the standard deviation are used to describe the noise.

3.9.1 NETD – *Thermal* category

The NETD is used for characterizing the noise of the imager. TRM assumes noise measurements carried out with measurement equipment that does not significantly limit the frequency spectrum of the noise.

The NETD is defined as the temperature difference between a large uniform target and a uniform background that results in a SNR of one at the measurement point in a laboratory environment (i.e. no atmospheric absorption).

There are several different locations in the signal processing chain where measurements or calculations of the NETD can be done:

- NETD after analog signal processing (1st generation imagers),
- NETD after FPA read-out (2nd generation imagers),
- NETD after digital processing (imagers sampled in both directions), and
- NETD after video signal processing (imagers with analog video feed).

The calculation of the NETD at these locations requires MTFs of all relevant components. If they are not known, no calculation is carried out and a message will be issued in the *System Performance Data* message box.

Note:	TRM4 takes into account noise folding caused by the sampling process.

NETD equation – 1st and 2nd generation imagers

The NETD measured at the detector output or in the pre-digital signal is given by

$$\mathrm{NETD}_{\mathrm{det}}\,(T_{\mathrm{scene}}) = \frac{4F_{\mathrm{eff_opt}}^{2}\cdot \Delta f_{\mathrm{DA}}^{1/2}}{\pi\cdot\tau_{\mathrm{opt_max}}\cdot\tau_{\mathrm{filter_max}}\cdot R_{\mathrm{max}}\cdot(A_{\mathrm{det}}\cdot N_{\mathrm{TDI}})^{1/2}\cdot D^{*}_{\mathrm{peak}}(T_{\mathrm{scene}})\cdot W_{\mathrm{eff}}\,(T_{\mathrm{scene}})}, \quad (3.139)$$

where Δf_{DA} = noise bandwidth in [Hz] that depends on detector and imager (Section 3.9.4) and
N_{TDI} = number of detectors integrated in TDI for scanning imagers ($N_{\mathrm{TDI}} = 1$ for staring imagers).

The NETD in the digital or video signal is given by

$$\mathrm{NETD}_{\mathrm{dig|vid}}\,(T_{\mathrm{scene}}) = \frac{4F_{\mathrm{eff_opt}}^{2}\cdot \Delta f_{\mathrm{dig|vid}}^{1/2}}{\pi\cdot\tau_{\mathrm{opt_max}}\cdot\tau_{\mathrm{filter_max}}\cdot R_{\mathrm{max}}\cdot(A_{\mathrm{det}}\cdot N_{\mathrm{TDI}}\cdot \mathrm{FR}_{\mathrm{int}})^{\frac{1}{2}}\cdot D^{*}_{\mathrm{peak}}(T_{\mathrm{scene}})\cdot W_{\mathrm{eff}}\,(T_{\mathrm{scene}})}, \quad (3.140)$$

where Δf_{dig} = noise bandwidth in the digital signal (Section 3.9.5),
Δf_{vid} = noise bandwidth in the video signal (Section 3.9.6), and
$\mathrm{FR}_{\mathrm{int}}$ = frame integration factor.

The frame integration factor is defined as

$$\mathrm{FR}_{\mathrm{int}} = N_{\mathrm{frames}}\cdot\frac{1+\alpha}{1-\alpha}, \quad (3.141)$$

where α denotes the IIR Frame integration filter factor.

NETD equation – Uncooled detector

There are two fundamental assumptions in calculating the NETD of the imager device for *Uncooled* detectors.

First, it is assumed that the *detector NETD* is measured and the measurement data is made available to the TRM4 user.

Second, it is assumed that the noise of the uncooled detector measured in the laboratory is essentially internal and that the temperature of the uncooled detector is stabilized. Thus, the noise does not depend on the measurement setup.

The detected signal, however, depends on the parameters of the measurement setup and will change if the detector is used in a different measurement setup. Consequently, the NETD of the detector depends on the parameters of the measurement setup, too.

The relation between the measured detector NETD, $\mathrm{NETD_m}$, and the NETD for the parameters specified in the TRM project, $\mathrm{NETD_s}$, is given by:

$$\begin{aligned}\frac{\mathrm{NETD_s}}{\mathrm{NETD_m}} &= \frac{F^2_{\mathrm{eff_opt_s}}}{F^2_{\mathrm{eff_opt_m}}} \cdot \frac{\int_{\lambda_{1\mathrm{m}}}^{\lambda_{2\mathrm{m}}} \mathrm{d}\lambda \; R_{\mathrm{sys_m}}(\lambda) \cdot \left.\frac{\partial L(T,\lambda)}{\partial T}\right|_{T_{\mathrm{B_m}}}}{\int_{\lambda_{1\mathrm{s}}}^{\lambda_{2\mathrm{s}}} \mathrm{d}\lambda \; R_{\mathrm{sys_s}}(\lambda) \cdot \left.\frac{\partial L(T,\lambda)}{\partial T}\right|_{T_{\mathrm{B_s}}}} \\ &= \frac{F^2_{\mathrm{eff_{opt_s}}}}{F^2_{\mathrm{eff_{opt_m}}}} \cdot \frac{\int_{\lambda_{1\mathrm{m}}}^{\lambda_{2\mathrm{m}}} \mathrm{d}\lambda \, \tau_{\mathrm{opt_peak_m}} \, R_{\mathrm{DA_norm}}(\lambda) \cdot \left.\frac{\partial L(T,\lambda)}{\partial T}\right|_{T_{\mathrm{B_m}}}}{\int_{\lambda_{1\mathrm{s}}}^{\lambda_{2\mathrm{s}}} \mathrm{d}\lambda \, \tau_{\mathrm{opt}}(\lambda) \cdot \tau_{\mathrm{filter}}(\lambda) \cdot R_{\mathrm{DA_norm}}(\lambda) \cdot \left.\frac{\partial L(T,\lambda)}{\partial T}\right|_{T_{\mathrm{B_s}}}},\end{aligned} \tag{3.142}$$

where $\lambda_{1/2}$ are the cut-on/cut-off wavelength of the detector spectral response.

The subscripts *m* and *s* in Equation (3.142) denote parameters used in the detector measurement setup and parameters specified in the TRM4 project, respectively.

3.9.2 Temporal noise (*General* and *Visible* categories)

As stated above, there is no equivalent to the NETD in the *Visible* and *General* categories. For these categories, the noise is calculated directly without using any “noise equivalent” intermediaries.

The standard deviation of the noise, $\sigma_{\mathrm{det_int}}$, given by Equation (3.130) is governed by the noise sources, the integration of the noise over the detector area and the temporal integration.

In order to keep as much resemblance as possible to the thermal NETD, we shall call the standard deviation NES (it is an “equivalent signal” only where TDI and frame integration are involved).

The NES calculated at the detector output is given by

$$\mathrm{NES_{det}}(R) = \frac{\sigma_{\mathrm{det_int}}(R)}{N_{\mathrm{TDI}}^{1/2}} \tag{3.143}$$

The NES in the digital or video signal is given by

$$\mathrm{NES_{dig|vid}}(R) = \frac{\sigma_{\mathrm{det_int}}(R) \cdot (2 \cdot t_{\mathrm{int}})^{1/2} \cdot \Delta f_{\mathrm{dig|vid}}^{1/2}}{(N_{\mathrm{TDI}} \cdot \mathrm{FR_{int}})^{1/2}}. \tag{3.144}$$

The factor $1/(2 \cdot t_{\mathrm{int}})^{1/2}$ is the square root of the integration noise bandwidth (see Equation (3.150)).

For ICCD/ICMOS, the NES calculated at the detector output is simply:

$$\mathrm{NES_{det}}(R) = \sigma_{\mathrm{ICCD}}(R). \tag{3.145}$$

3.9.3 Noise bandwidth in the analog signal (1st generation *Thermal* imagers)

Noise filters can be divided into two groups:

- pre-sampling filters and
- post-sampling filters.

The pre-sampling filters are temporal in essence; they are translated to spatial filters in scanning imagers via the scan velocity v_{scan}.

This noise bandwidth applies to 1st generation thermal imagers. Unless specified via external data, the normalized noise power spectrum $S(\mathrm{r})$ at the output of the detector is assumed to be (the spatial frequency is used in order to be compatible with the rest of the paragraph):

$$S(r) = \left(1 + \left|\frac{r_{\mathrm{knee}}}{r}\right|\right) \cdot \mathrm{MTF}^2_{\mathrm{det_temp}}(r) \quad \text{for } 0 < r < \infty, \tag{3.146}$$

where r = spatial frequency,
r_{knee} = knee frequency of the $1/f$ noise, and
$\mathrm{MTF}_{\mathrm{det_temp}}$ = detector temporal MTF (Section 4.2.3).

The noise bandwidth in the analog signal (after analog signal processing filtering) is given by

$$\Delta f_{\mathrm{ana}} = v_{\mathrm{scan}} \cdot \int_0^{\infty} S(r) \cdot \mathrm{MTF}^2_{\mathrm{ana}}(r)\, \mathrm{d}r\,, \tag{3.147}$$

where $\mathrm{MTF}_{\mathrm{ana}}$ denotes the overall analog signal electronics MTF (Section 4.2.4).

3.9.4 Noise bandwidth in the pre-digital signal (integrating imagers and Uncooled detector)

Integrating imagers

Unless specified, the normalized noise power spectrum of the detector signals in a scanning imager is assumed to be

$$S(r) = \left(1 + \left|\frac{r_{\mathrm{knee}}}{r}\right|\right) \cdot \mathrm{MTF}^2_{\mathrm{det_temp}}(r) \cdot \mathrm{MTF}^2_{\mathrm{det_int}}(r) \quad \text{for } 0 < r < \infty, \tag{3.148}$$

where $\mathrm{MTF}_{\mathrm{det_int}}$ is the detector integration MTF (Section 4.2.3).

The noise bandwidth in the pre-digital signal of scanning imagers is given by

$$\Delta f_{\mathrm{DA}} = v_{\mathrm{scan}} \cdot \int_0^{\infty} S(r)\, \mathrm{d}r\,. \tag{3.149}$$

For staring or micro-scanned imagers, TRM assumes white noise. The noise bandwidth is then given by (the integration is done in the temporal frequency domain)

$$\Delta f_{\mathrm{DA}} = \int_0^{\infty} \mathrm{sinc}^2(t_{\mathrm{int}} \cdot \nu)\, \mathrm{d}\nu = \frac{1}{2 \cdot t_{\mathrm{int}}}. \tag{3.150}$$

ICCD/ICMOS

In case of ICCD/ICMOS, TRM4 assumes white noise at the photocathode. The photocathode signal is temporally filtered by the phosphor screen and integrated at the detector. The corresponding noise bandwidth is

$$\Delta f_{\mathrm{DA}} = \int_0^\infty \mathrm{MTF}_{\mathrm{ps}}^2(\nu) \cdot \mathrm{sinc}^2(\nu \cdot t_{\mathrm{int}})\ \mathrm{d}\nu. \tag{3.151}$$

The photocathode signal is not only integrated over the detector area, but also spatially filtered by the spatial transfer characteristics of the image intensifier and coupling element. The associated spatial bandwidth in *z*-direction is given by:

$$\Delta f_z = \frac{1}{f_{\mathrm{opt}_z}} \int_0^\infty \mathrm{d}r\ \mathrm{MTF}_{\mathrm{det_stat}_z}^2 \left(\frac{r}{M_{\mathrm{ICCD}}}\right) \cdot \mathrm{MTF}_{\mathrm{II}}^2(r) \cdot \mathrm{MTF}_{\mathrm{Coup}}^2(r)\,. \tag{3.152}$$

Uncooled detector

For uncooled detectors, TRM assumes that noise is dominated by white thermal and Johnson noise, while $1/f$ noise is suppressed by using, for instance an appropriate chopper. The corresponding noise bandwidth is therefore

$$\Delta f_{\mathrm{DA}} = \int_0^\infty \mathrm{MTF}_{\mathrm{uc}}^2(\nu)\ \mathrm{d}\nu = \frac{1}{4 \cdot t_{\mathrm{th}}}. \tag{3.153}$$

3.9.5 Noise bandwidth in the digital signal

1st generation *Thermal* category imagers with horizontal sampling

The noise bandwidth in the digital domain is given by

$$\Delta f_{\mathrm{dig}} = v_{\mathrm{scan}} \cdot \left(\int_0^\infty S(r) \cdot \mathrm{MTF}_{\mathrm{ana}}^2(r) \cdot \mathrm{MTF}_{\mathrm{adc}}^2(r) \cdot \mathrm{MTF}_{\mathrm{dig_h}}^2(r)\ \mathrm{d}r \right) \cdot H_{\mathrm{dig_v}}^2 \tag{3.154}$$

where $\mathrm{MTF}_{\mathrm{adc}}$ = MTF of the Analog-Digital converter (Section 4.2.5),
$\mathrm{MTF}_{\mathrm{dig_h}}$ = MTF of the horizontal digital filter (Section 4.2.6), and

$$H_{\mathrm{dig}_z}^2 = \frac{\frac{1}{2}\sum_{n=1}^{N} c_{\mathrm{dig},n_z}^2 + c_{\mathrm{dig},0_z}^2}{\left(\sum_{n=0}^{N} c_{\mathrm{dig},n_z}\right)^2}. \tag{3.155}$$

Integrating imagers

For integrating detectors, the detector signals are already sampled prior to Analog-Digital conversion.

The noise bandwidth in the digital domain for scanning imagers is given by

$$\Delta f_{\mathrm{dig}} = v_{\mathrm{scan}} \cdot \left(\int_0^\infty S(r) \cdot \mathrm{MTF}_{\mathrm{dig_h}}^2(r)\ \mathrm{d}r \right) \cdot H_{\mathrm{dig_v}}^2\,. \tag{3.156}$$

For staring or micro-scanned imagers (including ICCD/ICMOS) it is given by

$$\Delta f_{\mathrm{dig}} = \Delta f_{\mathrm{DA}} \cdot H_{\mathrm{dig_h}}^2 \cdot H_{\mathrm{dig_v}}^2. \tag{3.157}$$

Uncooled detector

The noise bandwidth in the digital signal is given by

$$\Delta f_{\mathrm{dig}} = \frac{1}{4 \cdot t_{\mathrm{th}}} \cdot H_{\mathrm{dig_h}}^2 \cdot H_{\mathrm{dig_v}}^2. \tag{3.158}$$

3.9.6 Noise bandwidth in the video signal

To simplify calculations, the symmetric noise power spectrum S_1 is introduced here. It is defined for frequencies between $-\infty$ and ∞ and given by

$S_1(r) = 0.5 \cdot S(r)$ with $\mathrm{MTF}(r) = \mathrm{MTF}(r)$ for $0 \le r < \infty$,

$S_1(r) = 0.5 \cdot S(-r)$ with $\mathrm{MTF}(r) = \mathrm{MTF}(-r)$ for $-\infty < r < 0$.

The factor of 1/2 is due to the conversion from the conventional spectrum description, which ranges from $r = 0$ to ∞, to the full frequency range between $-\infty$ and ∞.

1st generation *Thermal* category imagers with horizontal sampling

The noise bandwidth is given by

$$\Delta f_{\mathrm{vid}} = v_{\mathrm{scan}} \cdot \left(\int_{-\infty}^{\infty} S_{\mathrm{fold}}(r) \cdot \mathrm{MTF}_{\mathrm{po_h}}^2(r)\, \mathrm{d}r \right) \cdot H_{\mathrm{dig_v}}^2 , \tag{3.159}$$

where

$$S_{\mathrm{fold}}(r) = \sum_{i=-\infty}^{\infty} S_1\left(r - i \cdot r_{s_{\mathrm{h}}}\right) \cdot \mathrm{MTF}_{\mathrm{noise_pre}}^2\left(r - i \cdot r_{s_{\mathrm{h}}}\right) \tag{3.160}$$

with the horizontal noise prefilter MTF

$$\mathrm{MTF}_{\mathrm{noise_pre}}(r) = \mathrm{MTF}_{\mathrm{ana}}(r) \cdot \mathrm{MTF}_{\mathrm{adc}}(r) . \tag{3.161}$$

$\mathrm{MTF}_{\mathrm{po_h}}$ denotes the horizontal imager postfilter MTF without display

$$\mathrm{MTF}_{\mathrm{po_h}} = \mathrm{MTF}_{\mathrm{dig_h}} \cdot \mathrm{MTF}_{\mathrm{zoom_h}} \cdot \mathrm{MTF}_{\mathrm{dac}} \cdot \mathrm{MTF}_{\mathrm{vid}} \cdot \mathrm{MTF}_{\mathrm{crep}} , \tag{3.162}$$

where $\mathrm{MTF}_{\mathrm{zoom_h}}$ = horizontal electronic zoom MTF (Section 4.2.7),
$\mathrm{MTF}_{\mathrm{dac}}$ = Digital-Analog converter MTF (Section 4.2.7),
$\mathrm{MTF}_{\mathrm{vid}}$ = overall video signal processing MTF (Section 4.2.8), and
$\mathrm{MTF}_{\mathrm{crep}}$ = column replication MTF (Section 4.2.7).

Integrating scanning imagers

The noise bandwidth in the video signal of integrating scanning imagers is also given by Equation (3.159), but with the noise prefilter MTF, $\mathrm{MTF}_{\mathrm{noise_pre}}$, equal to unity. This means that for integrating scanning systems, the noise power spectrum in Equation (3.160) changes to

$$S_{\mathrm{fold}}(r) = \sum_{i=-\infty}^{\infty} S_1\left(r - i \cdot r_{s_{\mathrm{h}}}\right). \tag{3.163}$$

Integrating staring or micro-scanned imagers (including ICCD/ICMOS)

The noise bandwidth in the video signal is given by

$$\Delta f_{\mathrm{vid}} = \Delta f_{\mathrm{DA}} \cdot \alpha_{s_{\mathrm{h}}} \cdot \left(\int_{-\infty}^{\infty} \mathrm{MTF}_{\mathrm{po_h}}^2(r)\,\mathrm{d}r\right) \cdot H_{\mathrm{dig_v}}^2 . \tag{3.164}$$

Uncooled detector

The noise bandwidth in the video signal is given by

$$\Delta f_{\mathrm{vid}} = \frac{1}{4 \cdot t_{\mathrm{th}}} \cdot \alpha_{s_{\mathrm{h}}} \cdot \left(\int_{-\infty}^{\infty} \mathrm{MTF}_{\mathrm{po_h}}^2(r)\,\mathrm{d}r\right) \cdot H_{\mathrm{dig_v}}^2 . \tag{3.165}$$

3.10 Inhomogeneity equivalent signal and spatial noise

The IETD (Inhomogeneity Equivalent Temperature Difference) is the counterpart of NETD in the spatial noise domain. The IETD measurement is done in two steps: First, the spatial noise is measured on a uniform background and then the signal of a large target set against the same background is measured (like in NETD). The temperature difference between target and background for which the signal is equal to the standard deviation of the measured spatial noise is defined as IETD. It is a measured rather than calculated parameter and characterizes the residual fixed pattern noise contribution after non-uniformity correction.

As stated in the previous section, IETD is reserved to the category *Thermal*. There is no equivalent in the *Visible* and *General* categories and we use an “artificial” IES (Inhomogeneity Equivalent Signal) in order to be compatible with the IETD.

TRM4 deals with two types of non-uniformity:

- row-to-row non-uniformity (scanning imagers) and
- pixel non-uniformity (staring imagers).

3.10.1 Measured IETD and spatial non-uniformity

***Thermal* category**

The standard deviation of the measured (high-frequency) non-uniformity is specified in equivalent temperature difference units.

***General* and *Visible* categories**

Measured non-uniformity can be specified by two variables: the Photo Response Non-Uniformity (PRNU) and the Dark Signal Non-Uniformity (DSNU).

PRNU (multiplicative component)

PRNU is specified in [%] and the corresponding signal (carrier number) is calculated as

$$\begin{aligned} N_{\mathrm{PRNU}}(R) &= \frac{\mathrm{PRNU}}{100} \cdot \left(N_{\mathrm{carrier}}(R) - N_{\mathrm{dark}} \cdot E_{\mathrm{gain}} \cdot t_{\mathrm{int}}\right) \\ &= \frac{\mathrm{PRNU}}{100} \cdot j_{\mathrm{total_B}}(R) \cdot A_{\mathrm{det}} \cdot t_{\mathrm{int}} \cdot E_{\mathrm{gain}} \,. \end{aligned} \tag{3.166}$$

In case of ICCD/ICMOS, $j_{\mathrm{ICCD_total_B}}(R)$ is used instead of $j_{\mathrm{total_B}}(R)$.

DSNU (additive component)

DSNU is specified in [A] and the corresponding carrier number is given by

$$N_{\mathrm{DSNU}} = \frac{\mathrm{DSNU}}{e} \cdot t_{\mathrm{int}} \,. \tag{3.167}$$

The overall standard deviation of the non-uniformity is calculated from PRNU and DSNU as

$$\mathrm{IES}(R) = N_{\mathrm{NU}}(R) = \sqrt{N_{\mathrm{PRNU}}^2(R) + N_{\mathrm{DSNU}}^2} \,. \tag{3.168}$$

Note: It is important to remember that the N_{NU} is range-dependent, especially when using laboratory measurements for the characterization of non-uniformity

3.10.2 Spatial noise bandwidth

The spatial noise is assumed to be “white” at its origin, i.e. after non-uniformity correction and prior to any filtering.

All functions used in TRM that affect the noise are separable in horizontal and vertical directions. Therefore, the noise bandwidth can be written as

$$\Delta f_{\mathrm{I}} = \Delta f_{\mathrm{I_h}} \cdot \Delta f_{\mathrm{I_v}} \,. \tag{3.169}$$

Note: TRM does not deal with horizontal non-uniformities in scanning imagers.

3.10.3 Spatial noise bandwidths in the digital signal

Using the MTFs as defined in Sections 3.9.5 and 3.9.6, the spatial noise bandwidths in the digital signal of scanning and staring imagers are given by:

Scanning imagers

$$\Delta f_{\mathrm{I,dig_v}} = n_{\mathrm{inter}} \cdot \alpha_{\mathrm{s_v}} \cdot \int_0^{r_{\mathrm{s_v}}} \left(\frac{\mathrm{sinc}\left(n_{\mathrm{inter}} \cdot \alpha_{\mathrm{s_v}} \cdot r\right)}{\mathrm{sinc}\left(\alpha_{\mathrm{s_v}} \cdot r\right)} \right)^2 \cdot \mathrm{MTF}^2_{\mathrm{dig_v}}(r)\, \mathrm{d}r\,, \tag{3.170}$$

where $n_{\mathrm{inter}} \cdot \alpha_{s_{\mathrm{v}}}$ is the effective vertical (angular) detector pitch;

Staring imagers

$$\Delta f_{\mathrm{I,dig_z}} = N_{\mu\mathrm{step_z}} \cdot \alpha_{\mathrm{s_z}} \cdot \int_0^{r_{\mathrm{s_z}}} \left(\frac{\mathrm{sinc}\left(N_{\mu\mathrm{step_z}} \cdot \alpha_{\mathrm{s_z}} \cdot r\right)}{\mathrm{sinc}\left(\alpha_{\mathrm{s_z}} \cdot r\right)} \right)^2 \cdot \mathrm{MTF}^2_{\mathrm{dig_z}}(r)\, \mathrm{d}r\,, \tag{3.171}$$

where $N_{\mu\mathrm{step_z}} \cdot \alpha_{\mathrm{s_z}}$ is the effective (angular) detector pitch in z-direction and $N_{\mu\mathrm{step_z}} = 2$ for diagonal micro-scan.

3.10.4 Spatial noise bandwidths in the video signal

The noise bandwidths of the video signal are determined by:

Scanning imagers

$$\Delta f_{\mathrm{I,vid_v}} = \Delta f_{\mathrm{I,dig_v}}\,; \tag{3.172}$$

Staring imagers

$$\Delta f_{\mathrm{I,vid_h}} = N_{\mu\mathrm{step_h}} \cdot \alpha_{\mathrm{s_h}} \cdot \int_0^{r_{\mathrm{s_h}}} \left(\frac{\mathrm{sinc}\left(N_{\mu\mathrm{step_h}} \cdot \alpha_{\mathrm{s_h}} \cdot r\right)}{\mathrm{sinc}\left(\alpha_{\mathrm{s_h}} \cdot r\right)} \right)^2 \cdot \mathrm{MTF}^2_{\mathrm{po_h}}(r)\, \mathrm{d}r \tag{3.173}$$

with $N_{\mu\mathrm{step_h}} = 2$ for diagonal micro-scan and

$$\Delta f_{\mathrm{I,vid_v}} = \Delta f_{\mathrm{I,dig_v}}\,. \tag{3.174}$$

3.11 Minimum Temperature Difference Perceived (MTDP), Minimum Difference Signal Perceived (MDSP) and Minimum Contrast Perceived (MCP)

The Minimum Temperature Difference Perceived (MTDP) assesses the perception of the standard 4-bar test pattern by an observer. The standard 4-bar test pattern, shown in Figure 3.1 is a square pattern consisting of 7 bars or 3.5 line pairs. The MTDP is defined as the minimum temperature difference between the bars that is required to perceive the pattern (at

optimum phase, see Section 3.11.3). In contrast to the Minimum Resolvable Temperature Difference (MRTD) described in NATO STANAG 4347 [5], it is not required that all four bars of the test pattern are resolved by the observer.

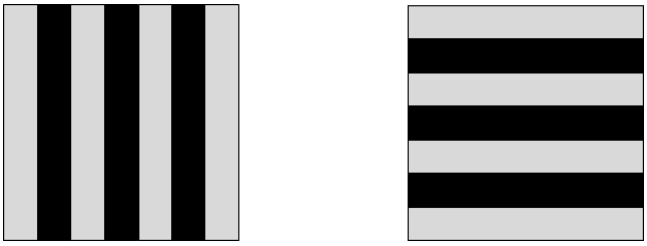


Figure 3.1: The standard 4-bar test pattern with vertical bars (left) is used for horizontal MTDP/MDSP/MCP, the test pattern with horizontal bars (right) for vertical MTDP/MDSP/MCP.

Modern staring thermal imagers usually are undersampled, which leads to artefacts in the displayed image. In this case, the four bars of the test pattern are resolved only over limited range of spatial frequencies, whereas at higher spatial frequencies the patterns are distorted (see Figure 3.2). Although MTDP was developed [1] to characterize staring undersampled thermal imagers, the concept can be applied to all kinds of imagers, whether undersampled or not.

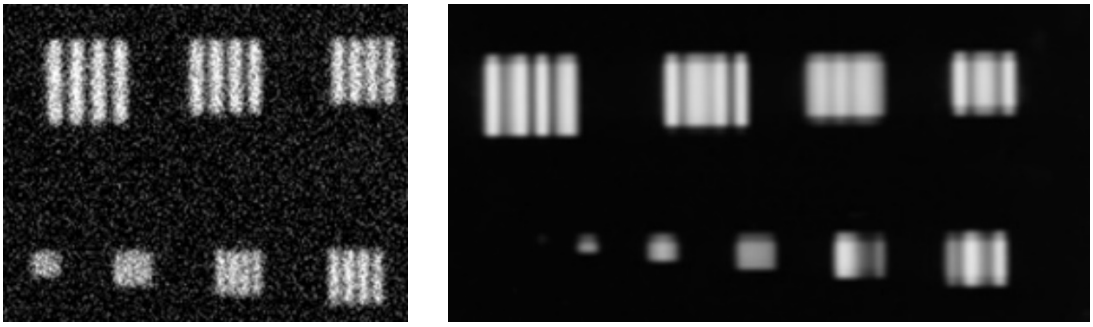


Figure 3.2: Reproduction of the standard 4-bar test pattern with a well sampled noisy imager (left) and an undersampled imager without noise (right).

The Minimum Difference Signal Perceived (MDSP) is a generalization of the MTDP concept. It is used in TRM4 calculations to characterize imagers of categories *Visible/NIR/SWIR* and *General* whose performance parameters are not expressed in temperature differences (NETD, IETD, MTDP, etc.) but in signals (NES, IES, MDSP, etc.).

In contrast to MTDP, MDSP cannot be measured in the laboratory. In order to have a better comparability to lab measurements for *Visible/NIR/SWIR* imagers, MDSP is expressed in terms of Minimum Contrast Perceived (MCP) in the TRM4 output if *Laboratory performance* option is selected by the user. MCP is measured in the same way as MTDP with the only difference that instead of the temperature difference between the bars, the modulation contrast between the bars is changed (for a fixed background radiance).

In the following, the MDSP will be used as the more general term and stands for both MDSP and MTDP. If differences exist in the treatment of these two concepts, it will be pointed out at the appropriate locations. The same holds for NES and IES, which for thermal systems must be replaced with NETD and IETD, respectively.

Note: One major difference between calculated MTDP (*Thermal* category) and MDSP (*Visible* and *General* categories) lies in the assumptions made in the *Thermal* category model, which result in a range-independent MTDP. Thus, the MTDP can be calculated for lab conditions and subsequently be used for range calculations. This is *not*, however, the case for the MDSP, since it is explicitly range-dependent. The displayed and presented MDSP is calculated at a reference range to be set by the user.

3.11.1 Overview of the TRM approach

Assessment of sampling

The MTDP/MDSP was originally conceived to overcome a limitation of the traditional approach to the range assessment of thermal imaging systems with respect to undersampled systems.

The classical MRTD approach did not consider spatial frequencies beyond the Nyquist frequency of the imager's detector. Experience showed, however, that a much better performance could be expected from undersampled systems than was predicted using the MRTD approach.

Assessment of imager noise

In the TRM approach temporal and spatial noise are assumed to be statistically independent. Thus, the filtered total system noise is the root sum square of the individual weighted noise components. Additionally, the TRM approach takes into account noise folding due to the sampling process.

Statistical nature of MTDP/MCP in case of fixed pattern noise

If fixed pattern noise is dominant, calculated and measured MTDP/MCP may differ. The measured MTDP/MCP depends on the actual position of the test pattern image with respect to the fixed pattern noise [13]. The resulting image impression might be favorable or unfavorable to the perception of the test pattern; thus, different results can be expected at different relative test pattern image locations in the image plane or for a different/new non-uniformity correction. The values calculated by TRM4 correspond to an average of measured MTDP/MCP.

Note: For MTDP/MCP measurements, the use of different test pattern locations within the FOV is recommended; then, the average of these measurements should be used as the figure of merit to assess range performance and/or for comparison to calculated values.

3.11.2 Equations for MTDP, MDSP and MCP

The MTDP in z-direction is given by

$$\mathrm{MTDP}_z(r_\mathrm{p}) = \frac{\frac{\pi}{2} \cdot \mathrm{SNR}_\mathrm{th} \cdot \Psi_z^{1/2}(r_\mathrm{p})}{\mathrm{AMOP}_z(r_\mathrm{p})}, \tag{3.175}$$

where r_p = spatial frequency of the standard 4-bar test pattern,
ψ_z = total perceived (system filtered) noise variance,
SNR_th =signal-to-noise threshold of the eye,
AMOP_z = Average Modulation at Optimum Phase given in Section 3.11.3.

MDSP is calculated by the same expression, however, the total perceived noise variance and AMOP depend on the target distance R. Therefore, MDSP is a function of R and is given by

$$\mathrm{MDSP}_z(R, r_\mathrm{p}) = \frac{\frac{\pi}{2} \cdot \mathrm{SNR}_\mathrm{th} \cdot \Psi_z^{1/2}(R, r_\mathrm{p})}{\mathrm{AMOP}_z(R, r_\mathrm{p})}. \tag{3.176}$$

For *Visible/NIR/SWIR* imagers and *Laboratory performance* specified, MCP is calculated and presented as TRM4 output. The MCP is determined by finding at given target and background radiances L_T and L_B, respectively, the spatial frequency r that solves

$$\frac{\mathrm{MDSP}_z(0, r)}{N_{\mathrm{total_}T}(0) + N_{\mathrm{total_}B}(0)} = \frac{L_T - L_B}{L_T + L_B} \tag{3.177}$$

and setting

$$\mathrm{MCP}_z(r) := \frac{L_T - L_B}{L_T + L_B}. \tag{3.178}$$

Here, *z* denotes either the horizontal direction, the vertical direction or the average value and $N_{\mathrm{total_}T/B}$ are the number of carriers in the storage gate of a detector element due to radiance from the target/background. Note that only the spectral radiance of the background, L_B, is defined by the user input (effective background reflectivity and illumination source) whereas the target radiance has the same spectral behavior and is set internally to different values for calculating the MCP curve.

Note: The MTDP/MDSP/MCP does not require that all four bars of the test pattern are resolved by the observer. If the imager is undersampled, only three or two bars can be resolved at spatial frequencies higher than approximately the Nyquist frequency[9]. The MDSP assesses imager performance also at this range of frequencies.

Note: In TRM4, the MTDP/MDSP is calculated for AMOP values greater than 0.1 % only. For smaller AMOP values, the MTDP/MDSP is assumed to be infinite.

[9] These bars are “apparent bars” that result from aliasing effects of the original bar pattern. They cover the whole target area and should not be confused with the actual bars on the target.

3.11.3 Average Modulation at Optimum Phase (AMOP)

The TRM approach uses the AMOP rather than the system MTF. The AMOP describes the maximum average modulation in the reproduced image of a standard 4-bar target.

In the classical MRTD approach (no longer supported in TRM4), the system MTF is used to calculate the modulation in the image of the standard 4-bar pattern, which is assumed to be given by $4/\pi \cdot \mathrm{MTF}_{\mathrm{sys}}$ at low spatial frequencies. An example for a "well behaved" system with no undersampling is shown in Figure 3.3 where the actual modulation in the test pattern (AMOP) is shown together with the approximated modulation given by $4/\pi \cdot \mathrm{MTF}_{\mathrm{sys}}$. For completeness, the system MTF is also shown.

If the imager is not well sampled, the four bars of the test pattern are resolved only at pattern frequencies lower than the Nyquist frequency. At higher pattern frequencies, a maximum of two or three (apparent) bars can be perceived. The modulation in the test pattern image can then no longer be characterized by a system MTF.

The modulation in the reproduced image depends on the relative position of the test pattern on the sampling matrix; this position is called the *phase*. It was shown [1] that for each pattern frequency an optimum phase can be found where the average modulation in the reproduced image is at a maximum. This modulation is defined as the AMOP for the given spatial frequency.

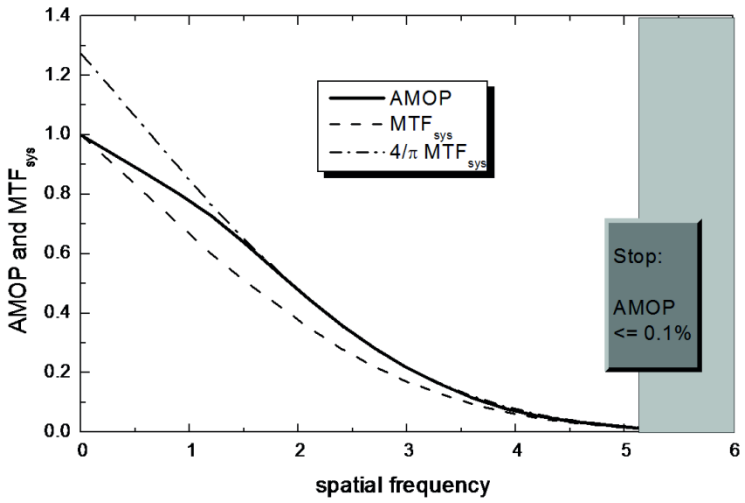


Figure 3.3: Actual modulation in the test pattern, system MTF and approximated modulation for a well sampled imager.

Note: AMOP cannot be calculated analytically. TRM determines AMOP by simulating the imaging of the 4-bar test pattern in both directions and calculating the average modulation in the image for each pattern frequency as function of phase. An example is shown in Figure 3.4, where the AMOP is compared to the system MTF.

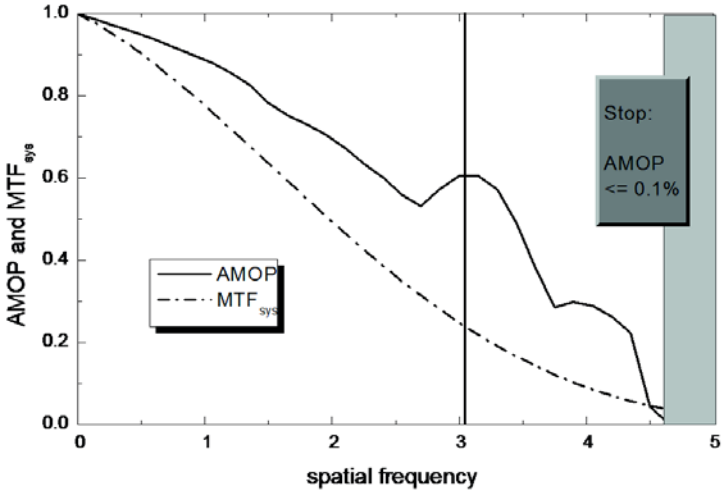


Figure 3.4: Average modulation in the test pattern at optimum phases (AMOP) for an under-sampled imager. For comparison, the "system MTF" is also shown. The vertical line marks the location of the Nyquist frequency of the detector.

3.11.4 Total perceived noise

The total perceived noise variance Ψ_z is the sum of the perceived temporal noise variance Ψ_z^{N} and the perceived fixed pattern noise variance Ψ_z^{I}:

$$\Psi_z(R, r_p) = \Psi_z^{\mathrm{N}}(R, r_{\mathrm{p}}) + \Psi_z^{\mathrm{I}}(R, r_{\mathrm{p}})\,. \tag{3.179}$$

Here, the subscript z stands for the MDSP direction.

Perceived temporal noise variance

The perceived temporal noise variance Ψ_z^{N} is given by

$$\Psi_z^{\mathrm{N}}(R, r_{\mathrm{p}}) = \frac{\mathrm{NES}^2(R)}{\Delta f \cdot N_{\mathrm{int_frames}}} \cdot E_{\mathrm{v}_z}^{\mathrm{N}}(r_{\mathrm{p}}) \cdot E_{\mathrm{h}_z}^{\mathrm{N}}(r_{\mathrm{p}}), \tag{3.180}$$

where
Δf = noise equivalent bandwidth (at the port where NES is measured or calculated) as described in Section 3.9,
$E_{\mathrm{v}_z}^{\mathrm{N}}(r_{\mathrm{p}})$ = eye/brain spatial integration of temporal noise in vertical direction (for MDSP in z-direction),
$E_{\mathrm{h}_z}^{\mathrm{N}}(r_{\mathrm{p}})$ = eye/brain spatial integration of temporal noise in horizontal direction (for MDSP in z-direction), and
$N_{\mathrm{int_frames}}$ = number of independent frames over which the eye integrates the noise.

In case of the eye integrating an integer number of video frames N_{eye} it can be shown that the number of independent integrated noise frames is given by

$$N_{\mathrm{int_frames}} = \frac{F_{\mathrm{r,imager}} \cdot t_{\mathrm{eye}}}{1 - \frac{2\alpha}{N_{\mathrm{eye}}} \cdot \frac{1 - \alpha^{N_{\mathrm{eye}}}}{1 - \alpha^2}} \tag{3.181}$$

with $N_{\mathrm{eye}} = F_{\mathrm{r,imager}} \cdot t_{\mathrm{eye}} / N_{\mathrm{frames}}$.

Note: Equation (3.180) assumes that there is no mismatch between imager and video frame rates and that the image is displayed and viewed without loss of information (Section 3.2.13). If this is not the case, the imager frame rate should be modified to obtain the appropriate value for $N_{\text{int_frames}}$.

In TRM, a more elaborate calculation is used that does not limit the eye integration to an integer number of video frames. The equations are more complex and not treated in this manual.

For ICCD/ICMOS systems, the expression for the perceived noise variance is more complex as noise from stemming from the image intensifier and the sensor array are filtered by different components. It reads

$$\Psi_z^{\mathrm{N,ICCD}}(R, r_\mathrm{p}) = \frac{\sigma_{\mathrm{II}}^2(R)}{N_{\text{int_frames_II}}} \cdot E_{\mathrm{v}_z}^{\mathrm{N,ICCD}}(r_\mathrm{p}) \cdot E_{\mathrm{h}_z}^{\mathrm{N,ICCD}}(r_\mathrm{p}) + \frac{\sigma_{\mathrm{CCD}}^2}{\Delta f_{\mathrm{DA}} \cdot N_{\text{int_frames}}} \cdot E_{\mathrm{v}_z}^{\mathrm{N}}(r_\mathrm{p}) \cdot E_{\mathrm{h}_z}^{\mathrm{N}}(r_\mathrm{p}), \tag{3.182}$$

where $\sigma_{\mathrm{II}}^2(R)$ and σ_{CCD}^2 are given by Equations (3.134) and (3.135), respectively, and $E_{z_{z'}}^{\mathrm{N,ICCD}}(r_\mathrm{p})$ are the eye/brain integrals for the noise from the photocathode signal given in Equation (3.203). In contrast to $N_{\text{int_frames}}$, the calculation of the number of independent frames over which the eye integrates the noise stemming from the image intensifier, $N_{\text{int_frames_II}}$, considers correlation between frames due to the temporal response behavior of the phosphor screen.

Perceived fixed pattern noise

The perceived fixed pattern noise, Ψ_z^{I}, depends on the type of imager. For scanning imagers, it is given by

$$\Psi_z^{\mathrm{I}}(R, r_\mathrm{p}) = \frac{\mathrm{IES}^2(R) \cdot g_{\mathrm{type}_z}}{\Delta f_{\mathrm{I_v}}} \cdot E_{\mathrm{v}_z}^{\mathrm{I}}(r_\mathrm{p}) \tag{3.183}$$

and for staring or micro-scanned imagers by

$$\Psi_z^{\mathrm{I}}(R, r_\mathrm{p}) = \frac{\mathrm{IES}^2(R) \cdot g_{\mathrm{type}_z}}{\Delta f_{\mathrm{I_h}} \cdot \Delta f_{\mathrm{I_v}}} \cdot E_{\mathrm{v}_z}^{\mathrm{I}}(r_\mathrm{p}) \cdot \mathrm{E}_{\mathrm{h}_z}^{\mathrm{I}}(r_\mathrm{p}), \tag{3.184}$$

where
Δf_{I_z} = noise equivalent bandwidth in z-direction (at the port where IES is measured or calculated) as presented in Section 3.10,
$E_{\mathrm{v}_z}^{\mathrm{I}}(r_\mathrm{p})$ = eye/brain spatial integration of spatial noise in vertical direction (for MDSP in z-direction),
$\mathrm{E}_{\mathrm{h}_z}^{\mathrm{I}}(r_\mathrm{p})$ = eye/brain spatial integration of spatial noise in horizontal direction (for MDSP in z-direction), and
g_{type_z} = additional weighting factor, which depends on the imager type.

The weighting factors g_{type_z} depend on the type of imager and were experimentally obtained in [1]. Their values are as follows:

- $g_{\mathrm{type_h}} = 0.0025$ for scanning imagers and horizontal MDSP,
- $g_{\mathrm{type_v}} = 0.04$ for scanning imagers and vertical MDSP, and

- $g_{\mathrm{type_v}} = g_{\mathrm{type_h}} = 1$ for staring/micro-scanned imagers.

3.11.5 Eye/brain spatial integration for scanning imagers

The MTFs used in this section are defined as follows:

Horizontal noise MTF

$$\begin{aligned}\mathrm{MTF}_{\Psi_\mathrm{h}} = \mathrm{MTF}_{\mathrm{ana}} \cdot \mathrm{MTF}_{\mathrm{adc}} \cdot \mathrm{MTF}_{\mathrm{dig_h}} \cdot \mathrm{MTF}_{\mathrm{zoom_h}} \cdot \mathrm{MTF}_{\mathrm{dac}} \cdot \mathrm{MTF}_{\mathrm{vid}} \\ \cdot \mathrm{MTF}_{\mathrm{dis_h}} \cdot \mathrm{MTF}_{\mathrm{add_post_h}} \cdot \mathrm{MTF}_{\mathrm{crep}}\,,\end{aligned} \tag{3.185}$$

Vertical noise MTF

$$\mathrm{MTF}_{\Psi_\mathrm{v}} = \mathrm{MTF}_{\mathrm{dig_v}} \cdot \mathrm{MTF}_{\mathrm{zoom_v}} \cdot \mathrm{MTF}_{\mathrm{dis_v}} \cdot \mathrm{MTF}_{\mathrm{add_post_v}} \cdot \mathrm{MTF}_{\mathrm{lrep}}\,. \tag{3.186}$$

Horizontal MDSP – Horizontal integration

Temporal noise

$$E_{\mathrm{h_h}}^{\mathrm{N}}(r_\mathrm{p}) = v_{\mathrm{scan}} \cdot \left[\int_{-\infty}^{\infty} S_{\mathrm{fold}}(r) \cdot \mathrm{MTF}_{\Psi_\mathrm{h}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{r}{2 \cdot r_\mathrm{p}}\right) \mathrm{d}r\right] \tag{3.187}$$

The definition of the folded noise power spectrum, $S_{\mathrm{fold}}(r)$, is given in Section 3.9.6. For 1st generation thermal imagers without horizontal sampling, $S_{\mathrm{fold}}(r)$ has to be replaced by the normalized noise power spectrum $S(r)$, which is defined in Section 3.9.4, and the integration should be carried out from 0 to ∞.

Horizontal MDSP – Vertical integration

Temporal noise

$$E_{\mathrm{v_h}}^{\mathrm{N}}(r_p) = \alpha_{s_\mathrm{v}} \cdot \left[\int_{-\infty}^{\infty} \mathrm{MTF}_{\Psi_\mathrm{v}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{7 \cdot r}{2 \cdot r_\mathrm{p}}\right) \mathrm{d}r\right] \tag{3.188}$$

Spatial noise

$$\begin{aligned}E_{\mathrm{v_h}}^{\mathrm{I}}(r_\mathrm{p}) = \alpha_{s_\mathrm{v}} \cdot n_{\mathrm{inter}} \\ \cdot \left[\int_{-\infty}^{\infty} \frac{\mathrm{sinc}^2\left(n_{\mathrm{inter}} \cdot \alpha_{s_\mathrm{v}} \cdot r\right)}{\mathrm{sinc}^2\left(\alpha_{s_\mathrm{v}} \cdot r\right)} \cdot \mathrm{MTF}_{\Psi_\mathrm{v}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{7r}{2r_\mathrm{p}}\right) \mathrm{d}r\right]\end{aligned} \tag{3.189}$$

Vertical MDSP – Horizontal integration

Temporal noise

$$\begin{aligned}E_{\mathrm{h_v}}^{\mathrm{N}}(r_\mathrm{p}) = v_{\mathrm{scan}} \\ \cdot \left[\int_{-\infty}^{\infty} S_{\mathrm{fold}}(r) \cdot \mathrm{MTF}_{\Psi_\mathrm{h}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{7 \cdot r}{2 \cdot r_\mathrm{p}}\right) \mathrm{d}r\right]\end{aligned} \tag{3.190}$$

For 1st generation thermal imagers without horizontal sampling, $S_{\mathrm{fold}}(r)$ has to be replaced by the normalized noise power spectrum $S(r)$ and the integration should be carried out from 0 to ∞.

Vertical MDSP – Vertical integration

Temporal noise

$$E_{\mathrm{v_v}}^{\mathrm{N}}(r_{\mathrm{p}}) = v_{\mathrm{scan}} \cdot \left[\int_{-\infty}^{\infty} \mathrm{MTF}_{\Psi_{\mathrm{v}}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{r}{2 \cdot r_{\mathrm{p}}}\right) \mathrm{d}r\right] \tag{3.191}$$

Spatial noise

$$E_{\mathrm{v_v}}^{\mathrm{I}}(r_{\mathrm{p}}) = \alpha_{\mathrm{s_v}} \cdot n_{\mathrm{inter}} \cdot \left[\int_{-\infty}^{\infty} \frac{\mathrm{sinc}^2(n_{\mathrm{inter}} \cdot \alpha_{\mathrm{s_v}} \cdot r)}{\mathrm{sinc}^2(\alpha_{\mathrm{s_v}} \cdot r)} \cdot \mathrm{MTF}_{\Psi_{\mathrm{v}}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{r}{2 \cdot r_{\mathrm{p}}}\right) \mathrm{d}r\right] \tag{3.192}$$

3.11.6 Eye/brain spatial integration for staring or micro-scanned imagers

The factor k_D used in the following equations is equal to one, except for imagers with diagonal micro-scan where $k_D = \sqrt{2}$.

The MTFs used in this section are defined as follows:

Horizontal noise MTF

$$\mathrm{MTF}_{\Psi_{\mathrm{h}}} = \mathrm{MTF}_{\mathrm{dig_h}} \cdot \mathrm{MTF}_{\mathrm{zoom_h}} \cdot \mathrm{MTF}_{\mathrm{dac}} \cdot \mathrm{MTF}_{\mathrm{vid}} \cdot \mathrm{MTF}_{\mathrm{dis_h}} \cdot \mathrm{MTF}_{\mathrm{add_post_h}} \cdot \mathrm{MTF}_{\mathrm{crep}} \tag{3.193}$$

Vertical noise MTF

$$\mathrm{MTF}_{\Psi_{\mathrm{v}}} = \mathrm{MTF}_{\mathrm{dig_v}} \cdot \mathrm{MTF}_{\mathrm{zoom_v}} \cdot \mathrm{MTF}_{\mathrm{dis_v}} \cdot \mathrm{MTF}_{\mathrm{add_post_v}} \cdot \mathrm{MTF}_{\mathrm{lrep}} \,. \tag{3.194}$$

Horizontal MDSP – Horizontal integration

Temporal noise

$$E_{\mathrm{h_h}}^{\mathrm{N}}(r_{\mathrm{p}}) = k_{\mathrm{D}} \cdot \alpha_{\mathrm{s_h}} \cdot \Delta f_{\mathrm{DA}} \cdot \left[\int_{-\infty}^{\infty} \mathrm{MTF}_{\Psi_{\mathrm{h}}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{r}{2 \cdot r_{\mathrm{p}}}\right) \mathrm{d}r\right] \tag{3.195}$$

Spatial noise

$$E_{\mathrm{h_h}}^{\mathrm{I}}(r_{\mathrm{p}}) = \alpha_{\mathrm{s_h}} \cdot N_{\mu\mathrm{step_h}} \cdot \left[\int_{-\infty}^{\infty} \frac{\mathrm{sinc}^2(N_{\mu\mathrm{step}_h} \cdot \alpha_{\mathrm{s_h}} \cdot r)}{\mathrm{sinc}^2(\alpha_{\mathrm{s_h}} \cdot r)} \cdot \mathrm{MTF}_{\Psi_{\mathrm{h}}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{r}{2 \cdot r_{\mathrm{p}}}\right) \mathrm{d}r\right] \tag{3.196}$$

Horizontal MDSP – Vertical integration

Temporal noise

$$E_{\mathrm{v_h}}^{\mathrm{N}}(r_{\mathrm{p}}) = k_{\mathrm{D}} \cdot \alpha_{\mathrm{s_v}} \cdot \left[\int_{-\infty}^{\infty} \mathrm{MTF}_{\Psi_{\mathrm{v}}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{7 \cdot r}{2 \cdot r_{\mathrm{p}}}\right) \mathrm{d}r\right] \tag{3.197}$$

Spatial noise

$$E_{\mathrm{v_h}}^{\mathrm{I}}(r_{\mathrm{p}}) = \alpha_{\mathrm{s_v}} \cdot N_{\mu\mathrm{step_v}} \cdot \left[\int_{-\infty}^{\infty} \frac{\mathrm{sinc}^2(N_{\mu\mathrm{step_v}} \cdot \alpha_{\mathrm{s_v}} \cdot r)}{\mathrm{sinc}^2(\alpha_{\mathrm{s_v}} \cdot r)} \cdot \mathrm{MTF}_{\Psi_{\mathrm{v}}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{7r}{2r_{\mathrm{p}}}\right) \mathrm{d}r\right] \quad (3.198)$$

Vertical MDSP – Horizontal integration

Temporal noise

$$E_{\mathrm{h_v}}^{\mathrm{N}}(r_{\mathrm{p}}) = k_{\mathrm{D}} \cdot \alpha_{\mathrm{s_h}} \cdot \Delta f_{\mathrm{DA}} \cdot \left[\int_{-\infty}^{\infty} \mathrm{MTF}_{\Psi_{\mathrm{h}}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{7 \cdot r}{2 \cdot r_{\mathrm{p}}}\right) \mathrm{d}r\right] \quad (3.199)$$

Spatial noise

$$E_{\mathrm{h_v}}^{\mathrm{I}}(r_{\mathrm{p}}) = \alpha_{\mathrm{s_h}} \cdot N_{\mu\mathrm{step_h}} \cdot \left[\int_{-\infty}^{\infty} \frac{\mathrm{sinc}^2(N_{\mu\mathrm{step_h}} \cdot \alpha_{\mathrm{s_h}} \cdot r)}{\mathrm{sinc}^2(\alpha_{\mathrm{s_h}} \cdot r)} \cdot \mathrm{MTF}_{\Psi_{\mathrm{h}}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{7r}{2r_{\mathrm{p}}}\right) \mathrm{d}r\right] \quad (3.200)$$

Vertical MDSP – Vertical integration

Temporal noise

$$E_{\mathrm{v_v}}^{\mathrm{N}}(r_{\mathrm{p}}) = k_{\mathrm{D}} \cdot \alpha_{\mathrm{s_v}} \cdot \left[\int_{-\infty}^{\infty} \mathrm{MTF}_{\Psi_{\mathrm{v}}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{r}{2 \cdot r_{\mathrm{p}}}\right) \mathrm{d}r\right] \quad (3.201)$$

Spatial noise

$$E_{\mathrm{v_v}}^{\mathrm{I}}(r_{\mathrm{p}}) = \alpha_{\mathrm{s_v}} \cdot N_{\mu\mathrm{step_v}} \cdot \left[\int_{-\infty}^{\infty} \frac{\mathrm{sinc}^2(N_{\mu\mathrm{step_v}} \cdot \alpha_{\mathrm{s_v}} \cdot r)}{\mathrm{sinc}^2(\alpha_{\mathrm{s_v}} \cdot r)} \cdot \mathrm{MTF}_{\Psi_{\mathrm{v}}}^2(|r|) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(|r|) \cdot \mathrm{sinc}^2\left(\frac{r}{2r_{\mathrm{p}}}\right) \mathrm{d}r\right] \quad (3.202)$$

3.11.7 Eye/brain spatial integration for ICCD/ICMOS

The eye/brain integrals for temporal noise take into account noise folding due to the sampling process and are given by:

$$E_{z_{z'}}^{\mathrm{N,ICCD}}(r_{\mathrm{p}}) = \frac{1}{2\Delta f_{\mathrm{z}} \cdot f_{\mathrm{opt}_z}} \int_{-\infty}^{\infty} \mathrm{d}r \sum_{n=-\infty}^{\infty} \mathrm{MTF}_{\mathrm{det_stat}_z}^2\left(\frac{r - n \cdot r_{s_z}}{M_{\mathrm{ICCD}}}\right) \cdot \mathrm{MTF}_{\mathrm{II}}^2(r - n \cdot r_{s_z}) \cdot \mathrm{MTF}_{\mathrm{Coup}}^2(r - n \cdot r_{s_z}) \cdot \mathrm{MTF}_{\Psi_z}^2(r) \cdot \mathrm{MTF}_{\mathrm{eye}}^2(r) \cdot \mathrm{sinc}^2\left(\frac{n_{z_{z'}} \cdot r}{2r_p}\right), \quad (3.203)$$

where $n_{z_{z'}} = 1$ if z = z′ and $n_{z_{z'}} = 7$ else. MTF_{Ψ_z} is defined for horizontal and verical directions by Equations (3.193) and (3.194).

3.11.8 Average MDSP

The following procedure is used to calculate the average MDSP:

1. For the given reference range[10], R_{ref}, the horizontal and vertical MDSP values are calculated for a set of spatial frequencies.

2. For each value of the horizontal $\mathrm{MDSP_h}(r_{\mathrm{h}}, R_{\mathrm{ref}})$, the spatial frequency r_{v} is determined which solves

$$\mathrm{MDSP_v}(r_{\mathrm{v}}, R_{\mathrm{ref}}) = \mathrm{MDSP_h}(r_{\mathrm{h}}, R_{\mathrm{ref}})\,. \tag{3.204}$$

   This results in pairs of spatial frequencies r_{h} and r_{v} with a commaon MDSP value.

3. For each such pair r_{h} and r_{v}, the geometric average, $\bar{r}$, is calculated and the average MDSP is defined by

$$\mathrm{MDSP_{avg}}(\bar{r}, R_{\mathrm{ref}}) := \mathrm{MDSP_v}(r_{\mathrm{v}}, R_{\mathrm{ref}}) = \mathrm{MDSP_h}(r_{\mathrm{h}}, R_{\mathrm{ref}})\,. \tag{3.205}$$

3.12 Range performance

The calculation of range performance in TRM is based on modified Johnson criteria. The original criteria were developed for the assessment of image intensifiers by J. Johnson [14].

3.12.1 Johnson criteria

The original Johnson criteria are as follows:

1. The actual target is replaced by an equivalent bar pattern, which is characterized by
   - Horizontal and vertical dimensions of the target and the equivalent bar pattern are identical.
   - The smaller target dimension is called the *critical* dimension. The equivalent bar pattern is periodic in the critical dimension.
   - The number of line pairs, $n_{\mathrm{lp_perc}}$, in the equivalent bar pattern depends on the perception task, which is typically detection, recognition, or identification of a target with a specified probability.
   - The contrast between the bars of the equivalent bar pattern equals the contrast between target and background.
2. The perception task is considered accomplished if the required number of line pairs is resolved in the equivalent bar pattern.

The application of the Johnson criteria in assessing imager range performance is done in three steps:

1. Considering a target at distance R, the spatial frequency of the bar pattern is given by

$$r_{n_{\mathrm{lp}}} = \frac{n_{\mathrm{lp_perc}}}{a_{\mathrm{tar_crit}}} \cdot R \tag{3.206}$$

[10] The MDSP is calculated for the reference range as specified in the *Modify Individual Parameters* dialog (see Section 2.4). For *Thermal* category calculations, the reference range is always zero.

with the critical target dimension $a_{\mathrm{tar_crit}}$ and the number of line pairs $n_{\mathrm{lp_perc}}$ along that dimension. From this, the task dependent version of the MDSP, the MNDS (Minimum Necessary Difference Signal) is derived (Section 3.12.4).

2. The *Effective Difference Signal,* EDS(R), between target and background is given by the difference signal at the target/background plane, $\mathrm{DS}(\lambda, 0)$ [Equation (3.128)], attenuated by the atmosphere:

$$\mathrm{EDS}(R) = N_{\mathrm{TDI}} \cdot \int_0^{\infty} \mathrm{DS}(\lambda, 0) \cdot \tau_{\mathrm{atm}}(\lambda, R)\, \mathrm{d}\lambda\,. \tag{3.207}$$

The *Effective Temperature Difference*, $\mathrm{ETD}(R)$, in the *Thermal* category is given by

$$\mathrm{ETD}(R) = N_{\mathrm{TDI}} \cdot \Delta T_0 \cdot \tau_{\mathrm{atm}}(R)\,, \tag{3.208}$$

where ΔT_0 is the temperature difference between target and background.

3. If the effective difference signal at range R is greater or equal to the MNDS at R, $n_{\mathrm{lp_perc}}$ line pairs or more can be resolved on the target. Thus, the maximum range R_0 is found from solving

$$\mathrm{EDS}(R_0) = \mathrm{MNDS}(R_0). \tag{3.209}$$

3.12.2 Modification of the Johnson criteria for two dimensions

In early thermal range performance models, the horizontal MRTD was used in assessing imager performance, irrespective of whether the critical dimension of the target was horizontal or vertical. More recent models use both the horizontal and vertical imager resolution to assess range performance. The usual procedure involves the replacement of the specified rectangular target by an equivalent square target with the same area as the original one, then calculating the MRTD for a horizontal and vertical 4-bar pattern. An averaged MRTD is calculated from the horizontal and vertical MRTD. The averaged MRTD is then used for range calculations.

TRM is using a somewhat different procedure, which is presented in Section 3.12.3.

3.12.3 Range performance for the *Thermal* category

TRM4 calculates the range performance of imagers of the *Thermal* category in accordance with STANAG 4347. The principle steps of the range calculation are:

1. The MTDP is calculated for both the horizontal and vertical directions.

2. The number of line pairs resolved on the target in each direction (as a function of range) is calculated using the directional MTDP_z and the actual target size in that direction a_{tar_z}. For each range R, the number of resolved line pairs, $n_{\mathrm{lp_res}_z}(R)$, is then found from the equation

$$\mathrm{ETD}(R) = \mathrm{MTDP}_z\left(\frac{n_{\mathrm{lp_res}_z}(R) \cdot R}{a_{\mathrm{tar}_z}}\right). \tag{3.210}$$

3. The average number of line pairs resolved on the target (as a function of range) is given by

$$\bar{n}_{\mathrm{lp_res}}(R) = \sqrt{n_{\mathrm{lp_res_h}}(R) \cdot n_{\mathrm{lp_res_v}}(R)}\,. \tag{3.211}$$

4. Finally, the range solution R_0 is found from solving

$$\bar{n}_{\mathrm{lp_res}}(R_0) = n_{\mathrm{lp_perc}}\,. \tag{3.212}$$

Thus, the range solution is defined as the target distance R at which the average number of line pairs resolved equals the specified number of line pairs for the task.

Average number of resolved line pairs

TRM4 calculates and displays the average number of resolved line pairs as a function of target distance. For each distance R, the number of line pairs that can be resolved in z-direction, $n_{\mathrm{lp_res}_z}$, is found by solving

$$\Delta T_0 = \frac{\mathrm{MTDP}\left(\frac{n_{\mathrm{lp_res}_z} \cdot R}{a_{\mathrm{tar}_z}}\right)}{\tau_{\mathrm{atm_eff}}(R)}. \tag{3.213}$$

The (directionally) averaged number of resolved line pairs is then calculated using Equation (3.211).

Number of resolved pixels

If $n_{\mathrm{lp_res_h}}$ line pairs are resolved in the horizontal and $n_{\mathrm{lp_res_v}}$ line pairs in the vertical direction (for the same ΔT_0), the number of *resolved pixels* is given by

$$n_{\mathrm{pix_res}} = n_{\mathrm{lp_res_h}} \cdot n_{\mathrm{lp_res_v}}\,. \tag{3.214}$$

Minimum Necessary Temperature Difference (MNTD)

The MNTD is defined as the minimal required temperature difference between target and background in order to be able to resolve an average number $\bar{n}_{\mathrm{lp}}$ of line pairs on the target. It is a function of target distance R. The range performance of the imager for a specified perception task with $n_{\mathrm{lp_perc}}$ line pairs is then given by the distance where the MNTD intersects the Effective Temperature Difference (ETD).

The MNTD is introduced to better understand the limiting factors that govern the range performance of the imager.

Unlike the MTDP, the MNTD is a task-dependent function. It is calculated using the following procedure and equations:

1. Directional MNTDs are calculated according to

$$\mathrm{MNTD}_z(R) = \mathrm{MTDP}_z\left(\frac{n_{\mathrm{lp_perc}} \cdot R}{a_{\mathrm{tar}_z}}\right). \tag{3.215}$$

2. For each ΔT, the distances R_{h} and R_{v} are calculated, at which

$$\mathrm{MNTD_h}(R_{\mathrm{h}}) = \mathrm{MNTD_v}(R_{\mathrm{v}}) = \Delta T \ . \tag{3.216}$$

3. The average distance is then calculated as

$$\bar{R}(\Delta T) = \sqrt{R_{\mathrm{h}}(\Delta T) \cdot R_{\mathrm{v}}(\Delta T)} \, . \tag{3.217}$$

4. The average MNTD at the distance $\bar{R}$ is then given by

$$\mathrm{MNTD_{avg}}(\bar{R}) = \Delta T \ . \tag{3.218}$$

Note: Range performance is calculated via the directional MTDP and MNTD. In the TRM4 output data the averaged MTDP is displayed; it is shown for informational purposes only.

3.12.4 Range performance for the *Visible* and *General* categories

For the *Visible* and *General* categories, the MDSP serves as a starting point to calculate the horizontal and vertical MNDS (Minimum Necessary Difference Signal) from which the average MNDS can be found. The intersection point of the average MNDS and the effective difference signal (EDS) then determines ranges.[11]

Unlike its *Thermal* category counterpart, MTDP, which depends only on the spatial frequency, MDSP depends on the spatial frequency as well as on the target distance. As indicated in Equation (3.219), AMOP depends directly (via Turbulence MTF) and indirectly (via spectral atmospheric transmission and spectral MTFs) on the target distance and the perceived noise variance Ψ is spatial frequency dependent via the 4-bar pattern target and range dependent via the signal from the background (affecting shot noise).

Minimum Necessary Difference Signal

The MNDS is defined as the minimal difference signal required to resolve an average number of line pairs, $\bar{n}_{\mathrm{lp}}$, on the target. It is a function of target distance R.

The MNDS may help to better understand the limiting factors that determine the range performance of the imager.

Unlike the MDSP, the MNDS is a task-dependent function. The one-dimensional MNDS in z-direction for a perception task requiring $n_{\mathrm{lp_perc}}$ line pairs to be resolved is given by

$$\mathrm{MNDS}_z(R) = \mathrm{MDSP}_z\left(R, r_{\mathrm{p_z}}\right) = \frac{\frac{\pi}{2} \cdot \mathrm{SNR_{th}} \cdot \Psi_z^{1/2}(\mathrm{R}, r_{\mathrm{p_z}})}{\mathrm{AMOP}_z(R, r_{\mathrm{p_z}})} \tag{3.219}$$

with the equivalent spatial frequency $r_{\mathrm{p}_z} = (n_{\mathrm{lp_perc}}/a_{\mathrm{tar}_z}) \cdot R$.

[11] In case the user specifies measured MCP/MRC values for imagers of category *Visible*, the rage performance is determined by the procedure discussed in Section 3.12.5.

Note: Unlike the case of the *Thermal* category, where the basic assumptions lead to a range-independent AMOP and detector noise (and hence range-independent MTDP), the AMOP for the *Visible* and *General* categories is range-dependent; this is due to the optional spectral atmospheric transmission and emission. Thus, the calculation of the MNDS cannot be accomplished by a simple transformation of the MDSP frequency to the MNDS's range. The AMOP has to be calculated for each and every combination of R and r_{p}, which is a computationally expensive task. Hence, range calculations for the *Visible* and *General* categories take much longer than that for the *Thermal* category.

Calculation of the MNDS at target distance R is done as follows:

1. Find the equivalent frequency r_{p} for the range R.
2. Calculate the perceived noise variance Ψ for r_{p} and the target distance R.
3. Calculate AMOP for the target distance R and the equivalent frequency. [12]
4. The average MNDS is then calculated by taking the geometric average of the ranges R_{h} and R_{v}, for which

$$\mathrm{MNDS_h}(R_{\mathrm{h}}) = \mathrm{MNDS_v}(R_{\mathrm{v}})\,. \tag{3.220}$$

5. Finally, the range solution for the task, R_0, is found by solving

$$\mathrm{MNDS_{avg}}(R_0) = \mathrm{EDS}(R_0). \tag{3.221}$$

3.12.5 Range performance based on measured MCP/MRC in the *Visible* category

In case measured MCP (or MRC) data is specified, the first step in determining the range performance is the calculation of the Minimum Necessary Contrast (MNC) for different target distances R:

$$\mathrm{MNC}_z(R) = \mathrm{MCP}_z(r = \frac{R \cdot n_{\mathrm{lp}}}{a_{\mathrm{tar}_z}}). \tag{3.222}$$

For CCD/CMOS and ICCD/ICMOS devices, the MNDS can be derived from the MNC using

$$\mathrm{MNDS}_z(R) = \mathrm{MNC}_z(R) \cdot [N_T(R) + N_B(R)]\,, \tag{3.223}$$

where $N_{T/B}$ are the number of carriers in the storage gate of a detector element due to radiance from the target/background. The MNC is used together with the Effective Contrast (EC) to determine the range performance using the same routines as for MNDS and EDS [Equations (3.220) and (3.221) in Section 3.12.4].

TRM4 calculates the Effective Contrast between target and background as

$$\mathrm{EC}(R) = \left|\frac{j_{\mathrm{total_T}}(R) - j_{\mathrm{total_B}}(R)}{j_{\mathrm{total_T}}(R) + j_{\mathrm{total_B}}(R)}\right| \tag{3.224}$$

for imager of category *Visible/NIR/SWIR* except for Direct View systems where

[12] AMOP is also a direct function of R since several MTFs are both range- and wavelength-dependent (see Chapter 4).

$$\mathrm{EC}(R) = \left| \frac{L_T^{\nu}(R) - L_B^{\nu}(R)}{L_T^{\nu}(R) + L_B^{\nu}(R)} \right| \tag{3.225}$$

with $L_{\mathrm{T/B}}^{\nu}(R)$ the luminance of the target/background signal at the eye.

3.12.6 Category *Thermal* vs. *General/Visible* – basic assumptions and consequences

There is an inherent difference between the MNTD calculated in the *Thermal* category and the MNDS calculated for the *Visible* and *General* categories in TRM4, due to different model assumptions. These assumptions are listed in Table 3.1.

	Thermal	**Visible/General**
Target/Background emissivity	Blackbody	any spectral function
Air temperature	same as background	any
Atmosphere model: transmission and emission	greybody (emissivity + transmissivity = 1)	greybody (emissivity + transmissivity = 1)
Atmospheric transmission	wavelength independent	spectral and range dependence
Target/background contrast	very small	any

Table 3.1: Basic assumptions for TRM4's categories.

The assumptions lead to the following results for *Thermal* category calculations only:

- The signal reaching the detector originating from the background together with the (emissive) path radiance is independent of distance; hence, detector noise is independent of target range.
- Due to the small contrast, the noise generated by the target and the background is almost identical.
- The difference signal is approximately proportional to ΔT.
- The spectral dependence of the difference signal is not a function of range.
- The MTFs and the AMOP are independent of range.

All that leads to the consequence that, for *Thermal* category calculations, the transformation from MTDP to MNTD is accomplished by a linear transformation (scaling) of the frequency axis (MTDP) to the spatial range axis (MNTD):

$$\mathrm{MNTD}_z(R) = \mathrm{MTDP}_z\left(\frac{\mathrm{n_{lp}}_z}{\mathrm{a_{tar}}_z} \cdot R\right). \tag{3.226}$$

The option of free specification of the parameters listed in Table 3.1 for the *Visible* and *General* categories results in a much more complex dependence of various variables on range and thus leads to costly calculations along with occasionally surprising results like contrast reversal, etc.

An illustrating example: a red target on a green background that cause the same number of carriers in the detector will be undetectable at zero range, but may become discernible at longer distances if the atmosphere is blocking e.g. the green component more strongly than the red one.

3.13 NIIRS calculation

The National Imagery Interpretability Rating Scale (NIIRS) is used to define the quality of aerial images. Therefore, NIIRS values provide means to compare and assess the performance of imagers in aerial imaging applications. NIIRS calculations in TRM4 are based on the General Image Quality Equation version 5 (GIQE 5) [15]:

$$\begin{aligned}\mathrm{NIIRS} = {} & 4.274 - 3.32 \cdot \log_{10}(\mathrm{GSD_w}) + 3.32 \cdot \left(1 - \mathrm{e}^{\frac{-1.9}{\mathrm{DSNR}}}\right) \cdot \log_{10}(\mathrm{RER_w}) \\ & -2 \cdot (\log_{10}(\mathrm{RER_w}))^4 - \frac{1.8}{\mathrm{DSNR}}\end{aligned} \tag{3.227}$$

where $\mathrm{GSD_w}$ = weighted Ground Sampled Distance in metric units,
$\mathrm{RER_w}$ = weighted Relative Edge Response, and
DSNR = difference signal to noise ratio.

In contrast to previous versions of the GIQE, GIQE 5 was developed having soft copy images in mind whose enhancement is interactively selected by the image analyst. Therefore, GIQE 5 predicts the quality of well enhanced images based on performance parameters of raw images.

Note: NIIRS calculations using Equation (3.227) are available in TRM4 for all categories and imager types except for 1st generation thermal imagers, night vision goggles, and direct view systems.

3.13.1 Ground sampled distance

The normal plane GSD (Ground Sampled Distance) in *z*-direction is given by:

$$\mathrm{GSD_{NP_z}} = \alpha_{s_z} \cdot R\,, \tag{3.228}$$

where α_{s_z} is the angular sampling distance and *R* is the slant range.

The ground plane GSD is calculated as:

$$\mathrm{GSD_{GP_z}} = \frac{\mathrm{GSD_{NP_z}}}{[\sin(\alpha_v)]^{1/2}}\,, \tag{3.229}$$

where α_v is the line-of-sight angle to the target in vertical direction (elevation angle).

As the elevation angle affects only the vertical direction, the weighted GSD is a geometrical mean of the normal-plane GSD in horizontal direction and the ground-plane GSD in the vertical direction. It reads

$$\mathrm{GSD_w} = (\mathrm{GSD_{NP_h}} \cdot \mathrm{GSD_{GP_v}})^{1/2} = \frac{(\mathrm{GSD_{NP_h}} \cdot \mathrm{GSD_{NP_v}})^{1/2}}{[\sin(\alpha_v)]^{1/4}}. \tag{3.230}$$

3.13.2 Relative Edge Response

The system edge response is defined by the system response at a certain position to a normalized edge in the scene (dark left side with value 0 and bright right side with value 1). The Relative Edge Response (RER) used in GIQE 5 is the difference between the system edge responses at two points located half a pixel pitch from the edge [16].

The RER can be calculated using:

$$\mathrm{RER}_z = 2 \cdot \alpha_{s_z} \int_0^{r_{c_z}} \mathrm{MTF}_{\mathrm{pre}_z}(r) \cdot \mathrm{sinc}\left(r \cdot \alpha_{s_z}\right) \mathrm{d}r \,, \tag{3.231}$$

where r_{c_z} is the cut-off frequency in *z*-direction, α_{s_z} is the angular sampling distance and $\mathrm{MTF}_{\mathrm{pre}_z}(r)$ includes contribution from turbulence, stabilization platform and imager device components.

Smear may affect MTF and RER in cases of moving imagers. This effect is in the moving direction only and may lead to asymmetric RER. In order to take into account the effect of smear, a weighted RER was defined in GIQE 5 [15]:

$$\mathrm{RER}_\mathrm{w} = (\mathrm{RER}_\mathrm{cs} \cdot {\mathrm{RER}_\mathrm{as}}^2)^{1/3} \,, \tag{3.232}$$

where RER_cs is the cross-scan RER and RER_as is the along-scan RER. In some cases, however, RER_cs might be smaller than RER_as, for example due to charge transfer efficiency effects. In order to accommodate the possibility that $\mathrm{RER}_\mathrm{cs} < \mathrm{RER}_\mathrm{as}$, TRM4 uses a more general form of the weighted RER [17]:

$$\mathrm{RER}_\mathrm{w} = (\mathrm{RER}_\mathrm{X} \cdot {\mathrm{RER}_\mathrm{Y}}^2)^{1/3} \,. \tag{3.233}$$

Here, RER_X is equal or larger than RER_Y and the X and Y directions can be either along-scan or cross-scan, depending on their relative RER values.

3.13.3 Difference signal to noise ratio

The DSNR used in NIIRS calculations is defined as the ratio of the signal difference between target and background divided by the standard deviation of the noise of a detector element sensing the scene radiation (including path radiance):

$$\mathrm{DSNR}(R) = \frac{\mathrm{ETD}(R)}{\sqrt{\mathrm{NETD}(R)^2 + \mathrm{IETD}(R)^2}} \tag{3.234}$$

for imagers of category *Thermal* and

$$\mathrm{DSNR}(R) = \frac{N_{\mathrm{Total_T}}(R) - N_{\mathrm{Total_B}}(R)}{\sqrt{\mathrm{NES}(R)^2 + \mathrm{IES}(R)^2}} \tag{3.235}$$

for imagers of the categories *Visible/NIR/SWIR* and *General*.

Note: It is assumed that the effect of non-uniformity noise on image interpretability is less important for scanning imagers and therefore non-uniformity noise is neglected in the DSNR calculation.

4 Calculation of MTFs

4.1 Calculation of MTFs

4.1.1 Definitions

The MTFs of the different components of the electro-optical imaging system are considered to be part of either the *prefilter* (components before sampling) or the *postfilter* (components after sampling).

4.1.2 MTF reference ranges

All MTFs shown in the MTF result windows are calculated for the reference range specified in the *Modify Individual Parameters* dialog (see Section 2.4). For laboratory calculations, a zero reference range is used.

> **Note:** Turbulence MTF at zero range is assumed to be equal to one (no MTF degradation).

4.1.3 Range and spectral calculations

Generally, an MTF can depend both on the distance between target and imager, R, and on the wavelength λ. To account for the spectral dependence, a weighted MTF is used, which is defined by

$$\mathrm{MTF}_{\mathrm{eff}}(r,R) = \left|\int_0^\infty w(\lambda,R)\cdot \mathrm{MTF}(\lambda,r,R)\,\mathrm{d}\lambda\right|, \tag{4.1}$$

where R is the target distance and r denotes the spatial frequency.

The *spectral weighting function* $w(\lambda,R)$ is given by

$$w(\lambda,R) = \frac{\mathrm{DS}_{\mathrm{in}}(\lambda,R)}{\int_0^\infty \mathrm{DS}_{\mathrm{in}}(\lambda,R)\mathrm{d}\lambda}, \tag{4.2}$$

with the input difference signal

$$\begin{aligned}\mathrm{DS}_{\mathrm{in}}(\lambda,R) = {} & \left[I(\lambda)\cdot\left(\rho_{\mathrm{T}}(\lambda)-\rho_{\mathrm{B}}(\lambda)\right)+\pi\left(L(T_{\mathrm{T}},\lambda)\epsilon_{\mathrm{T}}(\lambda)-L(T_{\mathrm{B}},\lambda)\epsilon_{\mathrm{B}}(\lambda)\right)\right] \\ & \cdot\tau_{\mathrm{opt}}(\lambda)\cdot\tau_{\mathrm{filter}}(\lambda)\cdot\tau_{\mathrm{atm}}(\lambda,R)\cdot R_{\mathrm{DA}}(\lambda)\,.\end{aligned} \tag{4.3}$$

4.1.4 Prefilter MTF

The prefilter MTF is the product of all MTFs that act on the signal prior to sampling. If the signal is not sampled, the prefilter MTF is considered to be the product of all MTFs prior to the detector signal read-out.

Two distinct prefilter MTFs are available as outputs in TRM4:

- Device prefilter, and
- System prefilter.

The *device prefilter* MTF includes all prefilter components which are an integral part of the imaging device and consists of the following MTFs:

$$\begin{aligned}\mathrm{MTF}_{\mathrm{pre_device}} = \mathrm{MTF}_{\mathrm{II}} \cdot \mathrm{MTF}_{\mathrm{coup}} \cdot \mathrm{MTF}_{\mathrm{opt}} \cdot \mathrm{MTF}_{\mathrm{det}} \cdot \mathrm{MTF}_{\mathrm{scan}} \cdot \mathrm{MTF}_{\mathrm{ana}} \cdot \\ \mathrm{MTF}_{\mathrm{adc}} \cdot \mathrm{MTF}_{\mathrm{add_pre}} \,.\end{aligned} \tag{4.4}$$

Some of these MTFs are themselves composed of several MTF sub-components (described in Section 4.2).

The *system prefilter* MTF includes all of the components of the device prefilter and two additional components which are not part of the device proper, namely stabilization and turbulence:

$$\mathrm{MTF}_{\mathrm{pre}} = \mathrm{MTF}_{\mathrm{pre_device}} \cdot \mathrm{MTF}_{\mathrm{stab}} \cdot \mathrm{MTF}_{\mathrm{turb}} \,. \tag{4.5}$$

4.1.5 Postfilter MTF

The postfilter MTF is the product of all MTFs which act on the signal after sampling or, if the signal is not sampled, after read-out.

Two postfilter MTFs are available as outputs in TRM4:

- Device postfilter, and
- System postfilter.

The *device postfilter* includes all components (after sampling) which are an integral part of the imager. It may consist of the following MTFs:

$$\mathrm{MTF}_{\mathrm{post_device}} = \mathrm{MTF}_{\mathrm{dig}} \cdot \mathrm{MTF}_{\mathrm{rec}} \cdot \mathrm{MTF}_{\mathrm{vid}} \cdot \mathrm{MTF}_{\mathrm{dis}} \cdot \mathrm{MTF}_{\mathrm{add_post}} \,. \tag{4.6}$$

Some of these MTFs may be the product of several individual MTFs as described in Section 4.2.

The *system postfilter* includes all of the components of the device postfilter and the MTF of the human eye, which naturally is not part of the device proper:

$$\mathrm{MTF}_{\mathrm{post}} = \mathrm{MTF}_{\mathrm{post_device}} \cdot \mathrm{MTF}_{\mathrm{eye}} \,. \tag{4.7}$$

4.1.6 Total device MTF and total system MTF

The total system and total device MTF are not used for range calculations in TRM4 as TRM4 uses the AMOP (Average Modulation at Optimal Phase) concept. However, these MTFs are made available as output to the batch mode user.

The total device MTF is defined as the product of the device pre- and postfilter MTFs:

$$\mathrm{MTF}_{\mathrm{dev}} = \mathrm{MTF}_{\mathrm{pre_device}} \cdot \mathrm{MTF}_{\mathrm{post_device}} \,. \tag{4.8}$$

The total system MTF is the product of the pre- and postfilter MTFs:

$$\mathrm{MTF}_{\mathrm{sys}} = \mathrm{MTF}_{\mathrm{pre}} \cdot \mathrm{MTF}_{\mathrm{post}} \,. \tag{4.9}$$

4.1.7 Average Modulation at Optimum Phase (AMOP)

In TRM, the system MTF is always replaced by the AMOP. The AMOP represents the modulation transfer function of a 4-bar rectangular target while the MTF is defined over purely infinite sinusoidal patterns. The assumption in common MRTD calculations is that only the first harmonic of the rectangular pattern is not blocked by the transfer function of the imager. The first harmonic component in a rectangular bar pattern has an amplitude of $\frac{4}{\pi}$. Hence, in the MTDP/MDSP equations, AMOP replaces $4/\pi \cdot \mathrm{MTF}_{\mathrm{sys}}$. For an undersampled system, the AMOP cannot be expressed analytically, and is thus found from a simulation of the imaging and sampling process [1].

4.2 MTF equations

All symbols not explicitly defined with the equations are listed and specified in Appendix D. Units of system component parameters and corresponding conversion factors are given in Appendix A.

MTF equations are given as either a function of the spatial frequency r, or of the temporal frequency ν. MTFs that act on electrical signals are given, in most cases, as a function of the temporal frequency.

TRM calculates and displays all MTFs with the spatial frequency referring to object space. Additionally, $\mathrm{sinc}(x)$ is defined as

$$\mathrm{sinc}(x) = \frac{\sin(\pi x)}{\pi x}. \tag{4.10}$$

4.2.1 Optics MTF

The MTF of the front optics consists of two components:

- Diffraction limited optics and
- Aberrations.

Diffraction MTF

TRM calculates diffraction MTFs, $\mathrm{MTF}_{\mathrm{diff}}$, for the following pupil geometries:

- Circular aperture,
- Circular aperture with circular central obscuration (cco),
- Circular aperture, truncated in one or both dimensions,
- Circular aperture, truncated in one or both dimensions with cco, and
- Rectangular aperture with or without central obscuration.

Both symmetrical and asymmetrical truncations are treated. MTF calculations are based on the overlap area of two displaced pupil functions [18]. Equations for the calculation of diffraction MTFs are given in Appendix B.

If the *automatic spectral weighting* option is selected, the diffraction MTF is calculated for each wavelength in the spectral band. The weighted MTF is then obtained using Equations (4.1) and (4.2).

Aberrations via the aberration parameter

In this case, the diffraction limited MTF is modified as follows:

$$\mathrm{MTF}_{\mathrm{opt}}(r) = [\mathrm{MTF}_{\mathrm{diff}}(r)]^{q}\,, \tag{4.11}$$

where $\mathrm{MTF}_{\mathrm{diff}}(r)$ is the diffraction MTF and q is the aberration parameter. For information and output purposes, the aberration MTF is then calculated via

$$\mathrm{MTF}_{\mathrm{abr_q}} = \frac{\mathrm{MTF}_{\mathrm{opt}}(r)}{\mathrm{MTF}_{\mathrm{diff}}(r)}. \tag{4.12}$$

Aberration via Gaussian blur

Instead of the aberration parameter q, another approach can be used to describe image degradation caused by aberrations: Gaussian blur. In this case, the MTF of the front optics is given by

$$\mathrm{MTF}_{\mathrm{opt}}(r) = \mathrm{MTF}_{\mathrm{diff}}(r) \cdot \mathrm{MTF}_{\mathrm{blur}}(r) \tag{4.13}$$

with

$$\mathrm{MTF}_{\mathrm{blur}}(r) = \exp\left\{-2\left(\pi \frac{\sigma_{\mathrm{blur}}}{f_{\mathrm{opt}}} r\right)^{2}\right\}, \tag{4.14}$$

where σ_{blur} denotes the standard deviation of the Gaussian blur disc.

4.2.2 Image intensifier and coupling element MTFs

Image intensifier MTF

TRM4 employs the following model [19] to calculate the image intensifier MTF:

$$\mathrm{MTF}_{\mathrm{II}}(r) = \exp\left\{-\left(\frac{r}{r_{\mathrm{c}}}\right)^{k}\right\}. \tag{4.15}$$

In case the user specifies the resolution limit of the image intensifier plus a standard set of MTF values, the parameter k and the spatial frequency r_{c} are determined from a nonlinear least square fit of the specified data. If only the resolution limit, $r_{\mathrm{res_limit}}$, is given by the user, TRM4 uses $k = 1.20$ and

$$r_c = \frac{r_{\text{res_limit}}}{\left(\ln\left[\frac{1}{0.045}\right]\right)^{1/1.2}}. \tag{4.16}$$

Coupling element MTF

The MTF of the coupling element between the image intensifier and the solid-state detector array, MTF_{coup}, has to be specified by external data. If no external data is provided, TRM4 assumes that $\text{MTF}_{\text{coup}} = 1$. For fiber optic tapers a Gaussian approximation is suggested for the coupling element MTF [20].

Phosphor screen temporal MTF

In case the user specifies the phosphor decay time, TRM4 assumes an exponential temporal response behavior of the phosphor screen. The associated MTF is

$$\text{MTF}_{\text{ps}}(\nu) = \frac{1}{\sqrt{1 + (\nu/\nu_0)^2}}, \tag{4.17}$$

with frequency ν_0 determinted by the specified phosphor decay time t_{decay}:

$$\nu_0 = \frac{\ln(10)}{2\pi \cdot t_{\text{decay}}}. \tag{4.18}$$

For the phosphor screen preset data, a double exponential decay as suggested in Ref. [21] was assumed, which results in the following MTF

$$\text{MTF}(\nu) = \left| \frac{a}{1 + i2\pi\nu t_1} + \frac{(1-a)}{1 + i2\pi\nu t_2} \right|. \tag{4.19}$$

The used parameters for the different preset data phosphor types were derived from available decay curves and are given in Table 4.1.

Phosphor type	a	t_1 [s]	t_2 [s]
P43	0.237	$3.455 \cdot 10^{-5}$	$6.205 \cdot 10^{-4}$
P46	0.139	$5.917 \cdot 10^{-7}$	$6.875 \cdot 10^{-8}$
P47	$5.470 \cdot 10^{-2}$	$3.662 \cdot 10^{-7}$	$4.200 \cdot 10^{-8}$
P20	0.914	$1.639 \cdot 10^{-3}$	$4.552 \cdot 10^{-2}$

Table 4.1: Parameters used in the calculation of the phosphor screen temporal MTF for phosphor screen preset data.

4.2.3 Detector MTF

Detector MTF components may be divided into two groups:

- “Static” or “Spatial” MTFs and
- "Dynamic" or "Image motion induced“ MTFs.

A special role plays the uncooled detector temporal MTF, which is associated with the exponential temporal response behavior of these detectors. It only contributes to the spatial signal degradation in case of a linear stabilization error of the platform.

Static detector MTF

MTFs contributing to the static detector MTF are a result of the geometrical properties of the detector and the way the signal is read out. MTF degradation due to the motion of the image on the focal plane is treated in the next paragraph.

The possible contributions to the static detector MTF are given by

$$\begin{aligned}\mathrm{MTF}_{\mathrm{det_stat}} = \mathrm{MTF}_{\mathrm{det_geo}} \cdot \mathrm{MTF}_{\mathrm{bin}} \cdot \mathrm{MTF}_{\mathrm{pseudo}} \cdot \mathrm{MTF}_{\mathrm{CTE}} \\ \cdot \mathrm{MTF}_{\mathrm{crosstalk}}\,,\end{aligned} \tag{4.20}$$

which are explained in detail below.

The detector static MTF depends on the shape and the response profile of the detector element. For SPRITE detectors, the static MTF in horizontal direction depends on the diffusion length, too.

Detector geometry MTF

The static MTFs of the detector geometries available in TRM4 are:

- Rectangular shape

$$\mathrm{MTF}_{\mathrm{det_geo}_z}(r) = \mathrm{sinc}\left(\frac{a_{\mathrm{det}_z}}{f_{\mathrm{opt}_z}} \cdot r\right)\ ; \tag{4.21}$$

- Rhombic shape

$$\mathrm{MTF}_{\mathrm{det_geo}_z}(r) = \mathrm{sinc}^2\left(\frac{a_{\mathrm{det}_z}}{2 \cdot f_{\mathrm{opt}_z}} \cdot r\right)\ ; \tag{4.22}$$

- Circular shape

$$\mathrm{MTF}_{\mathrm{det_geo}_z}(r) = \frac{f_{\mathrm{opt}_z}}{\pi \cdot d_{\mathrm{det}} \cdot r} \cdot J_1\left(2\pi \frac{d_{\mathrm{det}}}{f_{\mathrm{opt}_z}} \cdot r\right), \tag{4.23}$$

  where d_{det} is the diameter of a detector element and J_1 is the Bessel function of the first kind of order one;

- Circular aperture with loophole

$$\begin{aligned}\mathrm{MTF}_{\mathrm{det_geo}_z}(r) = \frac{2 f_{\mathrm{opt}_z}}{\pi d_{\mathrm{det}} \left(1 - \frac{d_{\mathrm{hole}}^2}{d_{\mathrm{det}}^2}\right) r} \left[J_1\left(\pi \frac{d_{\mathrm{det}}}{f_{\mathrm{opt}_z}} \cdot r\right) - \frac{d_{\mathrm{hole}}}{d_{\mathrm{det}}}\right. \\ \left. \cdot J_1\left(\pi \frac{d_{\mathrm{hole}}}{f_{\mathrm{opt}_z}} \cdot r\right)\right]\end{aligned} \tag{4.24}$$

  where d_{hole} is the diameter of the hole in the detector element;

- SPRITE detector MTF in scan direction

$$\mathrm{MTF}_{\mathrm{det_geo_h}}(r) = \mathrm{sinc}\left(\frac{a_{\mathrm{SPRITE}}}{f_{\mathrm{opt_h}}} \cdot r\right) \cdot \left[1 + \left(2\pi \frac{a_{\mathrm{diff}}}{f_{\mathrm{opt_h}}} \cdot r\right)^2\right]^{-1} \tag{4.25}$$

  The first term on the right-hand side of this equation is the read-out zone MTF, and the second term is the diffusion MTF. In the direction perpendicular to the scan direction the MTF is given by Equation (4.21).

Pixel binning MTF

Pixel binning in both horizontal and vertical direction causes a multiplicative MTF modification given by

$$\mathrm{MTF}_{\mathrm{bin}_z} = \frac{\mathrm{sinc}\left(n_{\mathrm{bin}_z} \frac{a_{\mathrm{pitch}_z}}{f_{\mathrm{opt}_z}} \cdot r\right)}{\mathrm{sinc}\left(\frac{a_{\mathrm{pitch}_z}}{f_{\mathrm{opt}_z}} \cdot r\right)}. \tag{4.26}$$

Electronic pseudo-interlace MTF

If electronic pseudo-interlace is used, the vertical detector MTF needs to be multiplied by the factor

$$\mathrm{MTF}_{\mathrm{pseudo_v}} = \cos\left(\pi \frac{a_{\mathrm{pitch_v}}}{f_{\mathrm{opt_v}}} \cdot r\right). \tag{4.27}$$

Charge transfer efficiency MTF

Finite change transfer efficiency leads to an additional multiplicative factor of

$$\mathrm{MTF}_{\mathrm{CTE}_z} = \exp\left\{-N_{\mathrm{transfer}_z}(1 - \mathrm{CTE}_z)\left[1 - \cos\left(\frac{2\pi r}{r_{\mathrm{stage}_z}}\right)\right]\right\}, \tag{4.28}$$

where N_{transfer_z} is the number of transfers between neighboring detector elements, CTE_z is the charge transfer efficiency between neighboring detector elements, and $r_{\mathrm{stage}_z} = f_{\mathrm{opt}_z}/a_{\mathrm{pitch}_z}$.

Detector crosstalk MTF

Diffusion of carriers between detector elements leads to spatial signal degradation. This can be described by

$$\mathrm{MTF}_{\mathrm{crosstalk}_z}(r) = \frac{1}{1 + \left(2\pi \mathrm{D}_{\mathrm{diff}_z} \cdot r\right)^2}, \tag{4.29}$$

with the diffusion constant, $\mathrm{D}_{\mathrm{diff}_z}$, given by

$$D_{\mathrm{diff}_z} = \frac{1}{2\pi r_{\mathrm{N}}} \sqrt{\frac{\mathrm{MTF}_{\mathrm{det_geo}_z}(r_{\mathrm{N}})}{\mathrm{MTF}_{\mathrm{det_spec}_z}(r_{\mathrm{N}})} - 1} \tag{4.30}$$

where $\mathrm{MTF}_{\mathrm{det_spec}_z}(r_\mathrm{N})$ is the MTF value specified by the user and the Nyquist frequency in *z*-direction defined as

$$r_\mathrm{N} = \frac{f_{\mathrm{opt}_z}}{2a_{\mathrm{pitch}_z}}. \tag{4.31}$$

Dynamic detector MTF

Image formation sometimes involves intentional moving of the image on the detector plane. This scanning process may degrade the device MTF. The corresponding MTF is, in general, a product of several contributions:

$$\mathrm{MTF}_\mathrm{scan} = \mathrm{MTF}_\mathrm{det_temp} \cdot \mathrm{MTF}_\mathrm{det_TDI} \cdot \mathrm{MTF}_\mathrm{det_int}\,. \tag{4.32}$$

Detector temporal MTF

TRM4 uses a first order Butterworth filter to model the carrier generation-recombination roll-off. It introduces the low-pass filter

$$\mathrm{MTF}_{\mathrm{det_temp_h}}(r) = \left[1 + \left(\frac{\nu_\mathrm{scan} \cdot r}{\nu_\mathrm{det_temp}}\right)^2\right]^{-1/2} \tag{4.33}$$

with the -3dB frequency $\nu_{\mathrm{det_temp}}$. Generally, this MTF component is important only in serial or serial/parallel imagers and is applied in the scanning direction.

TDI mismatch

The TDI mismatch MTF is given by

$$\mathrm{MTF}_{\mathrm{det_TDI_h}}(r) = \frac{\mathrm{sinc}\left(N_\mathrm{TDI} \cdot \frac{a_\mathrm{TDI}}{f_{\mathrm{opt_h}}} \cdot r\right)}{\mathrm{sinc}\left(\frac{a_\mathrm{TDI}}{f_{\mathrm{opt_h}}} \cdot r\right)}. \tag{4.34}$$

MTF due to integration while scanning - scanning imagers with integrating FPAs

In scanning imagers, the signal is collected while the image is moved over the focal plane. This "integrate while scan" mechanism introduces the following MTF component in the scanned direction:

$$\mathrm{MTF}_\mathrm{det_int}(r) = \mathrm{sinc}(t_\mathrm{int} \cdot \nu_\mathrm{scan} \cdot r). \tag{4.35}$$

MTF due to integration while scanning – staring imagers with micro-scan

In many respects, the micro-scan mechanism in staring imagers is analogous to the interlace mechanism in scanned imagers. Micro-scan does not introduce an MTF-degradation if there is no image motion while the detector signal is integrated.

This paragraph gives the MTF for cases where the signal is integrated while the image is in motion on the focal plane. Note that only the motion of the image due to the micro-scan is considered by the MTF.

- *Rectangular micro-scan (with integration during movement) MTF*

$$\mathrm{MTF}_{\mathrm{det_int}_z}(r) = \mathrm{sinc}\left(\frac{\eta_{\mathrm{cyc}} \cdot a_{\mathrm{pitch}_z}}{f_{\mathrm{opt}_z} \cdot N_{\mu\mathrm{step}_z}} \cdot r\right), \tag{4.36}$$

  where η_{cyc} is the integration duty cycle. It is assumed that the motion is continuous (i.e. no stop for change of direction) and linear.

- *Diagonal micro-scan (with integration during movement) MTF*

$$\mathrm{MTF}_{\mathrm{det_int}_z}(r) = \mathrm{sinc}\left(\frac{\eta_{\mathrm{cyc}} \cdot a_{\mathrm{pitch}_z}}{2 f_{\mathrm{opt}_z}} \cdot r\right), \tag{4.37}$$

  where η_{cyc} is the signal integration duty cycle. As for the rectangular micro-scan MTF, it is assumed that the motion is continuous (i.e. no stop for change of direction) and linear.

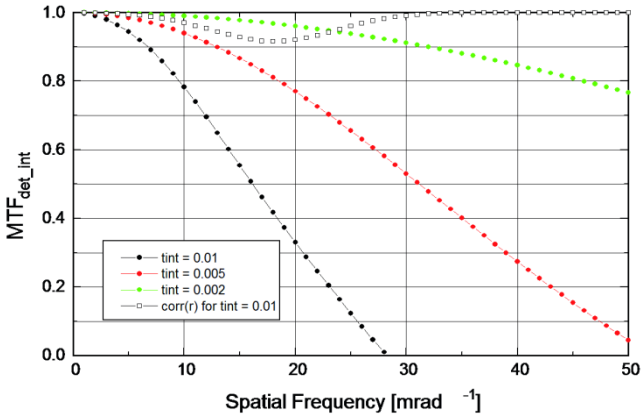


Figure 4.1: Example of the MTF caused by integration during movement for a circular scan (the integration time $t_{\mathrm{int}} = 0.01\ sec$ corresponds to a duty cycle $\eta_{\mathrm{cyc}} = 1$).

- *Circular (elliptical) micro-scan:* The motion pattern assumed in TRM4 is depicted in Figure 2.8. The corresponding MTF is given by

$$\mathrm{MTF}_{\mathrm{det_int}_z}(r) = \mathrm{sinc}\left[\frac{a_{\mathrm{pitch}_z}}{2 f_{\mathrm{opt}_z}} \cdot \sin\left(\pi \cdot t_{\mathrm{int}} \cdot F_{\mathrm{r,imager}} \cdot r\right)\right] \cdot \mathrm{Corr}(r)\,, \tag{4.38}$$

  where

$$\mathrm{Corr}(r) = 1 - 0.084\left(4 t_{\mathrm{int}} F_{\mathrm{r,imager}}\right)^2 \cdot \exp\left\{-12.5\left(0.64 - \sqrt{2} t_{\mathrm{int}} F_{\mathrm{r,imager}} \frac{a_{\mathrm{pitch}_z}}{f_{\mathrm{opt}_z}} r\right)^2\right\}. \tag{4.39}$$

  Figure 4.1 illustrates $\mathrm{MTF}_{\mathrm{det_int}}$ for circular scan. The imager is assumed to have a frame rate of 25 Hz and a spatial pitch of 0.1 mrad. The maximum integration time possible is 0.01 seconds ($\eta_{\mathrm{cyc}} = 1$). MTFs are presented for three different integration times (0.01 sec, 0.005 sec, and 0.002 sec, respectively). The correction factor $\mathrm{Corr}(r)$ for $t_{\mathrm{int}} = 0.01\ sec$ is shown, too.

Uncooled detector temporal MTF

TRM4 assumes that the change of the output signal of the uncooled detector due to a change in the incoming radiation follows an exponential behavior with thermal time constant t_{th}. The associated MTF is given by

$$\mathrm{MTF}_{\mathrm{uc}}(\nu) = \frac{1}{\sqrt{1 + (\nu/\nu_0)^2}} \tag{4.40}$$

with frequency $\nu_0 = 1/(2\pi \cdot t_{\mathrm{th}})$.

4.2.4 Analog signal processing MTF

Analog signal processing is performed in the scanned direction only. The analog processing MTF may be the product of three different contributions:

$$\mathrm{MTF}_{\mathrm{ana}} = \mathrm{MTF}_{\mathrm{ana_hp}} \cdot \mathrm{MTF}_{\mathrm{ana_lp}} \cdot \mathrm{MTF}_{\mathrm{ana_boost}} \cdot \tag{4.41}$$

High-pass filter (AC-coupling)

In AC-coupled imagers, the low frequency response is governed by a high-pass filter. Usually, it suppresses the low frequency noise component only (especially the 1/f noise) and has no significant effect on the signal. The high-pass filter is assumed to be a Butterworth filter of order m_{hp}. Its MTF is given by:

$$\mathrm{MTF}_{\mathrm{ana_hp}}(r) = \frac{\left(\frac{\nu_{\mathrm{scan}} \cdot r}{\nu_{\mathrm{hp}}}\right)^{m_{\mathrm{hp}}}}{\left[1 + \left(\frac{\nu_{\mathrm{scan}} \cdot r}{\nu_{\mathrm{hp}}}\right)^{2m_{\mathrm{hp}}}\right]^{\frac{1}{2}}}. \tag{4.42}$$

Low-pass filter (high frequency response)

The low-pass filter is assumed to be a Butterworth filter of order m_{lp}. Its MTF is given by:

$$\mathrm{MTF}_{\mathrm{ana_lp}}(r) = \left[1 + \left(\frac{\nu_{\mathrm{scan}} \cdot r}{\nu_{\mathrm{lp}}}\right)^{2m_{\mathrm{lp}}}\right]^{-\frac{1}{2}}. \tag{4.43}$$

High-frequency boost

High-frequency boost is used in some imagers in order to compensate for signal loss at high frequencies, but it increases the noise, too. The MTF of the high frequency boost is modelled by

$$\mathrm{MTF}_{\mathrm{ana_boost}} = 1 + \frac{K-1}{2}\left[1 - \cos\left(\frac{\pi \cdot \nu_{\mathrm{scan}} \cdot r}{\nu_{\mathrm{boost}}}\right)\right], \tag{4.44}$$

where K is the gain factor of the high frequency boost.

4.2.5 Analog-Digital converter MTF

Analog-Digital converters are applied in the horizontal direction only. It is assumed that the filter is given by a Butterworth filter of order m_{lp}. Its MTF is given by

$$\mathrm{MTF}_{\mathrm{adc}}(r) = \left[1 + \left(\frac{r}{\eta_{\mathrm{adc}} \cdot r_{\mathrm{s_h}}}\right)^{2m_{\mathrm{lp}}}\right]^{-\frac{1}{2}} \tag{4.45}$$

with the normalized cut-off frequency η_{adc} (normalization with respect to the horizontal sampling frequency of the device).

4.2.6 Digital filter MTF

TRM4 models digital filters as linear phase, symmetrical FIR filters (odd or even). The MTFs of these filters are:

Symmetrical odd filters

$$\mathrm{MTF}_{\mathrm{dig}_z}(r) = \frac{1}{\sum_{n=0}^{4} c_{\mathrm{dig},n_z}} \cdot \sum_{n=0}^{4} c_{\mathrm{dig},n_z} \cdot \cos\left(2\pi n \alpha_{\mathrm{s}_z} \cdot r\right), \tag{4.46}$$

Symmetrical even filters

$$\mathrm{MTF}_{\mathrm{dig}_z}(r) = \frac{1}{\sum_{n=1}^{4} c_{\mathrm{dig},n_z}} \cdot \sum_{n=1}^{4} c_{\mathrm{dig},n_z} \cdot \cos\left(2\pi \left[n - \frac{1}{2}\right] \alpha_{\mathrm{s}_z} \cdot r\right), \tag{4.47}$$

where α_{s_z} is the angular sampling distance of the device in z-direction and the c_{dig,n_z} are the specified filter coefficients.

4.2.7 Signal reconstruction MTF

This MTF describes the reconstruction process of the analog signal from the digital. The signal reconstruction function is the product of the MTFs of the Digital-Analog converter, zoom, and line and column replication.

For the horizontal direction the MTF is given by

$$\mathrm{MTF}_{\mathrm{rec_h}} = \mathrm{MTF}_{\mathrm{dac}} \cdot \mathrm{MTF}_{\mathrm{zoom_h}} \cdot \mathrm{MTF}_{\mathrm{diag_h}} \cdot \mathrm{MTF}_{\mathrm{crep}} \tag{4.48}$$

and for the vertical direction by

$$\mathrm{MTF}_{\mathrm{rec_v}} = \mathrm{MTF}_{\mathrm{zoom_v}} \cdot \mathrm{MTF}_{\mathrm{diag_v}} \cdot \mathrm{MTF}_{\mathrm{lrep}} \,. \tag{4.49}$$

MTF of the Digital-Analog converter

The digital signal may have been zoomed before read-out by the Digital-Analog converter (DAC). The DAC acts on signals with pixel distance $\alpha_{s_h}/(M_{zoom} \cdot M_{crep})$, where α_{s_h} is the angular sampling distance of the imager in the horizontal direction.

TRM has two options for the reconstruction function of the DAC: a rectangular reconstruction function (sample and hold) and a triangular one (linear interpolation). Their MTFs are given by:

Sample and Hold

$$MTF_{dac}(r) = \mathrm{sinc}\left(\frac{\alpha_{s_h}}{M_{zoom} \cdot M_{crep}} \cdot r\right), \tag{4.50}$$

Linear interpolation

$$MTF_{dac}(r) = \left[\mathrm{sinc}\left(\frac{\alpha_{s_h}}{M_{zoom} \cdot M_{crep}} \cdot r\right)\right]^2. \tag{4.51}$$

Electronic zoom

Two types of zoom can be specified in TRM: Zoom by pixel replication or zoom by linear interpolation. For pixel replication, the additional pixels are exact replicas of the neighboring pixels. For linear interpolation, the required additional pixels are obtained by linear interpolations of the data in the neighboring pixels. The MTFs of these zoom techniques are given by:

Pixel replication

$$MTF_{zoom_z}(r) = \frac{\mathrm{sinc}(\alpha_{s_z} \cdot r)}{\mathrm{sinc}\left(\frac{\alpha_{s_z} \cdot r}{M_{zoom}}\right)}, \tag{4.52}$$

Linear interpolation

$$MTF_{zoom_z}(r) = \left[\frac{\mathrm{sinc}(\alpha_{s_z} \cdot r)}{\mathrm{sinc}\left(\frac{\alpha_{s_z} \cdot r}{M_{zoom}}\right)}\right]^2. \tag{4.53}$$

Artificial zoom associated with diagonal micro-scan

An artificial “zoom” MTF is assumed to be applied to all diagonal micro-scanned devices. This MTF factor represents the “missing” data points in the horizontal and vertical directions. It imitates filling the missing data points with linear interpolation of their four nearest neighbors (see Figure 4.2).

The associated MTF is given by

$$MTF_{diag_z}(r) = \left[\frac{\mathrm{sinc}(\alpha_{s_z} \cdot r)}{\mathrm{sinc}\left(\frac{\alpha_{s_z} \cdot r}{2}\right)}\right]^2 = \cos^2\left(\pi \frac{\alpha_{s_z} \cdot r}{2}\right). \tag{4.54}$$

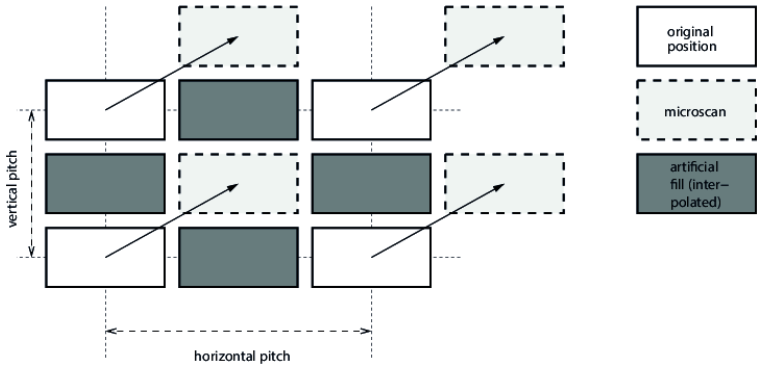


Figure 4.2: “Artificial zoom” for diagonal micro-scan.

Line and column replication

The line replication process acts only in vertical, the column replication process only in horizontal direction. Two types of line/column replication processes can be modelled in TRM: Pixel replication and linear interpolation. The respective MTFs are given by:

Pixel replication

$$\mathrm{MTF}_{\mathrm{crep|lrep}}(r) = \frac{\mathrm{sinc}\left(\alpha_{s_{h|v}} \cdot r\right)}{\mathrm{sinc}\left(\frac{\alpha_{s_{h|v}} \cdot r}{M_{\mathrm{crep|lrep}}}\right)}, \tag{4.55}$$

Linear interpolation

$$\mathrm{MTF}_{\mathrm{crep|lrep}}(r) = \left[\frac{\mathrm{sinc}\left(\alpha_{s_{h|v}} \cdot r\right)}{\mathrm{sinc}\left(\frac{\alpha_{s_{h|v}} \cdot r}{M_{\mathrm{crep|lrep}}}\right)}\right]^2. \tag{4.56}$$

Note: When both zoom and line replication are applied, the equations are somewhat different. These cases are not presented here.

4.2.8 Video signal processing MTF

The video signal is an analog signal obtained after signal reconstruction by a DAC. The video processing MTF may be the product of two different contributions:

$$\mathrm{MTF}_{\mathrm{vid}} = \mathrm{MTF}_{\mathrm{vid_lp}} \cdot \mathrm{MTF}_{\mathrm{vid_boost}}. \tag{4.57}$$

Low-pass filter

A Butterworth filter of order m_{lp} is assumed. Its MTF is given by

$$\mathrm{MTF}_{\mathrm{vid_lp}}(r) = \left[1 + \left(\frac{\mathrm{scf} \cdot r}{\nu_{\mathrm{vid_lp}}}\right)^{2m_{\mathrm{lp}}}\right]^{-\frac{1}{2}} \tag{4.58}$$

with the scaling factor scf.

High-frequency boost

The MTF of the high frequency boost in the video signal is given by

$$\mathrm{MTF}_{\mathrm{vid_boost}} = 1 + \frac{K-1}{2}\left[1 - \cos\left(\frac{\pi \cdot \mathrm{scf} \cdot r}{\nu_{\mathrm{vid_boost}}}\right)\right] \tag{4.59}$$

with the boost gain factor K.

4.2.9 Display MTF

TRM4 supports three display types: CRT, LED, and flat panel displays.

CRT display

TRM4 assumes a Gaussian spot profile on the display. The MTF is given by

$$\mathrm{MTF}_{\mathrm{dis}_z} = \exp\left[-2\pi^2 r^2 \left(\frac{\sigma_{\mathrm{CRT}_z}}{H_{\mathrm{picture}}} \cdot \frac{n_{\mathrm{active}} \cdot \mathrm{FOV_v}}{N_{\mathrm{FOV_v}}}\right)^2\right], \tag{4.60}$$

where $N_{\mathrm{FOV_v}}$ is the number of video lines covering the FOV in the vertical direction.

If the resolution is specified via the TV limiting resolution, the equivalent spot is assumed to be a circular Gaussian with its size given by

$$\sigma_{\mathrm{CRT}} = \frac{H_{\mathrm{picture}}}{1.18 \cdot N_{\mathrm{disp_res}}}. \tag{4.61}$$

LED display

The MTF of an LED display is assumed to be given by

$$\mathrm{MTF}_{\mathrm{dis}_z} = \mathrm{sinc}\left(\frac{a_{\mathrm{LED}_z} \cdot a_{\mathrm{pitch_v}}}{\mathrm{dis}_{\mathrm{pitch_v}} \cdot f_{\mathrm{opt}_z}} \cdot r\right). \tag{4.62}$$

Note: Do not mistake the LED here for a modern Flat panel LED display. The LED display specified herein is a simple device with few LEDs corresponding to the detector channels where the signals from the detectors were fed directly to the LEDs and a scanning mechanism was used to reproduce the image and display it to the observer. Modern Flat panel LEDs are represented by the *Flat panel* option.

Flat panel display

In TRM4, it is assumed that flat panel displays are monochromatic black and white with rectangular elements arranged in rows and columns. Another assumption is that the fill factor is unity, meaning that the element size is equal to the pitch between the elements. For such a display, the horizontal MTF is given by

$$\mathrm{MTF}_{\mathrm{dis_h}}(r) = \mathrm{sinc}\left(\frac{r}{r_{\mathrm{s_h}} \cdot M_{\mathrm{zoom}} \cdot M_{\mathrm{crep}}}\right) \tag{4.63}$$

and the vertical by

$$\mathrm{MTF}_{\mathrm{dis_v}}(r) = \mathrm{sinc}\left(\frac{r}{r_{\mathrm{s_v}} \cdot M_{\mathrm{zoom}} \cdot M_{\mathrm{lrep}}}\right). \tag{4.64}$$

Note: In the AMOP calculation, the footprint of flat panel display pixels is simulated in real space and not by means of the display MTF in spatial frequency space.

4.2.10 Stabilization error MTF

The stabilization error MTF, $\mathrm{MTF}_{\mathrm{stab}}$, is the product of the MTFs for different platform motion mechanisms:

$$\mathrm{MTF}_{\mathrm{stab}} = \mathrm{MTF}_{\mathrm{stab_lin}} \cdot \mathrm{MTF}_{\mathrm{stab_ran}} \cdot \mathrm{MTF}_{\mathrm{stab_sin}}\,. \tag{4.65}$$

Linear image motion

The overall linear motion MTF is given by

$$\mathrm{MTF}_{\mathrm{stab_lin}_z}(r) = \mathrm{MTF}_{\mathrm{stab_lin1}_z}(r) \cdot \mathrm{MTF}_{\mathrm{stab_lin2}_z}(r) \cdot \mathrm{MTF}_{\mathrm{stab_lin3}_z}(r). \tag{4.66}$$

The first MTF term in Equation (4.66) describes the image smear due to linear motion of the platform. This MTF is calculated for staring or micro-scanned imagers only (for other types of devices $\mathrm{MTF}_{\mathrm{stab_lin1}_z}(r) = 1$ for all r). It is given by

$$\mathrm{MTF}_{\mathrm{stab_lin1}_z}(r) = \mathrm{sinc}\left(t_{\mathrm{int}} \cdot \omega_{\mathrm{lin}_z} \cdot r\right), \tag{4.67}$$

where ω_{lin_z} is the angular velocity of the platform. An exception are ICCD/ICMOS and uncooled detectors, where the temporal response behavior of the imager result in different expressions for the MTF. For ICCD/ICMOS it reads

$$\mathrm{MTF}_{\mathrm{stab_lin1}_z}(r) = \mathrm{sinc}\left(t_{\mathrm{int}} \cdot \omega_{\mathrm{lin}_z} \cdot r\right) \cdot \mathrm{MTF}_{\mathrm{ps}}\left(\nu = \omega_{\mathrm{lin}_z} \cdot r\right) \tag{4.68}$$

and for the uncooled detector type

$$\mathrm{MTF}_{\mathrm{stab_lin1}_z}(r) = \mathrm{MTF}_{\mathrm{uc}}\left(\nu = \omega_{\mathrm{lin}_z} \cdot r\right). \tag{4.69}$$

The second MTF term in Equation (4.66) describes the image smear between consecutive interlaced fields and/or cases where frame integration is specified. This MTF depends on the type of imager and is given by:

Scanning imagers

$$\mathrm{MTF}_{\mathrm{stab_lin2}_z}(r) = \frac{\mathrm{sinc}\left(\frac{\omega_{\mathrm{lin}_z} \cdot N_{\mathrm{frames}} \cdot r}{F_{r,\mathrm{imager}}}\right)}{\mathrm{sinc}\left(\frac{\omega_{\mathrm{lin}_z} \cdot r}{F_{\mathrm{r,imager}} \cdot n_{\mathrm{inter}}}\right)}, \tag{4.70}$$

Staring imagers

$$\mathrm{MTF}_{\mathrm{stab_lin2}_z}(r) = \frac{\mathrm{sinc}\left(\frac{\omega_{\mathrm{lin}_z} \cdot N_{\mathrm{frames}} \cdot r}{F_{r,\mathrm{imager}}}\right)}{\mathrm{sinc}\left(\frac{\omega_{\mathrm{lin}_z} \cdot r}{F_{\mathrm{r,imager}} \cdot N_{\mu\mathrm{step_h}} \cdot N_{\mu\mathrm{step_v}}}\right)}. \tag{4.71}$$

In cases where IIR frame integration is specified, the MTFs given above are multiplied by the following IIR frame integration MTF:

$$\mathrm{MTF}_{\mathrm{stab_lin3}_z}(r) = \frac{1-\alpha}{\left|1-\alpha \cdot \exp\left\{\frac{-i \cdot 2\pi \cdot N_{\mathrm{frames}} \cdot \omega_{\mathrm{lin}_z} \cdot r}{F_{\mathrm{r,imager}}}\right\}\right|}. \tag{4.72}$$

Random image motion

The random vibration of the imager is described by a Gaussian MTF:

$$\mathrm{MTF}_{\mathrm{stab_ran}_z} = \exp\left[-2\pi^2 \sigma^2_{\mathrm{stab}_z} \cdot r^2\right], \tag{4.73}$$

with the standard deviation of the random movements σ_{stab_z}.

Sinusoidal image motion MTF

Sinusoidal image motion is usually due to high frequency resonances. The corresponding MTF is described by a first kind zero order Bessel function:

$$\mathrm{MTF}_{\mathrm{stab_sin}_z} = J_0\left(2\pi s_{\mathrm{amp}_z} \cdot r\right), \tag{4.74}$$

with the amplitude of the sinusoidal movement s_{amp_z}.

4.2.11 Additional imager MTFs

TRM has the option to add additional filters to both the pre- and postfilter chain. Additional MTFs can be specified via external data and/or as one of the following filters:

Sinc filter

$$\mathrm{MTF}_{\mathrm{add}_z}(r) = \mathrm{sinc}\left(\frac{r}{r_{\mathrm{zero}_z}}\right), \tag{4.75}$$

with the first zero of the MTF given by r_{zero_z}.

Sinc² filter

$$\mathrm{MTF}_{\mathrm{add}_z}(r) = \left[\mathrm{sinc}\left(\frac{r}{r_{\mathrm{zero}_z}}\right)\right]^2. \tag{4.76}$$

Gaussian filter

$$\mathrm{MTF}_{\mathrm{add}_z}(r) = \exp\left[-\frac{r^2}{2 \cdot \sigma^2_{\mathrm{gauss}_z}}\right], \tag{4.77}$$

with the standard deviation of the Gaussian distribution $\sigma_{\mathrm{gauss}_z}$.

Rectangular filter

This is an ideal flat low-pass filter, which cuts off all frequencies beyond r_{zero}. Its MTF is given by

$$\mathrm{MTF}_{\mathrm{add}_z}(r) = \begin{cases} 1 & \text{for } r \le r_{\mathrm{zero}_z}, \\ 0 & \text{for } r > r_{\mathrm{zero}_z}. \end{cases} \tag{4.78}$$

4.2.12 Observer (eye-brain) MTF

In TRM4, the eye MTF is calculated by using a simplified Kornfeld-Lawson eye model [22]. The eye MTF is given by

$$\mathrm{MTF}_{\mathrm{eye}}(r) = \exp\left[-\frac{\Gamma \cdot r}{M_{\mathrm{sys}}}\right], \tag{4.79}$$

where Γ is the width of the point spread function of the eye. The dependence of Γ on the ambient luminance is presented in Table 4.2.

Light level [$\log_{10}(\mathrm{fL})$]	Light level [cd/m²]	Γ
3	$3.426 \cdot 10^3$	0.81333
2	$3.426 \cdot 10^2$	0.9598
1	$3.426 \cdot 10^1$	1.0980
0	$3.426 \cdot 10^0$	1.4650
−1	$3.426 \cdot 10^{-1}$	1.8300
−2	$3.426 \cdot 10^{-2}$	2.2773

Table 4.2: Width of the eye point spread function, Γ, for various display light levels.

4.2.13 Turbulence MTF

The turbulence MTF differs from most of the other transfer functions described before in the following aspects:

- It is not a property of the electro-optical device or the carrying platform but caused externally in the atmosphere, and

- It depends on the current observation distance from imager to target.

This means that for every step in the range calculation, a new MTF has to be calculated.

Generally, the turbulence MTF at a target distance of R is given by

$$\mathrm{MTF}_{\mathrm{turb}}(r,R,\lambda) = \exp\left\{-\frac{1}{2}\phi(r,R,\lambda)\right\}, \tag{4.80}$$

where $\phi(r,R,\lambda)$ is the turbulence phase structure function, which depends on the turbulence model and exposure time (see below).

The turbulence MTF displayed in the output MTF dialog is spectrally weighted using Equations (4.1) and (4.2).

Long-exposure phase structure function

In the context of turbulence, *long-exposure* means that the time-scale of changes in the configuration of turbulence is much shorter than the integration time of the detector. In general, the time constant of turbulence depends on current target distance, prevailing wind speed, turbulence structure, etc. Typical values of the turbulence time constant are few tens of milliseconds or shorter, especially for horizontal line-of-sight applications. Therefore, most imaging systems operate on a frame-by-frame basis in the short-exposure regime. However, if the observer eye integrates over many frames during the eye integration time of about 100 to 200 ms, the observer will experience the long-exposure condition [16].

In the isotropic Kolmogorov type turbulence model [23], the long-exposure phase structure function is described by

$$\phi_{\mathrm{LE}}(r,R,\lambda) = 6.88 \cdot \left(\frac{\lambda \cdot |r|}{r_0}\right)^{5/3}, \tag{4.81}$$

where r_0 is the Fried parameter, describing the equivalent aperture diameter for which imaging will not be diffraction limited anymore due to turbulence effects.

The Fried parameter for finite target-observer distance, i.e. for spherical waves, can be determined from

$$r_0 = 0.185 \cdot \left[\frac{1}{\lambda^2}\int_0^R \left(\frac{R-z}{R}\right)^{5/3} \cdot C_n^2\big(h(z)\big)\,\mathrm{d}z\right]^{-3/5}, \tag{4.82}$$

where R is the current distance of target to imager, h is the altitude above ground level, and $C_n^2(h)$ is the altitude profile of the refractive turbulence structure constant.

Note: For constant C_n^2, the Fried parameter equation reduces to $r_0 = 0.185 \cdot (3/8 \cdot C_n^2 \cdot R/\lambda^2)^{-3/5}$.

Short-exposure phase structure function

We recommend to use the short-exposure phase structure function in case that the performance assessment is based on single frames (option *Still* in the *Modify Individual Parameters*

dialog) or when observational phenomena, like observer memory and so-called “lucky shots” [23], are expected to be important.

Effectively, the short-exposure structure function is the same as the long-exposure without the tip-tilt effects (i.e. the gross image motion in the image) and it is given by

$$\phi_{\mathrm{SE}}(r,R,\lambda) = \phi_{\mathrm{LE}}(r,R,\lambda)\left[1-\frac{1}{2}\left(\frac{\lambda\cdot|r|}{D_{\mathrm{opt}}}\right)^{1/3}\right], \tag{4.83}$$

where D_{opt} is the effective aperture diameter of the imager.

Simplified height dependence

TRM4 has a built-in simplified height dependence model of the turbulence profile, which can be used either for automatic scaling of measured structure constant results or for slant paths which do not exceed much more than 100 m in altitude above ground level. This height dependence is defined by [24]

$$C_n^2(h) = C_{n,\mathrm{ref}}^2 \cdot \left(\frac{h}{h_{\mathrm{ref}}}\right)^{-\delta}, \tag{4.84}$$

where h_{ref} is the reference height above ground level and δ is a constant that equals to $2/3$ and $4/3$ for nighttime and daytime scenarios, respectively.

4.3 External MTF data

Measured or otherwise obtained MTF data can be specified in TRM, see [4]. When specified, external MTF data replaces internally calculated MTFs.

External MTF data can be directly specified in one or two directions for the following components:

Optics:	$\mathrm{MTF}_{\mathrm{opt}}$
Image intensifier:	$\mathrm{MTF}_{\mathrm{II}}$
Coupling element:	$\mathrm{MTF}_{\mathrm{coup}}$
Detector geometry:	$\mathrm{MTF}_{\mathrm{det_geo}}$
Analog filter:	$\mathrm{MTF}_{\mathrm{ana}}$
Analog-Digital converter:	$\mathrm{MTF}_{\mathrm{adc}}$
Video processor:	$\mathrm{MTF}_{\mathrm{vid}}$
Display:	$\mathrm{MTF}_{\mathrm{dis}}$
Stabilization error:	$\mathrm{MTF}_{\mathrm{stab}}$
Additional prefilter:	$\mathrm{MTF}_{\mathrm{add_pre}}$
Additional postfilter:	$\mathrm{MTF}_{\mathrm{add_post}}$
Eye:	$\mathrm{MTF}_{\mathrm{eye}}$
Device prefilter:	$\mathrm{MTF}_{\mathrm{pre_device}}$

A. Scaling Factors

A.1 Introduction and basic system configuration

Most parameters needed to characterize system components (examples are shown in Figure A.1 and Figure A.2) are specified in units that refer to the component. For instance, detector dimensions are specified in μm, video frequencies in Hz. To calculate imager performance, all dimensions and frequencies are scaled to object space. Spatial dimensions are then given in mrad and spatial frequencies in mrad^{-1}.

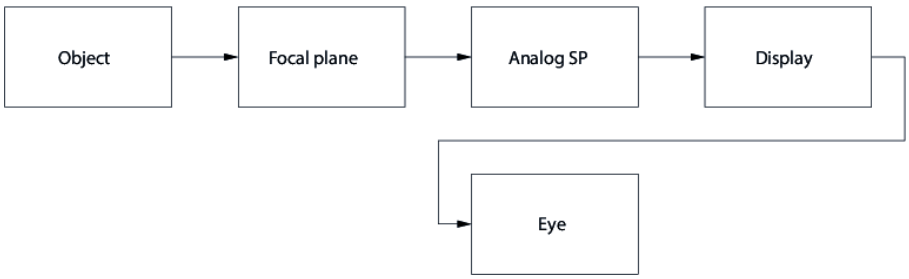


Figure A.1: Major components of a 1st generation IR imager without horizontal sampling.

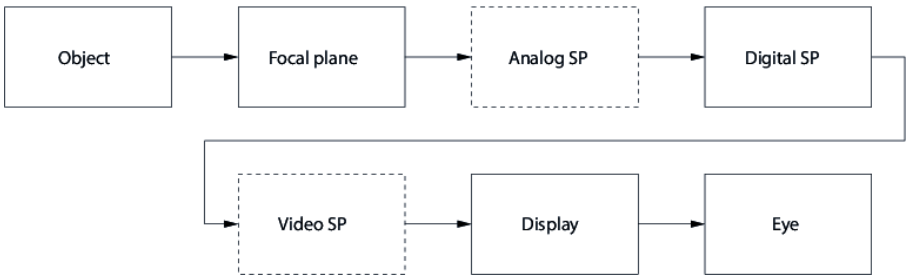


Figure A.2: Major components of imagers sampled in both directions. Analog signal processing applies to 1st generation imagers only. Video signal processing is applied only in case of CRT displays.

A.2 Scaling factors

The principle reference plane is object space. All calculated data is given in mrad and mrad^{-1}.

Focal plane units

$$\gamma_z\,[\mathrm{mrad}] = \frac{x\,[\mu\mathrm{m}]}{f_{\mathrm{opt}_z}\,[\mathrm{mm}]}, \tag{A.1}$$

where x denotes a dimension in the focal plane.

Analog signal processing units

$$\gamma_{\mathrm{h}}\,[\mathrm{mrad}] = v_{\mathrm{scan}}\,[\mathrm{mrad/sec}] \cdot t\,[\mathrm{sec}]\,, \tag{A.2}$$

where it is assumed that the signals are not multiplexed.

Digital memory units

$$\gamma\,[\mathrm{mrad}] = \frac{i \cdot \mathrm{FOV}_z\,[\mathrm{mrad}]}{N_{s_z}}, \tag{A.3}$$

where i designates any pixel numbering in the memory.

Video signal units

$$\gamma\,[\mathrm{mrad}] = t\,[\mu\mathrm{sec}] \cdot \mathrm{scf}\,[\mathrm{MHz\ mrad}]\,. \tag{A.4}$$

In case of standard video signals (CCIR/RS-170) and if the video line efficiency is specified, the scaling factor is given by

$$\mathrm{scf}\,[\mathrm{MHz\ mrad}] = \frac{\mathrm{FOV_h}\,[\mathrm{mrad}]}{\eta_{\mathrm{video}}\ \cdot\ t_{\mathrm{line_active}}\,[\mu\mathrm{sec}]}\,. \tag{A.5}$$

Display units

CRT displays:

$$\gamma\,[\mathrm{mrad}] = \frac{\mathrm{FOV_v}\,[\mathrm{mrad}]\ \cdot\ n_{\mathrm{active}}}{H_{\mathrm{picture}}\,[\mathrm{mm}]\ \cdot\ N_{\mathrm{FOV_v}}} \cdot x\,[\mathrm{mm}]\,, \tag{A.6}$$

where x designates a dimension in the display.

LED/Matrix displays:

$$\gamma\,[\mathrm{mrad}] = \frac{\mathrm{FOV_v}\,[\mathrm{mrad}]}{\mathrm{dis}_{\mathrm{pitch}}\,[\mathrm{mm}]\ \cdot\ N_{\mathrm{FOV_v}}} \cdot x\,[\mathrm{mm}]\,. \tag{A.7}$$

Eye space units

The conversion from object space to eye space dimensions is defined by the equation

$$\gamma_{\mathrm{eye}}\,[\mathrm{mrad}] = M_{\mathrm{sys}} \cdot \gamma_{\mathrm{obj}}\,[\mathrm{mrad}]\,, \tag{A.8}$$

where M_{sys} is the overall angular magnification of the system. For the different setups considered in TRM4 it is given by:

TV monitor, no eyepiece

$$M_{\mathrm{sys}} = \frac{H_{\mathrm{picture}}\,[\mathrm{mm}]\ \cdot\ N_{\mathrm{FOV_v}}}{\mathrm{FOV_v}\,[\mathrm{mrad}]\ \cdot\ D_{\mathrm{obs}}\,[\mathrm{m}]\ \cdot\ n_{\mathrm{active}}}, \tag{A.9}$$

Flat panel display, no eyepiece

$$M_{\mathrm{sys}} = \frac{\mathrm{dis}_{\mathrm{pitch}}\,[\mathrm{mm}] \cdot N_{\mathrm{FOV_v}}}{\mathrm{FOV_v}[\mathrm{mrad}]\ \cdot\ D_{\mathrm{obs}}\,[\mathrm{m}]}, \tag{A.10}$$

Observation of a TV monitor with an eye piece

$$M_{\mathrm{sys}} = \frac{H_{\mathrm{picture}}\,[\mathrm{mm}] \cdot N_{\mathrm{FOV_v}}}{\mathrm{FOV_v}\,[\mathrm{mrad}] \cdot f_{\mathrm{eyepiece}}\,[\mathrm{m}] \cdot n_{\mathrm{active}}}, \tag{A.11}$$

Observation of a flat panel monitor with an eyepiece

$$M_{\mathrm{sys}} = \frac{\mathrm{dis}_{\mathrm{pitch}}\,[\mathrm{mm}] \cdot N_{\mathrm{FOV_v}}}{\mathrm{FOV_v}\,[\mathrm{mrad}] \cdot f_{\mathrm{eyepiece}}\,[\mathrm{m}]}. \tag{A.12}$$

B. Calculation of Diffraction Limited MTFs

B.1 Pupil geometries

TRM4 calculates diffraction MTFs for the following pupil geometries:

- Circular aperture,
- Circular aperture with circular central obscuration (cco),
- Truncated circular apertures,
- Truncated circular apertures with cco,
- Rectangular apertures[13], and
- Rectangular apertures with cco.

TRM4 treats symmetric as well as asymmetric truncations. It calculates diffraction MTFs by auto-correlating the pupil function. All diffraction MTFs used in TRM were validated with the CODE V software.

This appendix describes the diffraction MTF of all available apertures, except for apertures with asymmetric truncations.

B.1.1 Definition of pupil dimensions

The pupil dimensions, see Figure B.1, are defined as follows:

D_{opt}	aperture diameter,
D_{obsc}	diameter of central obscuration,
W_{truncz}	dimension of truncated pupil in z-direction,
$W_{\mathrm{truncz'}}$	dimension of truncated pupil perpendicular to the z-direction.

B.1.2 Other terminology

The following TRM terminology is used here:

λ_{diff}	wavelength for diffraction MTF calculations,
r	spatial frequency,
r_{zero_z}	spatial frequency of the first zero in the MTF (no overlap between displaced pupil functions) in z-direction,
z	selected direction for the MTF calculation (either x or y),
z'	direction perpendicular to the z-direction,
A_{pup}	clear pupil area.

[13] Rectangular apertures are considered as a special case of truncated circular apertures.

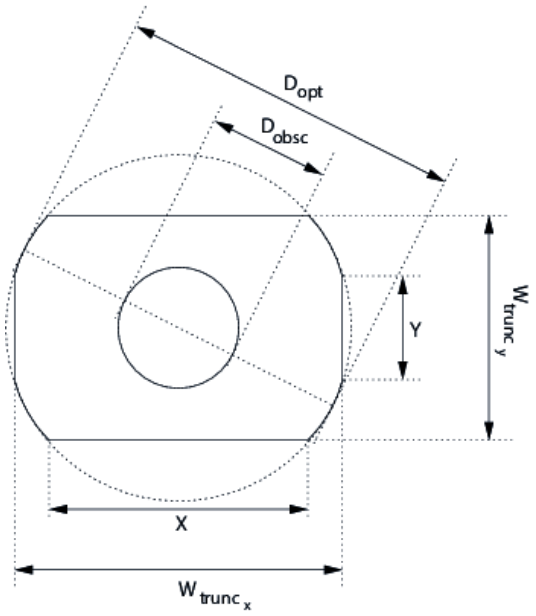


Figure B.1: Definition of pupil dimensions.

B.2 Definitions

Parameter, angle	Refer to
$Z = \sqrt{D_{\text{opt}}^2 - W_{\text{trunc}_{z'}}^2}$	Figure B.1
$\alpha = 2\cos^{-1}\left(\frac{\lambda_{\text{diff}} \cdot r}{D_{\text{opt}}}\right)$	Figure B.2
$\beta = 2\cos^{-1}\left[\frac{\frac{D_{\text{opt}}^2}{4} + (\lambda_{\text{diff}} \cdot r)^2 - \frac{D_{\text{obsc}}^2}{4}}{D_{\text{opt}} \cdot \lambda_{\text{diff}} \cdot r}\right]$	Figure B.3
$\gamma = 2\cos^{-1}\left[\frac{\frac{D_{\text{obsc}}^2}{4} + (\lambda_{\text{diff}} \cdot r)^2 - \frac{D_{\text{opt}}^2}{4}}{D_{\text{obsc}} \cdot \lambda_{\text{diff}} \cdot r}\right]$	Figure B.3
$\delta = 2\cos^{-1}\left(\frac{\lambda_{\text{diff}} \cdot r}{D_{\text{obsc}}}\right)$	Figure B.4
$\eta = 2\cos^{-1}\left(\frac{W_{\text{trunc}_z}}{D_{\text{opt}}}\right)$	Figure B.5
$\omega = 2\cos^{-1}\left(\frac{W_{\text{trunc}_{z'}}}{D_{\text{opt}}}\right)$	Figure B.5
$\phi = 2\cos^{-1}\left(\frac{2 \cdot \lambda_{\text{diff}} \cdot r - W_{\text{trunc}_z}}{D_{\text{obsc}}}\right)$	Figure B.6

Table B.1: Definition of variables and angles.

All parameters and angles used in the derivation are defined in Figure B.2 to Figure B.6 and in Table B.1.

For the horizontal MTF, the subscripts z and z' have to be replaced by the subscripts x and y, respectively. For the vertical MTF, the subscripts z and z' have to be replaced by the subscripts y and x, respectively.

All angles $2\cos^{-1}(\mathrm{ARG})$ in Table B.1 are confined to the following boundaries:

- The value of ARG must lie in the interval $[-1,1]$. ARG lies outside of this interval only in cases for which the corresponding expression is not used.
- The corresponding angles are mapped into the interval $[0,2\pi)$.

B.3 Equations

B.3.1 Method of MTF calculation

The calculations of the diffraction MTF are based on the overlap area of two displaced pupil functions [18]:

$$\mathrm{MTF}_{\mathrm{diff}}(r) = \frac{\text{overlap area of pupil funtions displaced by } (\lambda_{\mathrm{diff}} \cdot r)}{\text{clear area of pupil}}. \tag{B.1}$$

Spatial frequencies

The pupil displacement and the spatial frequency are related by

$$r = \frac{\text{pupil displacement}}{\lambda_{\mathrm{diff}}}. \tag{B.2}$$

Clear pupil area

In general, the clear pupil area is

$$A_{\mathrm{pup}} = \frac{D_{\mathrm{opt}}^2}{4}[\pi - (\eta - \sin\eta) - (\omega - \sin\omega)] - \frac{\pi}{4}D_{\mathrm{obsc}}^2. \tag{B.3}$$

For rectangular apertures it is given by

$$A_{\mathrm{pup}} = W_{\mathrm{trunc_x}} \cdot W_{\mathrm{trunc_y}} - \frac{\pi}{4}D_{\mathrm{obsc}}^2. \tag{B.4}$$

MTF cut-off frequency

Cut-off frequency in z-direction is

$$r_{\mathrm{zero}_z} = \begin{cases} \dfrac{D_{\mathrm{opt}}}{\lambda_{\mathrm{diff}}} & \text{for circular apertures,} \\ \dfrac{W_{\mathrm{trunc}_z}}{\lambda_{\mathrm{diff}}} & \text{for truncated or rectangular apertures.} \end{cases} \tag{B.5}$$

B.3.2 Diffraction limited MTF

The diffraction MTF in z-direction is given by

$$\mathrm{MTF}_{\mathrm{diff}_z}(r) = \begin{cases} \dfrac{2[(S_4 - S_3) - (S_2 - S_1) - S_5]}{A_{\mathrm{pup}}} & \text{for } r < \dfrac{Z}{\lambda_{\mathrm{diff}}}, \\ \dfrac{2[(S_4 - S_3) - (S_2 - S_1)]}{A_{\mathrm{pup}}} & \text{for } \dfrac{Z}{\lambda_{\mathrm{diff}}} \le r < \dfrac{W_{\mathrm{trunc}_z}}{\lambda_{\mathrm{diff}}}. \end{cases} \tag{B.6}$$

In this equation, the Z appearing in the boundary conditions refers to either X or Y shown in Figure B.1 and the variables $S_1, \ldots, S_5$ stand for different areas in the overlap of the pupil functions defined by:

$$S_1 = \begin{cases} \dfrac{D_{\mathrm{obsc}}^2}{8}(\delta - \sin\delta) & \text{for } r < \dfrac{D_{\mathrm{obsc}}}{\lambda_{\mathrm{diff}}} \\ 0 & \text{for } r \ge \dfrac{D_{\mathrm{obsc}}}{\lambda_{\mathrm{diff}}} \end{cases} \tag{B.7}$$

$$S_2 = \begin{cases} \dfrac{\pi \cdot D_{\mathrm{obsc}}^2}{4} & \text{for region i} \\ \dfrac{D_{\mathrm{obsc}}^2}{8}[\phi - \sin\phi) & \text{for region ii and iv} \\ \dfrac{D_{\mathrm{obsc}}^2}{8}(\gamma - \sin\gamma) & \text{for region iii} \\ 0 & \text{for region v} \end{cases} \tag{B.8}$$

$$S_3 = \begin{cases} \dfrac{D_{\mathrm{opt}}^2}{8}(\eta - \sin\eta) & \text{for regions i, ii, iv and v} \\ \dfrac{D_{\mathrm{opt}}^2}{8}(\beta - \sin\beta) & \text{for region iii} \end{cases} \tag{B.9}$$

$$S_4 = \frac{D_{\mathrm{opt}}^2}{8}(\alpha - \sin\alpha) \tag{B.10}$$

$$S_5 = \frac{D_{\mathrm{opt}}^2}{8}[(\alpha - \sin\alpha) - (\pi - \omega - \sin\omega)] - W_{\mathrm{trunc}_{z'}}\left(\frac{Z - r \cdot \lambda_{\mathrm{diff}}}{2}\right). \tag{B.11}$$

The spatial frequency domain is divided into the five regions as follows:

$$\text{i: } 0 < \mathrm{r} < \frac{W_{\mathrm{trunc}_z} - D_{\mathrm{obsc}}}{2 \cdot \lambda_{\mathrm{diff}}} \tag{B.12}$$

$$\text{ii: } \frac{W_{\mathrm{trunc}_z} - D_{\mathrm{obsc}}}{2 \cdot \lambda_{\mathrm{diff}}} < r < \frac{W_{\mathrm{trunc}_z} - \sqrt{W_{\mathrm{trunc}_z}^2 - D_{\mathrm{opt}}^2 + D_{\mathrm{obsc}}^2}}{2 \cdot \lambda_{\mathrm{diff}}} \tag{B.13}$$

$$\text{iii: } \frac{W_{\mathrm{trunc}_z} - \sqrt{W_{\mathrm{trunc}_z}^2 - D_{\mathrm{opt}}^2 + D_{\mathrm{obsc}}^2}}{2 \cdot \lambda_{\mathrm{diff}}} \le r < \frac{W_{\mathrm{trunc}_z} + \sqrt{W_{\mathrm{trunc}_z}^2 - D_{\mathrm{opt}}^2 + D_{\mathrm{obsc}}^2}}{2 \cdot \lambda_{\mathrm{diff}}} \tag{B.14}$$

$$\text{iv: } \frac{W_{\mathrm{trunc}_z} + \sqrt{W_{\mathrm{trunc}_z}^2 - D_{\mathrm{opt}}^2 + D_{\mathrm{obsc}}^2}}{2 \cdot \lambda_{\mathrm{diff}}} \le r < \frac{W_{\mathrm{trunc}_z} + D_{\mathrm{obsc}}}{2 \cdot \lambda_{\mathrm{diff}}} \tag{B.15}$$

$$\text{v:}\quad \frac{W_{\text{trunc}_z} + D_{\text{obsc}}}{2 \cdot \lambda_{\text{diff}}} \leq r < \frac{W_{\text{trunc}_z}}{\lambda_{\text{diff}}}. \tag{B.16}$$

If $W_{\text{trunc}_z}^2 - D_{\text{opt}}^2 + D_{\text{obsc}}^2 \leq 0$, then regions ii, iii, and iv are reduced to a single region which is defined by

$$\frac{W_{\text{trunc}_z} - D_{\text{obsc}}}{2 \cdot \lambda_{\text{diff}}} < r < \frac{W_{\text{trunc}_z} + D_{\text{obsc}}}{2 \cdot \lambda_{\text{diff}}}. \tag{B.17}$$

In this case, the definitions of S_2 and S_3 for regions ii and iv have to be used.

B.4 Figures

The following figures illustrate the parameters and angles used in this appendix. Figures refer to the calculation of the horizontal MTF unless otherwise specified.

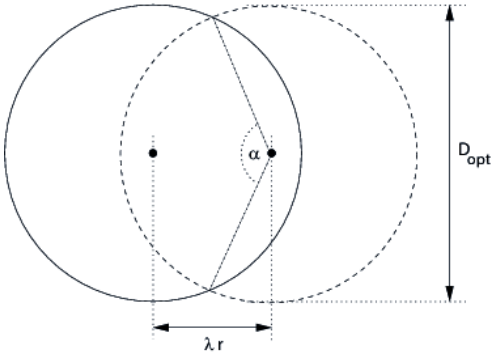


Figure B.2: Definition of the angle α for circular apertures.

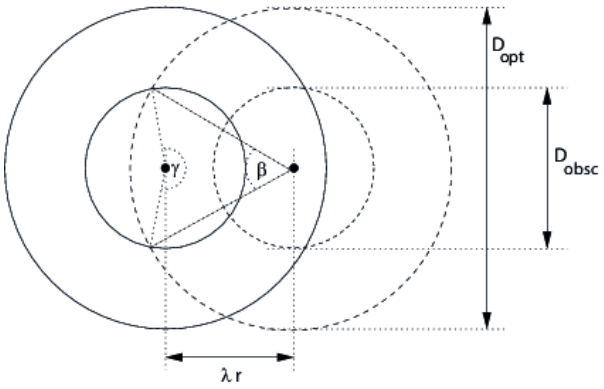


Figure B.3: Definition of the angles β and γ for apertures with central obscuration.

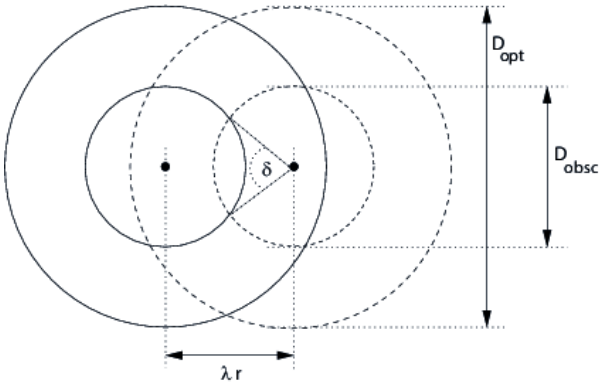


Figure B.4: Definition of the angle δ for an aperture with central obscuration.

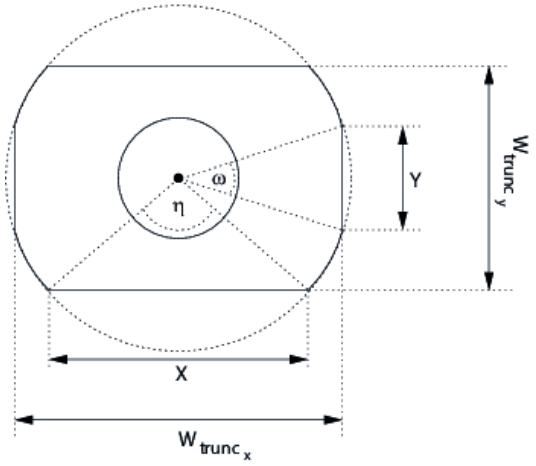


Figure B.5: Definition of the angles η and ω (example is for vertical MTF calculation).

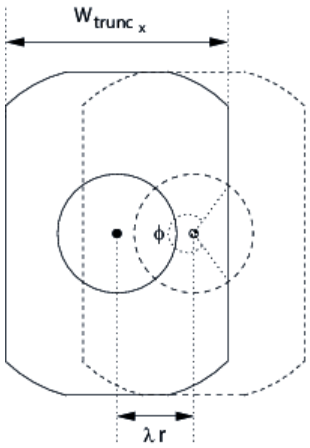


Figure B.6: Definition of the angle ϕ (for horizontal MTF only).

C. Preset Data Overview

C.1 Detector spectral quantum efficiencies

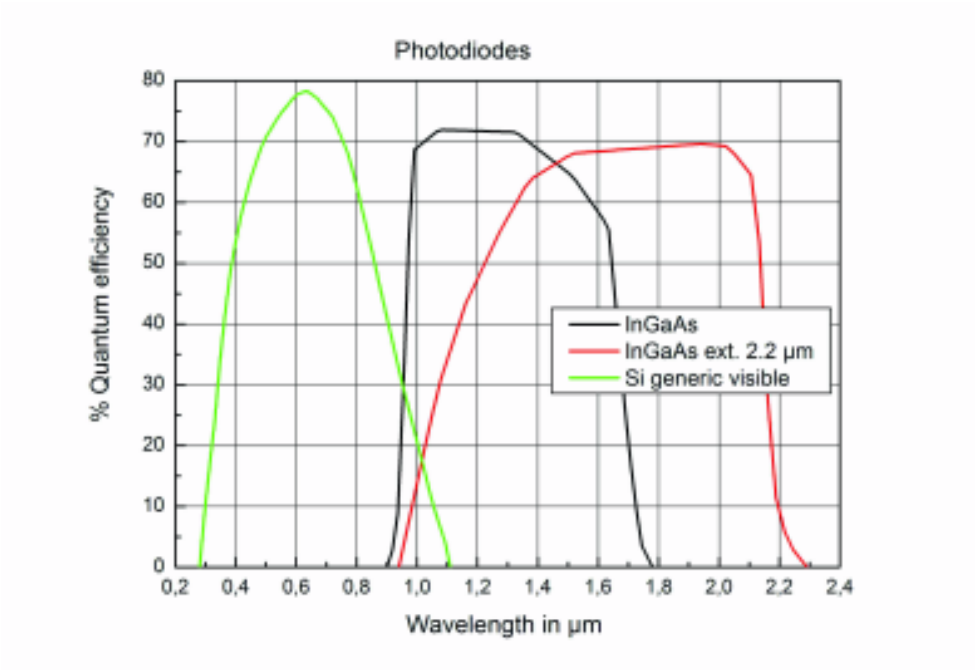
As a starting point for models, several generic spectral quantum efficiencies of typical detector types are available as presets.

Photodiode technology covers wavelengths from the ultra-violet to the short wavelength infrared range. The included types are a generic CCD sensor along with two different InGaAs sensors for the NIR and SWIR ranges, respectively.

For thermal imagers, the currently dominant sensor technologies are HgCdTe (or CMT, Cadmium Mercury Telluride) based detectors, and QWIPs (Quantum Well Infrared Photodetector), which are included for the MWIR and LWIR ranges.

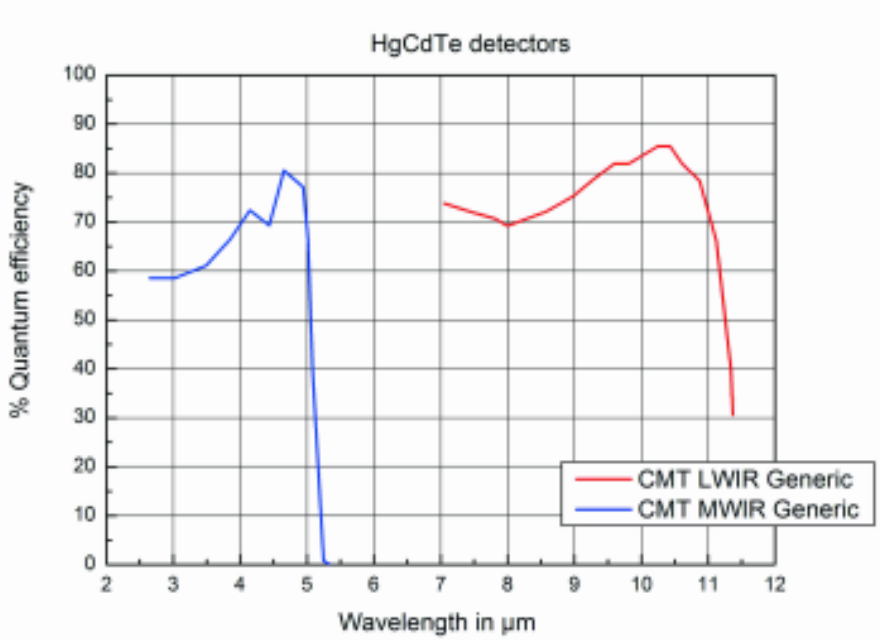
Below, graphs show the quantum efficiencies for the built-in detector presets, while their sensitive wavelength ranges and the TRM4 preset names are listed in two separate tables.

C.1.1 *Visible/NIR/SWIR* category

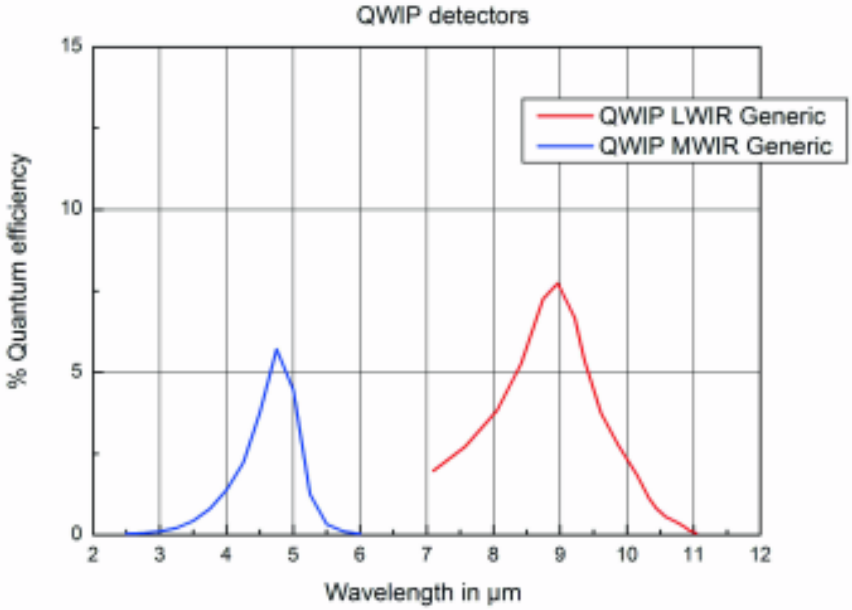


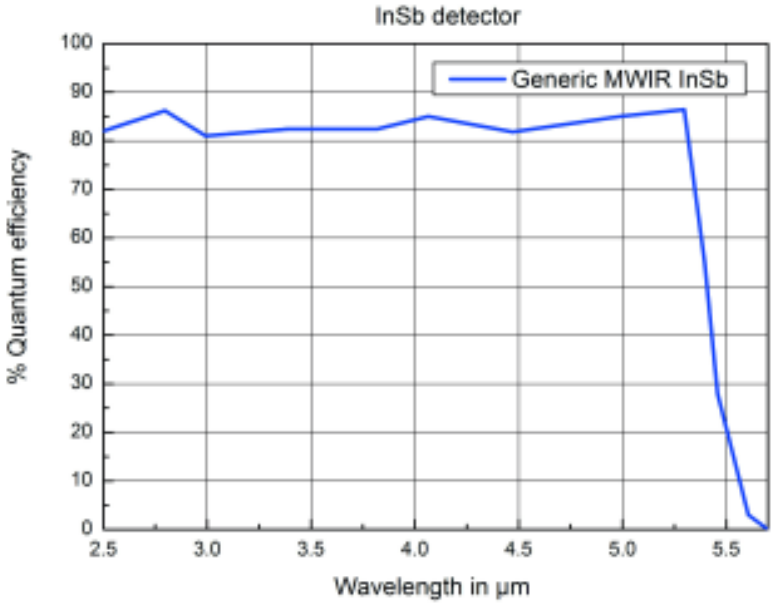
Type of Photodiode	TRM4 Preset Name	Wavelength Range [µm]
CCD Silicon	CCD Silicon generic	0.3 – 1.1
InGaAs	InGaAs generic	0.9 – 1.8
InGaAs extended 2.2 µm	InGaAs extended generic	1.1 – 2.3

C.1.2 *General* and *Thermal* categories



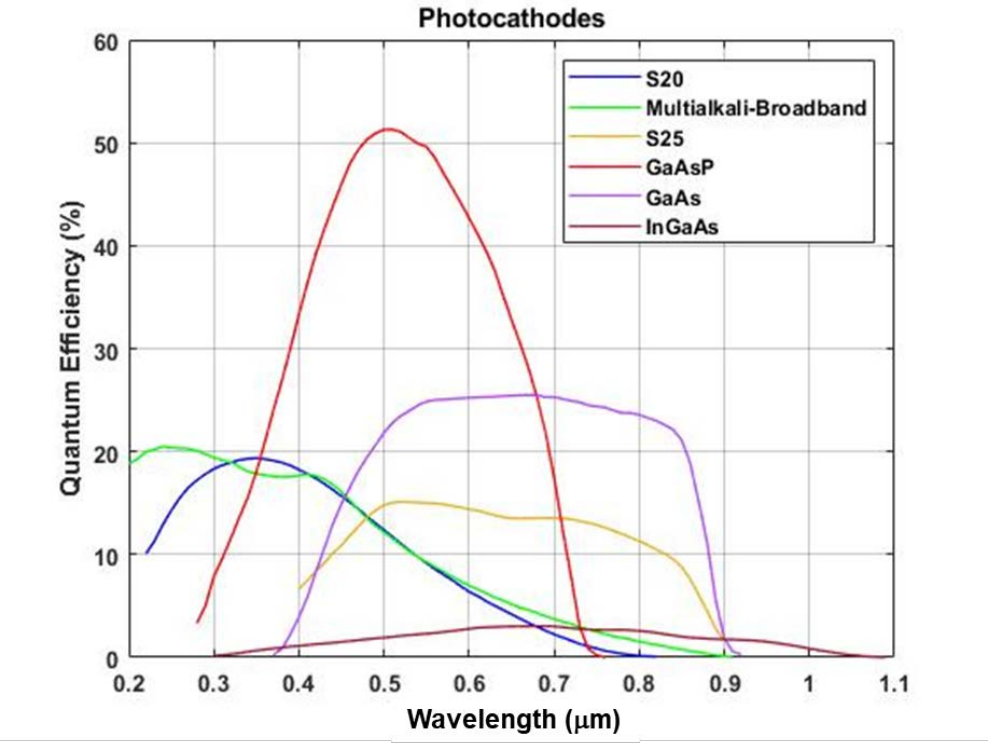
Type of Detector	TRM4 Preset Name	Wavelength Range [µm]
HgCdTe	CMT MWIR generic	2.6 – 5.3
QWIP	QWIP MWIR generic	2.5 – 6.0
InSb	InSb MWIR generic	2.5 – 5.7
HgCdTe	CMT LWIR generic	7.0 – 11.4
QWIP	QWIP LWIR generic	7.1 – 11.0




InSb detector
Generic MWIR InSb
% Quantum efficiency
Wavelength in µm
100
90
80
70
60
50
40
30
20
10
0
2.5
3.0
3.5
4.0
4.5
5.0
5.5

C.2 Photocathode spectral quantum efficiencies

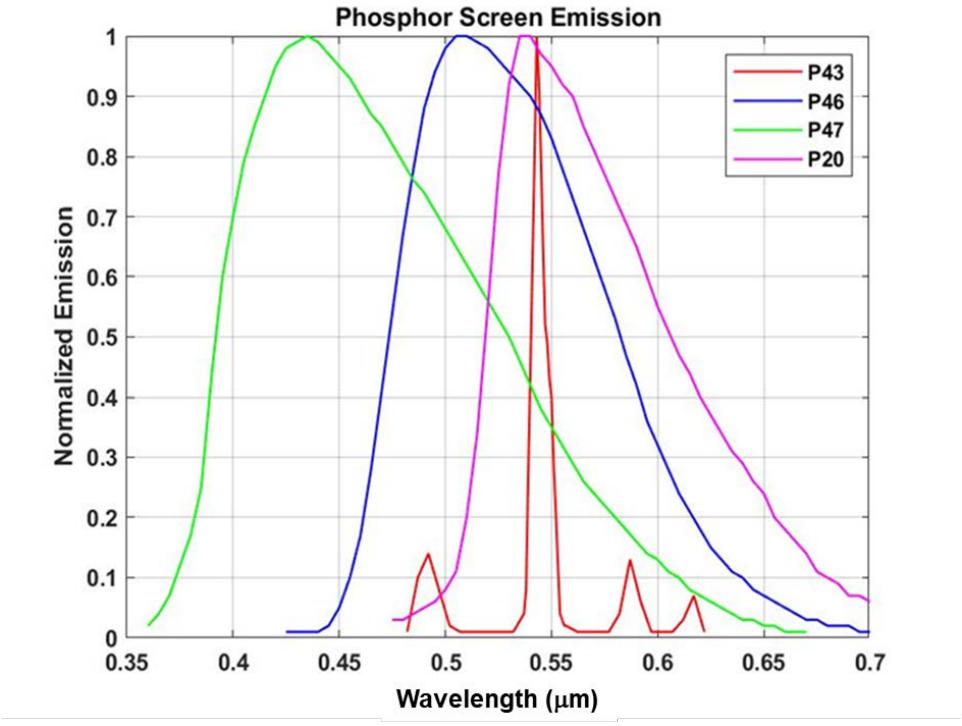
For ICCD/ICMOS and NVG imager types, quantum efficiencies of typical photocathode materials are available as presets. The photocathode materials included in TRM4 are sensitive from the ultra-violet to NIR spectral range, as shown in the graph and table below.



Photocathode Material	TRM4 Preset Name	Wavelength Range [μm]
S-20	S-20	0.22 – 0.82
MultiAlkali - Broadband	MultiAlkali-Broad	0.20 – 0.90
S-25	S-25	0.40 – 0.90
GaAsP	GaAsP	0.28 – 0.76
GaAs	GaAs	0.37 – 0.92
InGaAs	InGaAs	0.30 – 1.09

C.3 Phosphor screen normalized spectral emission

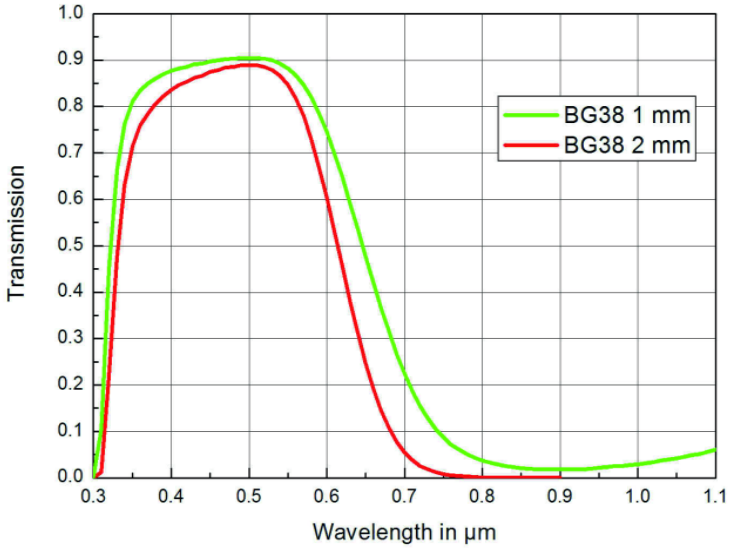
Several phosphor screen types are available as presets for ICCD/ICMOS imagers. The normalized spectral emission of the phosphor screens and the wavelength range are shown in the graph and table below. The temporal MTF of the phosphor screen presets (associated with the phosphor decay time) is calculated using Equation (4.19) and the parameters given in Table 4.1.



Phosphor Screen type	TRM4 Preset Name	Wavelength Range [μm]
P43	P43	0.48 – 0.62
P46	P46	0.42 – 0.70
P47	P47	0.36 – 0.67
P20	P20	0.47 – 0.70

C.4 IR cut-off filters

These presets are used for the definition of detector sensitivity via responsivity measurements in the *Visual* category.



Filter type	TRM4 Preset Name	Wavelength Range [µm]
Blue-Green Glass filter, 1 mm thickness	BG38 1 mm	0.3 – 1.1
Blue-Green Glass filter, 2 mm thickness	BG38 2 mm	0.3 – 0.9

C.5 Spectral reflectivity and emissivity of targets and backgrounds

All data presented here was taken from the publicly available ASTER Spectral Library [25][14]. Spectral data in ASTER gives emissivity values for a large number of different materials and objects, usually in the range from 0.3 – 14 μm.

In TRM4, systems modelled in the *Visible/NIR/SWIR* category expect reflectivity spectra, while those in the General and Thermal categories, mainly used for modelling in the MWIR and LWIR spectral regions, respectively, expect emissivity values. Thus, the ASTER spectra have been split at a wavelength of 2.5 μm, with data in the 2.5 – 14 μm region subsequently converted to emissivity values using the relation

$$\epsilon(\lambda) = 1 - \rho(\lambda), \tag{C.1}$$

where $\epsilon(\lambda)$ is the emissivity/emittance of the material/object at wavelength λ and $\rho(\lambda)$ is the reflectivity/reflectance at wavelength λ.

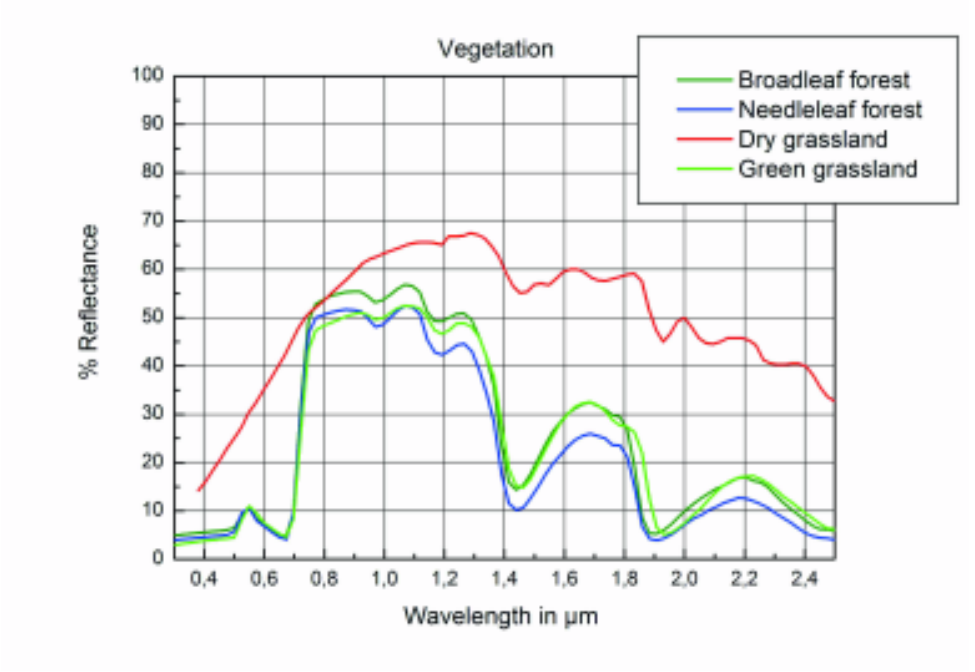
Additionally, all spectra were resampled to a lower resolution in order to adhere to the 100 sample limit of TRM4.

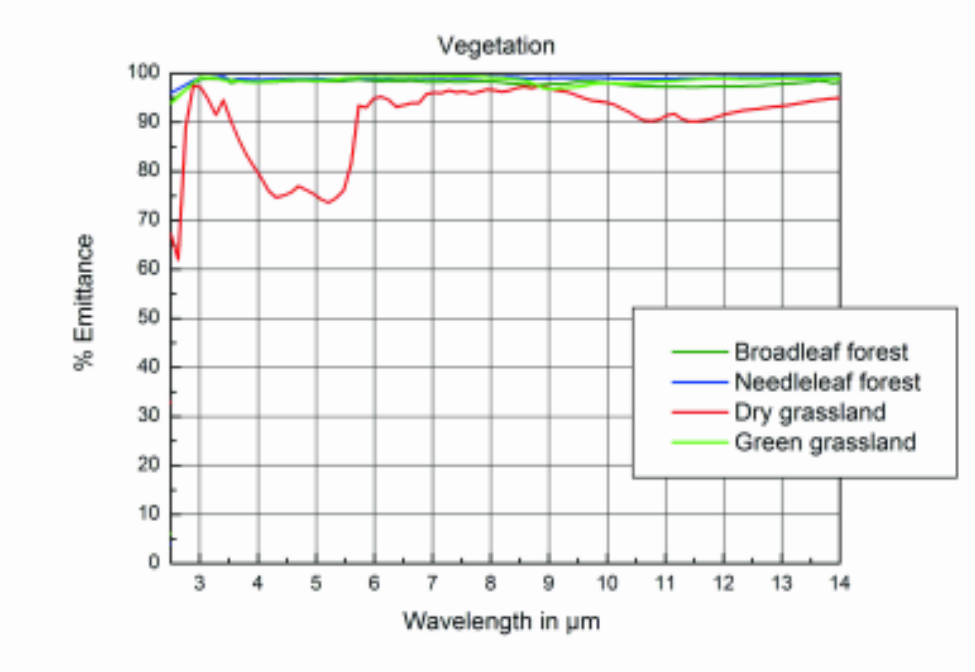
The following sections show the selected spectra for backgrounds and targets along with tables giving the ASTER library reference numbers.

[14] The library is accessible online at http://speclib.jpl.nasa.gov

C.5.1 Backgrounds

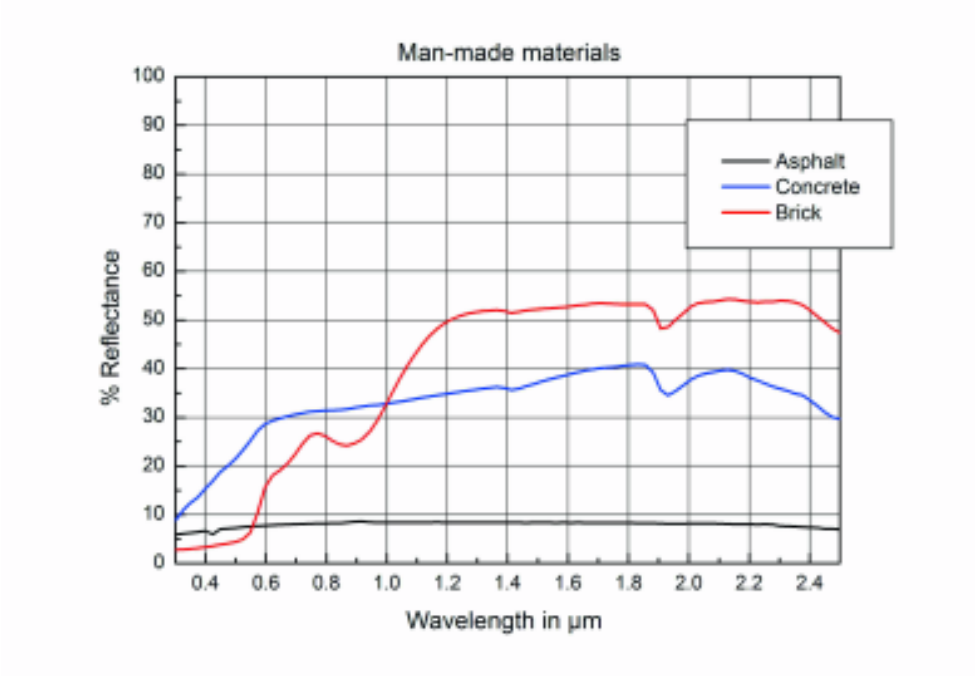
Vegetation

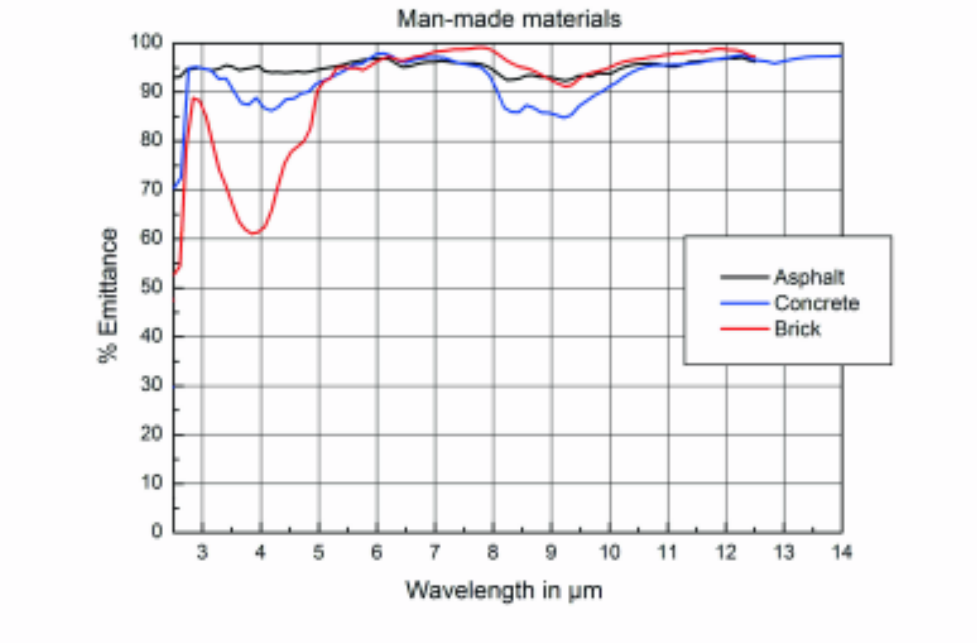




Aster Name/Subclass / Sample No.	TRM4 name
Dry grass / Dry grass / drygrass.doc	Dry grassland
Grass / Green grass / grass.doc	Green grassland
Confier / Conifers / conifers.doc	Needle leaf forest
Decidous / Deciduous / decidous.doc	Broad leaf forest

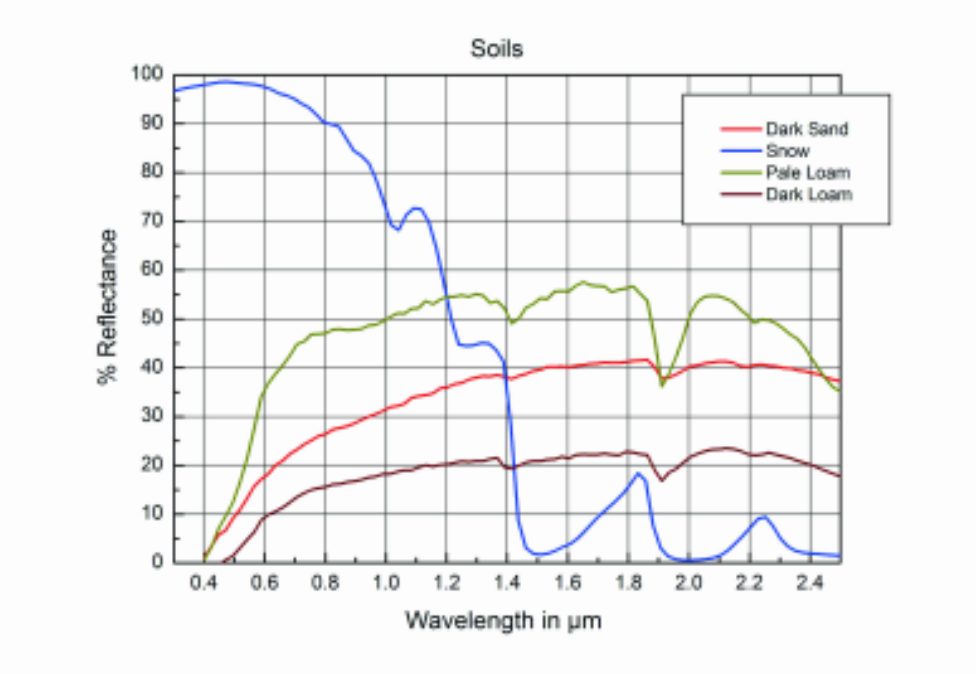
Man-made backgrounds

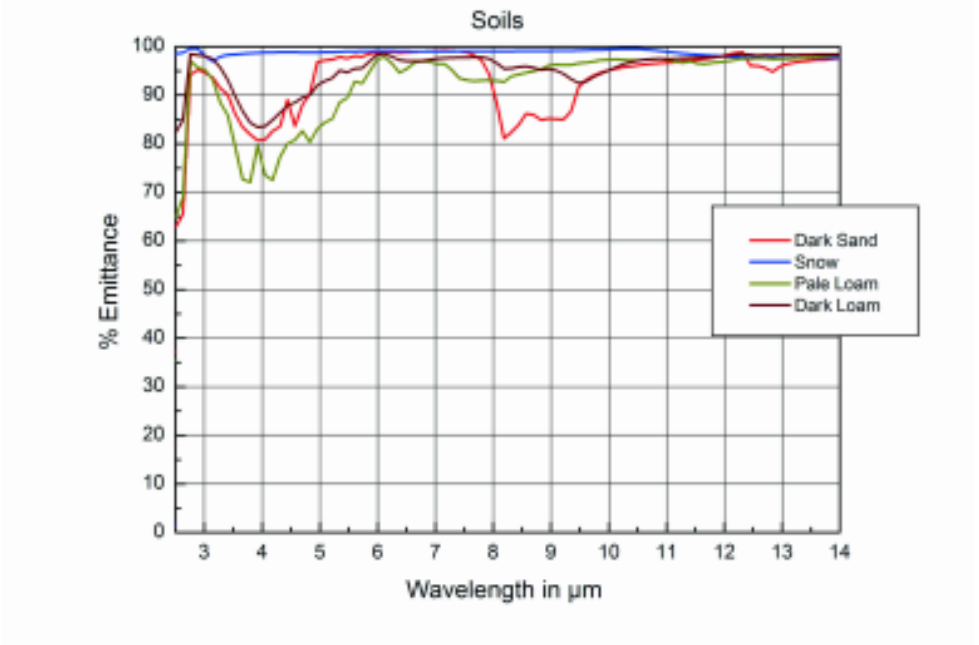




Aster Name/Subclass / Sample No.	TRM4 name
Construction Asphalt / Paving Asphalts / 0674UUUASP	Asphalt
Construction Concrete / Construction Concrete / 0598UUUCNC	Concrete
Weathered Red Brick / General Construction Materials / 0412UUUBRK	Brick

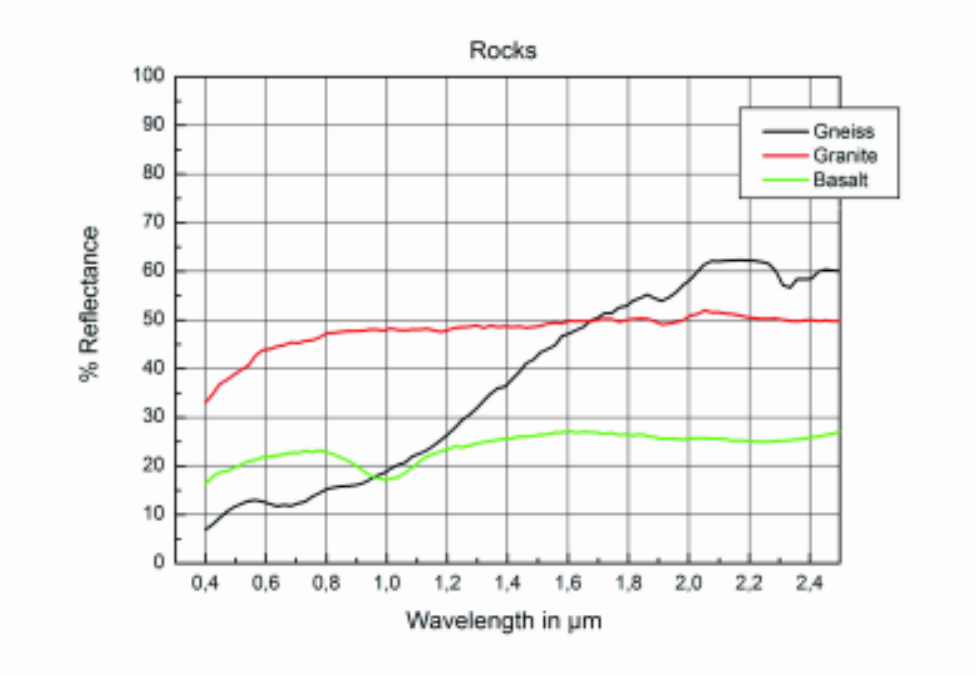
Soils

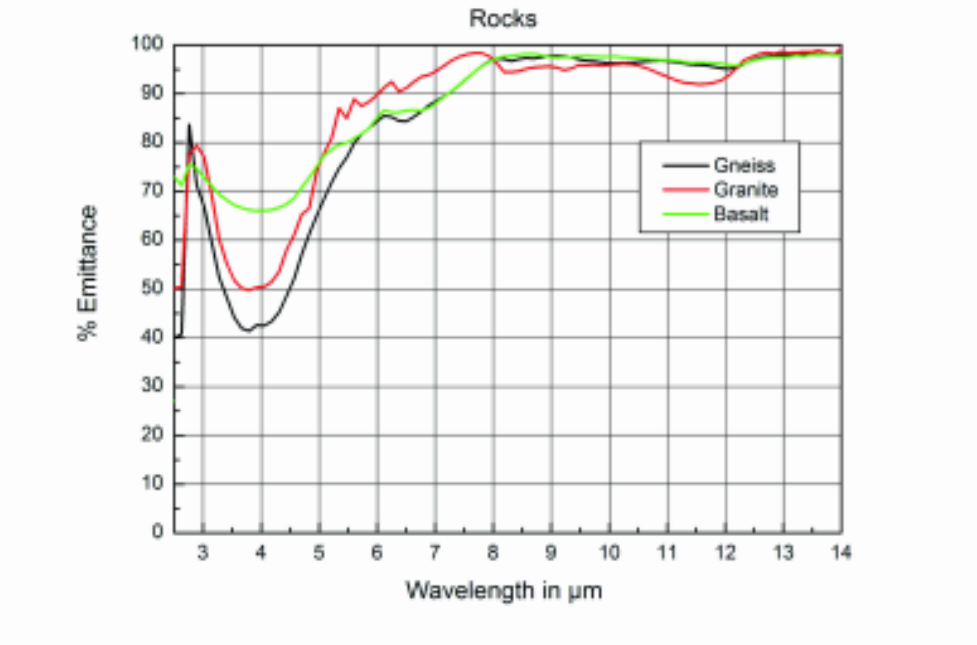




Aster Name/Subclass / Sample No.	TRM4 name
Brown to dark brown sand / Entisol / 87P706	Dark Sand
Medium Granular Snow / Medium Granular / COARSE.SNW	Snow
Very pale brown to brownish yellow interior dry gravelly silt loam / Aridisol / 82P2695	Pale Loam
Brown to dark brown gravelly fine sandy loam / Inceptisol / 87P325	Dark Loam

Rocks

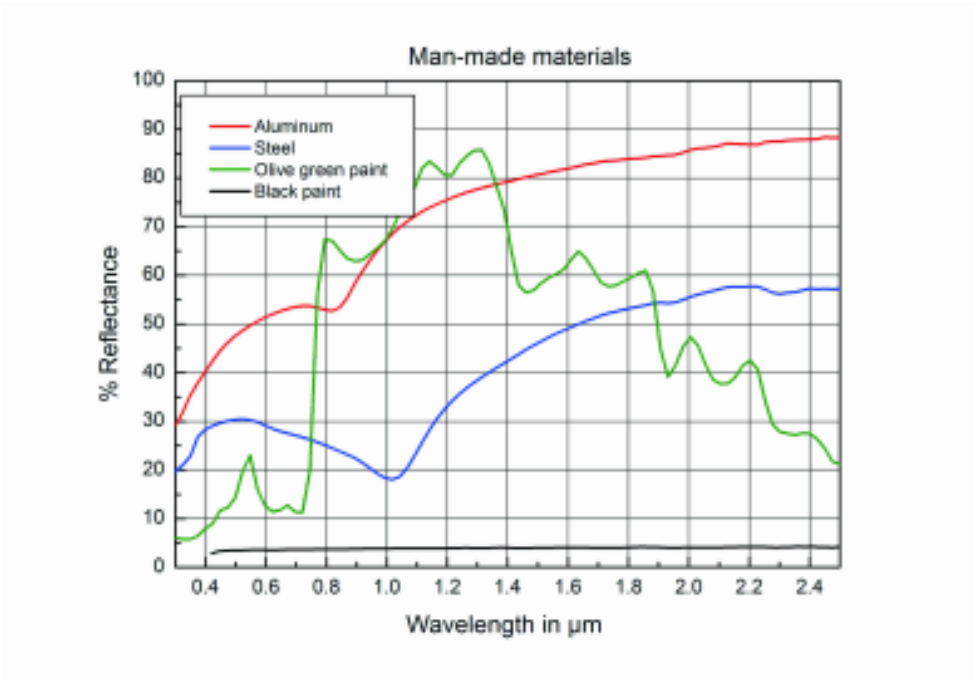


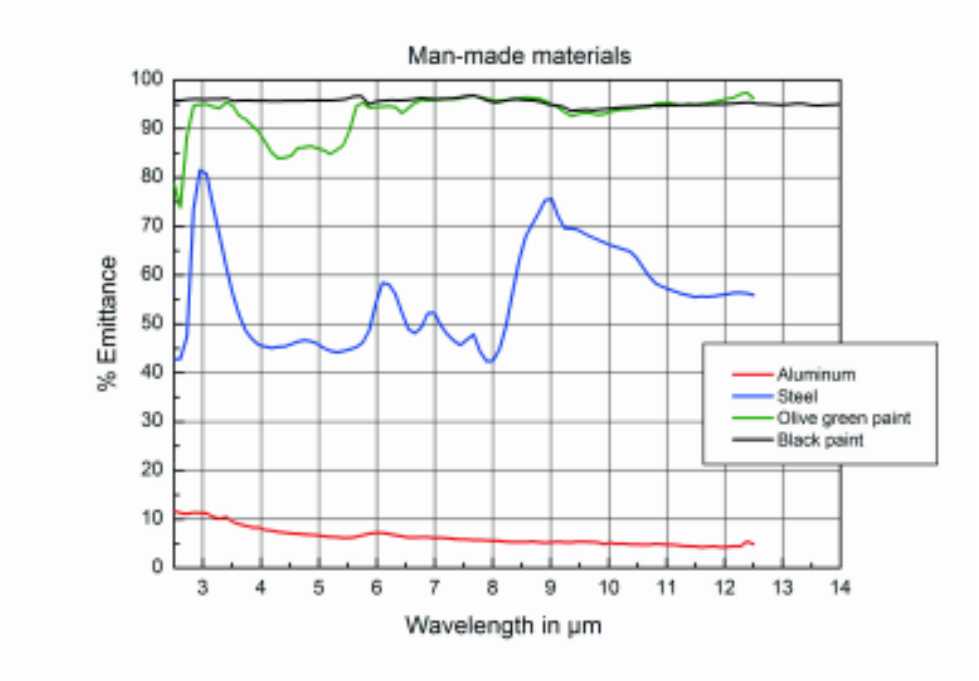


Aster Name/Subclass / Sample No.	TRM4 name
Diorite Gneiss / Gneiss / No. 464, Hunt and Salisbury Collection, U.S. Geological Survey	Gneiss
Granite / Felsic / Granite.H3	Granite
Basalt / Mafic / Basalt.7	Basalt

C.5.2 Targets

For targets, only man-made material presets are included in TRM4.





Aster Name/Subclass / Sample No.	TRM4 name
Aluminum Metal / Metal / 0384UUUALM	Aluminum
Galvanized Steel Metal / Metal / 0526UUUSTLb	Oxidized Steel
Olive Green Paint / Paints / 0407UUUPNT	Olive green paint
Black Paint / Paints / 0406UUUPNT	Black paint

C.6 Spectral irradiance of natural radiation sources

The data recommended in TRM4 are the same as those used in the US Army Night Vision and Electrons Sensors Directorate (NVESD) SSCAM / IICCD and IIMRC models (2002). They are typical for several different night-time conditions.

The data is normalized for use in TRM4. The peak irradiance values are listed in the Table C.1 below.

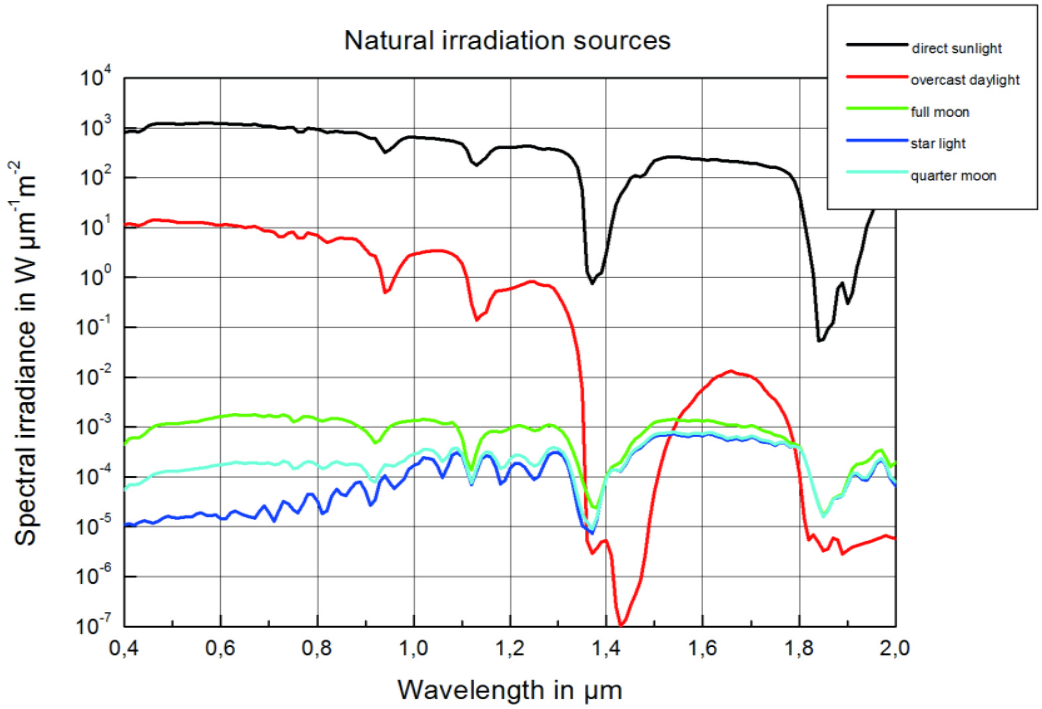


Figure C.1: Spectral irradiance of natural sources from visual to NIR.

Type of Irradiation	Peak Irradiance [W m^{-2} µm^{-1}]	TRM4 designation
Direct sunlight	1239	Direct sunlight; 0.4 – 2 µm
Overcast daylight	14.19	Overcast daylight; 0.4 – 2 µm
Full moon	$1.74 \cdot 10^{-3}$	Full moon; 0.4 – 2 µm
Quarter moon	$7.84 \cdot 10^{-4}$	Quarter moon; 0.4 – 2 µm
Star light	$7.04 \cdot 10^{-4}$	Star light; 0.4 – 2 µm

Table C.1: Peak irradiance of the various irradiation sources. Note that the external data is normalized and that the peak irradiance must be entered into the Irradiance dialog via the *Irradiance at peak* edit box.

C.7 Turbulence structure constant

C.7.1 Recommended values for constant C_n^2 and simple height scaling

Table C.2 lists values for the turbulence structure constant C_n^2 that were obtained from long-term measurements in arid, moderate, and subartic climate by Fraunhofer IOSB [26]. All measurements are scaled from the actual measurement height to a height of 2 m above ground; for daytime measurements, this was done using scaling proportional to $h^{-4/3}$, for nighttime measurements to $h^{-2/3}$.

Climate	Ground condition	Time of day	C_n^2 values at 2 m height above ground corresponding to cumulative frequency of occurrence [m$^{-2/3}$]			
			90 %		50 %	
			Winter	Summer	Winter	Summer
Arid (Mediterranean)	stony ground, no vegetation	Night Noon	$7.4 \cdot 10^{-14}$ $3.4 \cdot 10^{-13}$	$6.0 \cdot 10^{-14}$ $1.1 \cdot 10^{-12}$	$1.9 \cdot 10^{-14}$ $1.6 \cdot 10^{-13}$	$9.2 \cdot 10^{-15}$ $9.2 \cdot 10^{-13}$
Moderate (Central Europe)	grassland	Night Noon	$1.2 \cdot 10^{-13}$ $2.2 \cdot 10^{-13}$	$7.3 \cdot 10^{-14}$ $5.1 \cdot 10^{-13}$	$4.8 \cdot 10^{-14}$ $3.7 \cdot 10^{-14}$	$1.3 \cdot 10^{-14}$ $2.0 \cdot 10^{-13}$
Subartic (Northern Europe)	marshy ground, winter: snow	Night Noon	$2.2 \cdot 10^{-13}$ $1.5 \cdot 10^{-13}$	$3.8 \cdot 10^{-14}$ $3.8 \cdot 10^{-13}$	$6.4 \cdot 10^{-14}$ $4.6 \cdot 10^{-14}$	$6.0 \cdot 10^{-15}$ $7.6 \cdot 10^{-14}$

Table C.2: Recommended values of the turbulence structure constant for various climates and seasons.

As can be seen in Figure C.2, also taken from the publication [26], actual values for C_n^2 vary considerably during the day.

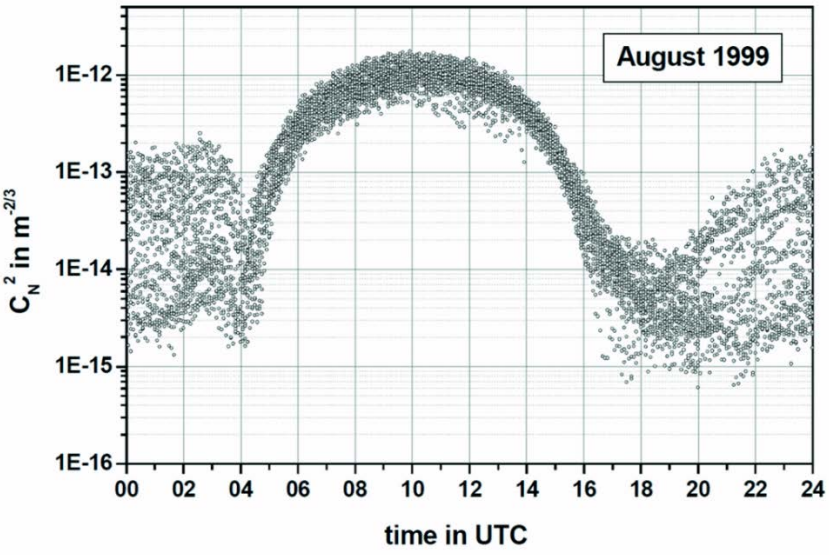


Figure C.2: Turbulence structure constant measurement results in arid climate during August 1999.

C.7.2 Preset height profiles of C_n^2

Figure C.3 shows the structure constant profiles available as presets in TRM4. The Hufnagel-Valley-Boundary (HVB)-profile uses standard parameter settings given in reference [27]. These profiles are suitable for air-to-surface and surface-to-air applications with long distances. It should be emphasized that these profiles tend to underestimate turbulence strength close to the ground. The same is true for the Greenwood (night) profile, also taken from [27].

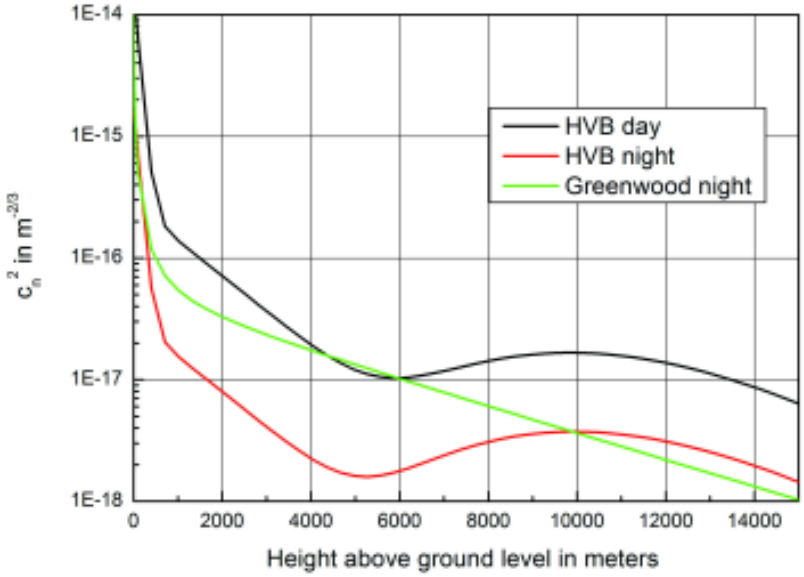


Figure C.3: Preset height profiles of the turbulence structure constant.

D. Conventions, Symbols, and Nomenclature

D.1 General conventions

It is assumed that scanning (if applicable) and signal read-out are always in horizontal direction. Systems with vertical scan and vertical read-out are not treated by TRM4.

Other names used for an imager are “device” or “system”.

The “categories” mentioned in the text refer to the different operational modes of TRM4 that model reflective, emissive, and mixed systems. The terms used are “*Visible* category” for purely reflective systems (which include NIR and SWIR systems), “*Thermal* category” for purely emissive systems, and “*General* category” for mixed emissive/reflective systems.

D.2 Symbols and nomenclature

Constants

c	speed of light in vacuum $2.997825 \cdot 10^{8}$ [m/s]
e	electron charge $1.6021 \cdot 10^{-19}$ [C]
h	Planck constant $6.6256 \cdot 10^{-34}$ [Js]
k	Boltzmann constant $1.38 \cdot 10^{-23}$ [J/K]
π	3.14159…

Sub- and superscripts

The most frequently used sub- and superscripts are:

h	horizontal direction,
v	vertical direction,
z	general direction (horizontal or vertical),
T	referring to the target,
B	referring to the background,
i	referring to target/background (replace by T or B),
N	sub-/superscript denoting temporal noise components (NETD/NES),
I	sub-/superscript denoting spatial noise components (IETD/IES).

D.3 Parameters and functions

Parameters and functions are listed in alphabetical order. First, notations in lowercase letters are listed, followed by capital letters.

No distinctions are made between the horizontal and vertical directions.

D.3.1 Latin letters

a_{det}	active detector element dimension (in Chapter 3 it may also refer to the binned or electronic pseudo interlace version, see *Note* in Section 3.2)
$a_{\mathrm{det_binned}}$	effective binned detector element dimension
$a_{\mathrm{det_epi}}$	effective binned detector element dimension with electronic pseudo interlace
a_{diff}	diffusion length of SPRITE detector
a_{LED}	active LED dimension
a_{pitch}	detector element pitch (in Chapter 3 it may also refer to the binned or electronic pseudo interlace version, see *Note* in Section 3.2)
$a_{\mathrm{pitch_binned}}$	effective detector element pitch after binning
$a_{\mathrm{pitch_epi}}$	effective detector element pitch with electronic pseudo interlace
a_{SPRITE}	read-out zone length of SPRITE detector
$a_{\mathrm{tar_crit}}$	critical target dimension
a_{tar_z}	effective target dimension in z-direction [Equations (3.83) and (3.84)]
a_{TDI}	TDI mismatch per detector element
A_{det}	sensitive area of a single detector element (in Chapter 3 it may also refer to the binned or electronic pseudo interlace version, see *Note* in Section 3.2)
$A_{\mathrm{det_binned}}$	effective sensitive area of a binned detector element
$A_{\mathrm{det_epi}}$	effective sensitive detector area with electronic pseudo interlace
$A_{\mathrm{front\|side\|top}}$	Effective area of the target front/side/top/bottom facet [Equation (3.80)]
A_{pup}	clear pupil area of the front optics
A_{tar}	effective area of the target [Equation (3.81)]
AMOP	Average Modulation at Optimum Phase
c_{dig,n_z}	n^{th} coefficient of a digital filter in a given direction
C_{f}	responsivity unit conversion factor
$C_n^2(h)$	height profile of the refractive structure constant of atmospheric turbulence
$C_{n,\mathrm{ref}}^2$	refractive structure constant of atmospheric turbulence at reference height h_{ref}
CTE_z	charge transfer efficiency between neighboring detector elements in a given direction
d_{det}	diameter of circular detector element
d_{hole}	hole diameter in loop-hole detector
$\mathrm{dis}_{\mathrm{pitch}}$	pitch between display elements (LEDs/pixels)
D	depth of the target
D_{dist}	geometrical distortion of the displayed image
D_{eff}	effective diameter of the front optics
D_{obs}	distance observer display
D_{obsc}	diameter of central obscuration of the front optics
D_{opt}	aperture diameter of the front optics
D_{peak}^*	peak detectivity of a (thermal) detector [Equations (3.63) and (3.64)]
D_{pc}	input diameter of the photocathode
DAS_z	detector angular subtense in z-direction
$\mathrm{DS}(R)$	difference signal [Equation (3.129)]
DSNR	difference signal to noise ratio [Equations (3.234) and (3.235)]
DSNU	dark signal non-uniformity
E_{gain}	electronic gain factor
$\mathrm{E}^{\mathrm{N,I}}(r_{\mathrm{p}})$	eye/brain spatial integration of temporal/spatial noise [Sections 3.11.5 and 3.11.6]

$\mathrm{E}^{\mathrm{N,ICCD}}(r_{\mathrm{p}})$	eye/brain spatial integration of temporal noise for ICCD/ICMOS [Section 3.11.7]
EBI	Equivalent Background Illumination
EC	Effective Contrast [Equations (3.224) and (3.225)]
EDS	Effective Difference Signal [Equation (3.207)]
ETD	Effective Temperature Difference [Equation (3.208)]
$\mathrm{ERI}_{\mathrm{facet}}(\lambda)$	emitted radiance from a facet of the target/background [Equation (3.85)]
f_{eyepiece}	focal length of the eyepiece
f_{opt}	focal length of the optics
$F_{\mathrm{eff_cs}}$	effective F-number of the cold shield
$F_{\mathrm{eff_opt}}$	effective F-number of the front optics
$F_{\mathrm{irr,T\|B}}$	target/background irradiative correction factor [Equations (3.96) to (3.98)]
F_{meas}	F-number of a measurement setup
F_{opt}	F-number for a circular aperture
F_{PR}	path radiance factor [Equation (2.31)]
$F_{\mathrm{r,imager}}$	imager frame rate
$F_{\mathrm{r,video}}$	frame rate of the display
$F\#$	F-number
FOV_z	imager field of view in z-direction
$\mathrm{FR}_{\mathrm{int}}$	frame integration factor [Equation (3.141)]
$\mathrm{GSD}_{\mathrm{w}}$	weighted Ground Sampled Distance [Equation (3.230)]
$\mathrm{G}_{\mathrm{ICCD}}$	effective gain of the ICCD/ICMOS
G_l	image intensifier luminous gain
G_{R}	image intensifier radiant gain at peak response
h_n	n^{th} impulse response coefficient of a digital filter
h_{ref}	reference altitude for turbulence structure coefficient
H	height of the target
H_{picture}	displayed image height
i_{dark}	dark current per detector element
i_{det}	detector output current for non-integrating detectors [Equation (3.136)]
$I(\lambda)$	spectral irradiance of the illuminating source at the target
$I_{\mathrm{phos}}(\lambda)$	normalized spectral emission of the phosphor screen
IETD	Inhomogeneity Equivalent Temperature Difference
IES	Inhomogeneity Equivalent Signal [Equation (3.168)]
j_{cs}	detector carrier flux density due to cold shield emission [Equation (3.106)]
j_{dark}	detector carrier flux density due to detector dark current [Equation (3.113)]
j_{device}	detector carrier flux density due to device emission [Equation (3.105)]
$j_{\mathrm{e/r_T\|B}}(R)$	detector carrier flux density due to emission/reflection from target/background [Equations (3.109) to (3.112)]
$j_{\mathrm{ICCD_dark}}$	detector carrier flux density due to detector dark current and photocathode dark emission [Equation (3.126)]
$j_{\mathrm{ICCD_total_}i}(R)$	detector carrier flux density due to target/background signal for ICCD/ICMOS [Equation (3.125)]
j_{optics}	detector carrier flux density due to optics emission [Equation (3.104)]
$j_{\mathrm{path_{E\|R}}}(R)$	detector carrier flux density due to emissive/reflective path radiance [Equations (3.107) and (3.108)]

$j_{\text{total_T\|B}}(R)$	total carrier flux density due to target/background [Equation (3.103)]
$J_{\text{back}}(\lambda)$	spectral irradiance incident on the background [Equations (3.93) and (3.94)]
$J_{\text{fp}}(\lambda, R)$	spectral irradiance on the focal plane [Equation (3.100)]
$J_{\text{front\|side\|top}}(\lambda)$	spectral irradiance incident on front/side/top/bottom of the target [Equation (3.92)]
K	gain factor of the frequency boost
$L(\lambda, T)$	spectral radiance of a blackbody at temperature T [Equation (3.57)]
$L(\lambda)$	radiance of broadband responsivity measurement source, defined by its color temperature
L_{max}^{v}	maximum output brightness of the phosphor screen
$L_{\text{phos_dark}}(\lambda)$	spectral radiance from phosphor screen due to photocathode dark emission
$L_{\text{phos_i}}(\lambda, R)$	spectral radiance from phosphor screen due to target/background
$L_{\text{sky}}(\lambda)$	sky radiance
L_{PR}	path radiance [Equation (2.30)]
$\text{LSF}_{\text{display}}(z)$	line spread function of the display in z-direction
$\text{LSF}_{\text{eye}}(z)$	line spread function of the eye in z-direction
m_{blocks}	number of blocks in serial/parallel scan
m_{hp}	order of a high-pass filter
m_{lp}	order of a low-pass filter
M_{crep}	column replication factor
M_{coup}	magnification of the coupling element of an ICCD/ICMOS
M_{II}	magnification of the image intensifier
M_{ICCD}	magnification between the photocathode input window and CCD/CMOS sensor of the ICCD/ICMOS
M_{lrep}	line replication factor
$\text{M}_{\text{raster,display}}$	CRT/LED display raster modulation [Equation (3.30)]
$\text{M}_{\text{raster,eye}}$	perceived raster modulation [Equation (3.32)]
M_{sys}	system angular magnification [Equations (A.9) to (A.12)]
M_{zoom}	zoom factor
MCP	Minimum Contrast Perceived [Equation (3.178)]
MDSP	Minimum Difference Signal Perceived [Equation (3.176)]
MNDS	Minimum Necessary Difference Signal [Equation (3.219)]
MNC	Minimum Necessary Contrast [Equation (3.222)]
MNTD	Minimum Necessary Temperature Difference [Equation (3.215)]
MTDP	Minimum Temperature Difference Perceived [Equation (3.175)]
n_z	number of available detector elements after binning in z-direction
n_{active}	number of active lines of a display
n_{bin_z}	number of binned detector elements in z-direction
n_{det_z}	number of (unbinned) detector elements in z-direction
n_{inter}	opto-mechanical scan interlace factor
n_{lp}	number of line pairs
$n_{\text{lp_perc}}$	number of line pairs required for a perception task
$n_{\text{lp_res}}$	number of resolved line pairs
$n_{\text{pix_res}}$	number of resolved pixels
$N_{\text{carrier_T/B}}$	number of carriers in a storage gate
N_{dark}	number of dark carriers in a detector element
$N_{\text{disp_res}}$	number of resolved TV lines per picture height (TV limiting resolution)

N_{DSNU}	signal due to DSNU [Equation (3.167)]
N_{floor}	noise floor
$N_{\mathrm{FOV_z}}$	number of display pixels required to cover the FOV (FPD) or number of video lines required to cover the vertical FOV (CRT)
N_{frames}	number of integrated frames
$N_{\mathrm{int_frames}}$	number of independent frames integrated by the eye [Equation (3.181)]
$\mathrm{N_{NU}}$	signal (number of carriers) due to non-uniformity [Equation (3.168)]
N_{PRNU}	signal due to PRNU [Equation (3.166)]
N_{quant}	quantization noise
N_{sam_z}	number of samples in a given direction
N_{scl}	number of scanned lines per frame
N_{stor}	storage capacity of a single detector element (in Chapter 3 it may also refer to the binned or electronic pseudo interlace version, see *Note* in Section 3.2)
$N_{\mathrm{stor_binned}}$	storage capacity of a binned detector element
$N_{\mathrm{stor_epi}}$	storage capacity of an effective detector element with electronic pseudo interlace
$N_{\mathrm{total_T/B}}$	Number of carriers in a detector element due to target/background
N_{transfer_z}	number of charge transfers between neighboring detector elements in a given direction
N_{TDI}	number of detectors integrated in TDI mechanisms for scanning imagers
$N_{\mu\mathrm{step}_z}$	number of micro-scan steps in a given direction
NES	Noise Equivalent Signal
$\mathrm{NES_{det}}$	NES at the detector output [Equation (3.143) or (3.145) for ICCD/ICMOS]
$\mathrm{NES_{dig/vid}}$	NES in the digital or video signal [Equation (3.144)]
NETD	Noise Equivalent Temperature Difference
$\mathrm{NETD_{det}}$	NETD at the detector output [Equation (3.139)]
$\mathrm{NETD_{dig/vid}}$	NETD in the digital or video signal [Equation (3.140)]
NF	noise figure
$\mathrm{NF_{II}}$	noise figure of the image intensifier
NIIRS	National Imagery Interpretability Rating Scale [Equation (3.227)]
p	detector storage gate fill (in percent) [Equation (3.138)]
PRNU	photo response non-uniformity
$\mathrm{PSF_{CRT}}$	spot point spread function of a CRT display [Equation (2.26)]
q	aberration parameter
r	spatial frequency
r_0	turbulence Fried parameter [Equation (4.82)]
r_{knee}	knee (or 3 dB) frequency
r_{max}	peak responsivity in units of [A/W]
$r_{n_{\mathrm{lp}}}$	spatial frequency of a bar pattern
r_{N}	spatial Nyquist frequency [Equation (3.44)]
r_{p}	spatial frequency of the standard 4-bar test pattern
r_{s}	spatial sampling frequency [Equation (3.43)]
r_{zero}	spatial frequency of the first zero in an MTF
R	target distance
R_0	range solution for a specified observation task
$R_{\mathrm{DA}}(\lambda)$	spectral responsivity of the detector assembly [Equation (3.60)]
$R_{\mathrm{DA_max}}$	peak responsivity of the detector assembly [Equation (3.62)]
$R_{\mathrm{DA_norm}}(\lambda)$	normalized spectral responsivity of the detector assembly [Equation (3.61)]
R_l	luminous responsivity of the photocathode

$R_{\max}$	normalization constant (used for spectral system response) [Equation (3.55)]
R_{ref}	reference range for calculation of MTF, MDSP, etc.
R_{sp}	specified detector responsivity (*Visual/NIR/SWIR* category)
$R_{\text{sys}}(\lambda)$	spectral imager response [Equation (3.53)]
$R_{\text{sys_norm}}(\lambda)$	normalized spectral imager response [Equation (3.54)]
RER_z	directional Relative Edge Response [Equation (3.231)]
RER_{w}	weighted Relative Edge Response [Equation (3.233)]
$\text{RRI}_{\text{facet}}(\lambda)$	reflected radiance from a facet of the target [Equation (3.86)]
$\text{RRI}_{\text{tar}}(\lambda)$	total reflected spectral radiance from target in the direction of the imager [Equation (3.96)]
s_{amp_z}	amplitude of sinusoidal vibration
scf	scaling factor of the video signal
$S(r)$	spatial frequency detector signal
$S_{\text{in}}(N)$	digital filter input signal for the Nth frame
$S_{\max}$	maximum signal (brightness) in a displayed line
$S_{\min}$	minimum signal (brightness) between two displayed lines
$S_{\text{out}}(N)$	digital filter output signal for the Nth frame [Equation (2.23)]
SNR_{th}	signal-to-noise threshold of the eye
SGR	sky-to-ground ratio [Equation (2.31)]
t_{decay}	phosphor decay time
t_{eye}	eye integration time
t_{int}	detector integration time [Equations (3.73) to (3.76)]
$t_{\text{line_active}}$	active line time of video standard
$t_{\text{line_FOV}}$	time to display the horizontal FOV in one video line
t_{th}	thermal time constant of uncooled detector
T_{cs}	temperature of the cold shield
T_{device}	temperature of the imager inner housing
$T_{\text{off_dev}}$	temperature offset of the inner housing of the imager
$T_{\text{off_opt}}$	temperature offset of the front optics
T_{opt}	temperature of the front optics
T_{scene}	background temperature of the scene
TDI	Time delayed integration
v_{scan}	scan velocity
V	meteorological visibility
$V(\lambda)$	CIE photopic luminous efficiency function
$w(\lambda, R)$	spectral weighting function [Equation (4.2)]
W	width of the target
$W_{\text{eff}}(T)$	effective differential radiance of a blackbody at temperature T [Equation (3.59)]
W_{trunc_z}	width of the truncated aperture in z-direction

D.3.2 Greek letters

α	IIR frame integration filter factor
α_{FOV}	aspect ratio of the field of view
α_{image}	aspect ratio of the displayed image

α_{s_z}	angular sampling distance in a given direction
α_z	line-of-sight angle to the target in a given direction
β_z	angle between irradiating beam and the normal to target front in given direction
γ_{coup}	coupling efficiency of the coupling element of ICCD/ICMOS
δ	exponent of turbulence height dependence
Δf	noise bandwidth
Δf_{DA}	pre-digital signal noise bandwidth
Δf_{dig}	noise bandwidth in the digital signal
Δf_{I_z}	non-uniformity bandwidth in a given direction
$\Delta f_{\mathrm{I,dig}}$	non-uniformity bandwidth in the digital signal
$\Delta f_{\mathrm{I,vid}}$	non-uniformity bandwidth in the video signal
Δf_{vid}	noise bandwidth in the video signal
Δf_{z}	spatial noise bandwidth of the ICCD/ICMOS in the pre-digital signal [Equation (3.152)].
ΔT_0	specified temperature difference between target and background
$\epsilon_i(\lambda)$	spectral emissivity of the target/background
$\eta(\lambda)$	spectral quantum efficiency of the detector in [%]
η_{adc}	Analog-Digital converter cut-off frequency normalized to device horizontal sampling frequency
η_{cs}	cold shield efficiency
η_{cyc}	integration duty cycle
$\eta_{\mathrm{pc}}(\lambda)$	spectral quantum efficiency of the photocathode [%]
η_{scan}	scan efficiency
η_{video}	video line efficiency [Equation (3.29)]
$\eta_{\lambda_{\mathrm{p}}}$	quantum efficiency at peak responsivity wavelength λ_{p}
Γ	width of the eye point spread function
λ_{diff}	wavelength for diffraction MTF calculations
λ_{p}	wavelength of peak responsivity
ν_{boost}	boost frequency
$\nu_{\mathrm{det_temp}}$	-3 dB frequency of detector responsivity roll-off
ν_{hp}	cut-on frequency of a high-pass filter
ν_{lp}	cut-off frequency of a low-pass filter
ν_{video}	frequency of the video signal
$\nu_{\mathrm{vid_boost}}$	frequency of max boost (video processor)
$\nu_{\mathrm{vid_lp}}$	cut-off frequency of a video processor low-pass filter
$\rho_i(\lambda)$	spectral reflectivity of the target/background [for TRM4: $\rho_i(\lambda) = 1 - \epsilon_i(\lambda)$]
σ_{atm}	effective atmospheric extinction coefficient
σ_{blur}	standard deviation of the Gaussian blur aberration disc
σ_{CRT}	CRT spot size
$\sigma_{\mathrm{det_int}}(R)$	(integrating) detector noise [Equation (3.130)]
$\sigma_{\mathrm{det_noi}}$	(non-integrating) detector noise [Equation (3.137)]
σ_{gauss}	standard deviation of a Gaussian MTF

$\sigma_{\mathrm{ICCD}}(R)$	standard deviation of detector noise for ICCD/ICMOS [Equation (3.133)]
σ_{stab}	standard deviation of the (random) stabilization error
$\tau_{\mathrm{atm}}(\lambda, \mathrm{R})$	atmospheric transmission
$\tau_{\mathrm{atm_eff}}(R)$	effective atmospheric transmission
$\tau_{\mathrm{atm_nul}}$	initial transmission loss in the atmosphere
$\tau_{\mathrm{filter}}(\lambda)$	spectral transmission of additional filters
$\tau_{\mathrm{filter_max}}$	maximum transmission of additional filters
$\tau_{\mathrm{filter_norm}}(\lambda)$	normalized spectral transmission of additional filters
$\tau_{\mathrm{IR_cut}}(\lambda)$	transmission of IR cut-off filter used for responsivity measurements
$\tau_{\mathrm{opt}}(\lambda)$	spectral transmission of the front optics
$\tau_{\mathrm{opt_max}}$	maximum transmission of the front optics
$\tau_{\mathrm{opt_norm}}(\lambda)$	normalized spectral transmission of the front optics
$\phi_{\mathrm{SE\|LE}}(\lambda, r, R)$	short exposure/long exposure phase structure function of turbulence [Equations (4.83) and (4.81)]
Ψ	total system filtered noise variance [Equation (3.179)]
ω_{lin}	angular velocity of the imager (stabilization)

D.4 Symbols/Nomenclature of MTFs

A list of individual and composite MTFs is given in this section. The MTFs are listed alphabetically in their indices.

D.4.1 List of individual MTFs

$\mathrm{MTF}_{\mathrm{abr_q}}$	aberration MTF for the q parameter
$\mathrm{MTF}_{\mathrm{adc}}$	MTF of the analog-digital converter
$\mathrm{MTF}_{\mathrm{add_post}}$	additional postfilter MTF
$\mathrm{MTF}_{\mathrm{add_pre}}$	additional prefilter MTF
$\mathrm{MTF}_{\mathrm{ana_boost}}$	boost MTF
$\mathrm{MTF}_{\mathrm{ana_hp}}$	analog high-pass filter MTF
$\mathrm{MTF}_{\mathrm{ana_lp}}$	analog low-pass filter MTF
$\mathrm{MTF}_{\mathrm{bin}}$	pixel binning MTF
$\mathrm{MTF}_{\mathrm{blur}}$	Gaussian blur aberration MTF
$\mathrm{MTF}_{\mathrm{coup}}$	coupling element MTF
$\mathrm{MTF}_{\mathrm{crep}}$	column replication MTF
$\mathrm{MTF}_{\mathrm{crosstalk}}$	detector element crosstalk MTF
$\mathrm{MTF}_{\mathrm{CTE}}$	charge transfer efficiency MTF
$\mathrm{MTF}_{\mathrm{dac}}$	Digital-Analog converter MTF
$\mathrm{MTF}_{\mathrm{det_geo}}$	detector geometry MTF
$\mathrm{MTF}_{\mathrm{det_int}}$	detector integration MTF
$\mathrm{MTF}_{\mathrm{det_TDI}}$	detector TDI MTF
$\mathrm{MTF}_{\mathrm{det_temp}}$	detector temporal MTF
$\mathrm{MTF}_{\mathrm{diag}}$	diagonal micro-scan MTF
$\mathrm{MTF}_{\mathrm{diff}}$	diffraction limited optics MTF

MTF_{dig} MTF of the digital filter
MTF_{dis} display MTF

MTF_{eye} MTF of the human eye

MTF_{II} image intensifier MTF

MTF_{lrep} line replication MTF

MTF_{opt} optics MTF

MTF_{ps} phosphor screen temporal MTF

MTF_{pseudo} electronic pseudo interlace MTF

MTF_{rec} signal reconstruction MTF

MTF_{stab_lin} MTF due to linear angular movement
MTF_{stab_ran} MTF due to random movement
MTF_{stab_sin} MTF due to sinusoidal movement

MTF_{uc} uncooled detector temporal MTF

MTF_{turb} turbulence MTF (range dependent)

MTF_{vid_boost} boost MTF in the video signal processing
MTF_{vid_lp} low pass filter in the video signal processing

MTF_{zoom} electronic zoom MTF

D.4.2 List of composite MTFs

MTF_{ana} overall electronics MTF for analog signal processing
$MTF_{ana} = MTF_{ana_hp} \cdot MTF_{ana_lp} \cdot MTF_{ana_boost}$

MTF_{det_stat} static detector MTF
$MTF_{det_stat} = MTF_{det_geo} \cdot MTF_{bin} \cdot MTF_{pseudo} \cdot MTF_{CTE} \cdot MTF_{crosstalk}$

MTF_{dev} device MTF
$MTF_{dev} = MTF_{pre_device} \cdot MTF_{post_device}$

MTF_{noise_pre} horizontal noise prefilter MTF (used in the NES bandwidth calculations)
$MTF_{noise_pre} = MTF_{ana} \cdot MTF_{adc}$

MTF_{po_h} imager postfilter MTF without display (used in the NES/IES bandwidth calculations)
$MTF_{po_h} = MTF_{dig_h} \cdot MTF_{zoom_h} \cdot MTF_{dac} \cdot MTF_{vid} \cdot MTF_{crep}$

MTF_{post} postfilter MTF
$MTF_{post} = MTF_{post_device} \cdot MTF_{eye}$

MTF_{post_device} device postfilter MTF

$$\mathrm{MTF}_{\mathrm{post_device}} = \mathrm{MTF}_{\mathrm{dig}} \cdot \mathrm{MTF}_{\mathrm{rec}} \cdot \mathrm{MTF}_{\mathrm{vid}} \cdot \mathrm{MTF}_{\mathrm{dis}} \cdot \mathrm{MTF}_{\mathrm{add_post}}$$

$\mathrm{MTF}_{\mathrm{pre}}$ prefilter MTF

$$\mathrm{MTF}_{\mathrm{pre}} = \mathrm{MTF}_{\mathrm{pre_device}} \cdot \mathrm{MTF}_{\mathrm{stab}} \cdot \mathrm{MTF}_{\mathrm{turb}}$$

$\mathrm{MTF}_{\mathrm{pre_device}}$ device prefilter MTF

$$\mathrm{MTF}_{\mathrm{pre_device}} = \mathrm{MTF}_{\mathrm{opt}} \cdot \mathrm{MTF}_{\mathrm{II}} \cdot \mathrm{MTF}_{\mathrm{coup}} \cdot \mathrm{MTF}_{\mathrm{det_stat}} \cdot \mathrm{MTF}_{\mathrm{scan}} \cdot \mathrm{MTF}_{\mathrm{ana}} \cdot \mathrm{MTF}_{\mathrm{adc}} \cdot \mathrm{MTF}_{\mathrm{add_pre}}$$

$\mathrm{MTF}_{\mathrm{scan}}$ scanning effects MTF

$$\mathrm{MTF}_{\mathrm{scan}} = \mathrm{MTF}_{\mathrm{det_temp}} \cdot \mathrm{MTF}_{\mathrm{det_TDI}} \cdot \mathrm{MTF}_{\mathrm{det_int}}$$

$\mathrm{MTF}_{\mathrm{stab}}$ overall stabilization MTF

$$\mathrm{MTF}_{\mathrm{stab}} = \mathrm{MTF}_{\mathrm{stab_lin}} \cdot \mathrm{MTF}_{\mathrm{stab_ran}} \cdot \mathrm{MTF}_{\mathrm{stab_sin}}$$

$\mathrm{MTF}_{\mathrm{sys}}$ overall system MTF

$$\mathrm{MTF}_{\mathrm{sys}} = \mathrm{MTF}_{\mathrm{pre}} \cdot \mathrm{MTF}_{\mathrm{post}}$$

$\mathrm{MTF}_{\mathrm{vid}}$ overall video signal processing MTF

$$\mathrm{MTF}_{\mathrm{vid}} = \mathrm{MTF}_{\mathrm{vid_lp}} \cdot \mathrm{MTF}_{\mathrm{vid_boost}}$$

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